Transit Infrastructure, Couples' Commuting Choices, and Gender Earnings Inequality

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Abstract.
I study how transit infrastructure affects gender inequality and aggregate GDP. Using data from Lima, I document three novel facts: women are more sensitive to commute times than men; spouses' commuting choices are interdependent; and relative earnings shape how couples trade off each spouse's commute. I build a quantitative spatial equilibrium model in which couples jointly choose where to live, where each spouse works, and how consumption is allocated within the household. I show how to use commuting data to identify the parameters that determine how much weight the household places on each spouse's utility. This allows me to measure welfare separately for husbands and wives. Accounting for interdependent commuting mutes the gains from transit improvements. The opening of a metro line and a bus rapid transit system increased GDP by 0.4 percent, compared with 1.3 percent in a standard model in which workers choose their workplaces as if they were single. Finally, transit improvements raised wives' earnings in dual-earner couples but widened the gender welfare gap by increasing consumption inequality within households.

Key words: gender gap; commuting; urban economics; collective household; Lima.
JEL classification: J16; J22; R23; R41.

1. Introduction

Governments worldwide invest heavily in transit infrastructure to reduce the costs of commuting in cities. With the global urban population expected to increase by more than 2.5 billion by 2050, these investments are likely to become increasingly important as cities grow and congestion worsens, particularly in middle- and low-income countries. But commuting costs do not affect all workers equally. Women tend to face tighter commuting constraints than men as they balance work with family and household responsibilities. This paper argues that understanding the aggregate consequences of transit infrastructure therefore requires accounting for gender differences in commuting constraints and, for couples, the fact that spouses make commuting decisions jointly.

I study how transit infrastructure affects gender inequality and aggregate economic activity in Lima, Peru, where the construction of a Bus Rapid Transit system (BRT) and Line 1 of the Metro substantially changed commuting times across the city. I combine georeferenced data on commuting, households, firms, wages, and transit infrastructure to document four facts. First, women's commuting flows fall more sharply with commute time than men's, and this gender difference is especially large among workers who live with a partner (Fact 1). Second, within dual-earner couples, where one spouse works helps predict where the other spouse works (Fact 2). Third, relative earnings shape how couples trade off each spouse's commute: when the wage ratio favors one spouse, couples are more willing to accept a longer commute for that spouse (Fact 3). Finally, the new transit infrastructure reduced gender earnings inequality in the locations most exposed to the new lines (Fact 4).

These facts point to a household dimension that is missing from standard models of commuting. For couples, workplace choices are made jointly: spouses share resources and trade off both partners' earnings and commuting costs. Accounting for these interactions can therefore change both the aggregate effects of transit and how its gains are distributed within the household. I ask how the aggregate and distributional effects of transit infrastructure change once commuting decisions and resource allocation are allowed to be interdependent within couples.

To quantify these mechanisms, I develop a new model in which couples jointly choose where each spouse works and how consumption is allocated within the household. Spouses pool resources imperfectly and choose a Pareto-efficient allocation, so each spouse's workplace choice depends not only on their own wage and commute, but also on the wage and commute of their partner.[2]

The household model makes explicit how couples trade off the earnings and commuting time of both spouses. Holding household welfare fixed, an increase in one spouse's commute must be compensated by a shorter commute for the other spouse or by sufficiently higher household earnings. How the household makes this tradeoff depends on each spouse's commuting costs and on the weight the household places on each spouse's utility. If workers

[2]Throughout, I use married households and couples interchangeably to refer to cohabiting partners, whether married or unmarried.

chose workplaces independently, improved access to a higher-paying destination would induce each worker to reallocate toward that destination whenever the earnings gain outweighs the additional commuting cost. With interdependent commuting, however, spouses share the budget. The household may therefore rebalance commutes across spouses in response to transit improvements. Rather than sending both spouses toward higher-paying jobs, it may allow one spouse to accept lower earnings while compensating that spouse with a shorter commute and a larger share of household consumption. This interdependence can therefore dampen the labor market response to better transit.

The model also separates earnings from welfare within the household. A transit improvement may allow the wife to reach a higher-paying job and increase her earnings relative to her husband's. But because spouses pool their earnings before allocating consumption, this does not necessarily increase her welfare relative to his. If the household reallocates a sufficiently large share of the resulting consumption gains toward the husband, the wife's earnings can rise relative to the husband's while her welfare falls relative to his. Thus, transit can narrow the gender earnings gap while widening the gender welfare gap.

I then embed the household commuting problem into a general equilibrium model of city structure in the spirit of Ahlfeldt et al. (2015) and Tsivanidis (2021). The general equilibrium model adds three elements. First, households choose where to live in addition to where working members commute. Singles choose a residence and a workplace. Couples choose a common residence and then choose between two arrangements: a dual earner arrangement, in which both spouses work and choose workplaces, and a male-breadwinner arrangement, in which only the husband works and chooses a workplace.[3] Second, firms in each location and sector produce using labor and commercial floorspace, combined through a Cobb–Douglas technology. The labor input is itself a composite of male and female labor, which are imperfect substitutes. This imperfect substitutability allows gender wage gaps to arise and vary across workplaces and sectors. Third, each location has a fixed stock of floorspace that is allocated between residential and commercial use. In equilibrium, wages, rents, commuting flows, labor force participation, and residential choices are jointly determined.

The calibration proceeds in four blocks. I first set standard production, housing, and expenditure parameters using household data and values from the literature. I then estimate commuting elasticities and the parameters that determine how much weight the household places on each spouse's utility, which I refer to as Pareto weights. The key intuition is that commuting choices identify these weights. In particular, when a household accepts a longer commute or a lower paying destination for one spouse, it reveals how it trades off commute and earnings across spouses. I discipline these Pareto weight parameters using commuting flows for singles, breadwinner households, and dual earner couples, building on Facts 1 and 3. Next, I recover male and female wages from labor market clearing and estimate the

[3] I assume that married households are comprised of a male individual, i.e., the husband, and a female individual, i.e., the wife. A simplifying assumption usually made in the literature on family economics is that the husband always works. Moreover, in the data, female breadwinner households constitute just 1.3% of all households. This is why I turn off the husband's participation margin: it does not seem quantitatively important but would add greater complexity to the model.

pooling friction that governs how strongly spouses' commuting choices are linked. I discipline this friction using the spousal interdependence in commuting choices documented in Fact 2. Finally, I estimate the female labor force participation elasticity and the residential location elasticities using Bartik instruments. These instruments use commuting links to weigh productivity changes across the city, leveraging the commuting structure implied by the model. Given these estimates, I invert the model fundamentals to rationalize employment, labor force participation, and residential patterns. As untargeted moments, the interdependent model better reproduces bilateral commuting flows for dual-earner couples than the independent benchmark.[4] The model also qualitatively matches the reduction in the gender earnings gap documented in Fact 4, with model estimates lying within the reduced-form confidence intervals over most of the spatial range.

To quantify the effects of transit infrastructure, I remove Line 1 of the Metro and the BRT by setting commute times back to their pre-construction values. I first evaluate this counterfactual in the baseline model, in which couples choose workplaces jointly. I also decompose the results into changes that come from wages at existing jobs and from reallocation across residences and workplaces.

The counterfactual shows that Line 1 and the BRT generate modest aggregate gains but large distributional effects. In the joint model, the infrastructure raises real GDP by about 0.4 percent and increases labor force participation by 2.64 percent. The income gains are concentrated among dual-earner couples. Wives' total earnings rise by 6.07 percent and their earnings per worker by 3.35 percent, while husbands' total earnings rise by 2.63 percent. As a result, the gender earnings gap narrows among dual-earner couples. Among singles, men's income rises slightly while women's income falls, so the gender earnings gap widens.

The welfare results show that higher earnings for wives do not imply their welfare increase relative to their husbands. Despite that the infrastructure improvements narrow the married earnings gap, realized welfare, measured as consumption net of commuting costs, rises more for husbands than for wives. This widens the welfare gap by about 22 percent. The reason is that, while wives use the new transit access to reach better-paying jobs, the household allocation shifts a larger share of household income toward husbands' consumption.

Comparing the interdependent model with the independent benchmark shows that household interdependence changes both the aggregate and distributional effects. The independent model predicts a much larger output gain: real GDP rises by 1.3 percent, compared with 0.4 percent under interdependence. Set against the cost of Line 1 and the BRT, these GDP gains cover the cost of capital up to an interest rate of about 66 percent under independence. Under interdependence, these GDP gains cover only about 20 percent. The reason is that workers are more mobile across workplaces when their choices are not tied to a spouse's commute. Therefore the aggregate labor supply elasticity is larger under independence.

Household interdependence also changes conclusions about inequality within the household. An independent model predicts relatively even income and welfare gains for both

[4] Although the calibration uses summary moments constructed from commuting flows, it does not target the bilateral commuting flows themselves.

spouses in dual-earner couples. But under interdependence, earnings and welfare diverge.[5] While the new transit lines allow married women to earn more, the shift in the household's commuting arrangement directs the realized consumption gains toward their husbands. Because welfare ultimately depends on who captures those consumption gains, the welfare gap between spouses actually widens. Thus, ignoring household interdependence not only overstates the aggregate output response, but misses who benefits more.

Line 1 and the BRT changed access unevenly across Lima, with the largest gains concentrated outside the center, where initially isolated locations gained better access to higher-paying jobs. Much of the earnings response comes from workers changing where they work rather than from higher wages at their existing jobs. Dual-earner wives, in particular, use the improved transit access to reach better-paying jobs, even when doing so requires longer commutes. Because workplace choices are joint within couples, when wives change their workplace destinations, husbands' workplace choices also adjust. As a result, husbands experience smaller earnings gains than they would if they chose their workplaces independently. The household partly compensates them by allocating a larger share of consumption to them. This helps explain why the earnings gap narrows while realized welfare moves in the opposite direction

This paper first contributes to the literature on economic geography and urban structure. Quantitative urban models have shown that transit infrastructure can have large effects on commuting, housing, employment, informality, crime, and welfare, including studies of BRT systems (Tsivanidis, 2021), metro systems (Severen, 2021; Zárate, 2021), steam railways (Heblich et al., 2020), and cable cars (Khanna et al., 2025).[6] This literature typically models workers as independent decision makers or abstracts from the fact that households often have more than one potential earner. I add this household dimension to a quantitative urban model of transit infrastructure. The paper shows that transit improvements affects access to jobs and the household's joint allocation of consumption and commuting. As a result, household interdependence changes both the aggregate gains from transit and the distribution of those gains across spouses and locations.

Second, this paper contributes to the literature on gender gaps, commuting costs, and job access. A growing body of work shows that women face tighter commuting constraints than men. These constraints help explain gender gaps in labor market outcomes. Le Barbanchon et al. (2021) show that unemployed women in France have a lower maximum acceptable commute than unemployed men, and estimate that gender differences in commute valuation account for 14 percent of the residualized gender wage gap. Liu and Su (2024) show that commuting and wage gaps are smaller near city centers in the United States, especially in occupations where high wage jobs are geographically concentrated. Black et al. (2014) document

[5]To compute individual welfare in the independent benchmark, one must assume a sharing rule for household consumption. I fix this allocation using the average consumption split among dual-earner couples. In the interdependent model, by contrast, the way couples split consumption is fully endogenous and estimated from commuting data, as I explain later.

[6]See also Ahlfeldt et al. (2015), Allen et al. (2015), Monte et al. (2018), Owens III et al. (2020), and Brinkman and Lin (2022). See Redding and Rossi-Hansberg (2017) for a review of spatial quantitative models.

that differences in commuting times help explain variation in married women's labor supply across U.S. cities. Evidence from South Korea and Pakistan further shows that improvements in mobility and connectivity can improve women's labor market outcomes (Kwon, 2022; Field and Vyborny, 2022). However, better connectivity does not always reduce gender inequality. Bütikofer et al. (2022) show that the opening of the Oresund Bridge increased commuting, wages, and gender inequality among Swedes living in Malmo, with especially large effects within couples.[7] My paper helps reconcile these findings. Better transit reduces gender gaps only when it connects women to better opportunities. Moreover, even when women get access to better jobs, it does not necessarily mean that their welfare will increase relative to their husbands.

Third, this paper contributes to household economics by showing that commuting choices provide information about how households value each spouse's welfare. Collective models of the household emphasize that couples make decisions as if they maximize a weighted objective, where the weights determine how much the household values each spouse's utility (Chiappori, 1988; Browning et al., 1994, 2014; Chiappori, 1992; Blundell et al., 2007; Chiappori et al., 2022). These weights are central for welfare analysis. However, they are difficult to discipline because many household choices are observed only at the household level and not at the spouses level. I show that joint commuting choices provide a new source of information. When a household chooses the workplace of each spouse, it reveals how the household trades off income and commuting across spouses. This allows me to estimate the household weights that govern the allocation of resources within households. Once we know how much each spouse consumes, then we can measure how transit improvements affects the welfare of each spouse.

2. Context

Lima is an ideal setting to perform this study for several reasons. First, as the political and economic hub of Peru, it comprises 30% of the population and 40% of the country's GDP. For many industries, Lima accounts for more than 50% of the sectoral GDP, such as in manufacturing, commerce, construction, and services. These industries often hire men and women with different intensities, which provides with valuable variation in labor opportunities across different locations within the city. Second, Lima's significant traffic congestion is worth noting, with buses serving as the primary mode of transportation for 69% of work trips in 2010 (before the new transit infrastructure were introduced), followed by taxis, 'colectivos', or motorbike taxis (11%), and personal cars (9%). Third, commuting patterns also differ by gender and civil status. According to time use surveys, men account for over 60% of total commute time. Instead, among married workers, men account for more than 70% of total commute time. This difference suggests that husbands specialize in commuting more often than wives. Finally, two transportation infrastructures were implemented in recent years, generating variation in transportation access across different locations within the city. These were the Line 1

[7] Kawabata and Abe (2018), Carta and De Philippis (2018), Gu et al. (2021), and Moreno-Maldonado (2023) provide related evidence on commuting, household location, and female labor supply.

of the Metro and the Bus Rapid Transit System (BRT).

The Line 1 of the Metro is a 34.4 km long transportation system that serves over 500,000 daily passengers. Construction started by the end of the 80s during the first presidency of Alan Garcia. However, construction stopped short after Alberto Fujimori became president in 1990. Almost 20 years later, construction resumed during the second presidency of Alan Garcia. The first 22 km were completed in 2011, with the remaining 12.4 km completed in 2014. Although the Metro does not directly cross through the Central Business District, it runs through high demand areas, including one of the most significant commercial centers in the city, the "Gamarra" market (see Figure 10). The Metro has significantly improved commuting times, reducing travel times from 2 hours and 45 minutes to just one hour for end-to-end trips. On average, work trips on the Metro take around 24 minutes (JICA, 2013).

The introduction of the BRT system in Lima occurred around 2010. Typically, a BRT system consists of a dedicated system of roadways for buses and has been successfully implemented in cities such as Bogota and Buenos Aires. The goal of the BRT system is to provide faster speeds while maintaining the simplicity of a bus system. As shown in Figure 10, Lima's BRT system connects the fringe to the modern municipalities, as well as the Central Business District (CBD). The system spans 26 km and serves approximately 700,000 passengers daily. A World Bank study found that commuting times were reduced by approximately 25-45% as a result (Oviedo et al., 2019).

3. Data and Facts

3.1. Data

The analysis requires data that identify where households live, where jobs are located, and how costly it is to commute between locations. I conduct the analysis at the zone level throughout, unless specified otherwise. In 2017, Lima was partitioned into more than 1,500 zones. This level preserves fine spatial variation while keeping the structural model computationally manageable and reducing concerns about excessive granularity.

I use several data sources to construct the residential and workplace sides of the model. On the residential side, I use the Population Censuses of 1993, 2007, and 2017 to obtain residence counts at the zone level for single men, single women, and cohabiting couples. To construct employment by block and industry, I use the 2008 Economic Census plus an administrative data set on firms' location and employment freely available for the year 2015. These data allow me to measure where households live and where employment is located before and after the expansion of Lima's transit network.

I then combine these data with information on commuting flows and travel times. The 2017 Population Census reports commuting flows at the municipality level.[8] I complement these flows with several commuting surveys collected in Lima between 2010 and 2018. To measure travel times between locations, I use data on the road network from OpenStreetMap and the Google Maps API. I complement these with the 2010 National Time Use Survey.

[8]There are 50 municipalities in Lima. Each municipality contains about 30 zones.

Finally, I use additional data sources to discipline workplace wages, rents, land use, and other calibration targets. Workplace mean wages come from the National Survey of Enterprises (ENE), waves 2015–2017. The survey records each firm's location at the district level, along with average gender specific pay across five occupational categories. I define $w_{g,d}$ as the gender specific average wage across firms located in workplace municipality d. I use the National Household Survey (ENAHO) for the 2007–2017 waves, which is precisely georeferenced, to analyze rents at the zone level. Data on land use come from the Metropolitan Institute of Planning of Lima.

Using the 2017 Population Census, I classify households into five groups: (i) *single females* and (ii) *single males*, who account for 31.9% and 35.9% of households in 2017; (iii) *male-breadwinner* households, where the wife stays at home while the husband works, which account for 13.3%; (iv) *female-breadwinner* households, which account for 1.3%; and (v) *dual earner* households, where both spouses work, which account for 17.6%.[9]

The Data Appendix provides additional details on variable construction, sample restrictions, and the mapping between the different geographic units used across datasets.

3.2. Stylized facts

I begin by documenting four stylized facts that motivate the model. First, women's commuting flows are more sensitive to commute times than men's. This gender gap widens among married workers. Second, in dual earner households, where one spouse works helps predict where the other spouse works, which suggests commuting choices are interdependent. Third, within dual earner households, relative earnings shape how strongly couples penalize each spouse's commute time. Fourth, the new transit infrastructure narrowed the male–female earnings gap in the more exposed locations. Together, these facts motivate a theory in which couples choose workplace locations jointly.

Fact 1: The commuting semi-elasticity differs by civil status and gender.

This fact establishes that women are more sensitive to commute times than men. This gap widens when women live with a partner. I document it by estimating the semi-elasticity of commuting probabilities to commute times in a standard gravity equation separately for the household groups I defined earlier (male and female singles, breadwinners, and dual earners).

Let $\pi^k_{d|o}$ denote the share of group-k workers who live in origin o and commute to destination d. Let t_{od} denote the commute time from o to d. For each group k I estimate

$$\log \pi^k_{d|o} = \gamma_k \, t_{od} + \alpha^k_o + \delta^k_d + \varepsilon^k_{od}, \tag{1}$$

where α^k_o and δ^k_d are origin and destination fixed effects. The slope γ_k is the *commuting semi-elasticity*. It measures how strongly group k avoids long commutes. A one-minute increase in t_{od} reduces the probability that a group-k worker living in o commutes to d by approximately $100\,|\gamma_k|$ percent. Thus, a more sensitive group has a more negative γ_k.

[9]The precise definition is in the Data Appendix of the online version of the paper.

I aggregate the data at the municipality level.[10] Commute times come from OpenStreetMap. I assign speeds matching documented average speeds in Lima and aggregate to the municipality level by taking the median across zones within each pair.[11]

Table 1 reports the PPML estimates. Reading across the three household-type panels, the female-to-male gap in $|\gamma_k|$ widens as women move from singlehood to marriage: 15% among singles (columns 1–2), 36% in breadwinner households (columns 3–4), and 40% in dual earner households (columns 5–6). Each pairwise difference is statistically significant at the 1% level. Figure 1 confirms the same picture non-parametrically. It plots the binscatter between residual log commuting probabilities and residual commute times after partialling out origin and destination fixed effects. Panel A does it for singles, while Panel B for dual-earner households. The relationship is log linear in both panels and steeper for women, with the gap widest among the married.

This pattern is consistent with women bearing larger non-monetary commuting costs once they live with a partner, e.g., child care responsibilities, household-production time, or differential safety concerns at long commute times. In this paper, I am agnostic about the specific mechanism behind this gap. What matters is that the gap is large enough to require gender specific commuting elasticities.[12]

Table 1: Commuting semi-elasticities by civil status and gender

	Single HH		Breadwinner HH		Dual-Earner HH	
	Males	Females	Males	Females	Males	Females
	(1)	(2)	(3)	(4)	(5)	(6)
Travel Time	-0.0565	-0.0651	-0.0466	-0.0634	-0.0480	-0.0673
	(0.0039)***	(0.0041)***	(0.0033)***	(0.0041)***	(0.0032)***	(0.0041)***
Origin FE	X	X	X	X	X	X
Destination FE	X	X	X	X	X	X
Gap (Female/Male-1)		15%		36%		40%
N	2500	2500	2500	2500	2500	2500

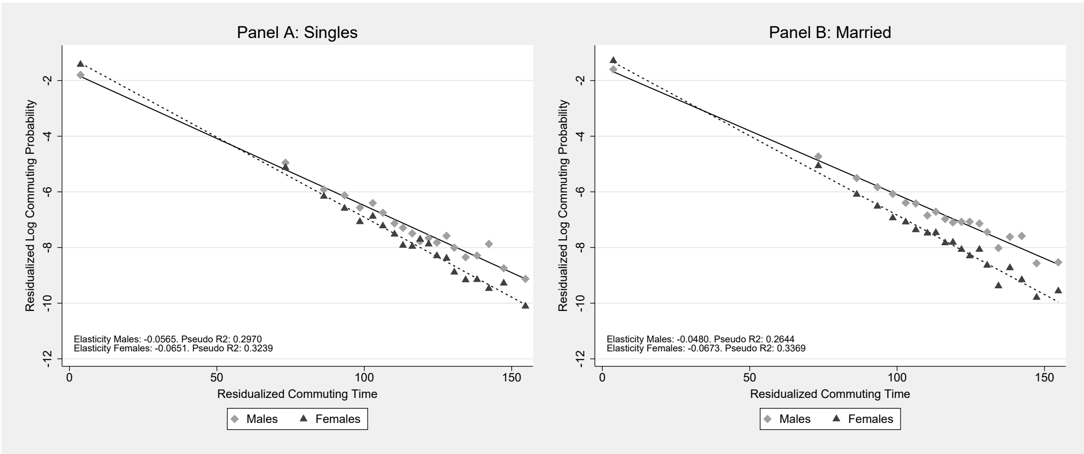
Notes: Observations are at the municipality-by-municipality level. The independent variable in all regressions is the median travel time t_{od} between origin o and destination d, with origin and destination fixed effects included. Column (1) uses the share of single male workers who reside in o and work in d as the dependent variable; column (2) the corresponding share for single female workers. Columns (3) and (4) use male-breadwinner and female-breadwinner households. Columns (5) and (6) use males and females in dual-earner households. The differences between columns (1)–(2), (3)–(4), and (5)–(6) are statistically significant at the 1 percent level.

[10] Origins are observed at the zone level, while destinations are only recorded at the municipality level.

[11] The Data Appendix in the online version shows that these computed times are well correlated with self-reported survey times available in and the 2010–2018 Lima Commuting Surveys.

[12] I do not interpret the married–single gap as the causal effect of marriage or attempt to model selection into marriage. I take the observed married and single populations as the relevant target groups and summarize their differences through group specific commuting elasticities. The counterfactual exercises then quantify the effects of new infrastructure conditional on these observed sensitivities.

Figure 1: Gravity in commuting flows by household type



Notes: Binscatter of residual log conditional commuting probabilities log $\hat{\pi}^k_{d|o}$ against residual commute times t_{od}, after partialling out origin and destination fixed effects from equation (1). Panel A shows single men and single women separately. Panel B shows men and women in dual-earner households separately. Fitted lines are OLS fits through the binned points. The elasticities reported in each panel are the corresponding PPML estimates of γ_k, as in Table 1.

In Appendix A, I rerun equation (1) with origin-by-destination controls (average education, age, and other covariates of route users). Women remain more sensitive to commute time than men after adding these controls. I also study how the elasticity changes with the presence of young children. Having young children at home makes women even more sensitive to commute times relative to men.

Relation to literature. These estimates speak to two literatures that use commuting data to discipline the cost of travel. First, in quantitative spatial models, authors routinely use commuting flows and commute times to discipline the tradeoff between workplace access and commuting costs. The richness of my Census data lets me characterize this tradeoff across gender and civil status for the first time in the gravity literature. My estimates are in the range reported elsewhere.[13] Second, a related literature on compensating differentials and commute valuation regresses wages on commute times, often exploiting quasi-experimental variation, to recover the wage premium a worker require to accept a longer commute.[14] This approach estimates an equilibrium gradient between wages and commute times. This gradient is informative about the monetary value of commuting time, but it also depends on the geography of wages and job attributes. If high wage jobs are concentrated in a few places, a reduction in commute times can give workers access to much better jobs. Wages then respond more strongly, and the wage–commute time gradient will be steeper. If high wage jobs are more dispersed, the same reduction in commute times changes the set of reachable jobs much less. Wages therefore respond less, and the estimated wage–commute time gradient will be flatter. In contrast, my approach estimates a primitive governing how workers trade off work-

[13]Point estimates of the semi-elasticity between commute times and commute flows include $[-0.0706, -0.0697]$ in Ahlfeldt et al. (2015), $[-0.042, -0.028]$ in Zárate (2021), $[-0.038, -0.029]$ in Warnes (2021), $[-0.046, -0.036]$ in Tsivanidis (2021), and $[-0.068, -0.039]$ in Khanna et al. (2025).

[14]Manning (2003) brought the gender dimension to this valuation problem. More recently, Le Barbanchon et al. (2021) use the joint distribution of reservation job attributes and realized job bundles to discipline this trade-off across genders.

place access and commuting costs, ceteris paribus, rather than an equilibrium relationship as in the literature on compensating differentials.

Fact 2: Where one spouse works depends on where the other spouse works.

This fact establishes that, in dual earner households, knowing where one spouse works carries information about where the other spouse works *beyond* what each spouse's own residence and own labor market access can explain. Therefore, this means the two commuting decisions are interlinked.

I use the complete household roster in dual earner households to test for interdependence. Let $\pi^{m|f}_{i|o,j}$ denote the probability that the husband works in i given that the household lives in o and the wife works in j. Let $\pi^{f|m}_{j|o,i}$ denote the analogous probability for the wife. I compute these conditional probabilities at the municipality level. Lima has 50 municipalities, so each conditional object has $50 \times 50 \times 50$ cells. For the husband, I estimate

$$\log \pi^{m|f}_{i|o,j} = \gamma\, t_{oi} + \alpha^{m|f}_{oj} + \delta^{m|f}_{ij} + \varepsilon^{(1)}_{oij}, \tag{2}$$

where t_{oi} is the husband's commute time, $\alpha^{m|f}_{oj}$ is an origin-by-wife-destination fixed effect, and $\delta^{m|f}_{ij}$ is a destination-pair fixed effect. The fixed effect $\alpha^{m|f}_{oj}$ absorbs all factors that affect the husband's labor market access from residence o, given that the wife works in j. The pair fixed effect $\delta^{m|f}_{ij}$ captures whether the pair (i, j) is especially likely or unlikely, beyond what can be explained by the husband's own commute time and the household's residence-by-wife-destination pair. The wife's regression mirrors this structure with i and j swapped.

The destination-pair fixed effect $\delta^{m|f}_{ij}$ is the object of interest. Under independent commuting within couples, the joint pair (i, j) should not predict the husband's commuting probability once I control separately for the husband's destination and the wife's destination. To formalize this, I run a second stage regression on the recovered pair fixed effects:

$$\log \pi^{m|f}_{i|j} = \psi_m\, \widehat{\delta}^{m|f}_{ij} + \mathrm{FE}_i + \mathrm{FE}_j + \varepsilon^{(2)}_{ij}, \tag{3}$$

where $\pi^{m|f}_{i|j}$ averages the husband's conditional probability over residences, and $\mathrm{FE}_i, \mathrm{FE}_j$ are husband- and wife-destination fixed effects.[15,16] Under independence, $\psi_m = 0$ by construction. A larger ψ_m indicates stronger interdependence in commuting choices: knowing where the wife works reveals more information about the husband's commuting choice, beyond what is captured by separate husband- and wife-destination dummies. The wife's analog ψ_f is constructed symmetrically.

Figure 2 plots residualized log conditional commuting probabilities against residualized destination-pair fixed effects. Panel A shows the results for husbands, while Panel B shows the results for wives. Both relationships are tight and strongly positive: $\widehat{\psi}_m = 0.87$ for men

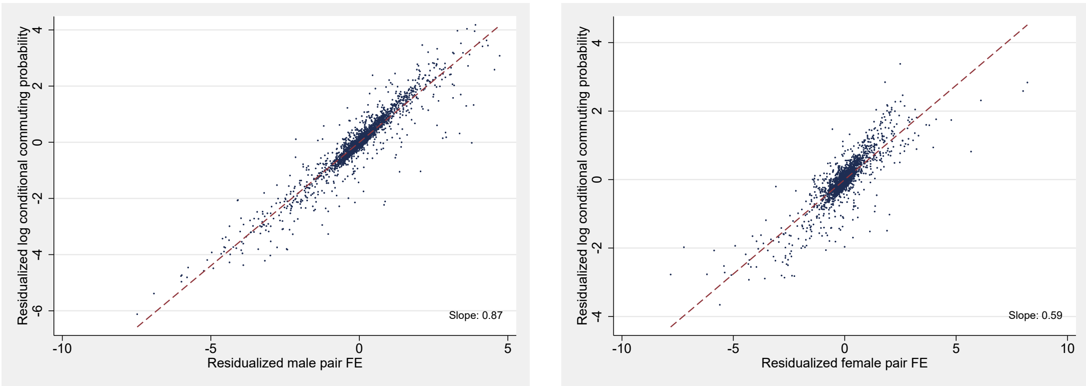
[15] I also control for t_{ij}, the commute time between the two workplaces, since couples may choose destinations partly based on the distance between their jobs.

[16] For simplicity, $\pi^{m|f}_{i|j}$ averages $\pi^{m|f}_{i|o,j}$ over residences, so it is the average probability that the husband works in i given that the wife works in j.

and $\widehat{\psi}_f = 0.59$ for women. This means that knowing where one spouse works is highly informative about where the other works.

Figure 2: Interdependence in commuting choices within dual earner households

Panel A: Husband Panel B: Wife



Notes: For each gender I estimate equation (2) on dual earner households and recover the destination-pair fixed effect $\widehat{\delta}_{ij}^{m|f}$ (husbands) or $\widehat{\delta}_{ij}^{f|m}$ (wives). I also control for t_{ij}, the commute time between the two workplaces. I then estimate the second stage regression (3). The figure plots the residualized log conditional commuting probabilities against the residualized destination-pair fixed effects. Panel A shows husbands; Panel B shows wives.

Taken together, the estimates $(\widehat{\psi}_m, \widehat{\psi}_f)$ show that commuting choices are interlinked in dual earner households. This motivates a model in which the household chooses workplace locations jointly. Later, I use these estimates to discipline the degree of interdependence in the structural model.

It is worth discussing the potential mechanisms behind this interdependence. Keep in mind that the workplace of workers is an important determinant of their income. The first mechanism is resource sharing. When couples share resources, one spouse's destination affects household income and therefore changes the value of the other spouse's workplace options. For example, a high paying job for one spouse can make the household more willing to accept a lower paying job with a shorter commute for the other spouse. Second, different destination pairs can change how income is distributed within the household. A destination pair that gives one spouse much higher earnings may also change that spouse's bargaining position inside the household. As a result, the household may place different weight on each spouse's commute across different workplace pairs. Fact 3 studies this second channel by asking whether commuting sensitivity varies with within household wage inequality.

<u>Relation to literature.</u> To the best of my knowledge, the result documented above is new to the literature. Prior work has tested for spousal interdependence in employment outcomes, such as labor force participation or hours worked, but not in workplace destination choice conditional on residence.[17] My approach differs in two ways. First, rather than studying how one spouse's commute affects the other spouse's employment probability or hours worked, I study how one spouse's workplace is related to the other spouse's workplace. The focus is

[17]The closest paper is Carta and De Philippis (2018), who show that increases in the husband's commute time, induced by office or plant relocations, reduce his wife's employment probability, with stronger effects for couples with children and for highly educated husbands.

therefore not only whether the spouse works, but where the spouse works. Second, I use conditional commuting probabilities to ask whether one spouse's workplace reveals information about the other's workplace. This requires observing enough dual earner households to construct conditional origin–destination probabilities by spouse. Standard commuting surveys are usually too small for this exercise. Instead, the Census data provide a complete household roster and enough observations to recover these conditional commuting probabilities.

Fact 3: The commuting semi-elasticity in dual earner households varies with within household wage inequality.

This fact shows that, in dual earner households, each spouse's sensitivity to commute time varies with relative earnings between the husband and wife. When the husband's workplace pays more than the wife's, couples are more willing to accept a longer commute for him and less willing to accept a longer commute for her. The opposite pattern appears when the wife's workplace pays more than the husband's. Thus, relative earnings shape how strongly couples respond to each spouse's commute time.

I document this pattern by extending equation (2) from Fact 2 to allow commuting slopes to vary with the wage ratio between the husband's and wife's workplaces. Let $Z_{ij} = w_{m,i}/w_{f,j}$ denote the husband's mean wage at destination i relative to the wife's mean wage at destination j. Throughout, a indexes the coefficient on the husband's commute t_{oi} and b indexes the coefficient on the wife's commute t_{oj}, with the superscript naming the conditional equation. For the husband, I estimate

$$\log \pi_{i|o,j}^{m|f} = a_t^{m|f}\, t_{oi} + a_{Zt}^{m|f}\, t_{oi} \log Z_{ij} + b_{Zt}^{m|f}\, t_{oj} \log Z_{ij} + \alpha_{oj}^{m|f} + \delta_{ij}^{m|f} + \varepsilon_{oij}^{m|f}, \tag{4}$$

where $\alpha_{oj}^{m|f}$ is the origin-by-wife-destination fixed effect and $\delta_{ij}^{m|f}$ is the destination-pair fixed effect. The wife's regression mirrors this structure with i and j swapped.

I report three specifications for each equation. Column 1 includes origin fixed effects and destination-pair fixed effects, but not origin-by-spouse-destination fixed effects. This allows me to include the spouse's commute time as a separate control, rather than absorbing it with the origin-by-spouse-destination fixed effect. Column 2 includes the origin-by-spouse-destination fixed effect but omits the destination-pair fixed effect. This lets me control for the distance between workplaces, t_{ij}, and the wage ratio, $\log Z_{ij}$, both of which vary only at the workplace-pair level. Column 3 includes both origin-by-spouse-destination fixed effects, and destination-pair fixed effects. Column 3 is my preferred specification because it is the most conservative regression, and can be mapped directly to equations in the model. Columns 4–6 repeat the same sequence for the wife's conditional equation, with i and j swapped.

Columns 1 and 4 show that the husband's commuting probability depends on both his own commute and his wife's commute, and symmetrically for the wife. The own-commute coefficient is negative and tightly estimated for both spouses: -0.0316 in the husband equation and -0.0517 in the wife's equation. This echoes the gravity pattern from Fact 1. The coefficient on the wife's commute in the husband equation is positive, $+0.0081$: holding the husband's commute time fixed, across destination pairs, a husband destination is more likely when the wife has a longer commute. Symmetrically, the coefficient on the husband's commute in the wife equation is also positive, $+0.0132$: holding the wife's commute time fixed, across destinations pairs, a wife destination is more likely when the husband has a longer commute. Thus, couples appear more likely to choose workplace pairs where one spouse takes the longer commute and the other has a shorter one.

The interactions with $\log Z$ show whether the commute time coefficients vary with the workplace wage ratio. In the husband equation, the interaction between his own commute and $\log Z$ is positive. Thus, when the husband's workplace pays more relative to the wife's workplace, the penalty on his commute becomes weaker. In the wife equation, the interaction between her own commute and $\log Z$ is negative. Thus, the same increase in the wage ratio makes the penalty on her commute stronger. In other words, as the wage ratio tilts toward the husband, couples become more willing to accept a longer commute for him and less willing to accept a longer commute for her. The cross-commute time interactions with $\log Z$ give the same message. In the husband equation, the cross-commute interaction is negative: when the husband's workplace pays more relative to the wife's, the husband's commuting probability becomes more sensitive to the wife's commute time. That is, as $\log Z$ increases, the household becomes less willing to choose husband destinations that come with a longer commute for the wife. In the wife equation, the cross-commute interaction is positive: when the husband's workplace pays more relative to the wife's, the wife's commuting probability becomes less sensitive to the husband's commute time. That is, as $\log Z$ increases, wife destinations become more likely when they are paired with a longer husband commute. Together, these interactions imply that, as relative pay shifts toward the husband's workplace, the longer commute is more likely to be attached to him rather than to his wife, and vice-versa.

Columns 2 and 5 drop the destination-pair fixed effect, which lets me include the cross-workplace commute time t_{ij} and the additive log wage ratio $\log Z_{ij}$. The coefficient on t_{ij} is negative in both equations: -0.0144 for the husband and -0.0126 for the wife. Holding both home-to-work commutes fixed, couples are less likely to choose workplace pairs that are far from each other. The log wage ratio enters with opposite signs across equations: $+0.5395$ in the husband equation and -0.3787 in the wife's equation. This means that husbands are more likely to work in destination pairs where the wage ratio favors the husband, while wives are more likely to work in pairs where the wage ratio favors the wife. Importantly, the interaction terms keep the same sign as in columns 1 and 4.

Table 2: Dual-earner commuting semi-elasticities and wage inequality within households

	Husband eq. ($\pi^{m\|f}_{i\|o,j'}$)			Wife eq. ($\pi^{f\|m}_{j'\|o,i}$)		
	(1)	(2)	(3)	(4)	(5)	(6)
Own commute t	-0.0316	-0.0280	-0.0334	-0.0517	-0.0437	-0.0548
	(0.0015)***	(0.0005)***	(0.0007)***	(0.0024)***	(0.0006)***	(0.0008)***
Spouse commute t'	0.0081	–	–	0.0132	–	–
	(0.0013)***			(0.0016)***		
Cross-workplace $t_{ij'}$	–	-0.0144	–	–	-0.0126	–
		(0.0006)***			(0.0006)***	
$\log Z$	–	0.5395	–	–	-0.3787	–
		(0.0622)***			(0.0507)***	
Own $t \times \log Z$	0.0018	-0.0000	0.0013	-0.0058	-0.0007	-0.0042
	(0.0015)	(0.0007)	(0.0010)	(0.0027)**	(0.0007)	(0.0013)***
Spouse $t' \times \log Z$	-0.0038	-0.0018	-0.0033	0.0050	0.0018	0.0024
	(0.0010)***	(0.0007)***	(0.0019)*	(0.0018)***	(0.0004)***	(0.0020)
Origin × spouse-dest FE		X	X		X	X
Spouse-pair FE	X		X	X		X
Origin FE only	X			X		
N	104944	91351	91351	104944	98826	98826

Notes: PPML estimates of bilateral commuting probabilities for dual-earner households. $Z_{jj'} = w_{m,j}/w_{f,j'}$ is the spousal wage ratio across destinations. Cols (1)/(4) include origin and spouse-pair fixed effects but not the origin-by-spouse-destination fixed effect, which lets the spouse's commute time enter as a separate control. Cols (2)/(5) include the origin-by-spouse-destination fixed effect but drop the spouse-pair fixed effect, so the cross-workplace travel time $t_{ij'}$ and the additive log Z become identifiable. Cols (3)/(6) include both fixed effects; these are the preferred specifications, since they correspond to the empirical equation that receives a structural interpretation in the model. Standard errors in parentheses, clustered by the absorbed FE dimensions. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Columns 3 and 6 add the destination-pair fixed effect on top of column 2, which absorbs t_{ij} and the additive log wage ratio and leaves only the own commute and the two interactions identified. These are my preferred specifications since they have a structural interpretation, as we will see later. The four interaction coefficients keep the same sign pattern as in earlier columns: $a^{m|f}_{Zt} = +0.0013$ and $b^{m|f}_{Zt} = -0.0033$ for the husband, $b^{f|m}_{Zt} = -0.0042$ and $a^{f|m}_{Zt} = +0.0024$ for the wife.

One way to summarize these patterns is that the household behaves as if it aggregates the two spouses' commuting costs with weights that depend on the wage ratio. If a spouse receives a higher weight, it means that a spouse's commute time accounts for a larger share of the household's overall commuting burden, so the household is more sensitive to an extra minute of commuting for that spouse. The model in Section 5.1 shows how this weight interpretation emerges naturally in a collective household model with commuting choices. I use the two own-commute coefficients plus the four interaction coefficients in columns 3 and 6 to discipline this collective household margin in the structural model.

Relation to literature. To my knowledge, the result documented above is new to the literature. The collective and bargaining literature studies how relative earnings shape allocations within the household, including consumption, leisure, and time use, but usually takes workplace locations as given (Chiappori, 1992; Chiappori et al., 2022). Quantitative spatial models, in contrast, study workplace choices and commuting flows, but typically treat the commuting semi-elasticity as a worker specific parameter that does not vary with the spouse's commute time or with the relative pay inside the household (Ahlfeldt et al., 2015; Tsivanidis, 2021). My approach connects these two ideas. I show that, in dual earner households, the commute sensitivity attached to each spouse varies with the wage ratio between their workplaces.

Fact 4: New transit infrastructure reduced the gender earnings gap in the more exposed locations

I study the reduced form effect of Line 1 of the Metro and the BRT on the male–female earnings gap. To address the concern that stations were not placed at random, I define treatment locations as those within a buffer of any Line 1 or BRT station, and control locations as those within a buffer of any Line 2 or Line 4 station—planned at the same time as Line 1 and the BRT but built only later.

Improvements in access from the new transit infrastructure were heterogeneous across the city. For example, improvements in access to central destinations were concentrated in fringe locations, which were initially the most isolated, and were smaller in locations already close to the center. I therefore compare treated locations to control locations separately by their position in the city. Using the 2007–2017 georeferenced waves of the National Household Survey, I let the effect of infrastructure vary with distance to the CBD using a rolling-window specification. For each distance c from the CBD, I keep observations in locations whose distance to the CBD lies within h kilometers of c. In the baseline figure, I set $h = 3.0$ kilometers. Within each window, I estimate a difference-in-differences specification for labor income that interacts treatment with an indicator for women, controlling for individual characteristics, zone fixed effects, and year fixed effects. For the BRT, the relative earnings response of women was small in central locations. Beyond a distance to the CBD of roughly 3 km, however, women's labor income increased more than men's labor income, narrowing the male–female earnings gap. This pattern, while still similar, is less robust for the Metro. Details are available in Appendix B.[18]

I later use these reduced form estimates as untargeted moments. I test whether the calibrated model can match earnings responses across space and gender.

Relation to literature. A growing literature documents that improvements in mobility raise women's labor market and human capital outcomes more than men's: Martinez et al. (2020) on public transit in Lima, Kwon (2022) on transport infrastructure in South Korea, Field and Vyborny (2022) on subsidized rides in Pakistan, Abou Daher et al. (2025) on the lifting of the Saudi female driving ban, Fiala et al. (2025) on bicycle access for girls in Zambia, and Alba (2024) on transit and access to college in Lima. Relative to this literature, I study directly the response on wages (among other outcomes).

[18]Appendix B reports robustness checks that vary the rolling-window bandwidth and the distances used to define treated and control locations. It also complements the household survey evidence with reduced form results from the 1993, 2007, and 2017 Population Censuses on changes in population and household composition near the new transit infrastructure.

I now turn to the household decision problem that rationalizes the facts I have documented in this section.

4. A New Theory of Interdependent Commuting

The facts in Section 3 showed that commuting choices within couples cannot be represented as two independent workplace choices. They point instead to a household decision problem in which each spouse's commute is evaluated jointly with the other spouse's commute and the earnings attached to both workplaces. This section builds that decision problem and then studies how it changes the predicted effects of commuting-cost reductions.

In Section 4.1, I develop a partial equilibrium model of commuting with married households that delivers interdependent commuting choices. Households are modeled as collective, with spouses pooling resources imperfectly and choosing Pareto efficient allocations. Relative to the standard commuting model, in which each spouse trades off wages against their own commuting time, a model with interdependent commuting implies that these tradeoffs are shaped by several mediating forces within the household. First, they depend on how effectively spouses can pool their income. Second, they depend on the relative weight the household assigns to each spouse's utility. Third, they depend on both the level and the relative strength of each spouse's response to commuting times. Together, these forces determine how wages and commuting times are traded off not only within each spouse, but also across spouses. A key implication is that these channels create a dampening force. When one spouse benefits from better access to high wage destinations, the other spouse shares part of that gain through household income and may therefore have less incentive to reallocate as strongly.

In Section 4.2, I study comparative statics in a simplified geography under both independent and interdependent commuting. The main insight is that the gains from lower commuting costs depend not only on how much transit improvements relax commuting frictions, but also on where those gains are located relative to the geography of wages. Under independent commuting, this is summarized by the covariance between wages and route specific transit gains. Under interdependent commuting, an additional composition channel appears, because improvements in one spouse's commute can also affect the other spouse's commuting choices. The effect of this additional composition channel depends on whether the induced reallocations across destinations covary differently with husbands' and wives' wages.

Finally, in Section 5, I embed this choice structure into a spatial model of city structure to endogenize labor demand and account for the general equilibrium.

4.1. Microfoundation of interdependent commuting with collective households

Environment. The city consists of N married households distributed in S locations. Locations are indexed by o, i, j. Households are endowed with two spouses, a husband and a wife, each with their private utility.

Following the literature on family economics (Browning et al., 2011), I assume that households pool their resources and choose Pareto efficient allocations. In other words, each spouse maximizes their utility subject to delivering a given utility level to the other spouse. As is standard in this literature, I do not model the bargaining process that supports this allocation. Instead, I take Pareto efficiency as a reduced form description of household decision-making.

However, I assume that resource pooling is imperfect. One can think of this friction as arising from a coordination problem or a technological constraint that prevents households from fully sharing resources. For example, spouses may be willing to share resources for joint household needs, but less willing to finance the other spouse's purely private consumption.

Preferences. Conditional on where they live and work, spouses need to choose private consumption and residential floorspace, which is a public good within the household. Individual utility is

$$V^m_{oi} = \log\left(\frac{C^{\beta^m}_{m,oij}(H^R_{oij})^{1-\beta^m}}{d^m_{oi}}\right), \qquad V^f_{oj} = \log\left(\frac{C^{\beta^f}_{f,oij}(H^R_{oij})^{1-\beta^f}}{d^f_{oj}}\right), \tag{5}$$

where: $C_{m,oij}$ is the husband's private consumption of the final good, $C_{f,oij}$ is that of the wife, H^R_{oij} is residential floorspace, and $d^m_{oi} = \exp(\kappa^m t_{oi})$ and $d^f_{oj} = \exp(\kappa^f t_{oj})$ are iceberg commuting costs. Notice that conditional on origin residence o, the husband works in destination i and the wife works in destination j. From here onward, I will assume that spouses agree on how much they should spend on the public good, so $\beta^m = \beta^f = \beta$.

Budget constraint. Because spouses pool their income within the household, total expenditure on private consumption and housing must equal the sum of the husband's and wife's labor earnings:

$$PC_{m,oij} + PC_{f,oij} + r_{R_o}H_{R_{oij}} = w_{m,i} + w_{f,j}, \tag{6}$$

where P is the price of the final good, which does not depend on the origin location as the final good is freely traded within the city. r_{R_o} is the rental value of the residential floorspace. Finally, $w_{m,i}$ is the wage received by the husband when working at i, whereas $w_{f,j}$ is the wage received by the wife when working at j.

Partial pooling. Under partial pooling, private consumption must additionally satisfy

$$PC_{m,oij} \leq v w_{m,i}, \qquad PC_{f,oij} \leq v w_{f,j}. \tag{7}$$

These restrictions limit the extent to which one spouse's earnings can be used to finance the other spouse's private consumption by pooling resources. Here, higher values of v correspond to lesser frictions in the household's ability to pool resources. When $v \to \infty$, spouses can fully pool resources to finance either spouse's private consumption. When $v \to 1$, resource pooling can only be used to finance housing expenditure. Private consumption must be financed out of each spouse's own income.

Working under partial pooling is helpful in at least two ways. First, as I show below, greater pooling makes spouses' commuting decisions more interdependent on aggregate. In this sense, v governs the aggregate degree of interdependence in commuting. In particular, one can show that as the housing expenditure share $1-\beta \to 0$ and $v \to 1$, commuting becomes fully separable across spouses, so that one spouse's commuting decision no longer affects the other's. By contrast, when $v > 1$, the pooling constraints are slack for a subset of destination pairs, so household resources are shared more broadly across spouses' commuting choices. As v rises, this set expands, and commuting decisions become more interdependent in the aggregate. I will later estimate v using commuting data from dual earner households by matching Fact 2. Second, partial pooling makes commuting choices better behaved: when one spouse's wage at a given destination goes to zero, that spouse no longer wishes to commute there. Under perfect income pooling with Fréchet shocks, by contrast, there is always a positive mass commuting even to zero wage locations.

Setting up the collective household problem. Conditional on residential and workplace locations, for any given vector of prices and wages $(P, r_{R_o}, w_{m,i}, w_{f,j})$ and location characteristics (d^m_{oi}, d^f_{oj}), an allocation $(C_{m,oij}, C_{f,oij}, H_{R_{oij}})$ is efficient if there exists a feasible promised utility $\bar{V}^f_{oj}(P, r_{R_o}, w_{m,i}, w_{f,j})$ such that $(C_{m,oij}, C_{f,oij}, H_{R_{oij}})$ solves

$$\begin{aligned} \max_{C_{m,oij}, C_{f,oij}, H_{R_{oij}}} \quad & V^m_{oi}(C_{m,oij}, H_{R_{oij}}) \\ \text{s.t.} \quad & PC_{m,oij} + PC_{f,oij} + r_{R_o} H_{R_{oij}} = w_{m,i} + w_{f,j}, \\ & V^f_{oj}(C_{f,oij}, H_{R_{oij}}) \geq \bar{V}^f_{ij}(w_{m,i}, w_{f,j}), \\ & PC_{m,oij} \leq v w_{m,i}, \qquad PC_{f,oij} \leq v w_{f,j}. \end{aligned} \tag{8}$$

Equivalently, any efficient allocation can be represented as the solution to a weighted household problem with Pareto weight $\zeta_{ij} \in (0,1)$ on the wife's utility. Re-expressing the problem in expenditure units, $x_{m,oij} \equiv PC_{m,oij}$, $x_{f,oij} \equiv PC_{f,oij}$, $z_{oij} \equiv r_{R_o} H_{R_{oij}}$, and $y_{oij} \equiv w_{m,i} + w_{f,j}$, the household problem becomes

$$\begin{aligned} \max_{x_{m,oij}, x_{f,oij}, z_{oij}} \quad & \widetilde{W}_{oij} \equiv (1-\zeta_{ij}) V^m_{oi}\left(\frac{x_{m,oij}}{P}, \frac{z_{oij}}{r_{R_o}}\right) + \zeta_{ij} V^f_{oj}\left(\frac{x_{f,oij}}{P}, \frac{z_{oij}}{r_{R_o}}\right) \\ \text{s.t.} \quad & x_{m,oij} + x_{f,oij} + z_{oij} = y_{oij}, \\ & x_{m,oij} \leq v w_{m,i}, \\ & x_{f,oij} \leq v w_{f,j}. \end{aligned} \tag{9}$$

The term ζ_{ij} determines which allocation on the Pareto frontier is selected by the household. When the household places greater weight on the husband's utility, the chosen allocation tends to favor him; when it places greater weight on the wife's utility, the chosen allocation tends to favor her. I allow ζ_{ij} to depend on (i,j) because the commuting-destination pair determines the spouses' relative wages. More specifically, in the collective household problem, the promised utility $\bar{V}^f_{oj}(w_{m,i}, w_{f,j})$ may vary with the earnings opportunities associated with the workplace pair (i,j), and this variation is inherited by the Pareto weight mapping $\zeta_{ij} = \zeta(w_{m,i}/w_{f,j})$.

In the standard setting in family economics, higher wages for one spouse may strengthen that spouse's bargaining position by improving their outside option, for example the utility they could attain if living alone, which is embodied in the utility promise $\bar{V}^{f}_{oj}(w_{m,i}, w_{f,j})$. In the present context, however, the relevant object is the promised utility associated with a given workplace arrangement (i, j). Thus, the Pareto weight reflects the utility promise needed to make a given workplace pair acceptable within the marriage, rather than the compensation required to keep the spouse in the marriage. For this reason, a higher husband wage relative to the wife's does not necessarily imply a lower Pareto weight on the wife's utility. If a workplace pair with high male earnings is attractive for the household but costly for the wife to accept (e.g., because it requires her to work in a lower wage destination), sustaining that arrangement may require a larger utility promise to her. In that case, the Pareto weight on the wife's utility, ζ_{ij}, may increase rather than decrease with the husband's wage relative to the wife's.

Solving the collective household problem. Let ρ denote the multiplier on the budget constraint, and let $\psi_m, \psi_f \geq 0$ denote the multipliers on the partial-pooling constraints. The first order conditions are

$$\frac{\beta(1-\zeta_{ij})}{x_{m,oij}} = \rho + \psi_m, \qquad \frac{\beta\zeta_{ij}}{x_{f,oij}} = \rho + \psi_f, \qquad \frac{1-\beta}{z_{oij}} = \rho, \tag{10}$$

together with complementary slackness,

$$\psi_m\big(vw_{m,i} - x_{m,oij}\big) = 0, \qquad \psi_f\big(vw_{f,j} - x_{f,oij}\big) = 0. \tag{11}$$

Since both caps cannot bind simultaneously when household income is positive, there are three relevant regimes. Define the thresholds

$$\underline{a}_{ij} \equiv \frac{\beta(1-\zeta_{ij})}{v - \beta(1-\zeta_{ij})}, \qquad \bar{a}_{ij} \equiv \frac{v - \beta\zeta_{ij}}{\beta\zeta_{ij}}.$$

<u>*Case 1: interior allocation.*</u> If neither cap binds, the optimal allocation is

$$\big(x^{I}_{m,oij},\, x^{I}_{f,oij},\, z^{I}_{oij}\big) = \Big(\beta(1-\zeta_{ij})y_{oij},\, \beta\zeta_{ij}y_{oij},\, (1-\beta)y_{oij}\Big). \tag{12}$$

Thus, the household spends a share β of income on the final good and a share $1-\beta$ on housing, while the split of final good expenditure across spouses is governed by ζ_{ij}. The interior allocation is feasible if and only if

$$\underline{a}_{ij}\, w_{f,j} \;\leq\; w_{m,i} \;\leq\; \bar{a}_{ij}\, w_{f,j}.$$

Intuitively, the interior region requires spouse incomes not to be too asymmetric.

<u>*Case 2: husband is constrained.*</u> If the husband reaches the cap, the optimal allocation is

$$\big(x^{K}_{m,oij},\, x^{K}_{f,oij},\, z^{K}_{oij}\big) = \left(vw_{m,i},\, \frac{\beta\zeta_{ij}}{1-\beta+\beta\zeta_{ij}}\big(w_{f,j} + (1-v)w_{m,i}\big),\, \frac{1-\beta}{1-\beta+\beta\zeta_{ij}}\big(w_{f,j} + (1-v)w_{m,i}\big)\right). \tag{13}$$

This regime applies when

$$w_{m,i} < \underline{a}_{ij}\, w_{f,j}.$$

In this region, the husband spends the maximum amount allowed by the partial-pooling constraint, and the remaining resources are split between the wife's private consumption and housing.

Case 3: wife is constrained. Symmetrically, if the wife reaches the cap, the optimal allocation is

$$\left(x^{K'}_{m,oij},\, x^{K'}_{f,oij},\, z^{K'}_{oij}\right) = \left(\frac{\beta(1-\zeta_{ij})}{1-\beta\zeta_{ij}} \left(w_{m,i} + (1-v) w_{f,j}\right),\, v w_{f,j},\, \frac{1-\beta}{1-\beta\zeta_{ij}} \left(w_{m,i} + (1-v) w_{f,j}\right) \right). \tag{14}$$

This regime applies when

$$w_{m,i} > \bar{a}_{ij}\, w_{f,j}.$$

In this region, the wife spends the maximum amount allowed by the partial-pooling constraint, and the residual resources are allocated between the husband's private consumption and housing.

Household welfare. We can now derive the household's total utility of living in o and commuting to the pair i, j. Substituting the optimal allocation into the household utility and exponentiating yields

$$\mathcal{W}_{oij} \equiv \exp(\widetilde{W}_{oij}) = \frac{\Xi_{ij}}{P^{\beta} (r^R_o)^{1-\beta} (d^m_{oi})^{1-\zeta_{ij}} (d^f_{oj})^{\zeta_{ij}}}. \tag{15}$$

Here, Ξ_{ij} is the regime specific wage aggregator implied by the household problem:

$$\Xi_{ij} = \begin{cases} K_m\, w_{m,i}^{\beta(1-\zeta_{ij})} \left(w_{f,j} + (1-v) w_{m,i}\right)^{1-\beta+\beta\zeta_{ij}}, & \text{if } w_{m,i} < \underline{a}_{ij}\, w_{f,j}, \\ K_I\, \left(w_{m,i} + w_{f,j}\right), & \text{if } \underline{a}_{ij}\, w_{f,j} \le w_{m,i} \le \bar{a}_{ij}\, w_{f,j}, \\ K_f\, \left(w_{m,i} + (1-v) w_{f,j}\right)^{1-\beta\zeta_{ij}} w_{f,j}^{\beta\zeta_{ij}}, & \text{if } w_{m,i} > \bar{a}_{ij}\, w_{f,j}. \end{cases}$$

The constants K_I, K_K, and $K_{K'}$ collect terms needed to keep welfare continuous at the thresholds.[19]

The object Ξ_{ij} summarizes the household resources that are available for allocation once the partial-pooling constraints are taken into account. In the interior region, pooling is sufficiently loose that the household can allocate the full sum of both spouses' incomes, so the relevant wage aggregator collapses to $w_{m,i} + w_{f,j}$. By contrast, in the constrained regions one spouse first reaches the maximum expenditure allowed by the cap, and the remainder of the allocation problem is solved using the residual resources left after satisfying that cap. For this

[19]Specifically, $K_I \equiv [\beta(1-\zeta_{ij})]^{\beta(1-\zeta_{ij})} (\beta\zeta_{ij})^{\beta\zeta_{ij}} (1-\beta)^{1-\beta}$, $K_m \equiv v^{\beta(1-\zeta_{ij})} \dfrac{(\beta\zeta_{ij})^{\beta\zeta_{ij}} (1-\beta)^{1-\beta}}{(1-\beta+\beta\zeta_{ij})^{1-\beta+\beta\zeta_{ij}}}$, and $K_f \equiv v^{\beta\zeta_{ij}} \dfrac{[\beta(1-\zeta_{ij})]^{\beta(1-\zeta_{ij})} (1-\beta)^{1-\beta}}{(1-\beta\zeta_{ij})^{1-\beta\zeta_{ij}}}$.

reason, the wage aggregator no longer depends on total household income alone, but instead on a nonlinear combination of the constrained spouse's own wage and the residual resources available to finance the unconstrained spouse's consumption and housing.

Marginal rate of substitution between spouses' commuting times and wages. Income pooling and differences in bargaining positions within the household introduce new tradeoffs that are absent in the model of independent commuting and that have important implications on how responsive workers are to commuting cost reductions. First, the level $\zeta_{ij}\kappa^f$ governs the household's willingness to give up the wife's wage in exchange for a shorter commute for her. Under income pooling, this tradeoff also depends on total household income. A higher husband wage $w_{m,i}$ makes the household more willing to accept a lower wife wage in exchange for reducing her commute, because the household can partly insure the loss in her earnings through the husband's income. Second, the ratio $\frac{\zeta_{ij}\kappa^f}{(1-\zeta_{ij})\kappa^m}$ governs the rate at which the household substitutes the wife's commute time against the husband's. I derive both expressions and discuss their implications in Appendix C. These expressions show that interdependence can dampen responses to commuting-cost reductions. When one spouse gains access to a high wage destination, the other spouse shares part of the income gain and may therefore have less need to adjust workplace choices.

Commuting probabilities with Fréchet distributed shocks. Up to this point I have held constant commuting destinations. I now allow households to choose commuting destinations endogenously. I assume that there is a continuum of households indexed by ω. I introduce idiosyncratic preference shocks over commute destinations, $\epsilon_{m,i}(\omega)$ and $\epsilon_{f,j}(\omega)$, which are iid Fréchet with shape parameters θ^m and θ^f, respectively.

For a given destination pair (i, j), household welfare can be written as

$$\mathcal{W}_{oij}(\omega) = \mathcal{W}_{oij}\, \epsilon_{m,i}(\omega)\epsilon_{f,j}(\omega).$$

I model dual earner commuting choices as a sequential problem. Conditional on residence o, the wife chooses destination j first, anticipating the market access associated to the husband's later choice of destination i. I adopt this sequence for tractability, as it delivers closed form commuting probabilities while allowing spouses to differ in their commuting elasticities, which is consistent with Fact 1.[20]

[20]There are two natural alternatives to the baseline formulation. First, one could model the two idiosyncratic shocks as simultaneous rather than sequential. In general, when the two Fréchet shape parameters differ, that formulation does not deliver closed form unconditional commuting probabilities, which is why I do not choose it. Note, however, if $\theta_m = \theta_f$, the sequential formulation is equivalent to a simultaneous choice over destination pairs and therefore also admits a closed form representation of unconditional commuting probabilities. Second, one could keep the sequential structure but reverse the ordering, so that the husband chooses first and the wife chooses conditional on his destination. I use the current ordering as the baseline because an empirical feature I want to capture is that the husband's commuting responds to the wife's destination. A simple way to think about this is that the wife's workplace helps pin down the household's effective geography—how the couple coordinate schedules, childcare logistics, who is closer to home, etc. The husband then chooses his commute conditional on that environment. Note that the main point is that spouses' commuting choices are interdependent through household utility, and that does not depend on the particular ordering.

Conditional on the wife choosing j, the husband chooses i. Since P and r_o^R are common across the husband's alternatives conditional on j, they cancel from the conditional choice probability. Standard Fréchet algebra then implies

$$\pi^m_{i|o,j} = \frac{\left(\frac{\Xi_{ij}}{(d^m_{oi})^{1-\zeta_{ij}}(d^f_{oj})^{\zeta_{ij}}}\right)^{\theta^m}}{\Phi_{oj}}, \tag{16}$$

where

$$\Phi_{oj} \equiv \sum_l \left(\frac{\Xi_{lj}}{(d^m_{ol})^{1-\zeta_{lj}}(d^f_{oj})^{\zeta_{lj}}}\right)^{\theta^m}. \tag{17}$$

Intuitively, conditional on the wife commuting to j, a destination pair (i, j) is more attractive to the husband when it delivers a larger effective wage bundle Ξ_{ij} and lower overall commuting costs. The relative importance of the husband's and wife's commute costs is governed by $1 - \zeta_{ij}$ and ζ_{ij}, respectively. The term Φ_{oj} is a market access term: it summarizes how attractive the full set of possible male destinations is for a household living in o, conditional on the wife commuting to j.

The wife chooses j anticipating this market access term. Hence the wife's commuting probability is

$$\pi^f_{j|o} = \frac{\Phi_{oj}^{\theta^f/\theta^m}}{\sum_n \Phi_{on}^{\theta^f/\theta^m}}. \tag{18}$$

Because ζ_{ij} varies with the destination pair, the wife's commuting cost is already embedded in the market access term Φ_{oj}, so it cannot in general be factored out as a separable term.

Finally, the husband's unconditional commuting probability is obtained by integrating over the wife's destination choice:

$$\pi^m_{i|o} = \sum_j \pi^m_{i|o,j}\, \pi^f_{j|o}. \tag{19}$$

Notice that by Bayes' rule, the corresponding conditional commuting probability for the wife is also pinned down: $\pi^f_{j|o,i} = \frac{\Pi_{ij|o}}{\pi^m_{i|o}} = \frac{\pi^m_{i|o,j}\,\pi^f_{j|o}}{\sum_n \pi^m_{i|o,n}\,\pi^f_{n|o}}$.

Substitution patterns. Pairs that share the wife's destination can act as either substitutes or complements depending on the relative size of θ^m and θ^f, while pairs associated with different wife destinations always act as substitutes. Appendix C works out the elasticities.

4.2. Comparative statics in a simplified geography under independent and interdependent commuting

To build further intuition, I now consider a simplified linear geography in which

$$\tau_{od} = \tau_{\max}\frac{|o-d|}{S-1}.$$

and where I hold the Pareto weight $\zeta_{ij} = \zeta$ constant for all pairs (i, j) and assume that $v \to \infty$. A fall in $\tau_{\max}$ proportionally shrinks commuting times on all routes. For spouse $k \in \{m, f\}$, define the transit gain on route $o \to d$ as

$$g_{od}^k \equiv \frac{\partial \log d_{od}^k}{\partial \log \tau_{\max}} = \kappa^k \tau_{\max} \Delta_{od}, \qquad \Delta_{od} \equiv \frac{|o - d|}{S - 1}.$$

Figure 3 illustrates this idea. Holding residence fixed at $o = A$, a proportional reduction in $\tau_{\max}$ generates larger transit gains on longer routes, so the gain on link AE is larger than the gain on link AB. Thus, routes that are farther from residence o experience larger proportional gains when $\tau_{\max}$ falls.

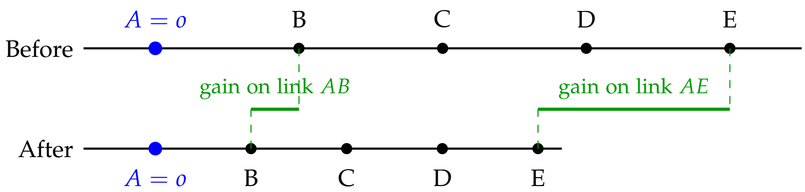


Figure 3: Transit gains in a linear geography.

Under independent commuting, define the average gain $\bar{g}_o^{k,I} \equiv \sum_d \pi_{d|o}^{k,I} g_{od}^k$. Under interdependent commuting, conditional on the wife commuting to j, define the husband's average gain $\bar{g}_{oj}^m \equiv \sum_i \pi_{i|o,j}^m g_{oi}^m$, the household level gain associated with wife destination j, $H_{oj} \equiv (1 - \zeta)\bar{g}_{oj}^m + \zeta g_{oj}^f$, and the average household level gain $\bar{H}_o \equiv \sum_j \pi_{j|o}^f H_{oj}$.

Labor supply. By labor supply, I mean the allocation of commuting probabilities across workplace destinations. Under independent commuting, a reduction in $\tau_{\max}$ reallocates each spouse toward destinations with larger own transit gains:

$$\frac{\partial \log \pi_{d|o}^{k,I}}{\partial \log \tau_{\max}} = -\theta_k \left(g_{od}^k - \bar{g}_o^{k,I} \right).$$

Thus, when commuting costs fall, spouse k shifts toward destinations whose transit gains are above the average gain of the destinations currently chosen from residence o.

Under interdependent commuting, the same logic applies, but now the husband's commute is affected by the wife's commute:

$$\frac{\partial \log \pi_{i|o,j}^m}{\partial \log \tau_{\max}} = -\theta_m (1 - \zeta) \left(g_{oi}^m - \bar{g}_{oj}^m \right), \qquad \frac{\partial \log \pi_{j|o}^f}{\partial \log \tau_{\max}} = -\theta_f \left(H_{oj} - \bar{H}_o \right).$$

Conditional on the wife's destination, the husband reallocates toward male destinations with larger-than-average gains. The wife, by contrast, reallocates toward destinations with high household level gains H_{oj}, because choosing j changes both her own commuting burden and the husband's market access. Thus, the total effect on the husband's unconditional commuting probability combines these two margins:

$$\frac{\partial \pi_{i|o}^m}{\partial \log \tau_{\max}} = -\sum_j \pi_{i|o,j}^m \pi_{j|o}^f \left[\theta_m (1 - \zeta) \left(g_{oi}^m - \bar{g}_{oj}^m \right) + \theta_f \left(H_{oj} - \bar{H}_o \right) \right].$$

Income. To summarize the income channel, it is useful to distinguish average gains from wage weighted gains. Under independent commuting, define $\tilde{g}_o^{k,I} \equiv \sum_d \tilde{\pi}_{d|o}^{k,I} g_{od}^k$, and $\tilde{\pi}_{d|o}^{k,I} \propto w_{k,d}\pi_{d|o}^{k,I}$. Then

$$-\frac{\partial \log y_o^{k,I}}{\partial \log \tau_{\max}} = \theta_k \left(\tilde{g}_o^{k,I} - \bar{g}_o^{k,I} \right).$$

Hence, under independent commuting, lower commuting costs raise expected income when destinations with larger transit gains are also high wage destinations.

Under interdependent commuting, define the husband's conditional expected wage $\mu_{oj}^m \equiv \sum_i w_{m,i}\pi_{i|o,j}^m$ and the husband's conditional wage weighted gain $\tilde{g}_{oj}^m \equiv \sum_i \tilde{\pi}_{i|o,j}^m g_{oi}^m$, where $\tilde{\pi}_{i|o,j}^m \propto w_{m,i}\pi_{i|o,j}^m$.

The wife's expected income responds according to

$$-\frac{\partial \log y_o^f}{\partial \log \tau_{\max}} = \theta_f \frac{\sum_j w_{f,j}\pi_{j|o}^f (H_{oj} - \bar{H}_o)}{\sum_j w_{f,j}\pi_{j|o}^f},$$

while the husband's expected income responds according to

$$-\frac{\partial \log y_o^m}{\partial \log \tau_{\max}} = \frac{\sum_j \pi_{j|o}^f \mu_{oj}^m \left[\theta_f (H_{oj} - \bar{H}_o) + \theta_m (1-\zeta)(\tilde{g}_{oj}^m - \bar{g}_{oj}^m) \right]}{\sum_j \pi_{j|o}^f \mu_{oj}^m}.$$

These expressions make clear that lower commuting costs raise income when they shift probability toward destinations that both gain more from the transit improvement and pay higher wages. Under interdependence, the husband's gains reflect two channels. The first is an indirect composition effect across wife destinations, captured by $\theta_f(H_{oj} - \bar{H}_o)$: lower commuting costs make some wife destinations more attractive than others, and those destinations imply different conditional expected wages for the husband, $\pi_{j|o}^f \mu_{oj}^m$. The second is a direct reallocation effect across the husband's own destinations conditional on j, captured by $\theta_m(1-\zeta)(\tilde{g}_{oj}^m - \bar{g}_{oj}^m)$: holding the wife's destination fixed, lower commuting costs may shift the husband toward higher wage male destinations. The overall effect is a weighted average of these two forces across wife destinations, where the weights $\pi_{j|o}^f \mu_{oj}^m$ reflect how important each wife destination is for the husband's expected income.

Gender earnings gap (GEG). Let $\mathcal{G}_o \equiv \log y_o^m - \log y_o^f$ denote the gender earnings gap at residence o. Define $E_o^I \equiv -\frac{\partial \mathcal{G}_o^I}{\partial \log \tau_{\max}}$, and $E_o^D \equiv -\frac{\partial \mathcal{G}_o^D}{\partial \log \tau_{\max}}$. Under independent commuting,

$$E_o^I = \theta_m \left(\tilde{g}_o^{m,I} - \bar{g}_o^{m,I} \right) - \theta_f \left(\tilde{g}_o^{f,I} - \bar{g}_o^{f,I} \right).$$

Thus, the gender earnings gap narrows or widens depending only on whether lower commuting costs shift men toward high wage destinations more strongly than they shift women toward high wage destinations.

Under interdependent commuting, define the weights $\omega_{oj}^m \equiv \frac{\pi_{j|o}^f \mu_{oj}^m}{\sum_n \pi_{n|o}^f \mu_{on}^m}$, and $\omega_{oj}^f \equiv \frac{\pi_{j|o}^f w_{f,j}}{\sum_n \pi_{n|o}^f w_{f,n}}$. Then

$$E_o^D = \theta_m (1-\zeta) \sum_j \omega_{oj}^m \left(\tilde{g}_{oj}^m - \bar{g}_{oj}^m \right) + \theta_f \sum_j \left(\omega_{oj}^m - \omega_{oj}^f \right) (H_{oj} - \bar{H}_o).$$

The first term is the direct male reallocation effect: conditional on the wife's destination, lower commuting costs may shift the husband toward higher wage male destinations. The second term is an indirect composition effect: lower commuting costs may also reallocate households across wife destinations, and those destinations may be relatively more favorable for the husband's conditional income than for the wife's own wage, or vice versa. For this reason, lower commuting costs may narrow the gender gap less under interdependence than under independence, and in some cases may even widen it.

A useful way to read these expressions is to think of them in terms of covariances. In the independent model, the effect of lower commuting costs on the gender earnings gap depends only on whether the destinations that gain the most from cheaper commuting are also the destinations that pay higher wages for men relative to women. In other words, the gap responds to the difference between the covariance of transit gains and wages on the male side and the corresponding covariance on the female side.

Under interdependent commuting, the first term preserves this covariance logic, but now on the male side it is computed *conditional on the wife's destination* and then averaged across wife destinations. The second term is conceptually different. It is not a covariance between wages and gains within a spouse, but a covariance between household level transit gains and the relative importance of each wife destination for male versus female earnings. In other words, the second term asks whether the wife destinations that become especially attractive when commuting costs fall are destinations that matter more for the husband's conditional expected wage or for the wife's own wage. This is the new force introduced by interdependence: transit improvements affect the gender earnings gap not only through direct access to better paid jobs, but also through the household's reallocation across wife destinations that reshape the husband's opportunity set indirectly.

The Pareto weight deserves a mention. I have held it constant throughout this comparative-statics exercise, thus the effects above arise solely from resource pooling. If commuting destination pairs were allowed to change the Pareto weight by altering the wage distribution within the household, there would be additional effects coming from the collective perspective of the household, as changes in destination pairs would also change the weight the household places on each spouse's utility. Whether this channel reinforces or offsets the pooling mechanism depends on how relative earnings affect the Pareto weight. If higher male earnings shift weight toward the husband and higher female earnings shift weight toward the wife, then the collective channel may reinforce the initial reallocation patterns. If the opposite is true, the collective channel may dampen or even reverse those patterns.

Examples with different geographies of wages To illustrate how the geography of wages interacts with the model's predictions, I solve the model on a simple linear city under four representative wage geographies and trace the response of commuting flows, the gender earnings gap, and aggregate GDP to a uniform reduction in commuting costs. The exercise shows that the same transit improvement can have very different effects on the gender earnings gap depending on where gender wage inequality is concentrated in space, and that aggregate

GDP gains are systematically smaller under interdependent commuting than under independent commuting. The full set of examples, decomposition, and figures is in Appendix D.

5. Bridging Data and Model

This section starts by summarizing the main components of the quantitative model. Afterwards, it describes the calibration and estimation procedures. Finally, it shows the model's performance when matching untargeted moments.

5.1. Full Model

The model has three pieces: households, production, and housing. Households make residence, participation, and commuting decisions. Firms combine male and female labor with floorspace via CES technologies. Each location's fixed floorspace stock is allocated between residential and commercial use. The subsubsections that follow develop each in turn and close with the definition of equilibrium. Full derivations are in Appendix E.

5.1.1. Environment

The city consists of a finite set of locations indexed by $o, d = 1, \ldots, S$. I use o for residence, d for a generic workplace, i for the husband's workplace, and j for the wife's workplace. Commuting times between locations are denoted by $\{t_{od}\}$. There are three household groups: single men, single women, and married couples. A mass N^m of single men and a mass N^f of single women choose where to live and where to work. A mass N^{mf} of married couples chooses a common residence. Within married households, women endogenously choose whether to participate in the labor market, giving rise to dual earner households, in which both spouses work, and male-breadwinner households, in which only the husband works. Locations are endowed with a fixed stock of floorspace $\{\bar{H}_o\}$ and differ in amenities $\{u_o^m, u_o^f, u_o^{mf}\}$, household-production productivity $\{\xi_{o0}\}$, industrial productivities $\{A_{os}\}$, and a land use wedge $\{\iota_o\}$.

5.1.2. Households

Singles. For singles of gender $g \in \{m, f\}$, let

$$\delta_{od}^g \equiv \exp(\varkappa^g t_{od})$$

denote iceberg commuting costs between residence o and workplace d. Conditional on the residence-workplace pair (o, d), singles spend constant shares of income on the final good and residential floorspace:

$$PC_{od}^g = \beta^g w_{g,d}, \qquad r_{R_o} H_{R,od}^g = (1 - \beta^g) w_{g,d}.$$

Conditional on residence o, workplace choice follows a Fréchet gravity equation,

$$\varpi_{d|o}^g = \frac{\left(\frac{w_{g,d}}{\delta_{od}^g}\right)^{\vartheta^g}}{\Psi_o^g}, \qquad \Psi_o^g \equiv \sum_{\ell=1}^{S} \left(\frac{w_{g,\ell}}{\delta_{o\ell}^g}\right)^{\vartheta^g},$$

where ϑ^g governs the sensitivity of commuting choices to wage and commuting differences. Define the corresponding residential market access term

$$\Omega_o^g \equiv (\Psi_o^g)^{1/\vartheta^g}.$$

Before workplace shocks are realized, singles choose residence according to

$$\lambda_o^g = \frac{\left(\dfrac{u_o^g \Omega_o^g}{P^{\beta^g} r_{R_o}^{1-\beta^g}}\right)^{\eta^g}}{\sum_{n=1}^{S}\left(\dfrac{u_n^g \Omega_n^g}{P^{\beta^g} r_{R_n}^{1-\beta^g}}\right)^{\eta^g}},$$

where η^g controls the dispersion of residential tastes.

Married couples: dual earners. The dual earner collective problem was developed in Section 4.1. Here I collect the objects that enter general equilibrium, introducing a notation convention: I append a trailing $,1$ subscript to indicate that the couple is in the dual earner arrangement (the $,0$ counterpart for male-breadwinner households is introduced below).

Conditional on residence o and destination pair (i, j), the deterministic welfare $\mathcal{W}_{oij}$ is given by (15), the husband's conditional commuting probability $\pi^m_{i|o,j}$ by (16), and the associated market access term Φ_{oj} by (17). The wife's commuting probability is $\pi^f_{j|o,1}$: the same object as the wife's marginal in (18), with the $,1$ subscript made explicit.

The new objects are the joint dual earner commuting probability over the pair (i, j), the husband's unconditional dual earner commuting probability, and the ex ante dual earner market access term:

$$\Pi_{ij|o} \equiv \pi^m_{i|o,j}\,\pi^f_{j|o,1}, \qquad \pi^m_{i|o,1} = \sum_{j=1}^{S} \Pi_{ij|o}, \qquad \Omega_{o1} \equiv \left(\sum_{j=1}^{S} \Phi_{oj}^{\theta^f/\theta^m}\right)^{1/\theta^f}. \tag{20}$$

Ω_{o1} will enter the female labor force participation decision below.

Throughout the married household block, I allow the Pareto weight to depend on spouses' relative wages through a mapping $\zeta_{ij} = \zeta(w_{m,i}/w_{f,j})$. I make the parametric form of this mapping explicit in the calibration.

Male-breadwinner households. If the wife does not work, the household derives utility from the husband's labor income and from household production with productivity ξ_{o0}. The deterministic payoff from sending the husband to i is

$$\mathcal{W}_{oi,0} = \xi_{o0}\frac{w_{m,i}}{P^{\beta} r_{R_o}^{1-\beta} d^m_{oi}}.$$

The husband's commuting probability in breadwinner households is therefore

$$\pi^m_{i|o,0} = \frac{\left(\dfrac{w_{m,i}}{d^m_{oi}}\right)^{\theta^m}}{\Phi_{o0}}, \qquad \Phi_{o0} \equiv \sum_{\ell=1}^{S}\left(\frac{w_{m,\ell}}{d^m_{o\ell}}\right)^{\theta^m}.$$

Define the corresponding breadwinner market access term as

$$\Omega_{o0} \equiv \Phi_{o0}^{1/\theta^m}.$$

Female labor force participation. Conditional on residence o, the household compares the dual earner value Ω_{o1} against the breadwinner value $\xi_{o0}\Omega_{o0}$. Let the preference shock over the wife's participation state be iid Fréchet with shape parameter ν. Then the probability that a couple living in o is dual earner is

$$\mu_o = \frac{\Omega_{o1}^{\nu}}{\Omega_{o1}^{\nu} + (\xi_{o0}\Omega_{o0})^{\nu}}.$$

The corresponding ex ante marital market access term is

$$\Omega_o^{mf} = [\Omega_{o1}^{\nu} + (\xi_{o0}\Omega_{o0})^{\nu}]^{1/\nu}.$$

Residential choice of couples. Before participation and commuting shocks are realized, couples draw an idiosyncratic residential taste shock. Their residence probability is

$$\lambda_o^{mf} = \frac{\left(\dfrac{u_o^{mf}\Omega_o^{mf}}{P^{\beta} r_{R_o}^{1-\beta}}\right)^{\eta^{mf}}}{\sum_{n=1}^{S}\left(\dfrac{u_n^{mf}\Omega_n^{mf}}{P^{\beta} r_{R_n}^{1-\beta}}\right)^{\eta^{mf}}}.$$

Couples therefore choose residence by trading off amenities, housing costs, the option value of dual earner commuting, and the option value of specializing in household production.

5.1.3. Labor supply, residential floorspace demand, and final good demand

Let

$$N_o^m \equiv N^m \lambda_o^m, \qquad N_o^f \equiv N^f \lambda_o^f, \qquad N_o^{mf} \equiv N^{mf} \lambda_o^{mf}$$

denote the resident population at o.

Aggregate labor supplied to workplace location d by men is

$$L_{m,d} = \sum_{o=1}^{S} N_o^m \varpi_{d|o}^m + \sum_{o=1}^{S} N_o^{mf}(1-\mu_o)\pi_{d|o,0}^m + \sum_{o=1}^{S} N_o^{mf}\mu_o \sum_{j=1}^{S} \Pi_{dj|o},$$

and aggregate labor supplied to workplace location d by women is

$$L_{f,d} = \sum_{o=1}^{S} N_o^f \varpi_{d|o}^f + \sum_{o=1}^{S} N_o^{mf}\mu_o \pi_{d|o,1}^f.$$

Per household residential floorspace demand—$H_{R,o}^g$ for singles of gender g, $H_{R,o}^{mf,1}$ for dual earner couples, and $H_{R,o}^{mf,0}$ for male-breadwinner couples—is derived in Appendix E (equations (83), (84), (85)). Summing across resident population, total residential floorspace demand at o is

$$H_{R,o}^D = N_o^m H_{R,o}^m + N_o^f H_{R,o}^f + N_o^{mf}\left[\mu_o H_{R,o}^{mf,1} + (1-\mu_o) H_{R,o}^{mf,0}\right]. \tag{21}$$

Analogously, per household final good consumption—C_o^g, $C_o^{mf,1}$, $C_o^{mf,0}$—is derived in the appendix ((87), (88), (89)). Total final good consumption at location o is

$$C_o^D = N_o^m C_o^m + N_o^f C_o^f + N_o^{mf}\left[\mu_o C_o^{mf,1} + (1-\mu_o) C_o^{mf,0}\right], \tag{22}$$

and aggregate household demand for the final good in the city is $C^D = \sum_{o=1}^{S} C_o^D$.

Following an Armington assumption, a representative final good producer aggregates the set of location-by-sector varieties with elasticity of substitution $\sigma_D > 1$. Cost minimization by the final good producer implies the standard Armington demand system. See (93)–(94) in Appendix E for the full expressions.

5.1.4. Production

I model the production side using standard assumptions in the literature. There are $s \in \{1, \ldots, K\}$ industries producing varieties differentiated by location under perfect competition. Following Tsivanidis (2021), firms at location o and industry s produce using a Cobb-Douglas technology over labor and commercial floorspace, $Q_{os} = A_{os}\, M_{os}^{\alpha_s} H_{F,os}^{1-\alpha_s}$, where $M_{os} = \left(\sum_g \alpha_{sg} (L_{g,os}^D)^{(\sigma-1)/\sigma}\right)^{\sigma/(\sigma-1)}$. Industries aggregate male and female workers with a CES of elasticity σ.[21] The total labor share is the sum of each gender's share, $\alpha_s = \sum_g \alpha_{sg}$. Here A_{os} is the productivity of location o in industry s. Industries differ in the intensity with which they use each gender, captured by α_{sg}. Full derivation of the production function, labor composite, unit costs, and factor demands is in Appendix E, equations (95)–(100).

5.1.5. Aggregate factor demand

Aggregating across sectors, total labor demand by gender and total commercial floorspace demand at location o are

$$L_{g,o}^D \equiv \sum_{s=1}^{K} L_{g,os}^D, \qquad H_{F,o}^D \equiv \sum_{s=1}^{K} H_{F,os}^D.$$

5.1.6. Housing and land use

The total floorspace stock at location o is fixed at $\bar{H}_o$. A share χ_o is allocated to residential use and a share $1-\chi_o$ to commercial use. Residential and commercial market clearing therefore require

$$\chi_o \bar{H}_o = H_{R,o}^D, \qquad (1-\chi_o)\bar{H}_o = H_{F,o}^D.$$

Landowners allocate floorspace to the most profitable use. A unit of residential floorspace earns r_{R_o}, whereas a unit of commercial floorspace earns $(1-\iota_o) r_{F_o}$ because of location specific regulations or wedges. Therefore,

$$\chi_o = \begin{cases} 1, & \text{if } r_{R_o} > (1-\iota_o) r_{F_o}, \\ (0,1), & \text{if } r_{R_o} = (1-\iota_o) r_{F_o}, \\ 0, & \text{if } r_{R_o} < (1-\iota_o) r_{F_o}. \end{cases}$$

[21] Hence in equilibrium a gender wage gap may arise in each location. I also assume that single and married men are perfect substitutes, and likewise single and married women, so conditional on a destination workers of the same gender receive the same wage regardless of marital status. Expected incomes may still differ because different groups commute to different destinations.

5.1.7. Definition of equilibrium

Definition 1. Given the model's parameters for singles $\{\beta^g, \varkappa^g, \vartheta^g, \eta^g\}$, for couples $\{\beta, \kappa^m, \kappa^f, \theta^m, \theta^f, v, \nu\}$, and for production $\{\alpha_{sg}, \alpha_s, \sigma, \sigma_D\}$; the mapping from spouses' wage ratios into Pareto weights $\{\zeta_{ij}(w_{m,i}/w_{f,j})\}_{i,j=1,...,S}$; city population by household group $\{N^m, N^f, N^{mf}\}$; and exogenous location specific characteristics, namely floorspace stock $\{\bar{H}_o\}$, amenities $\{u^m_o, u^f_o, u^{mf}_o\}$, household-production productivity $\{\xi_{o0}\}$, industrial productivities $\{A_{os}\}$, commuting times $\{t_{od}\}$, and land use wedge $\{\iota_o\}$, the competitive equilibrium of the model is the vector of labor supplies $\{L_{m,o}, L_{f,o}\}$, labor demands $\{L^D_{m,o}, L^D_{f,o}\}$, floorspace demands $\{H^D_{R,o}, H^D_{F,o}\}$, wages $\{w_{m,o}, w_{f,o}\}$, rents $\{r_{R_o}, r_{F_o}\}$, and land use shares $\{\chi_o\}$ such that households solve their residence, commuting, and labor force participation problems; firms minimize costs and choose factor demands optimally; labor markets clear, $L_{m,o} = L^D_{m,o}$ and $L_{f,o} = L^D_{f,o}$; floorspace markets clear; and the final good market clears.

5.2. Calibration

I calibrate the model in four sequential blocks. Parameters estimated in one block are held fixed when estimating the next block. Block 0 sets parameters that come from the literature or directly from ENAHO aggregates. Block 1 estimates the commuting elasticities and the Pareto weight parameters by GMM, matching the gravity moments from Facts 1 and 3. Block 2 inverts male and female wages from labor market clearing given the pooling friction v, and selects v so the model reproduces the gender gap in the Fact 2 interdependence coefficients, $\psi_m - \psi_f$. Block 3 estimates the female labor force participation elasticity ν and the residential location elasticities $\{\eta^m, \eta^f, \eta^{mf}\}$ using Bartik instruments. Then inverts location–industry productivities, the household-productivity shifter and amenities that rationalizes employment, female participation, and residents by location.

5.2.1. Block 0: Parameters from literature and data

From the National Household Survey (ENAHO, 2010–2018 pooled) I estimate industry-by-gender labor income shares $\{\alpha_{sg}\}$ using earnings data aggregated to seven sectors, and household-type housing expenditure shares $\{1-\beta^m, 1-\beta^f, 1-\beta\}$ using expenditure data for single men, single women, and married couples. Three parameters are calibrated from the literature: the male–female elasticity of substitution σ (Johnson and Keane, 2013), the final good demand elasticity of substitution σ_D (Feenstra et al., 2018), and the industry labor input share α_s (Ahlfeldt et al., 2015). Values and sources are summarized in Panels A and B of Table 3.

5.2.2. Block 1: Commuting elasticities and the Pareto weight

The goal of Block 1 is to recover the commuting parameter vector $\Theta \equiv (\vartheta^m, \vartheta^f, \theta^m, \theta^f, \kappa^f, \zeta_0, \zeta_1)$. These parameters govern how singles, breadwinner households, and dual earner households trade off wages against commute time and, for dual earners, how the Pareto weight varies with spouses' relative earnings. The overall strategy is a GMM estimator that stacks moments from three groups of gravity PPML regressions: single-gender flows, breadwinner flows, and dual earner flows. The appendix provides the full derivation of the estimating equations and the details of the GMM algorithm.

I impose a few restrictions to map these moments to a parameter vector. First, the commute-cost scale is common within gender across household types, so $\varkappa^g = \kappa^g$ for $g \in \{m, f\}$. Second, among married individuals, male commuting parameters are common across male-breadwinner and dual earner households, and female commuting parameters are common across female-breadwinner and dual earner households. So, one (κ^m, θ^m) applies to husbands in either arrangement and one (κ^f, θ^f) applies to wives in either arrangement. Moreover, following Ahlfeldt et al. (2015), Tsivanidis (2021), and Zárate (2021), I normalize $\kappa^m = 0.01$. For dual earner households, I allow the Pareto weight to depend on spouses' relative earnings, following the collective household literature in which wages shape the relative weight of spouses' utility (Fortin and Lacroix, 1997; Chiappori et al., 2002). I parameterize it as a logit function of the spouses' log wage ratio, $\zeta_{ij} = \Lambda\Big(\zeta_0 + \zeta_1 \log\Big(\frac{w_{m,i}}{w_{f,j}}\Big)\Big)$.

Identification comes from three sets of moments. First, PPML regressions on single-men and single-women flows, estimated as part of Fact 1, pin down the reduced form slopes $\gamma_m^S = -\vartheta^m \varkappa^m$ and $\gamma_f^S = -\vartheta^f \varkappa^f$. Second, analogous regressions on male- and female-breadwinner flows, also from Fact 1, deliver $\gamma_m^{MB} = -\theta^m \kappa^m$ and $\gamma_f^{FB} = -\theta^f \kappa^f$. "I exclude female-breadwinner households from the counterfactuals, but I keep them in the estimation sample because their flows identify $\theta^f \kappa^f$ without the dual earner cross spouse spillovers. Third, dual earner commuting flows provide the moments that identify the remaining three unknowns $(\kappa^f, \zeta_0, \zeta_1)$. In particular, the coefficient on own commute time discipline the average level of the Pareto weight, while the coefficient on the interaction terms between commute times and the log wage ratio discipline how the Pareto weight tilts with relative earnings. Those same interaction terms also carry information about κ^f through the wife's commute-cost component inside the dual earner flow equations.

A complication is that the dual earner system is nonlinear in commute times and wage ratios through ζ_{ij}, so the estimated coefficients in Fact 3 cannot be mapped linearly to structural parameters in $\zeta_{ij} = \Lambda\Big(\zeta_0 + \zeta_1 \log\Big(\frac{w_{m,i}}{w_{f,j}}\Big)\Big)$. I therefore proceed in three steps. In the first step, I estimate auxiliary PPML regressions on the data and recover the associated fixed effects, which turn out to be the husband's empirical market access term $\widehat{\Phi}_{oj}$. In the second step, for any candidate Θ, I build the implied Pareto weight $\zeta_{ij}(\Theta)$. I use these Pareto weights together with the recovered fixed effects and data on commute times to construct model-based commuting flows for dual earner households. In the third step, In the third step, In the third step, I run the same PPML regressions on these model implied flows. The coefficients I obtain are the model's prediction for the Fact 3 dual earner moments. Stacking them with the single and breadwinner moments gives the model moment vector $m(\Theta)$. I choose $\widehat{\Theta}$ to minimize the GMM criterion $g(\Theta)'Wg(\Theta)$, where $g(\Theta) = m(\Theta) - \widehat{m}$.

The numerical search is much smaller than the dimension of Θ suggests. At each candidate $(\kappa^f, \zeta_0, \zeta_1)$, the four commuting shape parameters $(\vartheta^m, \vartheta^f, \theta^m, \theta^f)$ are recovered analytically from the single and breadwinner slopes. The GMM search therefore only goes over $(\kappa^f, \zeta_0, \zeta_1)$.

I nevertheless keep the single and breadwinner slopes inside the stacked moment vector so that their sampling variation propagates into the standard errors of $\widehat{\Theta}$. Appendix F provides the full system of moments, the auxiliary regressions, the weighting matrix, and the numerical algorithm.

I report the Block 1 estimates in Panel C of Table 3. The female commute-cost scale is $\widehat{\kappa}^f \approx 0.03$. The Pareto weight parameters are $\widehat{\zeta}_0 \approx -0.38$ and $\widehat{\zeta}_1 \approx 0.50$: at the average wage ratio the household places slightly more weight on the wife, and her weight rises further as the husband's relative wage rises. At first sight, this seems to contradict the family economics literature, where a higher male wage typically translates into a higher male Pareto weight (and vice-versa for the wife). The discrepancy comes from what ζ_{ij} represents in this model. In the standard literature, the Pareto weight is anchored to the outside option of leaving the marriage: a higher outside wage for one spouse raises the value of being single, and the within marriage utility promise to that spouse must rise to keep them in the marriage. In my setting, the utility promise depends on the workplace pair (i, j), so ζ_{ij} is the share of household welfare required to sustain the specific arrangement (i, j), not the marriage itself. The signs then make sense. When the husband's workplace pays much more than the wife's, sustaining (i, j) typically forces the wife to accept a longer commute or a lower wage destination relative to alternative arrangements such as a different workplace pair. To keep her willing to take that arrangement, the household must promise her a larger share of household utility, hence $\widehat{\zeta}_1 > 0$. Fact 3 already points in the same direction: the wife's own-commute semi-elasticity becomes more negative as the wage ratio tilts toward the husband, indicating that the household weighs her commute more heavily in those arrangements.

All seven elements of $\widehat{\Theta}$ are precisely estimated and statistically different from zero. Appendix F.7 reports the diagnostics related to the model's fit and expands the discussion.

5.2.3. Block 2: Wage inversion and pooling friction

Block 2 recovers the wage vectors that make the model consistent with observed workplace employment and estimates the degree of pooling frictions.

Given the parameters from Blocks 0–1, I impose labor market clearing separately for men and women at each location. For any candidate wage vector, the residence population and participation shares from the data determine the number of workers available by household type. The commuting model then maps these workers into workplace labor supply. On the demand side, observed location-industry employment pins down total labor demand by industry, and firms' CES cost minimization conditions split this total into male and female labor demand. I choose wages for each gender so that predicted labor supply equals labor demand in every location.

I solve this inversion separately for 2007 and 2017, and under both the independent and interdependent commuting models. In the independent model, dual earner spouses choose workplaces separately. In the interdependent model, workplace choices depend on the household problem, so the wage inversion must be solved for a given value of the pooling friction v.

I estimate v by matching the interdependence in spouses' commuting choices documented in Fact 2. For each candidate v, I solve the interdependent wage inversion, construct the model implied conditional commuting probabilities, aggregate them from zones to municipalities to match the empirical moments, and run the same two-stage procedure used in the data. This gives model implied coefficients $\{\psi_m(v), \psi_f(v)\}$. I choose v to match the gender gap in interdependence, $\psi_m - \psi_f$, minimizing $\left[(\psi_m(v) - \psi_f(v)) - (\widehat{\psi}_m - \widehat{\psi}_f)\right]^2$. Targeting the gap lets the single pooling parameter discipline the differential strength of spousal interdependence between husbands and wives. Appendix G provides the wage-inversion equations, the construction of the ψ moments, and the numerical algorithm.

The estimation yields $\widehat{v} \approx 2.55$ (SE 0.90; Panel D of Table 3), well above the lower bound $v = 1$ at which only housing is pooled across spouses. At this estimate the model implies interdependence coefficients $\psi_m \approx 0.96$ and $\psi_f \approx 0.67$, whose gap $\psi_m - \psi_f \approx 0.28$ reproduces the Fact 2 gap in the data, $\widehat{\psi}_m - \widehat{\psi}_f \approx 0.28$.[22] Intuitively, each spouse can finance private consumption up to $\widehat{v}$ times their own earnings, drawing the remainder from the partner's wage through household pooling.[23] A detailed discussion of the estimate and the identification diagnostics is provided in Appendix G.8.

5.2.4. Block 3: Female labor force participation and location elasticities

Block 3 recovers the remaining location level objects in the model. Conditional on the wages and pooling friction from Block 2, I first invert location–industry productivities $\{A_{os}\}$ from observed workplace employment and floorspace rents using the production-side first order conditions. These productivities summarize the production residuals needed to rationalize observed employment and rents.

I then estimate the female labor force participation Fréchet shape ν. This parameter governs how strongly couples switch between the breadwinner and dual earner arrangements when the market access gap between those two states changes. The identification concern is that unobserved changes in local household productivity can move both the dual earner share and the wage geography that enters the market access gap. I instrument the change in the gap with a productivity-side Bartik: for each location, I aggregate realized productivity changes at other locations, weighted by pre-period commuting accessibility, and combine across sectors using sector employment shares. The Bartik sum at each location excludes its closest commuting neighbors, so the variation comes from productivity changes farther away in the city. I further restrict the estimation sample to zones at least one kilometer from the new Metro and BRT stations, where reduced form evidence in Fact 4 suggests that local unobservables respond to proximity to the new transit infrastructure. Including those zones would risk contaminating the IV exclusion restriction, since the same local unobservables that respond to the new infrastructure can comove with the productivity Bartik. I report the robustness of $\widehat{\nu}$ to the number of excluded neighbors and to the distance cutoff from the new stations in

[22] The two levels ψ_m and ψ_f are not separately targeted, since the estimator matches only their gap. Even so, the model reproduces each level approximately, somewhat overstating both relative to the data ($\widehat{\psi}_m \approx 0.87$, $\widehat{\psi}_f \approx 0.59$).

[23] This describes the split at the binding cap $PC_k = v w_k$, not for pairs whose constraint is slack.

Appendix H.2.[24]

Next, I recover the household-productivity shifter $\{\xi_{o0}\}$. Given $\widehat{\nu}$, wages, market access terms, and observed dual earner shares, this inversion assigns each location the value of the breadwinner arrangement, embodied in $\{\xi_{o0}\}$, needed to rationalize observed female labor force participation.

I then estimate the residential location elasticities $\{\eta^m, \eta^f, \eta^{mf}\}$. These parameters govern how strongly each household group reallocates residence across zones when residential market access changes. The identification concern is similar to the one for ν, but here the endogenous variable is residential utility. Thus, the concern now is the correlation between changes in market access and changes in amenities. I use the same productivity-side Bartik construction, instrument, and sample restriction as in the ν regression. I report the robustness of $\{\eta^m, \eta^f, \eta^{mf}\}$ to the number of excluded neighbors and to the distance cutoff from the new stations, in addition to sectoral Bartik shocks in Appendix H.2.[25]

I report the Block 3 estimates in Panel E of Table 3. The female-participation Fréchet shape is $\widehat{\nu} \approx 1.05$ (SE 0.20). A 10-percent widening of the market access gap between the dual earner and breadwinner arrangements raises the dual earner share by about 11 percent. Residential elasticities are $\widehat{\eta}^f \approx 1.00$ (SE 0.16) for single women,[26] $\widehat{\eta}^m \approx 4.26$ (SE 0.85) for single men, and $\widehat{\eta}^{mf} \approx 2.00$ (SE 0.37) for couples. First stage F-statistics are 227 for ν, 132 for η^f, 51 for η^m, and 68 for η^{mf}. Among the three groups, single men respond most strongly to changes in residential utility, single women the least, and couples sit in between.

Finally, given all parameters calibrated up to this point, I invert residential amenities $\{u_o^m, u_o^f, u_o^{mf}\}$ from the residence-choice equations. These amenities are the residual location characteristics that rationalize the observed residential distribution of each household group. Appendix H provides the equations I use to invert household-productivities and amenities, the Bartik instruments, the IV specifications, and the numerical algorithm.

5.3. Untargeted Moments

In this subsection, I examine the extent to which the model can account for the observed variation in bilateral commuting probabilities, an outcome not targeted for calibration. I compare predictions from the model with interdependent commuting against an alternative model where commuting is independent within couples. To give the alternative model the best chance to replicate the data, I back out exogenous characteristics, assuming this is the true model. I keep parameter values as estimated for Table 3.

In Appendix I, I compare model and Census commuting probabilities at the municipality

[24] An alternative shift share would use Metro- and BRT-induced changes in commute times in place of productivity shocks. The amenity response documented in Fact 4 enters this instrument with the opposite sign, biasing it toward zero; empirically, the resulting IV estimates are zero or negative.

[25] The employment-weighted aggregate yields residential-elasticity estimates above the bound $\eta > 1$ imposed by the Fréchet assumption for all three household groups, with first stages above the rule-of-thumb threshold of 10 (Staiger and Stock, 1997). The seven sector specific shifters are reported as alternatives in the robustness tables.

[26] The point estimate for single women sits just above the Fréchet bound $\eta > 1$. For the calibration I set $\eta^f = 1.01$ for numerical stability.

Table 3: Calibrated and estimated parameters

Parameter	Description	Value (SE)	Source
Panel A: Block 0 – estimated from aggregate data (ENAHO)			
$\{\alpha_{sg}\}$	Input share by industry (7 sectors), female shown	$\{0.15, 0.33, 0.33, 0.27,$ $0.20, 0.21, 0.14\}$	ENAHO earnings
$\{1-\beta^m, 1-\beta^f, 1-\beta\}$	Housing expenditure shares by household type	$\{0.274, 0.270, 0.232\}$	ENAHO expenditure
Panel B: Block 0 – calibrated from literature			
σ	Male–female elasticity of substitution	2.0000	Johnson and Keane (2013)
σ_D	Final-good demand elasticity of substitution	5.0000	Feenstra et al. (2018)
α_s	Industry labor input share	0.8000	Ahlfeldt et al. (2015)
κ^m	Male iceberg-disutility normalization	0.0100	Ahlfeldt et al. (2015)
Panel C: Block 1 – commuting elasticities and the Pareto weight			
ϑ^m	Male commuting Fréchet shape (singles)	5.6533 (0.0297)	GMM (Fact 1 + 3)
ϑ^f	Female commuting Fréchet shape (singles)	2.1783 (0.0272)	GMM (Fact 1 + 3)
θ^m	Male commuting Fréchet shape (couples)	4.6614 (0.0301)	GMM (Fact 1 + 3)
θ^f	Female commuting Fréchet shape (couples)	2.1214 (0.0255)	GMM (Fact 1 + 3)
κ^f	Female commuting-cost scale	0.0299 (0.0004)	GMM (Fact 1 + 3)
ζ_0	Pareto-weight intercept	-0.3846 (0.0170)	GMM (Fact 1 + 3)
ζ_1	Pareto-weight wage-ratio slope	0.5025 (0.0160)	GMM (Fact 1 + 3)
Panel D: Block 2 – pooling friction			
v	Pooling friction	2.5500 (0.8974)	GMM (Fact 2)
Panel E: Block 3 – extensive margin and location elasticities			
ν	Female-participation Fréchet shape	1.0508 (0.1954)	Bartik IV
η^f	Residential elasticity, single women	1.0020 (0.1616)	Bartik IV
η^m	Residential elasticity, single men	4.2600 (0.8540)	Bartik IV
η^{mf}	Residential elasticity, couples	1.9993 (0.3695)	Bartik IV

Note: The choice of $\kappa^m = 0.01$ is consistent with values used in Tsivanidis (2021) and Zárate (2021). The seven aggregated sectors for $\{\alpha_{sg}\}$ are, in order: (1) manufacturing (other); (2) manufacturing (textiles); (3) services; (4) financial and business services; (5) wholesale; (6) retail trade; (7) transportation.

level, separately by household type. For singles, the fit is slightly worse. Single workers face the same commuting problem in both models, so the change comes entirely from the different wages each model implies. For dual earner couples, the interdependent model fits commuting flows better for both spouses.

6. Counterfactuals: The Impact of the Metro Line

I use the calibrated model to measure how Line 1 of the Metro and the BRT changed the city. I solve the same counterfactual under two commuting structures. In the interdependent model, spouses choose commutes jointly, as in the model developed above. In the independent benchmark, each spouse chooses a workplace by trading off wages against their own commute time, without accounting for the partner's commute. Comparing the two versions shows how much within household commuting interdependence matters for aggregate output, the distribution of earnings, and welfare incidence.[27]

The section proceeds as follows. Section 6.1 reports the aggregate effects on GDP, income, welfare, and the gender earnings and welfare gaps, and decomposes these effects under both commuting structures. Section 6.2 studies how the effects vary across locations, with a focus on commute times, earnings, and the within couple consumption split. Section 6.3 runs the model through the same difference-in-differences design as Fact 4 and compares its predictions with the reduced form evidence. Section 6.4 explains why aggregate output rises by less under interdependent commuting, through the elasticity of labor supply.[28]

6.1. Aggregate Impact

Aggregate and distributional effects. The interdependent model predicts modest aggregate gains but sharp differences across households. Panel A of Table 4 shows that Line 1 of the Metro and the BRT raise GDP by 0.23 percent and total labor income by 0.41 percent. Deflating nominal GDP by the Cobb-Douglas price index implies a real GDP gain of about 0.4 percent.[29] Total labor income rises even though labor income per worker falls slightly, because the metro also raises labor force participation by 2.64 percent.

The income gains are concentrated among dual earner couples. Wives in these couples increase total earnings by 6.07 percent and earnings per worker by 3.35 percent. The gap between these two numbers reflects the entry of additional wives into paid work. Husbands in dual earner couples increase total earnings by 2.63 percent, but their earnings per worker are essentially unchanged. Singles experience much smaller changes: single men's income rises by 0.24 percent, while single women's income falls by 0.57 percent. The metro therefore narrows the gender earnings gap among dual earner couples, but widens it among singles.

The independent model gives a different picture in Panel B of Table 4. Aggregate gains

[27] I recalibrate amenities and productivities so that the independent model also matches the observed equilibrium.

[28] Section K studies the counterfactual completion of the planned Metro network.

[29] Deflating the 0.23 percent nominal GDP growth by the Cobb-Douglas price deflator $P^{\bar{\beta}}r^{1-\bar{\beta}}$, with $\bar{\beta} \approx 0.80$ the average consumption share, implies real GDP growth of about 0.4 percent.

are larger: GDP rises by 1.41 percent and total labor income by 1.42 percent with a real GDP growth of 1.3. The distributional effects are also flatter. Earnings rise by similar amounts for men and women within singles and within dual earner couples, so the gender gap within each group barely changes. Comparing the two models shows that commuting interdependence dampens the aggregate output response and makes the incidence of the metro more uneven across household types.

Table 4: Effect of the transit infrastructure on aggregate outcomes and income by group.

	Interdependent		Independent	
	Δ% total	Δ% p.c.	Δ% total	Δ% p.c.
Panel A: Aggregates				
GDP	+0.23		+1.41	
Total labor income	+0.41	−0.14	+1.42	+0.52
Gender gap, aggregate	−2.02	−0.69	−2.16	−0.00
Rents	−0.10		+1.38	
Price index	−0.15		−0.19	
Labor force participation	+2.64		+4.31	
Panel B: Income by group				
Single females	−0.57	−0.57	+0.89	+0.89
Single males	+0.24	+0.24	+0.75	+0.75
Gender gap, singles	+0.82	+0.82	−0.14	−0.14
Male-breadwinner	−3.05	+0.31	−4.28	+1.06
Dual-earner wife	+6.07	+3.35	+5.25	+0.91
Dual-earner husband	+2.63	−0.01	+5.23	+0.88
Gender gap, dual-earner	−3.25	−3.25	−0.03	−0.03

Notes. Effect of Line 1 of the Metro and the BRT at 2017, the proportional change between the calibrated economy with and without the new infrastructure (positive means the infrastructure raised the outcome). Panel A reports aggregate outcomes and Panel B income by group. The total column is the change in the total and the per capita (p.c.) column is the change per worker; the two coincide for singles and differ on the married rows by the labor force margin. GDP, rents, the price index, and labor force participation carry a total column only. The gender gap rows are the change in male-to-female earnings ($y_{\text{male}}/y_{\text{female}} - 1$), with a negative value meaning the gap narrowed.

To understand where the earnings gains come from, I decompose the change in earnings per worker into two pieces. The first is a wage effect: it holds residence and workplace choices fixed and asks how pay changes at the jobs workers already hold. The second is a reallocation effect: it holds wages fixed and asks how earnings change because workers move across residences and workplaces. I split this reallocation effect into three margins: residence, the worker's own commute, and the spouse's commute.[30]

The own-commute margin captures movement across the worker's own workplaces. It is

[30]The decomposition is exact and follows directly from writing average earnings per worker as a share-weighted average over all residence–workplace combinations. Let $s = (o, i, j)$ index a household's residence o, the husband's workplace i, and the wife's workplace j. Average earnings per worker can be written as $A = \sum_s W_s y_s$, where W_s

positive when the metro lets a worker move toward higher paying jobs. The spouse-commute margin captures the interaction between the two workplace choices. It is negative for a wife, for example, when her husband's workplace adjustment pulls the household toward choices that lower her earnings, and positive when his adjustment instead raises her earnings. This term is zero for singles and for male-breadwinner couples, where only one spouse works. Table 5 reports the decomposition for both commuting structures.

In the interdependent model, dual earner wives gain because they move toward better paying jobs. Their earnings per worker rise by 3.35 percent, almost entirely through the own-commute margin, which contributes 3.47 percentage points. The cross spouse channel is substantial: the spouse-commute term adds 0.90 percentage points, about 30 percent of the total gain, reflecting that husbands' workplace adjustments reinforce wives' gains. Husbands also move toward better paying jobs through their own commute margin, which contributes 2.62 percentage points. But the wife's workplace move pushes the joint household choice toward lower paying jobs for him, so his spouse-commute term is -1.65 percentage points. These forces offset each other, leaving husbands' earnings per worker essentially unchanged. Wages at their original jobs are essentially flat, except for husbands in male breadwinner households and singles. The residence margin is negative for every group.

Moving from the interdependent model to the independent model involves two changes. First, the direct cross spouse channel is shut down: in the independent model, one spouse's commuting decision no longer responds to the other's, so the spouse-commute term disappears. Second, because I reinvert the independent model to match the same equilibrium, it

is the share of workers associated with choice s and y_s is the worker's earnings at that choice. Consider a change from an initial equilibrium (0) to a counterfactual equilibrium (1). The total change in earnings can be written as $\Delta A = \sum_s (W_s^1 y_s^1 - W_s^0 y_s^0)$. Writing $\bar{W}_s = \frac{1}{2}(W_s^0 + W_s^1)$ and $\bar{y}_s = \frac{1}{2}(y_s^0 + y_s^1)$ for the midpoint share and earnings, the change splits exactly as

$$\Delta A = \underbrace{\sum_s \bar{W}_s\,(y_s^1 - y_s^0)}_{\text{wage effect}} + \underbrace{\sum_s \bar{y}_s\,(W_s^1 - W_s^0)}_{\text{reallocation effect}},$$

an identity because the remaining cross terms cancel. The first term is the wage effect. It keeps shares at their midpoint values and measures how earnings change when pay changes at the choices workers already make. The second term is the reallocation effect. It keeps earnings at their midpoint values and measures how earnings change as workers move across choices. I split the reallocation effect into margins using the product structure of the choice share, $W_s = \phi_o\,\pi^{\text{own}}\,\pi^{\text{sp}}$, where ϕ_o is the residence's share of the group's workers, π^{own} is the conditional probability of the worker's own workplace, and π^{sp} that of the spouse's workplace (the spouse term is absent for singles and the breadwinner). A change in a product is not additive across its factors, so I weight each factor's log change by the logarithmic mean $L(a,b) = (a-b)/(\ln a - \ln b)$, with $L(a,a) = a$. This weight satisfies the exact identity $W_s^1 - W_s^0 = L(W_s^1, W_s^0)\,(\ln W_s^1 - \ln W_s^0)$. Taking logs turns the product into a sum, so $\ln W_s^1 - \ln W_s^0 = \Delta \ln \phi_o + \Delta \ln \pi^{\text{own}} + \Delta \ln \pi^{\text{sp}}$ leaves no interaction term behind. The reallocation effect then splits into three additive margins,

$$\sum_s \bar{y}_s\,(W_s^1 - W_s^0) = \underbrace{\sum_s \bar{y}_s\,L_s\,\Delta \ln \phi_o}_{\text{residence}} + \underbrace{\sum_s \bar{y}_s\,L_s\,\Delta \ln \pi^{\text{own}}}_{\text{own commute}} + \underbrace{\sum_s \bar{y}_s\,L_s\,\Delta \ln \pi^{\text{sp}}}_{\text{spouse commute}},$$

with $L_s \equiv L(W_s^1, W_s^0)$ and $\Delta \ln x \equiv \ln x^1 - \ln x^0$. These three margins and the wage effect sum to ΔA with no residual. For reference, the first decomposition is the Bennet (arithmetic) indicator decomposition. The second decomposition, of the reallocation effect, is the log-mean Divisia index (LMDI) decomposition.

recovers a different set of baseline commuting shares. These baseline shares determine exposure to the new infrastructure: a household gains more when its no-metro commute lies along the routes improved by the Metro and BRT.

This shift in baseline exposure helps explain why the own-commute terms also change in Panel B. For dual earner wives, the own-commute contribution falls from 3.47 percentage points in the interdependent model to 1.87 in the independent model. For dual earner husbands, it falls from 2.62 to 1.28. In other words, the independent model alters both the cross spouse channel and the distribution of exposure to the infrastructure.

Table 5: Decomposition of the transit infrastructure's effect on per-worker earnings, by spouse.

Panel A: Interdependent model

	Changing shares				At fixed shares	Total
	Residential choice	Own commute	Spouse commute	Subtotal	Wage	
Single female	−0.78	+0.03	+0.00	−0.75	+0.18	−0.57
Single male	−0.65	+0.28	+0.00	−0.37	+0.61	+0.25
Male-breadwinner husband	−0.44	+0.21	+0.00	−0.23	+0.55	+0.32
Dual-earner wife	−1.02	+3.47	+0.90	+3.35	+0.00	+3.35
Dual-earner husband	−1.15	+2.62	−1.65	−0.18	+0.17	−0.01

Panel B: Independent model

	Changing shares				At fixed shares	Total
	Residential choice	Own commute	Spouse commute	Subtotal	Wage	
Single female	−0.85	+1.65	+0.00	+0.80	+0.09	+0.89
Single male	−0.78	+0.98	+0.00	+0.20	+0.55	+0.75
Male-breadwinner husband	−0.59	+1.20	+0.00	+0.61	+0.45	+1.06
Dual-earner wife	−1.05	+1.87	+0.00	+0.82	+0.09	+0.91
Dual-earner husband	−0.77	+1.28	+0.00	+0.52	+0.37	+0.88

Notes. Panels A and B decompose the change in each spouse's average earnings per worker into a Wage term and a Changing-shares block that sum to the Total, which equals the per-worker (Δ% p.c.) column of Table 4. The Wage term is the change in pay at the spouse's current jobs, holding all choice probabilities fixed. The Changing-shares block is the earnings effect of the choice probabilities shifting, attributed to the residential choice, the own commute, and the spouse commute. The male-breadwinner wife earns nothing and is omitted.

Welfare. The income results do not tell us who enjoys the resources generated by the metro. This is especially important for couples. A key advantage of my approach is that the interdependent model allows me to recover how resources are shared within the household since the Pareto weights determine how the household divides consumption between spouses. This lets me compute welfare separately for the wife and the husband. The independent model does not have a comparable within household allocation rule. I therefore use it only as a benchmark. For singles and dual earner couples, I assume each individual consumes what they earn. For male-breadwinner couples, I split the husband's income using the average Pareto weight implied by the interdependent model.

I report two welfare measures. The first is *ex ante welfare*. This is expected utility before residential, participation, and commuting shocks are realized. This measure values the option of reaching the residences and jobs made accessible by the metro. For spouse g in a couple,

Appendix E.3.6 shows that

$$\mathcal{U}^g = \frac{\gamma_E}{\eta^{mf}} + \log \mathcal{R}^{mf} + \sum_o \lambda_o^{mf} \left(\mathcal{U}_o^g - \log R_o \right), \qquad g \in \{m, f\}.$$

Here, $\mathcal{R}^{mf}$ is overall residential market access for the couple in the city. R_o is access to jobs from residence o. λ_o^{mf} is the probability of living in o. $\mathcal{U}_o^g$ is spouse g's expected utility conditional on living there. Singles are defined analogously in equation (33).

The second measure is *deterministic welfare*. This is consumption net of the commute cost actually paid. Unlike ex ante welfare, this measure does not value the option of accessing additional jobs or residences. It only evaluates the consumption, housing, and commuting bundle associated with the choices households make. For spouse g in a dual earner couple living in o,

$$\mathcal{V}_o^g = \frac{1}{P^\beta r_{R_o}^{1-\beta}} \sum_j \pi_{j|o}^f \sum_i \pi_{i|o,j}^m \frac{x_{g,oij}^\beta z_{oij}^{1-\beta}}{d_{oij}^g}.$$

Table 6 shows that wives' higher earnings do not close the welfare gap. This is true under both welfare measures. Under ex ante welfare, married women and men gain almost the same amount, 12.29 and 12.35 percent, so the married gender welfare gap is essentially unchanged. Under deterministic welfare, dual earner husbands gain 21.93 percent, while dual earner wives gain only 0.12 percent. Thus, even in the group where women experience relatively stronger income gains, these gains do not translate into higher welfare relative to men.

For singles, the two welfare measures yield different implications for the gender gap. Ex ante welfare rises for both single women and single men, and single women gain more: 13.81 percent against 5.77 percent. This reflects an access effect: the metro improves commuting possibilities, allowing individuals to reach locations and jobs they prefer, and therefore expands the set of options available before choices are made. The singles welfare gap therefore narrows under ex ante welfare. Deterministic welfare, however, is based only on the realized outcome—consumption net of commuting—after choices are made. Under this measure, welfare falls for both groups, and more so for single women: by 4.78 percent for women compared with 0.37 percent for men. The reason is that singles use the cheaper commute to move toward jobs and residences they like, but these choices do not raise their realized consumption net of commuting, so deterministic welfare falls.

The independent model gives a useful benchmark for interpreting the married results. Under ex ante welfare, the married gender gap widens by 3.02 percent in the independent model, compared with almost no change in the interdependent model. Under deterministic welfare, the gap widens less in the independent benchmark: by 2.78 percent among dual earner couples, compared with 21.79 percent in the interdependent model. The fact that the deterministic welfare gap increases only modestly in the independent model, but very strongly in the interdependent model, shows that the large gap arises from how the household jointly allocates resources and choices when spouses' decisions are linked.

Taken together, these results highlight two related facts. First, even though married women experience larger income gains than men in dual earner couples, these gains do not raise their

Table 6: Effect of the transit infrastructure on welfare, by group.

	Interdependent	Independent
Panel A: Ex-ante welfare		
Single female	+13.81	+12.83
Single male	+5.77	+5.27
Gender gap, singles	−7.06	−6.70
Married woman	+12.29	+6.47
Married man	+12.35	+9.68
Gender gap, married	+0.05	+3.02
Panel B: Deterministic per-spouse welfare		
Single female	−4.78	−4.09
Single male	−0.37	−0.20
Gender gap, singles	+4.63	+4.05
Male-breadwinner wife	−1.47	−0.17
Male-breadwinner husband	+4.37	+0.34
Gender gap, male-breadwinner	+5.92	+0.52
Dual-earner wife	+0.12	−5.13
Dual-earner husband	+21.93	−2.49
Gender gap, dual-earner	+21.79	+2.78
Average Pareto weight	−1.49	*N.A.*

Notes. Percentage change in welfare. Panel A is the ex-ante (nested-Fréchet) utility of each group before any commuting, participation, or residential shock, and the married rows are the husband's and wife's ex-ante utilities. Panel B is the change in deterministic per-spouse welfare. Gender gap rows are the change in the male-to-female ratio, where negative means the gap narrowed. In the independent model the Pareto weight is undefined, so dual-earner spouses are valued at their own earnings and breadwinner consumption uses the average weight. Table 7 decomposes the Panel B changes.

welfare relative to their husbands. Under ex ante welfare the two spouses gain in step, so the married gap barely moves. Under deterministic welfare the wife's realized gain is almost fully eroded while the husband's rises sharply, so the gap widens. Second, this widening reflects a within-household reallocation that shifts realized consumption toward the husband. Because the wife's workplace move lowers her Pareto weight, her share of household consumption falls even as her earnings rise, and this reallocation is large enough to offset both her higher earnings and her improved access to desirable locations. In other words, although the metro expands the set of opportunities available to married women, the way resources are allocated within the household prevents these opportunities from raising their welfare relative to their husbands.

To understand why wives' income gains do not become welfare gains, I decompose deterministic welfare. The decomposition separates two forces. To measure the first force, I hold residence and workplace choices fixed and ask how welfare changes because consumption, housing, commute costs, the price index, and rents change at those choices. To measure the second force, I hold those objects fixed and ask how welfare changes because households move across residences and workplaces. I split this reallocation term further into three margins: residence, the worker's own workplace, and the partner's workplace.[31] The partner-workplace term captures how one spouse's welfare changes when the other spouse changes workplace. For dual-earner couples, it is zero in the independent model by construction.[32] Table 7 reports the decomposition.

The first result is that, holding residence and jobs fixed, the metro raises deterministic welfare for every group. The metro's first order effect is to make the existing commute cheaper, so the commute term is by far the largest in this block, while consumption, housing, prices, and rents move much less. This direct commute saving is largest for single women and dual earner wives, whose fixed-choice commute terms are 22.15 and 23.07 percentage points. Husbands gain less at fixed choices: 13.81 percentage points for dual earner husbands, 8.99 for male breadwinners, and 7.84 for single men. At this stage, women appear to benefit the most: their baseline commutes receive larger time savings, consistent with the higher commute elasticity estimated for women.

The second result is that reallocation reverses much of this first order gain for women. Among singles, reallocation works only through residence and the worker's own workplace. A change in workplace can affect welfare through income and distance. A job farther from home lowers welfare when commute times are fixed, given the longer commute. A lower

[31] Welfare is a share-weighted average over residence and workplace choices, $\mathcal{V}^g = \sum_s \pi_s a_s$, where s indexes a residence and a pair of workplaces, and $a_s = x_g^{\beta} z^{1-\beta} / (d^g P^{\beta} r^{1-\beta})$ is spouse g's payoff at that choice. I use logarithmic-mean weights to write the change in welfare as an exact sum of a fixed-choice component and a reallocation component. Since the choice probability factors into a residence probability and conditional workplace probabilities, the reallocation component can be split into residence, own-workplace, and partner-workplace terms.

[32] The male-breadwinner wife is the one exception. Having no earnings of her own, her consumption in the independent benchmark is imputed as the average Pareto-weight share of the husband's income, so her welfare still moves with the husband's workplace reallocation and her partner term stays nonzero. Dual-earner spouses instead each consume their own earnings, so their welfare does not depend on the partner's workplace.

Table 7: Decomposition of the transit infrastructure's effect on deterministic welfare, by group.

Panel A: Interdependent model

	Reallocation				At fixed shares						Total
	Residence	Own	Partner	Subtotal	Consump.	Housing	Commute	Price	Rent	Subtotal	
Single female	−7.01	−20.20	+0.00	−27.22	+0.06	+0.02	+22.15	+0.10	+0.09	+22.43	−4.78
Single male	−1.41	−7.47	+0.00	−8.89	+0.38	+0.14	+7.84	+0.11	+0.05	+8.52	−0.37
Male-breadwinner wife	−2.35	+0.00	+1.38	−0.97	−0.85	+0.09	−0.00	+0.11	+0.15	−0.50	−1.47
Male-breadwinner husband	−1.61	−7.40	+0.00	−9.01	+4.02	+0.09	+8.99	+0.11	+0.16	+13.37	+4.37
Dual-earner wife	−1.77	−20.36	−1.21	−23.33	+0.01	+0.03	+23.07	+0.11	+0.23	+23.45	+0.12
Dual-earner husband	−0.16	−3.45	+11.62	+8.01	−0.30	+0.03	+13.81	+0.12	+0.25	+13.92	+21.93

Panel B: Independent model

	Reallocation				At fixed shares						Total
	Residence	Own	Partner	Subtotal	Consump.	Housing	Commute	Price	Rent	Subtotal	
Single female	−6.01	−19.60	+0.00	−25.61	−0.00	−0.00	+21.74	+0.14	−0.35	+21.53	−4.09
Single male	−1.06	−7.19	+0.00	−8.25	+0.31	+0.11	+7.94	+0.14	−0.45	+8.05	−0.20
Male-breadwinner wife	−2.74	+0.00	+2.34	−0.40	+0.21	+0.06	−0.00	+0.15	−0.19	+0.23	−0.17
Male-breadwinner husband	−1.79	−6.69	+0.00	−8.48	+0.21	+0.06	+8.60	+0.15	−0.19	+8.83	+0.34
Dual-earner wife	−3.30	−22.80	+0.00	−26.10	+0.02	+0.01	+20.99	+0.14	−0.18	+20.97	−5.13
Dual-earner husband	−5.04	−6.15	+0.00	−11.19	+0.21	+0.06	+8.46	+0.15	−0.18	+8.69	−2.49

Notes. Panels A and B decompose the interdependent and independent welfare change of Table 6 into a Reallocation block and an At-fixed-shares block that sum to the Total. The At-fixed-shares block holds the choice probabilities fixed and collects Consumption, Housing, Commute, Price, and Rent. The Reallocation block is the welfare effect of the choice probabilities shifting, attributed to the residence, the own workplace, and the partner workplace. In Panel B the partner term is zero for every group except the male-breadwinner wife, whose consumption is imputed as the average Pareto-weight share of the husband's income and therefore still responds to his workplace reallocation. Each block sums to its Subtotal, and the two Subtotals sum to the Total, which equals the Panel B change in Table 6. The dual-earner rows drop the outlier couple residences, with their weight redistributed proportionally, as in Panel B.

paying job lowers welfare if the cut in commute cost does not offset the loss in income. For single women, the own-workplace term is -20.20 percentage points and the residence term is -7.01, for a total reallocation effect of -27.22. This more than offsets their first order gains of 22.43, so their deterministic welfare falls by 4.78 percent. Single men also lose from reallocation, but much less: their own-workplace term is -7.47 and their residence term is -1.41, leaving their welfare almost unchanged at -0.37 percent. The interpretation is that single women use the new commuting technology to move toward jobs that are better in access terms, but these moves do not raise income and consumption enough to offset the higher commuting burden in the deterministic measure.[33]

A similar pattern emerges for wives in dual earner couples, but it is even more pronounced. Dual earner wives receive the largest direct commute saving in the table, 23.07 percentage points, yet their total welfare barely changes. The reason is that reallocation almost fully offsets the first order gains. Their own-workplace term is -20.36 percentage points, and the partner-workplace term is also slightly negative, at -1.21. This is informative because the wife's own workplace term contains more than her wage and distance. Her workplace choice also changes the household's Pareto weight, that is, her share of consumption. Thus, even though wives move toward better paying jobs, the induced household allocation leaves them with almost no deterministic welfare gain.

Husbands in dual earner couples are the opposite case. Their own direct commute saving is smaller than their wives', at 13.81 percentage points, and their own-workplace term is negative, at -3.45. But they gain through the partner-workplace margin. The wife's workplace adjustment raises the husband's welfare by 11.62 percentage points, either because it increases household resources, shifts consumption toward him, or both. This partner term turns the husband's reallocation component positive, at 8.01 percentage points, and raises his total deterministic welfare by 21.93 percent. This is why the deterministic welfare gap widens even though wives' earnings rise more than husbands' earnings.

Panel B repeats the exercise for the independent model. First, for singles the two panels are close but not identical: single women lose 4.09 percent against 4.78, and single men 0.20 against 0.37. These small gaps come from the model inversion, since I recover different amenities and productivities in each model to match the data. Second, setting the partner term to zero in Panel A does not reproduce Panel B, because the decomposition within a panel is only an accounting exercise. However, moving from Panel A to Panel B is a causal comparison, since the independent model shuts the cross spouse channel down everywhere and resolves the equilibrium. Third, the dual earner couples are where the differences between Panels are more substantial. Take the first order term, the commute saving at fixed shares. It depends on the commuting shares before the metro. Since the two models recover different shares, it comes out smaller in the independent model for both spouses, +8.46 against +13.81 for the husband and +20.99 against +23.07 for the wife. The reallocation terms are different as well. Under independence, the husband's own-workplace term is more negative (-6.15 against

[33] I interpret the residence and workplace terms as accounting objects, not as separate causal mechanisms, since residence choices may matter precisely because they position households for different commutes.

-3.45), and with the partner term now gone, his reallocation subtotal flips from a gain to a loss (+8.01 to -11.19). His welfare falls 2.49 percent, against a gain of 21.93 in the interdependent model. For the wife, the smaller commute saving and the missing partner term push her welfare down further, to 5.13 percent.

There are four takeaways from this section. First, the metro raises real GDP by 0.4 percent and concentrates the income gains in dual earner couples, and the gender earnings gap narrows for married couples and widens for singles. Second, the income gains favor wives for two reasons: the joint commute choice pushes husbands toward slightly lower paying jobs, and women's baseline commutes are more exposed to the routes improved by the Metro and BRT. Third, the welfare results show that wives' earnings gains do not translate into larger welfare gains. Ex ante welfare rises for everyone, but married women and men gain almost equally. Deterministic welfare shifts the gains toward husbands, especially in dual earner couples. Fourth, the decomposition explains why. Wives receive large commute time savings and move toward better paying jobs, but the household allocation gives husbands a higher share of the realized consumption gain.

6.2. Heterogeneity

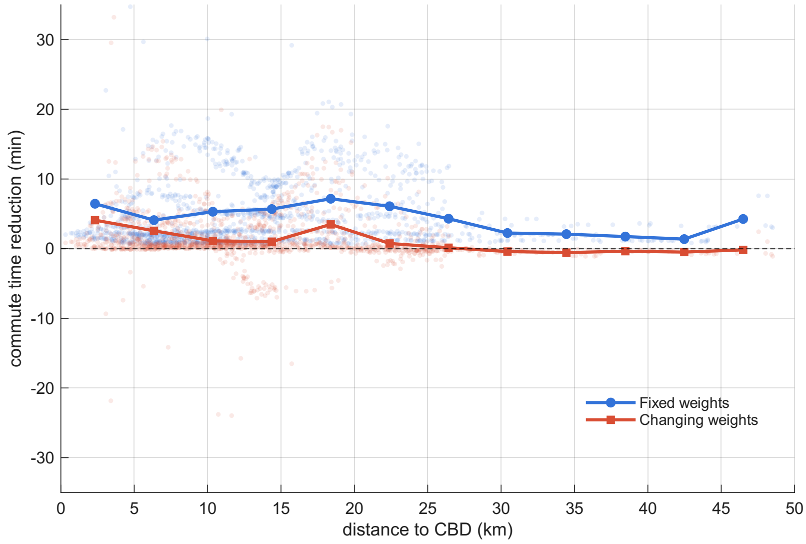
The aggregate results hide large spatial differences. The metro changed access much more in some parts of the city than in others, so the aggregate effects mask who was actually exposed to the new lines. Leveraging the fine spatial resolution of the data, I study these effects in two steps. I first trace how commute times, earnings, and the within couple consumption split change across locations. I then compare the model's spatial predictions with the reduced form evidence from Fact 4.

How did commute times change across locations? I begin with the metro's most direct effect: travel times. I measure commute time changes in two ways. The first measure holds residence and workplace choices fixed at their no-metro values. It captures the mechanical time saving on the routes households were already using. The second measure lets households reoptimize. It captures the realized change in commute time after households adjust where they live and work. The gap between the two measures shows how much of the direct time saving households keep after reallocation.

Figure 4 plots the average commute time reduction against distance to the CBD. The fixed-weights line is positive throughout and is largest at intermediate distances, rising to a peak around 18–20 kilometers before declining toward the fringe, which shows that the new lines created large direct time savings outside the core. The changing-weights line is lower because households use the new infrastructure to adjust their residence and workplace choices. Where this line remains positive, households still commute less than before. Where it turns negative, households give up the time saving and move toward locations that require longer commutes.

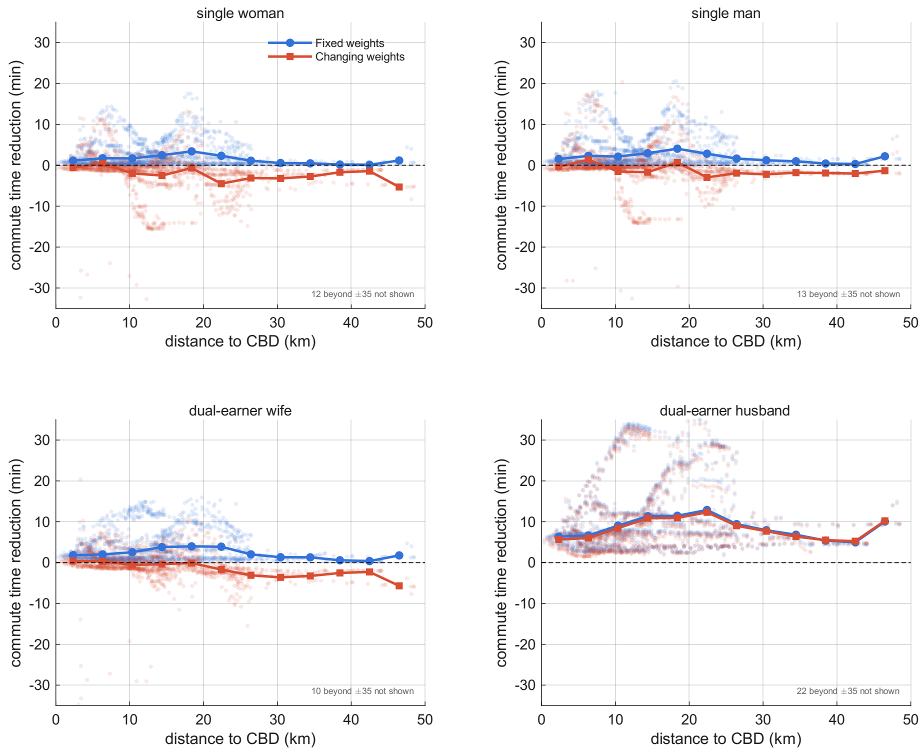
Figure 5 shows that this adjustment differs sharply across groups. Dual earner husbands keep almost all of the direct time saving: their fixed-weights and changing-weights lines are close and remain positive across space. Single men use most of the time saving to reach farther jobs, so their commute time remains as in the pre-metro environment. For single women and dual earner wives, however, the changing-weights line turns negative away from the CBD, so they relocate to jobs distant enough that their commute time ends up above its pre-Metro level. Whether those more distant jobs also pay more is the question I take up next.

Figure 4: Commute time reduction across space



Notes: Population-weighted average reduction in commute time against distance to the CBD, under two measures: fixed weights and changing weights. Points are locations. Lines are population-weighted binned means. The vertical axis is restricted to ±35 minutes. Interdependent model.

Figure 5: Commute time reduction across space, by group



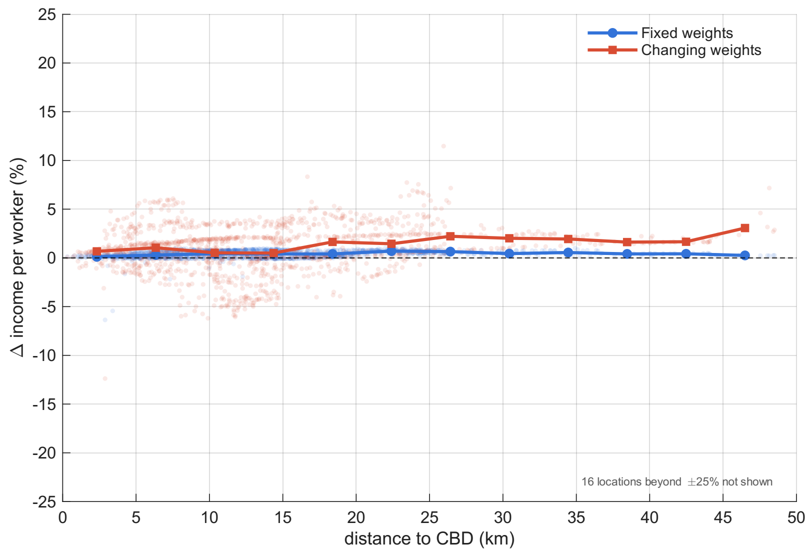
Notes: Reduction in average commute time against distance to the CBD, by group, under two measures: fixed weights and changing weights. Points are locations. Lines are population-weighted binned means. The vertical axis is restricted to ±35 minutes and locations outside this range are omitted, with their number shown in each panel. Interdependent model.

How did income change across locations? I next ask whether the longer commutes taken by women lead to higher earnings. I use the same two measures as before. The fixed-weights measure holds residence and workplace choices at their no-metro values and lets only wages change. The changing-weights measure lets households reoptimize across residences and jobs. The gap between the two measures captures the income gained or lost through reallocation.

Figure 6 shows that the fixed-weights line is close to zero across the city. This means that the metro did not substantially raise earnings at the jobs workers already held via general equilibrium responses. The changing-weights line is much higher, especially outside the center, which shows that the income gains come from moving toward better paying jobs rather than from wage growth at existing jobs.

Figure 7 shows that this reallocation channel is concentrated among women. For single women and dual earner wives, the changing-weights line rises above the fixed-weights line and increases with distance from the CBD. These are the same groups whose realized commutes lengthened in the previous figures. The model therefore points to a clear tradeoff: women use the new transit infrastructure to reach better paying jobs, even when doing so requires longer commutes. For single men and dual earner husbands, the changing-weights line remains close to the fixed-weights line, so reallocation does little to their earnings.

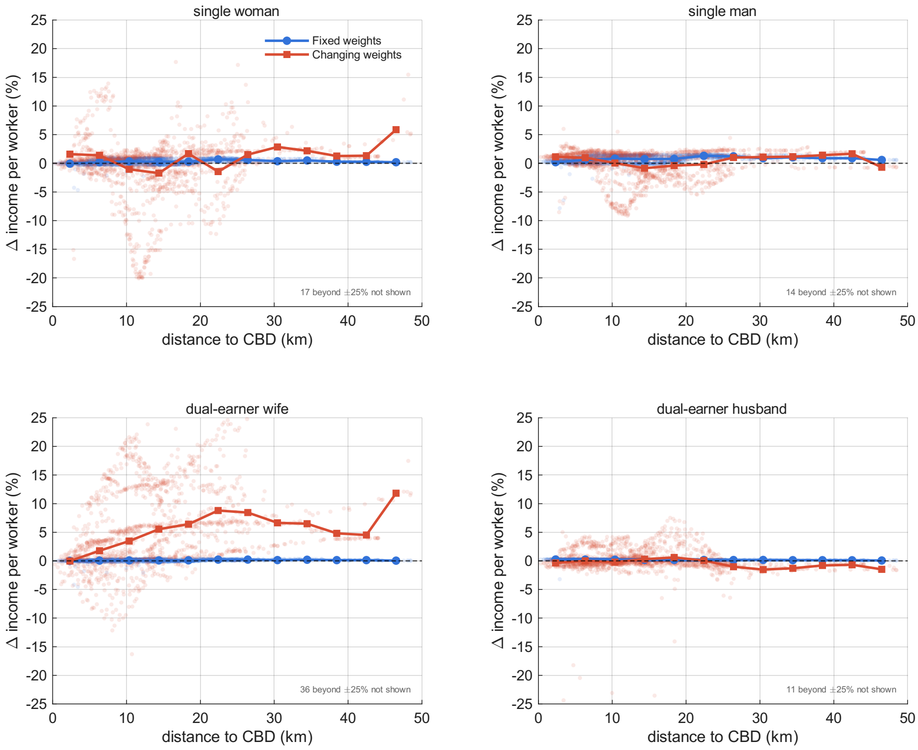
Figure 6: Change in earnings per worker across space



Notes: Population-weighted average change in earnings per worker against distance to the CBD, under two measures: fixed weights and changing weights. Points are locations. Lines are population-weighted binned means. The vertical axis is restricted to ±25 percent and locations outside this range are omitted, with their number shown in the figure. Interdependent model.

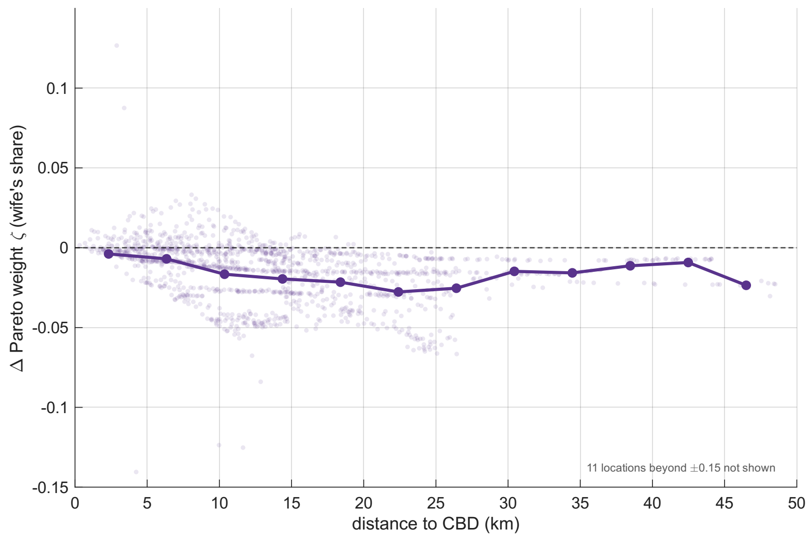
Who captured the gains? The consumption split. The income results show that dual earner wives move toward better paying jobs, especially outside the center. I now ask whether those gains translate into a larger share of household consumption. Figure 8 plots the change in the commute-flow-weighted Pareto weight on the wife, which determines her share of household consumption. The change is negative in most locations and is largest around 20 to 25 kilometers from the CBD. This is the same part of the city where wives' commutes lengthen and their earnings rise. Thus, the spatial pattern mirrors the aggregate welfare result: the metro improves wives' access to better paying jobs, but the within household allocation shifts consumption toward husbands.

Figure 7: Change in earnings per worker across space, by group



Notes: Change in earnings per worker against distance to the CBD, by group, under two measures: fixed weights and changing weights. Points are locations. Lines are population-weighted binned means. The vertical axis is restricted to ±25 percent and locations outside this range are omitted, with their number shown in each panel. Interdependent model.

Figure 8: Change in the within couple Pareto weight across space



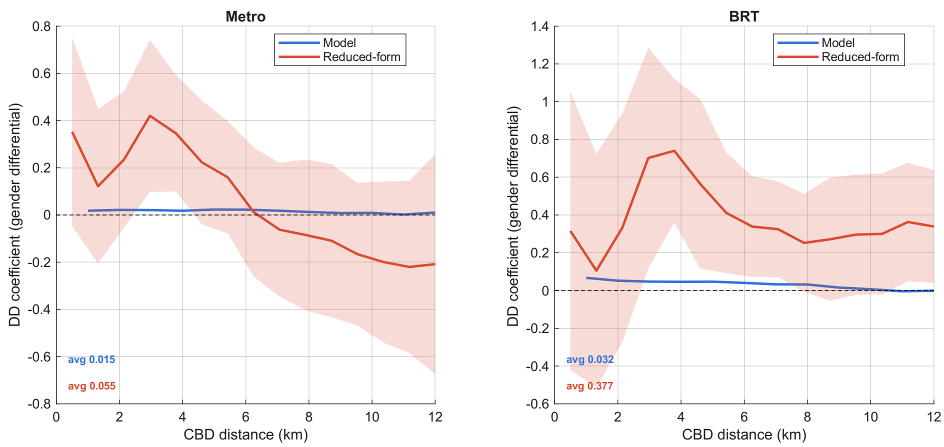
Notes: Change in the commute-flow-weighted Pareto weight on the wife, her share of household consumption, against distance to the CBD, across couple locations. A negative value means the average wife's consumption share falls. Points are locations. The line is the binned mean. The vertical axis is restricted to ±0.15 and locations outside this range are omitted, with their number shown in the figure. Interdependent model.

6.3. Reconciling with the reduced form evidence

I next compare the model with the reduced form evidence from Fact 4. I run the model through the same difference-in-differences design, comparing treated locations near constructed stations with control locations near stations that had not yet been built, separately by distance to the CBD. The outcome with a direct reduced form counterpart is the gender income gap.

Figure 9 shows that the model explains part, but not all, of the reduced form decline in the gender gap in income per worker. For Line 1, the model's average effect between 0 and 12 kilometers from the CBD is 0.015, compared with 0.055 in the reduced form. For the BRT, the model's average effect is 0.032, compared with 0.377 in the reduced form. The commuting channel therefore accounts for a meaningful share of the Metro effect, but a much smaller share of the BRT effect. The remaining gap leaves room for mechanisms outside the model, such as gendered changes in informality, neighborhood amenities, or safety around stations. At the same time, the reduced form estimates are noisy. The confidence bands are wide and contain the model line over most of the plotted range, so the model is broadly consistent with the reduced form evidence even though it does not match the point estimates exactly.

Figure 9: Gender gap in income per worker: model vs reduced form



Notes: DD coefficient on the gender gap in income per worker against distance to the CBD, rolling window $h = 3$ km, for Line 1 (Metro) and the BRT. The shaded band is the 95% confidence interval.

6.4. Why is GDP growth lower under interdependent commuting?

The aggregate results above show a large difference between the two commuting structures: the same metro raises GDP by 0.23 percent under interdependent commuting and by 1.41 percent under independent commuting. This difference comes from the local labor supply elasticity. When workers can move easily across workplaces, a local productivity gain attracts more labor and raises output more. When workers are tied to a spouse's workplace choice, labor moves less freely, and the same productivity gain produces a smaller output response.

I measure this elasticity numerically with the model. I raise productivity at one workplace by one percent, solve for the equilibrium, and record how much male and female employment move into that workplace. The shock raises productivity by the same one percent in every sector of that workplace, and I divide the employment response by the wage change the shock induces there, so the object I report is a labor supply elasticity. I average the location specific elasticities using employment weights, separately by gender. Under interdependent commuting, the elasticity is 1.22 for women and 3.02 for men. Under independent commuting, both are larger, at 1.77 for women and 4.49 for men.[34] The lower elasticities under interdependence reflect the household constraint: spouses choose workplaces jointly, so each worker's response to a local opportunity is limited by the partner's commute.

This distinction between singles and married households matters beyond just transit infrastructure. The local labor supply elasticity determines how any place-based shock maps into employment, wages, and output. It is the margin behind the local demand shocks studied by Greenstone et al. (2010), the place-based policies studied by Kline and Moretti (2014), and the local employment elasticities in the quantitative spatial models of Monte et al. (2018) and Redding and Rossi-Hansberg (2017). Treating commuting as an individual choice makes workers appear more mobile across workplaces than they are when household constraints matter. As a result, the independent model overstates the output response to spatially targeted shocks.

Do the output gains pay for the project? The lower output response under interdependence also weakens the project's cost–benefit case. Line 1 and the BRT cost about US$1.7 billion, and Lima's regional output is about US$84 billion, so in real terms the 0.4 percent gain under interdependence is about US$0.34 billion per year, against about US$1.1 billion under the independent benchmark.[35] Treating the gain as a permanent flow, it covers the interest on the construction cost as long as the interest rate stays below the ratio of the annual gain to the cost—the maximum viable rate—which household interdependence cuts from about 66 percent to about 20 percent.[36] Peru's Ministry of Economy and Finance evaluates public investment at an 8 percent social discount rate, so the project clears the bar under both models, but interdependence leaves far less margin. These figures count only construction and ignore the operating and maintenance cost of running the service, which is typically covered at a loss, so the returns here are upper bounds.

[34]To keep the computation tractable, I compute these averages on a random five percent sample of locations rather than shocking every workplace.

[35]The real GDP gains are 0.4 and 1.3 percent, after netting out the change in prices and rents, and I take Lima's output as 40 percent of Peru's 2017 national GDP of about US$211 billion (World Bank, current US$). Investment figures are the as-built civil-works costs: Line 1 Tramos 1 and 2, liquidated at US$519 and US$885 million (Congreso de la República del Perú, 2018), and the Metropolitano at about US$262 million (Global Infrastructure Hub, 2019).

[36]Write I for the construction cost and G for the permanent annual output gain. The present value of the gain, G/r, covers the cost when the interest rate r stays below the maximum viable rate G/I. With a cost of about US$1.67 billion (Line 1 civil works US$1.40 billion plus the Metropolitano US$0.26 billion) and annual gains of about US$0.34 billion under interdependence and US$1.10 billion under independence, this rate is $0.34/1.67 \approx 0.20$ and $1.10/1.67 \approx 0.66$. Undiscounted payback, I/G, is inversely proportional to the gain, so it is about $1.3/0.4 \approx 3$ times longer under interdependence. At the 8 percent rate, the discounted payback horizon n, solving $I = G\left[1-(1+r)^{-n}\right]/r$, is about 1.7 years under independence and 6.5 years under interdependence.

7. Conclusion

I develop a quantitative model of urban transit in which spouses make commuting decisions jointly rather than as independent workers. This household dimension matters for two reasons. First, it changes how we measure the aggregate gains from a transit improvement. When one spouse gets better access to jobs, the household can reallocate work, commuting, and consumption. The total response can therefore look quite different from independent responses. Second, the household dimension changes how these gains are distributed, both across locations and within the household.

One important lesson is that earnings do not tell the whole story about who gains. The male-female earnings gap is informative about women's labor market outcomes, but it is not a sufficient statistic for women's welfare. Couples also reallocate resources within the household, and this means that the welfare gap between spouses can move in the opposite direction from the earnings gap.

Several questions remain open. In the model, the Pareto weights that determine consumption allocations within couples are recovered from the cross section of joint commuting choices. Panel data on commuters would allow these weights to be estimated under fewer assumptions. Commuting choices may also provide a way to identify these weights nonparametrically, in the spirit of the identification results pursued in the family economics literature. The location of each spouse's job reveals how the household trades off their earnings and commuting costs, and may therefore contain useful information about within-household allocations.

Finally, the household margin matters beyond transit infrastructure. Joint commuting decisions also shape labor supply elasticities, a key object in the analysis of place based policies. Any policy that changes the local return to work will have effects that depend on how households jointly adjust labor supply. Accounting for interdependent commuting is therefore likely to matter for the evaluation of a much wider class of policies.

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A. Fact 1 Extended

Heterogeneity. A child at home can plausibly raise sensitivity to commute times, since every minute outside the home is more costly when one has childcare responsibilities. Following the classification in Section 3.2, I split the 2017 Population Census into the same five household groups: single females, single males, male-breadwinner households, female-breadwinner households, and dual earner households (separating men and women in the last group). I further split each by whether at least one child below 8 years old lives in the household, giving twelve groups in total. For each group k I reconstruct the commuting share $\pi^k_{d|o}$ and run the same gravity equation as in Section 3.2,

$$\log \pi^k_{d|o} = \gamma_k \, t_{od} + \alpha^k_o + \delta^k_d + \varepsilon^k_{od}, \tag{23}$$

where t_{od} is the commute time from origin o to destination d, α^k_o and δ^k_d are origin and destination fixed effects, and γ_k is the commuting semi-elasticity.

Table 8 reports the results. Panel A shows commuting semi-elasticities for households without children. Panel B does it for those with at least one child. Two takeaways stand out. First, women in breadwinner and dual earner households are much more sensitive to commute times than men whether or not a child is at home. Among singles, the gap appears only when there is a child at home. Second, having a child at home raises sensitivity to commute times for all women and only marginally for men. The exercise reinforces a central observation of the paper: under current social norms, balancing family and work responsibilities is more costly for women than for men.

Robustness. In Section 3.2 I estimated commuting semi-elasticities from a gravity equation connecting flows from origins o to destinations d for each household group. The Census data also let me control for the average characteristics of individuals using each route, which addresses the concern that, for example, single and married households differ in age, education, and other characteristics.

If differences in sensitivity to commute times between men and women persist after introducing controls, that is informative about one of the main mechanisms of the paper: under current social norms, balancing family and work responsibilities is more costly for women, especially when married. To check this, I re-estimate equation (1) with the average characteristic at the (o, d) pair as a control: age, education, religion, language, recent moving history, and dwelling characteristics such as the number of rooms and whether sewage is available.

Table 9 reports the results. Even with origin-destination controls, women remain much more sensitive to commute times than men, especially when married. Compared to my baseline results, the female-to-male gap in sensitivity narrows but stays sizeable. For example, women in breadwinner households are now 35% more sensitive than men, against 36% in the baseline. Women in dual earner households are now 31% more sensitive than men, against 40% in the baseline.

Table 8: Commuting semi-elasticity by presence of children

	Single HH		Breadwinner HH		Dual-Earner HH	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: No children	Males	Females	Males	Females	Males	Females
Travel Time	-0.0566	-0.0620	-0.0454	-0.0612	-0.0477	-0.0636
	(0.0038)***	(0.0039)***	(0.0034)***	(0.0040)***	(0.0032)***	(0.0039)***
Gap (Female/Male-1)		10%		35%		33%
Panel B: At least one child	Males	Females	Males	Females	Males	Females
Travel Time	-0.0565	-0.0756	-0.0479	-0.0709	-0.0482	-0.0723
	(0.0043)***	(0.0047)***	(0.0032)***	(0.0047)***	(0.0033)***	(0.0046)***
Gap (Female/Male-1)		34%		48%		50%
Gap (Kid/No Kid - 1)	0%	22%	6%	16%	1%	14%
Origin FE	X	X	X	X	X	X
Destination FE	X	X	X	X	X	X
N	2500	2500	2500	2500	2500	2500

Notes: Observations are at the municipality-by-municipality level. The independent variable in all regressions is the median travel time t_{od} between origin o and destination d, with origin and destination fixed effects included. Column (1) uses the share of single male workers who reside in o and work in d as the dependent variable; column (2) the corresponding share for single female workers. Columns (3) and (4) use male-breadwinner and female-breadwinner households. Columns (5) and (6) use males and females in dual-earner households. The differences between columns (1)–(2), (3)–(4), and (5)–(6) are statistically significant at the 1 percent level, except for single men and women in Panel A.

Table 9: Commuting semi-elasticity with origin-destination controls

	Single HH		Breadwinner HH		Dual-Earner HH	
	Males	Females	Males	Females	Males	Females
	(1)	(2)	(3)	(4)	(5)	(6)
Travel Time	-0.0429	-0.0494	-0.0369	-0.0499	-0.0390	-0.0511
	(0.0027)***	(0.0028)***	(0.0022)***	(0.0039)***	(0.0025)***	(0.0031)***
Origin-Destination Controls	X	X	X	X	X	X
Origin FE	X	X	X	X	X	X
Destination FE	X	X	X	X	X	X
Gap (Female/Male-1)		15%		35%		31%
N	2500	2500	2500	2500	2500	2500

Notes: Observations are at the municipality-by-municipality level. The independent variable in all regressions is the median travel time t_{od} between origin o and destination d, with origin and destination fixed effects included. I also control for the average characteristic at the (o,d) level: age, education, religion, language, recent moving history, and dwelling characteristics such as the number of rooms and whether sewage is available. Column (1) uses the share of single male workers who reside in o and work in d as the dependent variable; column (2) the corresponding share for single female workers. Columns (3) and (4) use male-breadwinner and female-breadwinner households. Columns (5) and (6) use males and females in dual-earner households. The differences between columns (1)–(2), (3)–(4), and (5)–(6) are statistically significant at the 1 percent level.

B. Fact 4 Extended

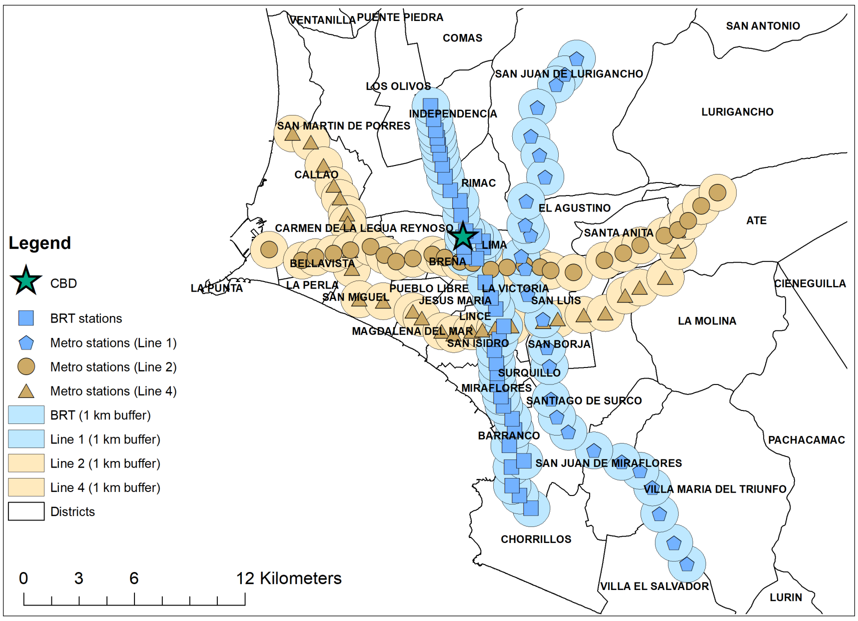
This appendix provides the reduced form evidence behind Fact 4. The main analysis uses the georeferenced National Household Survey to study whether the new transit infrastructure changed the male–female earnings gap. I then report complementary results from the Population Censuses on population and household composition near the new infrastructure. I use these results as benchmarks for comparing the responses generated by the structural model.

B.1. Planned stations

I perform the reduced form analysis by exploiting the planned network of Metro lines. I compare locations close to Line 1 of the Metro or the BRT, which were built during the sample period, to locations close to Line 2 or Line 4, which were planned at the same time but built only later.[37] The idea is that planned-but-not-yet-constructed stations provide a more natural comparison group than locations far from the planned transit network. These locations were also selected for future transit investments, but they did not experience the same improvement in transit access during the period I study.

I define treated locations as those within 2 km to any Line 1 or BRT station. Moreover, I define control locations as those within 2 km to any Line 2 or Line 4 station, excluding locations that are also close to Line 1 or BRT stations. This restriction reduces contamination from the treated network. The results are robust to alternative buffer definitions. Figure 10 shows the constructed and planned stations, together with stylized buffers of 1 km for each group.

Figure 10: Constructed and planned Metro and BRT stations.



[37] I leave out Line 3 because it overlaps with many of the same locations as the BRT.

B.2. Empirical strategy

The 2007–2017 waves of the National Household Survey. The main object is the effect of the new infrastructure on women's labor income relative to men's labor income. I estimate this effect for the Metro and the BRT.

The average effects of these infrastructure projects are not particularly informative. This is because improvements in access varied across the city. Fringe locations, initially more isolated, experienced larger gains in access to central employment destinations thanks to the new infrastructure. In contrast, locations closer to the CBD were already near many jobs, so gains in access were smaller. For this reason, I let the effect of infrastructure to vary with distance to the CBD using a rolling-window specification. For each distance c from the CBD, I keep observations in locations whose distance to the CBD lies within h kilometers of c. In the baseline figure, I set $h = 3.0$ kilometers.

Let $q \in \{Metro, BRT\}$, and let $Treat_l^q$ indicate whether location l is treated by mode q. Let $Post_t$ equal one for years after 2009. Within each window, I estimate

$$\begin{aligned} \log y_{ilt} = & \sum_{q \in \{Metro, BRT\}} \beta_q(c) \left(Treat_l^q \times Post_t \times Female_i \right) + \sum_{q \in \{Metro, BRT\}} \mathcal{L}_{ilt}^q \\ & + X_{ilt}' \Gamma(c) + Z_{lt}' \Pi(c) + \alpha_l(c) + \delta_t(c). \end{aligned} \tag{24}$$

Here, y_{ilt} is labor income, $Female_i$ is an indicator for women, $\alpha_l(c)$ are zone fixed effects, and $\delta_t(c)$ are year fixed effects. The terms $\mathcal{L}_{ilt}^q$ collect the lower-order interactions among treatment status, the post period, and gender for each mode q. The vector X_{ilt} includes individual controls: age, years of education, marital status, mother's language, and whether the individual moved from another municipality. The vector Z_{lt} includes zone level and dwelling controls, including elevation, slope, dwelling characteristics, and their interactions with gender and year. I restrict the sample to working age adults between 25 and 65. I estimate equation (24) by Poisson pseudo-maximum likelihood and cluster standard errors at the zone level.

The coefficients $\beta_{Metro}(c)$ and $\beta_{BRT}(c)$ measure the change in women's labor income relative to men's labor income in treated locations around distance c from the CBD, relative to locations close to planned-but-not-yet-built stations. A positive value means that women's labor income increased relative to men's labor income, reducing the male–female earnings gap. Plotting $\beta_q(c)$ across values of c shows where the new infrastructure reduced the gender earnings gap.

I report results using $h = 3.0$ kilometers as the baseline bandwidth. I also report estimates using alternative bandwidths to show that the spatial pattern is not driven by this choice. I also report additional household level outcomes using the same rolling-window logic but without the interaction with $Female_i$. These specifications estimate the effect of the new infrastructure on poverty, rents, income, and expenditure at the household level. I leave the pre-trend analysis to the Census data because the household survey estimates become noisy once I split the sample by event time.

Population Censuses of 1993, 2007 and 2017. I use the Population Censuses as complementary evidence. The Census data allow me to study population and household composition at a fine spatial scale, but they do not contain labor income. The outcomes are measured at the block level and include total population, the singles-to-married ratio, the females-to-males ratio, and the dual-earner-to-male-breadwinner odds among couples. I use the same buffer definitions and bandwidth as in the ENAHO baseline: treatment is defined by a 2 km buffer around constructed Line 1 or BRT stations, while control areas are defined by a 2 km buffer around planned Line 2 or Line 4 stations outside the treatment buffer. The rolling-window bandwidth is $h = 3$ km.

I start with an event study specification using the subset of locations that I can match across 1993, 2007, and 2017.[38] After showing that pre-trends are parallel in this smaller panel, I use the larger 2007–2017 matched panel to estimate difference-in-differences specifications that better exploit the cross-sectional variation in the data.

First, within each window I estimate the event study specification

$$\log y_{lt}(c) = \sum_{q\in\{Metro,BRT\}} \sum_{k\in\{1993,2017\}} \beta_k^q(c) \left(Treat_l^q \times \mathbb{1}\{t=k\}\right) \\ + Z_{lt}'\Pi(c) + \alpha_l(c) + \delta_t(c) + \varepsilon_{lt}, \tag{25}$$

where y_{lt} is a Census outcome at block l in year t, $Treat_l^q$ equals one for blocks within 2 km of a constructed Line 1 station ($q = Metro$) or BRT station ($q = BRT$), $\alpha_l(c)$ are block fixed effects, and $\delta_t(c)$ are year fixed effects. The omitted year is 2007, so $\beta_{1993}^q(c)$ checks whether treated and control blocks were already on different trends before construction, while $\beta_{2017}^q(c)$ measures the post-construction differential change for mode q. The vector Z_{lt} collects block level controls — elevation, slope, the log of block area, and the distance to the closest relevant station — each interacted with year (with 2007 as the omitted year). For each window center c, I keep blocks whose distance to the CBD lies within $h = 3$ kilometers of c, and I restrict the sample to blocks observed in all three Census years (1993, 2007, 2017). I estimate equation (25) by Poisson pseudo-maximum likelihood and cluster standard errors at the block level.

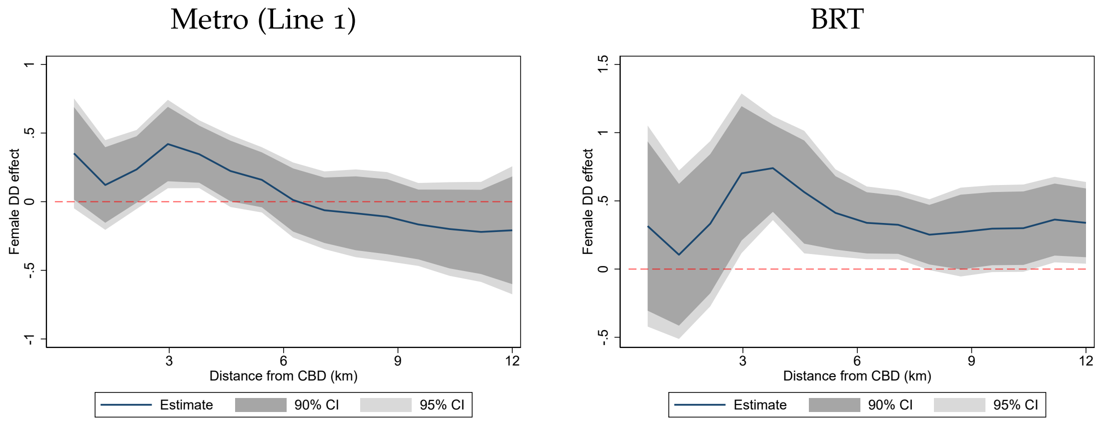
I then use the larger 2007–2017 matched panel to estimate difference-in-differences specifications for Census outcomes. The specification mirrors equation (25) but replaces the event time interactions with $Treat_l^q \times Post_t$, where $Post_t$ equals one in 2017, and restricts the sample to 2007 and 2017. I keep the same rolling-window structure over distance to the CBD with $h = 3$ km, the same block fixed effects, and the same clustering at the block level.

[38] To estimate the event study specification, I need a panel of locations observed in 1993, 2007, and 2017. The match between the 1993 and 2007 geographies is imperfect, partly because many locations observed in 2007 did not exist in 1993. Out of 605 locations in the treatment or control group, I can match 125 across the three Census years. Matched locations tend to be at lower altitude and closer to the CBD than unmatched locations, as expected. However, there are no statistically significant differences in location size, slope, or overall treatment status. While this matched sample is selected and may introduce a selection concern, it is still useful for checking pre-trends. I restrict the event study grid to distances up to 9 km from the CBD because the matched-location panel becomes thin at larger distances. The 2007–2017 DD specification uses the larger matched panel and extends the rolling-window grid to 12 km..

B.3. Results on the male female earnings gap

Baseline results. Figure 11 reports the rolling-window estimate of $\beta_q(c)$ from equation (24) after splitting the treatment indicator by transit mode $q \in \{Metro, BRT\}$. The bandwidth is $h = 3.0$ kilometers. The horizontal axis is the distance to the CBD, while the vertical axis is the difference-in-differences estimate of the female-treat-post coefficient at that distance. The shaded areas are the 90% and 95% confidence intervals.

Figure 11: Rolling-window DD on female labor income relative to male.



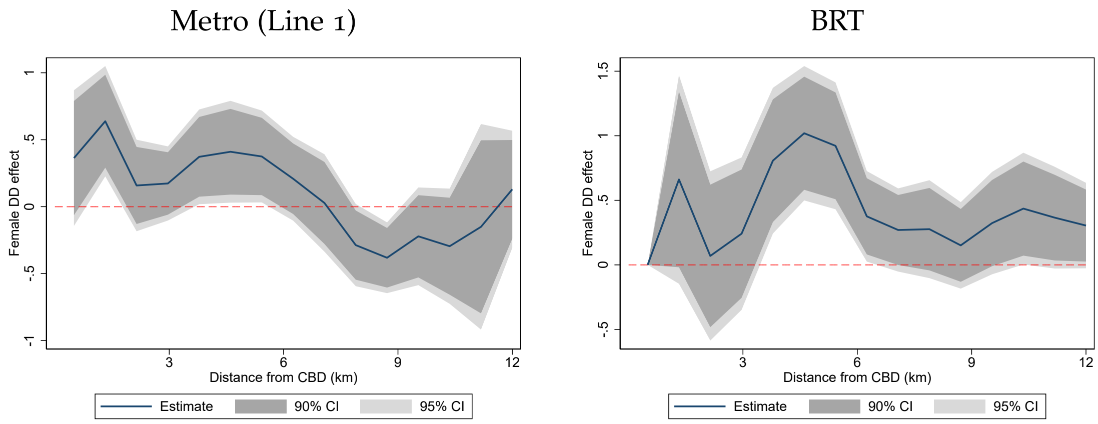
Notes. Each panel plots $\beta_q(c)$ from equation (24): the rolling-window DD on the female–treat–post interaction in mode q, at distance c from the CBD. Rolling-window bandwidth: $h = 3$ km, so each window keeps observations whose distance to the CBD lies within h of c. Treatment is defined as blocks within 2 km of a constructed Line 1 station (left panel) or BRT station (right panel). The control group is blocks within 2 km of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Sample restricted to working age adults (25–65) in the 2007–2017 ENAHO. Estimated by Poisson pseudo-maximum likelihood, with zone fixed effects, year fixed effects, and standard errors clustered at the zone level. Individual level controls include age (and its square), years of education (and its square), marital status, an indicator for non-Spanish mother tongue, and an indicator for whether the individual moved from another municipality. Block- and dwelling level controls include elevation and slope each interacted with year, the four dwelling characteristics (electricity access, dwelling type, floor material, and number of rooms), and all of these interacted with gender. Shaded areas are 90% and 95% confidence intervals.

The estimates show that the effect of the new infrastructure on the gender earnings gap was not uniform across the city (Figure 11). For the BRT, the relative earnings response of women was small in central locations. Beyond a threshold of roughly 3 km, however, women's labor income increased more than men's labor income, narrowing the male–female earnings gap. For the Metro, the pattern was similar, but some of the female interaction estimates were negative in the fringe, suggesting that the gender earnings gap widened, although not significantly. One possible explanation is that the two systems improved access to different types of employment destinations. The BRT connected locations to workplaces that were relatively more female intensive, including San Isidro, the modern employment center of the city. The Metro, by contrast, connected locations more directly to the old historical center, including Gamarra's market, which is more male intensive.

These reduced form patterns are useful beyond documenting the direct effect of the infrastructure because they provide an external check on the mechanism in the structural model.

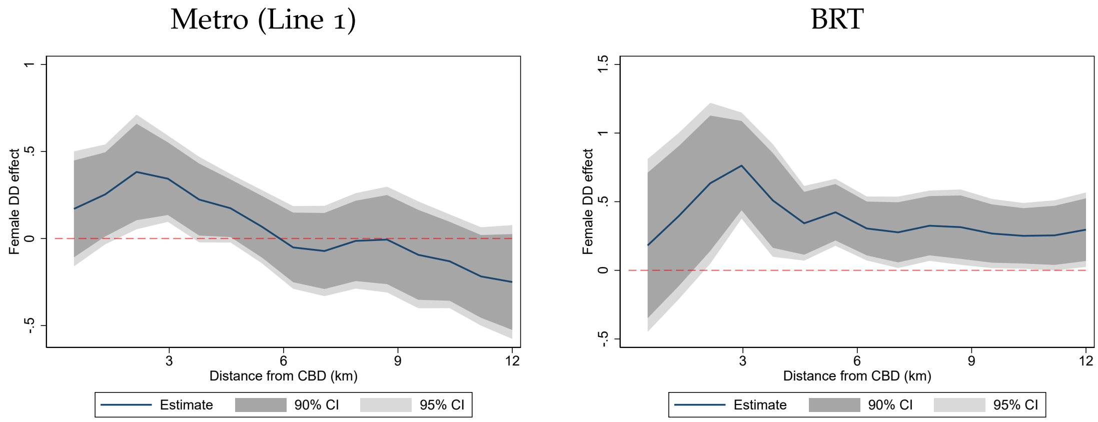
Robustness to the rolling-window bandwidth. Figures 12 and 13 repeat the split rolling-window DD in Figure 11 using a narrower window ($h = 2$ km) and a wider window ($h = 4$ km). Narrower windows show more local detail but produce noisier estimates, while wider windows trade local resolution for tighter confidence bands. I find the same qualitative patterns across bandwidth choices: the BRT produced larger reductions in the gender earnings gap beyond 3 km, while the Metro produced some reductions in the gap at around 3 km, with some gaps widening in fringe locations.

Figure 12: Rolling-window DD on female labor income relative to male, narrower window $h = 2$ km.



Notes. Each panel plots $\beta_q(c)$ from equation (24): the rolling-window DD on the female–treat–post interaction in mode q, at distance c from the CBD. Rolling-window bandwidth: $h = 2$ km, so each window keeps observations whose distance to the CBD lies within h of c. Treatment is defined as blocks within 2 km of a constructed Line 1 station (left panel) or BRT station (right panel). The control group is blocks within 2 km of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Sample restricted to working age adults (25–65) in the 2007–2017 ENAHO. Estimated by Poisson pseudo-maximum likelihood, with zone fixed effects, year fixed effects, and standard errors clustered at the zone level. Individual level controls include age (and its square), years of education (and its square), marital status, an indicator for non-Spanish mother tongue, and an indicator for whether the individual moved from another municipality. Block- and dwelling level controls include elevation and slope each interacted with year, the four dwelling characteristics (electricity access, dwelling type, floor material, and number of rooms), and all of these interacted with gender. Shaded areas are 90% and 95% confidence intervals.

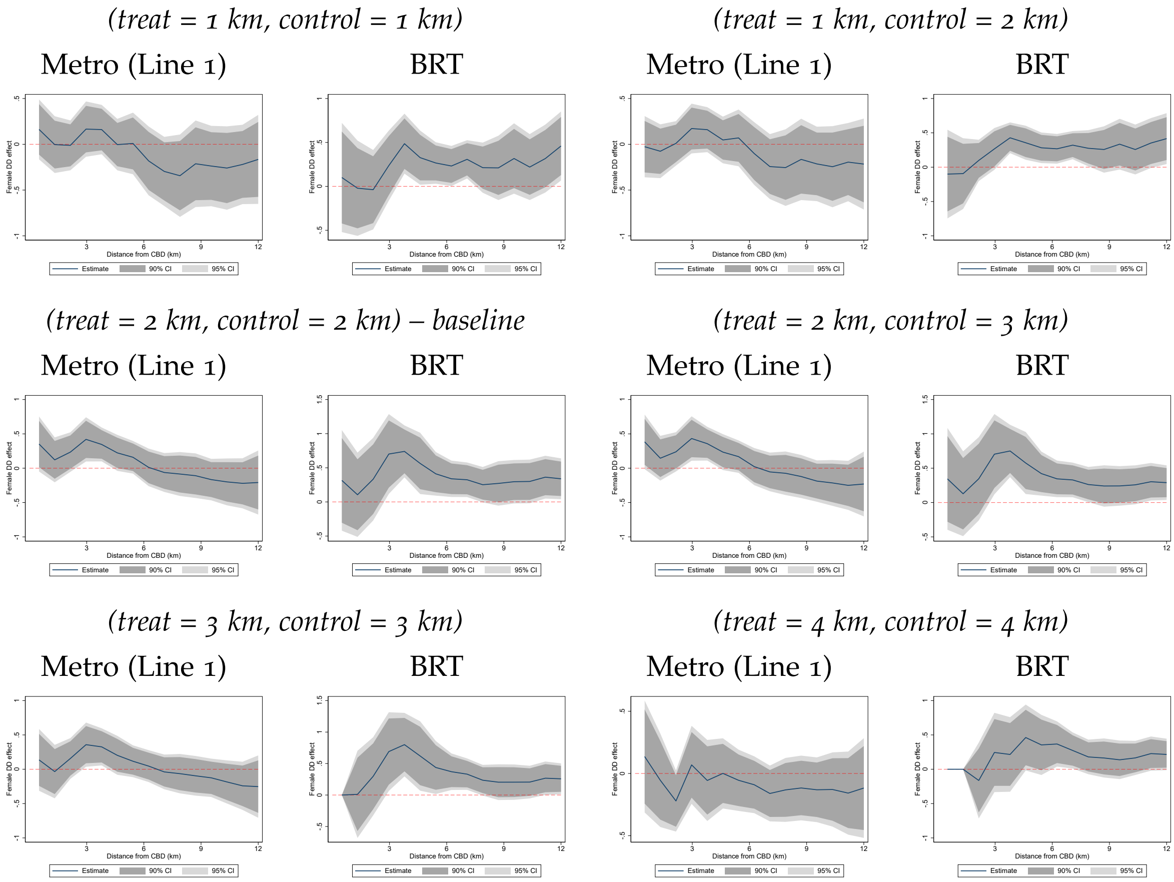
Figure 13: Rolling-window DD on female labor income relative to male, wider window $h = 4$ km.



Notes. Each panel plots $\beta_q(c)$ from equation (24): the rolling-window DD on the female–treat–post interaction in mode q, at distance c from the CBD. Rolling-window bandwidth: $h = 4$ km, so each window keeps observations whose distance to the CBD lies within h of c. Treatment is defined as blocks within 2 km of a constructed Line 1 station (left panel) or BRT station (right panel). The control group is blocks within 2 km of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Sample restricted to working age adults (25–65) in the 2007–2017 ENAHO. Estimated by Poisson pseudo-maximum likelihood, with zone fixed effects, year fixed effects, and standard errors clustered at the zone level. Individual level controls include age (and its square), years of education (and its square), marital status, an indicator for non-Spanish mother tongue, and an indicator for whether the individual moved from another municipality. Block- and dwelling level controls include elevation and slope each interacted with year, the four dwelling characteristics (electricity access, dwelling type, floor material, and number of rooms), and all of these interacted with gender. Shaded areas are 90% and 95% confidence intervals.

Robustness on treatment and control groups definition. In the baseline, I define treatment and control groups as locations within 2 km and 2 km from treatment and control stations, respectively. Here, I show results under different definitions, keeping the $h = 3$ km rolling window. Figure 14 repeats the split rolling-window DD of Figure 11 for six buffer pairs: the baseline (2 km, 2 km) and five alternatives (1 km, 1 km), (1 km, 2 km), (2 km, 3 km), (3 km, 3 km), and (4 km, 4 km). Each row of the figure shows two different buffer pairs side by side, with the Metro and BRT panels for each pair.

Figure 14: Robustness to the treatment and control buffer definitions. Rolling-window DD on female labor income relative to male, by transit mode.

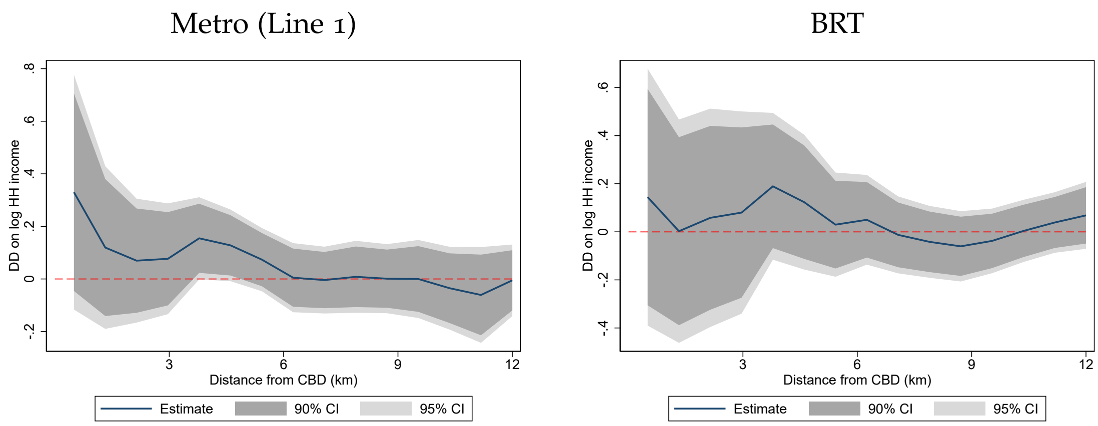


Notes. Each panel plots $\beta_q(c)$ from equation (24): the rolling-window DD on the female–treat–post interaction in mode q, at distance c from the CBD. The rolling-window bandwidth is held fixed at $h = 3$ km, so each window keeps observations whose distance to the CBD lies within h of c. Each pair of panels corresponds to a different (treatment buffer, control buffer) combination, listed above each pair in italics. The $(2\text{ km}, 2\text{ km})$ pair is the baseline. Treatment is defined as blocks within the listed treatment radius of a constructed Line 1 station (Metro panel) or BRT station (BRT panel). Control is blocks within the listed control radius of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Sample restricted to working age adults (25–65) in the 2007–2017 ENAHO. Estimated by Poisson pseudo-maximum likelihood, with zone fixed effects, year fixed effects, and standard errors clustered at the zone level. Individual level controls include age (and its square), years of education (and its square), marital status, an indicator for non-Spanish mother tongue, and an indicator for whether the individual moved from another municipality. Block- and dwelling level controls include elevation and slope each interacted with year, the four dwelling characteristics (electricity access, dwelling type, floor material, and number of rooms), and all of these interacted with gender. Shaded areas are 90% and 95% confidence intervals.

Overall, the same spatial pattern emerges across all six buffer definitions. If anything, the reduction in the gender earnings gap for the Metro is not statistically significant when we define a very narrow treatment buffer of 1 km, or a very wide buffer of 4 km. The possibility that the Metro widened the gap in fringe locations also appears in some specifications although the estimates are not statistically significant. By contrast, the reduction in the gender earnings gap for the BRT is robust across all specifications.

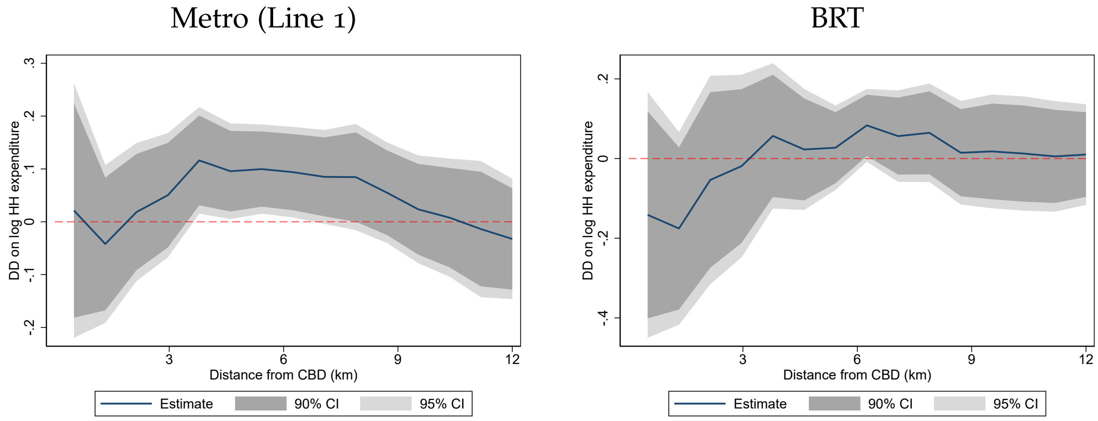
Additional outcomes. I run the same rolling-window on four aggregate household outcomes—total household income, household expenditure, rents, and a poverty indicator—without the female interaction. Sample is restricted to one observation per household. The first three outcomes use Poisson pseudo-maximum likelihood. The binary poverty indicator uses a linear probability model with the same fixed-effect structure. Figures 15–18 report results at $h = 3$ km.

Figure 15: Rolling-window DD on log household income.



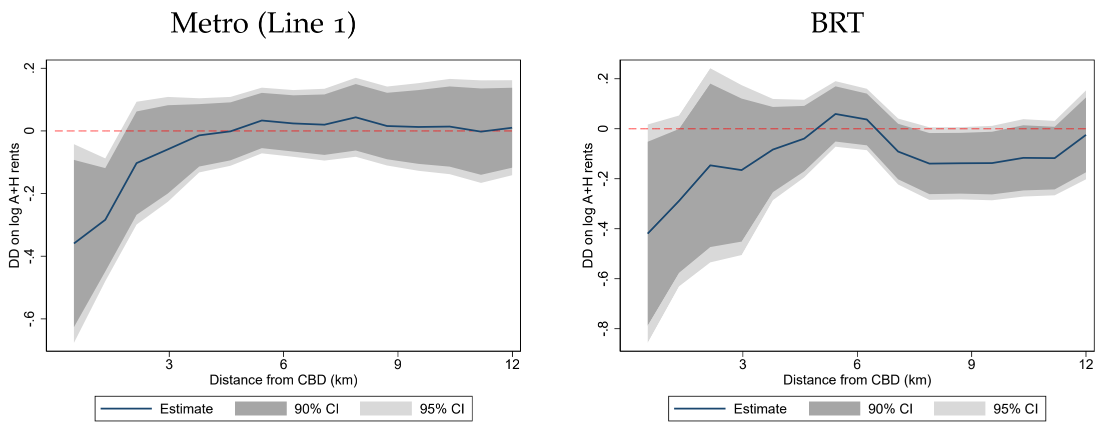
Notes. Each panel plots the rolling-window DD coefficient on the treatment-by-post interaction in mode q at distance c from the CBD. The outcome is log total household income. Rolling-window bandwidth: $h = 3$ km. Treatment is defined as blocks within 2 km of a constructed Line 1 station (left panel) or BRT station (right panel). The control group is blocks within 2 km of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Sample restricted to one observation per household in the 2007–2017 ENAHO. Estimated by Poisson pseudo-maximum likelihood, with zone fixed effects, year fixed effects, and standard errors clustered at the zone level. Household-head controls include age (and its square), years of education (and its square), marital status, an indicator for non-Spanish mother tongue, and an indicator for whether the head moved from another municipality. Block- and dwelling level controls include elevation and slope each interacted with year, plus the four dwelling characteristics (electricity access, dwelling type, floor material, and number of rooms). No gender interactions are included in this specification. Shaded areas are 90% and 95% confidence intervals.

Figure 16: Rolling-window DD on log household expenditure.



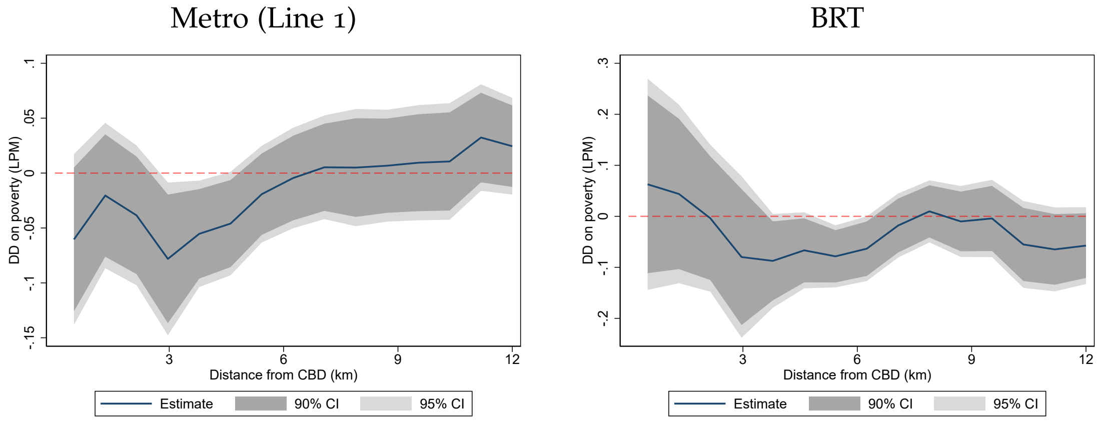
Notes. Each panel plots the rolling-window DD coefficient on the treatment-by-post interaction in mode q at distance c from the CBD. The outcome is log total household expenditure. Rolling-window bandwidth: $h = 3$ km. Treatment is defined as blocks within 2 km of a constructed Line 1 station (left panel) or BRT station (right panel). The control group is blocks within 2 km of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Sample restricted to one observation per household in the 2007–2017 ENAHO. Estimated by Poisson pseudo-maximum likelihood, with zone fixed effects, year fixed effects, and standard errors clustered at the zone level. Household-head controls include age (and its square), years of education (and its square), marital status, an indicator for non-Spanish mother tongue, and an indicator for whether the head moved from another municipality. Block- and dwelling level controls include elevation and slope each interacted with year, plus the four dwelling characteristics (electricity access, dwelling type, floor material, and number of rooms). No gender interactions are included in this specification. Shaded areas are 90% and 95% confidence intervals.

Figure 17: Rolling-window DD on log rents.



Notes. Each panel plots the rolling-window DD coefficient on the treatment-by-post interaction in mode q at distance c from the CBD. The outcome is log monthly rent. Rolling-window bandwidth: $h = 3$ km. Treatment is defined as blocks within 2 km of a constructed Line 1 station (left panel) or BRT station (right panel). The control group is blocks within 2 km of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Sample restricted to one observation per household in the 2007–2017 ENAHO. Estimated by Poisson pseudo-maximum likelihood, with zone fixed effects, year fixed effects, and standard errors clustered at the zone level. Household-head controls include age (and its square), years of education (and its square), marital status, an indicator for non-Spanish mother tongue, and an indicator for whether the head moved from another municipality. Block- and dwelling level controls include elevation and slope each interacted with year, plus the four dwelling characteristics (electricity access, dwelling type, floor material, and number of rooms). No gender interactions are included in this specification. Shaded areas are 90% and 95% confidence intervals.

Figure 18: Rolling-window DD on the poverty rate.



Notes. Each panel plots the rolling-window DD coefficient on the treatment-by-post interaction in mode q at distance c from the CBD. The outcome is a binary indicator for being in poverty. Rolling-window bandwidth: $h = 3$ km. Treatment is defined as blocks within 2 km of a constructed Line 1 station (left panel) or BRT station (right panel). The control group is blocks within 2 km of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Sample restricted to one observation per household in the 2007–2017 ENAHO. Estimated by a linear probability model, with zone fixed effects, year fixed effects, and standard errors clustered at the zone level. Household-head controls include age (and its square), years of education (and its square), marital status, an indicator for non-Spanish mother tongue, and an indicator for whether the head moved from another municipality. Block- and dwelling level controls include elevation and slope each interacted with year, plus the four dwelling characteristics (electricity access, dwelling type, floor material, and number of rooms). No gender interactions are included in this specification. Shaded areas are 90% and 95% confidence intervals.

These complementary outcomes provide context for interpreting the gender earnings results. Figures 15 and 16 show some positive household-resource responses, but these responses are spatially uneven and mostly statistically insignificant. For the Metro, household expenditure and income increases at intermediate distances before falling back toward zero in the fringe. For the BRT, the effects on household resources is generally imprecisely estimated.

Figure 17 shows that rent responses are also localized. Rents fall near the CBD for both modes, especially close to BRT stations, and then move closer to zero at larger distances, with some negative rent responses for the BRT at intermediate-to-outer distances. Poverty responses in Figure 18 are similarly heterogeneous. Around Metro stations, poverty falls at intermediate distances from the CBD, for both the Metro and BRT.

In conclusion, as discussed above, women's relative earnings position improves, but this improvement does not translate mechanically into higher household income, higher expenditure, higher rents, or lower poverty in every part of the city. If anything, the evidence points to higher household resources (higher income, higher expenditure, lower poverty rates) mainly at intermediate distances from the CBD.

B.4. Results on population and household composition

Figures 19–22 report the results. For each outcome, the first column shows the pre-period comparison between 1993 and 2007 using the balanced event study panel. The second column shows the post-treatment comparison between 2007 and 2017 using the same smaller panel, so it can be read directly against the pre-trend estimates. The third column reports the 2007–2017 difference-in-differences using the larger matched panel. I therefore use the second column to compare the post-period pattern with the pre-period check, and the third column as a more holistic description of the 2007–2017 change.

The pre-period panels are mostly reassuring. For total population and the dual-earner-to-male-breadwinner odds, the estimates are statistically indistinguishable from zero across distance bins. The female-to-male ratio also shows no statistically significant pre-trend, although the Metro estimates increase away from the CBD and are close to significance at some distances. The main exception is the singles-to-married ratio: farther from the CBD, treated locations experienced a relative decline in singles compared with married households before the systems opened, for both the Metro and the BRT. Importantly, these pre-period differences occur mostly away from the distance bins where I find the largest reductions in the gender earnings gap, which are concentrated around 3 km from the CBD.

Turning to the post-treatment event study estimates and the DD estimates, the results for total population are similar across the two specifications. Population falls around treated stations for both modes, especially closer to the CBD, and the decline becomes smaller with distance. The DD estimates are more precise and show the same pattern in the larger 2007–2017 panel: negative effects at close and intermediate distances, followed by estimates that move back toward zero in the outer part of the city.

For the singles-to-married ratio, the two specifications also point to a similar pattern. The ratio increases closer to the CBD and falls farther out. The DD estimates make this sign reversal clearer, with more precisely estimated declines in the outer windows. Thus, treated locations farther from the CBD experienced a relative increase in married households compared with singles.

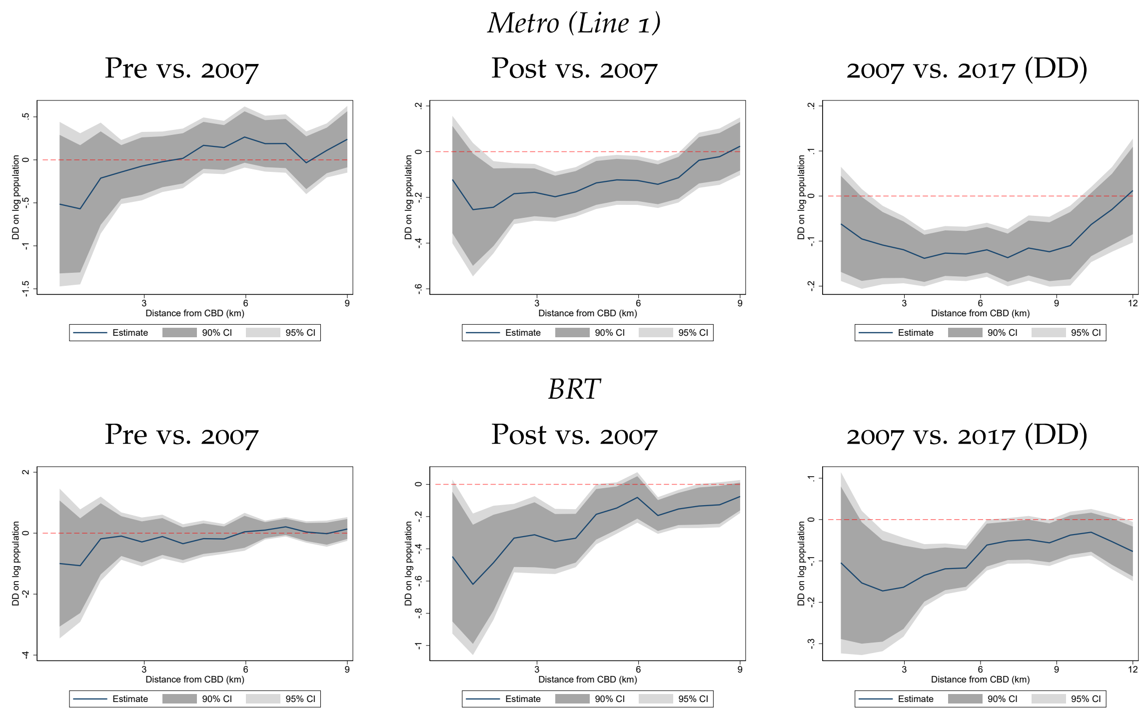
For the female-to-male ratio, neither specification shows a broad increase in the share of women living near treated stations. The estimates are small and often imprecise. If anything, the ratio rises slightly close to the CBD in some windows and becomes negative at intermediate or outer distances, especially in the DD estimates.

The dual-earner-to-male-breadwinner odds show the main difference between the two specifications. In the event study panels, the estimates are noisier and do not show a clear near-CBD increase. For the BRT they become positive only after the first few kilometers, while for the Metro they are close to zero or only weakly positive outside the center. In the larger 2007–2017 DD panel, however, the odds increase around treated stations, especially near the CBD, and then fade toward zero with distance.

In sum, the Census results show that the reduction in the gender earnings gap was accompanied by broader equilibrium changes. In a spatial equilibrium, population movements are informative about the net attractiveness of locations, after accounting for changes in commuting access, rents, amenities, and other local conditions. Treated areas experienced relative population declines, even though female earnings increased in many distance bins. This suggests that the infrastructure did not uniformly raise net welfare for all residents near stations, despite increases in commuting access and may have even decreased the quality of exogenous amenities in these areas. At the same time, the composition results point to heterogeneous effects across household types. Farther from the CBD, married households become relatively more common than singles, suggesting that these locations may have become relatively more attractive to married households. At intermediate distances, the female-to-male ratio falls in some specifications, even though female earnings improve, which suggests that men may have benefited relatively more along other margins in those areas. Finally, the increase in dual-earner-to-male-breadwinner odds is concentrated mainly near the CBD, while the reduction in the gender earnings gap is more spatially widespread.

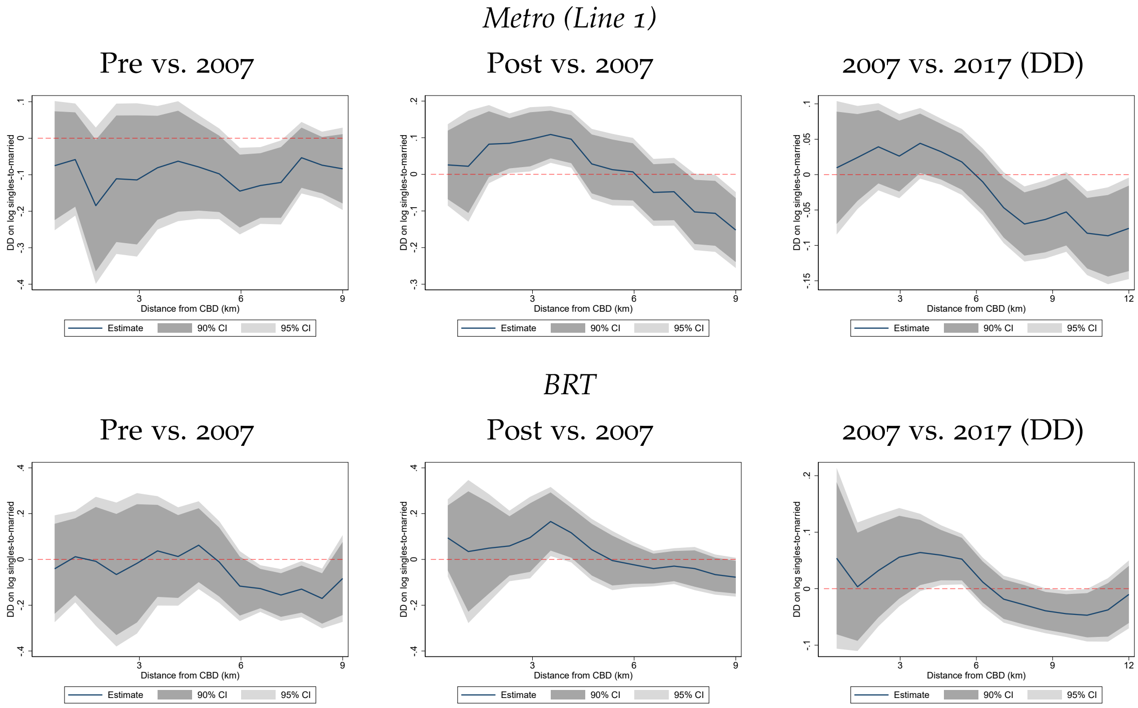
These results highlight the need of the structural analysis to separate the role of commuting access from changes in amenities and rents by removing the transit infrastructure while holding these margins fixed in alternative counterfactuals. These results also suggest that one needs to be careful when planning to use the variation produced by the infrastructure rollout as instruments in the structural model since changes in commute times may be correlated with changes in residential amenities.

Figure 19: Census reduced form: total population.



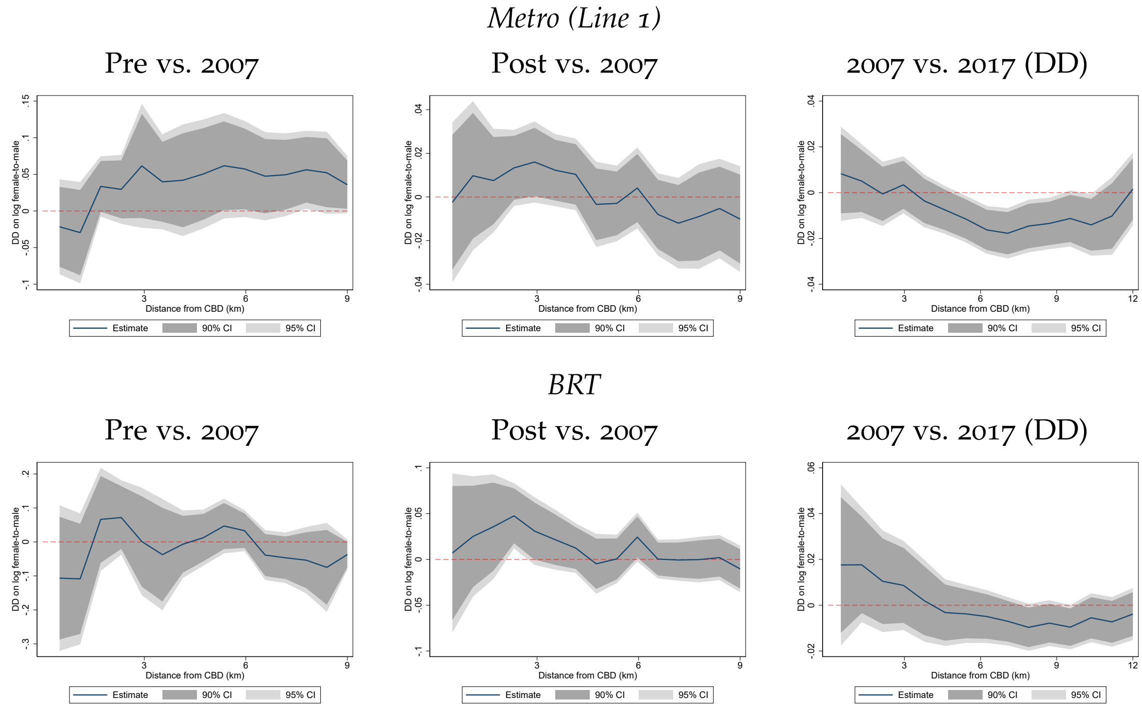
Notes. Rolling-window bandwidth: $h = 3$ km, so for each window center c the sample is restricted to blocks whose distance to the CBD lies within h of c. Treatment is defined as blocks within 2 km of a constructed Line 1 station (top row) or BRT station (bottom row). The control group is blocks within 2 km of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Columns 1 and 2 use the 125 blocks that can be matched across the 1993, 2007, and 2017 Censuses. Column 3 uses the larger 2007–2017 matched panel. Column 1 plots the pre-period placebo coefficient $\beta^q_{1993}(c)$ from equation (25), column 2 plots the post-period coefficient $\beta^q_{2017}(c)$ from the same regression, and column 3 plots the 2007–2017 DD coefficient on $Treat^q_l \times Post_t$. Estimated by Poisson pseudo-maximum likelihood with block fixed effects. Standard errors are clustered at the block level. Block level controls include elevation, slope, the log of block area, and the distance to the closest relevant station, each interacted with year (with 2007 as the omitted year). The matched-panel DD column additionally interacts log distance to the CBD with year. Shaded areas are 90% and 95% confidence intervals.

Figure 20: Census reduced form: singles-to-married ratio.



Notes. Rolling-window bandwidth: $h = 3$ km, so for each window center c the sample is restricted to blocks whose distance to the CBD lies within h of c. Treatment is defined as blocks within 2 km of a constructed Line 1 station (top row) or BRT station (bottom row). The control group is blocks within 2 km of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Columns 1 and 2 use the 125 blocks that can be matched across the 1993, 2007, and 2017 Censuses. Column 3 uses the larger 2007–2017 matched panel. Column 1 plots the pre-period placebo coefficient $\beta^q_{1993}(c)$ from equation (25), column 2 plots the post-period coefficient $\beta^q_{2017}(c)$ from the same regression, and column 3 plots the 2007–2017 DD coefficient on $Treat^q_l \times Post_t$. Estimated by Poisson pseudo-maximum likelihood with block fixed effects. Standard errors are clustered at the block level. Block level controls include elevation, slope, the log of block area, and the distance to the closest relevant station, each interacted with year (with 2007 as the omitted year). The matched-panel DD column additionally interacts log distance to the CBD with year. Shaded areas are 90% and 95% confidence intervals.

Figure 21: Census reduced form: females-to-males ratio.

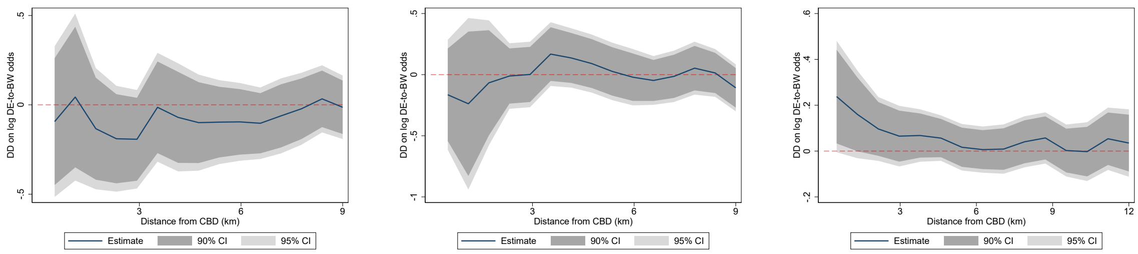


Notes. Rolling-window bandwidth: $h = 3$ km, so for each window center c the sample is restricted to blocks whose distance to the CBD lies within h of c. Treatment is defined as blocks within 2 km of a constructed Line 1 station (top row) or BRT station (bottom row). The control group is blocks within 2 km of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Columns 1 and 2 use the 125 blocks that can be matched across the 1993, 2007, and 2017 Censuses. Column 3 uses the larger 2007–2017 matched panel. Column 1 plots the pre-period placebo coefficient $\beta^q_{1993}(c)$ from equation (25), column 2 plots the post-period coefficient $\beta^q_{2017}(c)$ from the same regression, and column 3 plots the 2007–2017 DD coefficient on $Treat^q_l \times Post_t$. Estimated by Poisson pseudo-maximum likelihood with block fixed effects. Standard errors are clustered at the block level. Block level controls include elevation, slope, the log of block area, and the distance to the closest relevant station, each interacted with year (with 2007 as the omitted year). The matched-panel DD column additionally interacts log distance to the CBD with year. Shaded areas are 90% and 95% confidence intervals.

Figure 22: Census reduced form: dual-earner-to-male-breadwinner odds.

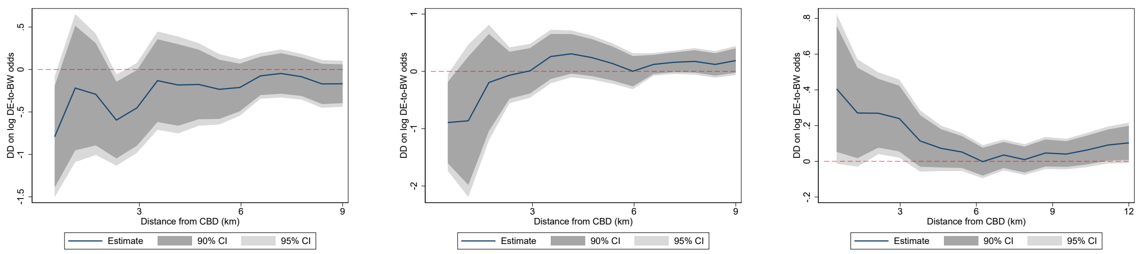
Metro (Line 1)

Pre vs. 2007 | Post vs. 2007 | 2007 vs. 2017 (DD)



BRT

Pre vs. 2007 | Post vs. 2007 | 2007 vs. 2017 (DD)



Notes. Rolling-window bandwidth: $h = 3$ km, so for each window center c the sample is restricted to blocks whose distance to the CBD lies within h of c. Treatment is defined as blocks within 2 km of a constructed Line 1 station (top row) or BRT station (bottom row). The control group is blocks within 2 km of a planned Line 2 or Line 4 station that lie outside the treatment buffer. Columns 1 and 2 use the 125 blocks that can be matched across the 1993, 2007, and 2017 Censuses. Column 3 uses the larger 2007–2017 matched panel. Column 1 plots the pre-period placebo coefficient $\beta^q_{1993}(c)$ from equation (25), column 2 plots the post-period coefficient $\beta^q_{2017}(c)$ from the same regression, and column 3 plots the 2007–2017 DD coefficient on $Treat^q_l \times Post_t$. Estimated by Poisson pseudo-maximum likelihood with block fixed effects. Standard errors are clustered at the block level. Block level controls include elevation, slope, the log of block area, and the distance to the closest relevant station, each interacted with year (with 2007 as the omitted year). The matched-panel DD column additionally interacts log distance to the CBD with year. Shaded areas are 90% and 95% confidence intervals.

C. Implications of Joint Commuting on Within Household Trade offs

C.1. Marginal rate of substitution between spouses' commuting times and wages

In this section, I explain the implications of this choice structure while holding commuting destinations constant. Specifically, I ask two questions. First, holding the husband's commute time constant, how much of the wife's wage would the household be willing to give up to reduce her commute time by one minute? Second, suppose the wife's commute time from origin o to destination j falls by one minute. By how much can the husband's commute time increase while keeping household utility $\widetilde{W}_{oij}$ constant?

To answer the first question, I take a total derivative of household utility with respect to the wife's commuting time and wage, and set it equal to zero.[39] In the interior region, to decrease the wife's commute time by one minute, the household would be willing to give up effective household income $w_{m,i} + w_{f,j}$ by $\zeta_{ij}\kappa^f$ log points:

$$d \log(w_{m,i} + w_{f,j}) = \zeta_{ij}\kappa^f \, dt_{oj}.$$

This feature is also present in the class of spatial models with iceberg costs and constant-returns utility functions. The difference in this setting is that I generalize it in two ways. First, income depends on the wages of both spouses, so each spouse's commuting choice helps determine household resources. Second, the value of reducing the wife's commute is scaled by the Pareto weight ζ_{ij}, which captures how much weight the household places on her utility, and by the elasticity of commuting costs with respect to commuting time, κ^f.

From $d \log(w_{m,i} + w_{f,j}) = \frac{1}{w_{m,i}+w_{f,j}} \, dw_{f,j}$, it follows that

$$\frac{dw_{f,j}}{w_{f,j}} = \zeta_{ij}\kappa^f \frac{w_{m,i} + w_{f,j}}{w_{f,j}} \, dt_{oj}.$$

Thus, as the husband's wage $w_{m,i}$ increases, the household's willingness to give up the wife's wage in exchange for a shorter commute for her also rises, proportionally to $\zeta_{ij}\kappa^f$.

To answer the second question, I take a total derivative of household utility with respect to the spouses' commute times and set it equal to zero. Holding everything else constant, this yields

$$(1 - \zeta_{ij})\kappa^m \, dt_{oi} + \zeta_{ij}\kappa^f \, dt_{oj} = 0,$$

so that

$$dt_{oi} = -\frac{\zeta_{ij}\kappa^f}{(1 - \zeta_{ij})\kappa^m} \, dt_{oj}.$$

This result means that the household can substitute time freed up from the wife's commute with $\frac{\zeta_{ij}\kappa^f}{(1-\zeta_{ij})\kappa^m}(-dt_{oj})$ additional minutes in the husband's commute. Thus, $\frac{\zeta_{ij}\kappa^f}{(1-\zeta_{ij})\kappa^m}$ is the exchange rate at which the wife's commuting time is traded off against the husband's. The more weight the household places on the wife's utility, via $\frac{\zeta_{ij}}{1-\zeta_{ij}}$, the more valuable it is to reduce her commute relative to the husband's. Similarly, the more costly it is for the wife to commute

[39]For simplicity, I hold the Pareto weight constant. Note, however, that changing the wife's wage would also affect the relative position of the husband and wife.

relative to the husband, via $\frac{\kappa^f}{\kappa^m}$, the more the household values reductions in her commute relative to his.

Thus, the level of $\zeta_{ij}\kappa^f$ determines the tradeoff between the wife's wage and her commute, whereas the ratio $\frac{\zeta_{ij}\kappa^f}{(1-\zeta_{ij})\kappa^m}$ determines the tradeoff between the wife's commuting time and the husband's commuting time.

An important implication from this section is that income pooling can make households less responsive to reductions in commuting costs. In an independent commuting model, a reduction in the cost of reaching a high wage destination affects only the worker facing that commuting cost. The worker then reallocates toward that destination whenever the wage gain is large enough relative to the remaining commute cost. In the household model, by contrast, each spouse evaluates workplace choices through household resources. If lower commuting costs make a high wage destination more attractive for the husband, the wife also benefits from the higher household income. This pooled-income gain reduces the incremental value of sending the wife to a farther or higher wage destination as well. The same logic applies in the opposite direction when the wife's workplace generates the income gain. As a result, the household may respond less strongly to commuting-cost reductions than two independent workers would, because part of the gain from improved access is already shared within the household.

C.2. Substitution patterns

In this section, I explain how commuting probabilities respond when one destination pair becomes more attractive. Specifically, suppose the value of one pair (i, j) rises, so that $\mathcal{W}_{oij}$ increases while the value of every other pair stays fixed. This is a pair-specific increase in attractiveness, not a route-level change in commute times, which would shift every pair that shares the same route (studied in the comparative statics of Section 4.2). In that case, the probability that households choose that exact pair rises:

$$\frac{\partial \log \Pi_{ij|o}}{\partial \log \mathcal{W}_{oij}} = \theta^m (1 - \pi^m_{i|o,j}) + \theta^f \pi^m_{i|o,j} (1 - \pi^f_{j|o}) > 0.$$

The interesting question is what happens to the probabilities of other destination pairs.

Now consider another pair that shares the same wife's destination j, but has a different husband's destination, say (l, j) with $l \neq i$. In this case, two forces arise. On the one hand, conditional on the wife commuting to j, the husband substitutes toward destination i and away from destination l. This pushes the probability of (l, j) down. On the other hand, when (i, j) becomes more attractive, the market access term Φ_{oj} rises, which makes wife's destination j more attractive in the first stage. This pushes the probability of all pairs associated with j up, including (l, j). Therefore, pairs that share the same wife's destination can behave either as substitutes or as complements. Which force dominates depends on the relative size of θ^m and θ^f: θ^m governs substitution across the husband's destinations within a given wife's destination, whereas θ^f governs how strongly households reallocate across the wife's destinations in response to changes in the husband's market access term.

$$\frac{\partial \log \Pi_{lj|o}}{\partial \log \mathcal{W}_{oij}} = \pi^m_{i|o,j} \left[-\theta^m + \theta^f \left(1 - \pi^f_{j|o} \right) \right], \qquad l \neq i.$$

Finally, consider a pair associated with a different wife's destination, say (l, n) with $n \neq j$. In this case, the effect is unambiguously negative. When (i, j) becomes more attractive, households become more likely to select wife's destination j in the first stage, which pulls probability mass away from other wife's destinations such as n. Thus, pairs associated with different wife's destinations are always substitutes in this sequential structure.

$$\frac{\partial \log \Pi_{ln|o}}{\partial \log \mathcal{W}_{oij}} = -\theta^f \pi^m_{i|o,j} \pi^f_{j|o} < 0, \qquad n \neq j.$$

In sum, under this sequence of shocks, destination pairs that share the same wife destination can behave either as complements or substitutes, whereas destination pairs associated with different wife destinations are always substitutes.[40]

[40]Reversing the order would give an analogous representation with the roles of husbands and wives reversed.

D. Examples with Different Geographies of Wages

To illustrate the model further, as in the comparative statics above, I study a uniform reduction in commuting costs by lowering $\tau_{\max}$ by 20%. This exercise raises access most strongly for locations on the fringe, especially to destinations near the center. Moreover, the magnitude of this access gain is larger when the initial commuting frictions, summarized by $\tau_{\max}$, are higher. I assume that households are uniformly distributed across all locations, and assume a constant Pareto weight of 0.5, effectively turning off the collective household framework, while still allowing for income pooling.[41]

Figure 23 reports four examples. In all cases, male wages are monocentric and peak at the center. What changes across panels is the geography of female wages, and therefore the geography of the gender earnings gap. The first row is a benchmark monocentric case in which male and female wages have the same spatial pattern. The second row keeps the same monocentric shape for both spouses, but imposes a constant proportional gender gap at all locations. The third row makes the gender gap smallest at the center and larger in the periphery. The fourth row instead lets female wages peak in the periphery, so the gender gap reverses sign outside the center.

The first and second rows provide a useful benchmark. In the first row, male and female wages share the same broad monocentric geography, so lower commuting costs mainly improve access to the same high wage central location for both spouses. For that reason, the effect on the gender earnings gap remains close to zero. In fact, if male and female commuting elasticities were also identical, the effect would be exactly zero in this benchmark. At low values of $\tau_{\max}$, the independent model predicts a small increase in the gender earnings gap, whereas the interdependent model predicts a small decrease. As commuting frictions rise, the two models gradually diverge. In the independent model, cheaper commuting eventually reduces the gap because women, who are more responsive in this calibration, reallocate more strongly toward the high wage center. In the interdependent model, by contrast, that female reallocation becomes weaker when initial commuting frictions are high. The reason is that the household does not evaluate the wife's move toward the center based only on her own wage gain. Once the husband can more easily access the central high wage location, the marginal value of sending the other spouse there also falls, because the household is already benefiting from higher earnings on the husband's side. Since it is more costly to send the wife given her higher sensitivity to commute times, households more often than not send the husband instead. That dampening matters especially for the wife at high commuting frictions, $\tau_{\max}$. As a result, the gap-closing force coming from female reallocation is weaker in the interdependent model than in the independent model. In the second row, where the proportional gender gap is constant across locations, the same basic logic continues to hold, although the divergence between the two models becomes more visible as $\tau_{\max}$ rises. In this case, the gen-

[41] The calibration considers a city with five locations. The commuting-cost elasticity is set to $\kappa^m = \kappa^f = 0.05$, the Fréchet shape parameters are set to $\theta^f = 6$ and $\theta^m = 5$, the expenditure share on the final good is $\beta = 0.8$, and the Pareto weight is held constant at $\zeta = 0.5$. I set $v = 10$, so the economy is effectively close to perfect pooling in these examples.

der gap further weakens the household's incentive to send the wife to the center, since the return to paying the additional commuting cost is smaller for her than for the husband.

The remaining rows show that what matters is not only the size of the gender gap, but also where that gap is located in space. In the third row, where the gender gap is smallest at the center and larger in the periphery, the difference becomes stark: the independent model predicts that lower commuting costs increase the gap over a wide range, whereas the interdependent model predicts the opposite. Intuitively, the central location is both the highest wage location and the most equal one. When commuting frictions are high, reducing them disproportionately improves access to that center. In the dependent model, this matters not only through direct access to higher wages, but also because the household composition channel reallocates spouses' choices more strongly toward destination pairs associated with that central location. As a result, the dependent model places more weight on the high wage, low-gap center, which makes the effect on the gender earnings gap more negative. In the fourth row, where female wages are relatively stronger in the periphery, the sign reverses: lower commuting costs now tend to increase the gender earnings gap in both models, especially at $\tau_{\max}$. The reason is that cheaper commuting makes it easier for husbands to reach the high wage center, while wives already have strong earnings opportunities in the periphery. This effect is stronger in the independent model, where the husband reallocates more aggressively toward the center. In the dependent model, that force is dampened because the wife's peripheral earnings partly relax the household's incentive to send the husband to the far central job.

Taken together, these examples show that the effect of lower commuting costs on the gender earnings gap depends not only on the overall level of gender inequality in wages, but also on how that inequality is distributed across space relative to the geography of commuting costs reductions.

Finally, these differences in household reallocation also matter in the aggregate. In all four example geographies, aggregate GDP rises less under interdependent commuting than under independent commuting after a uniform reduction in commuting costs (Appendix Figure 25). The reason is that within household reallocation dampens each spouse's responsiveness to changes in commuting frictions, so cheaper commuting translates into smaller shifts toward high wage locations. This overstatement by the independent model is especially large when initial commuting frictions are high, that is, when $\tau_{\max}$ is large. This result underscores the importance of accounting for dual earner households when evaluating the aggregate welfare effects of new transit infrastructure.

D.1. Decomposition across geographies

This section reports the decomposition of the gender earnings gap response from Figure 23 into the direct male reallocation term and the indirect composition term derived in Section 4.2.

D.2. GDP response across geographies

Under interdependent commuting, dual earner households are less responsive to changes in commuting frictions than they are under independent commuting, because the household

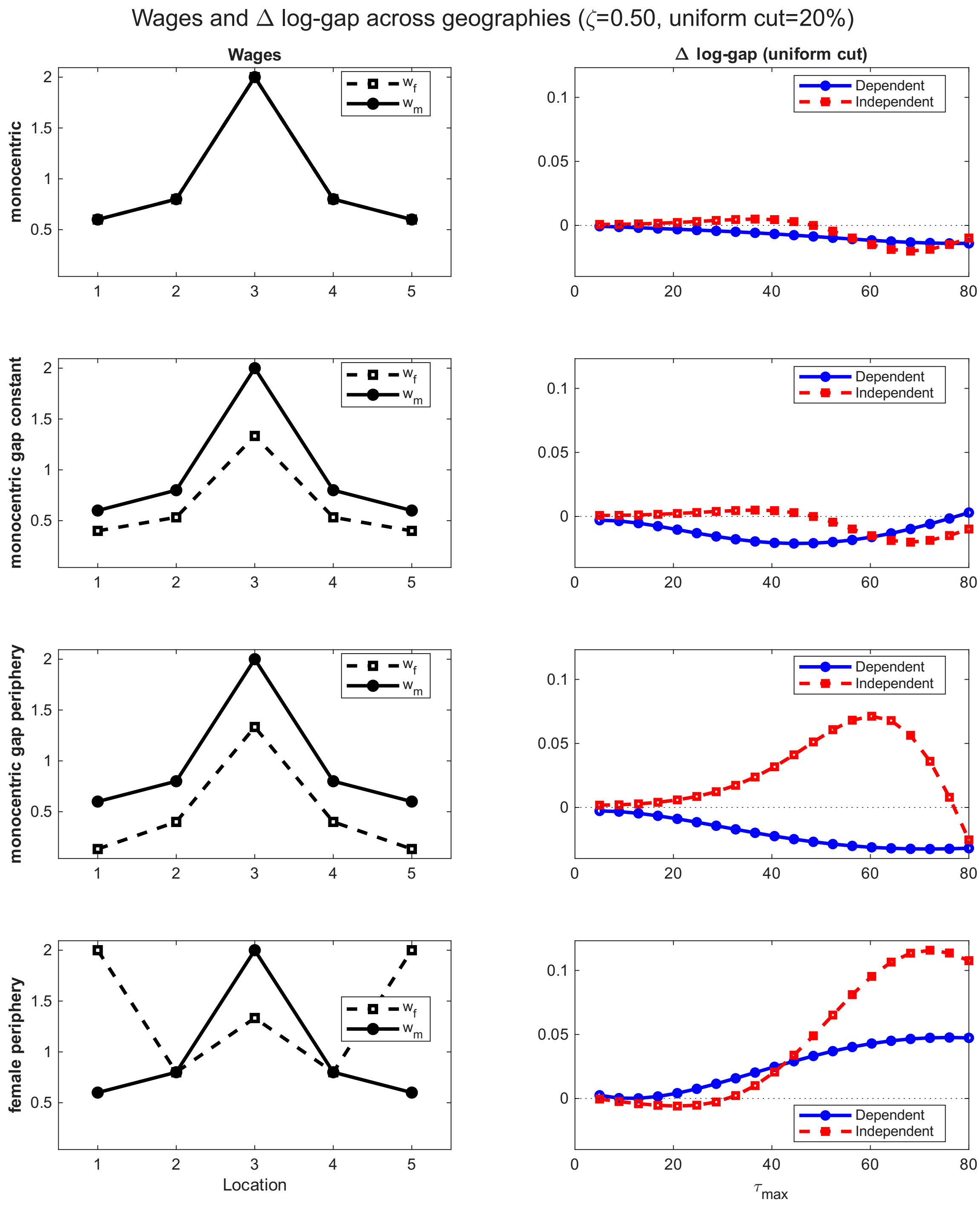


Figure 23: Wage geographies and the gender earnings gap response to a uniform commuting-cost cut.

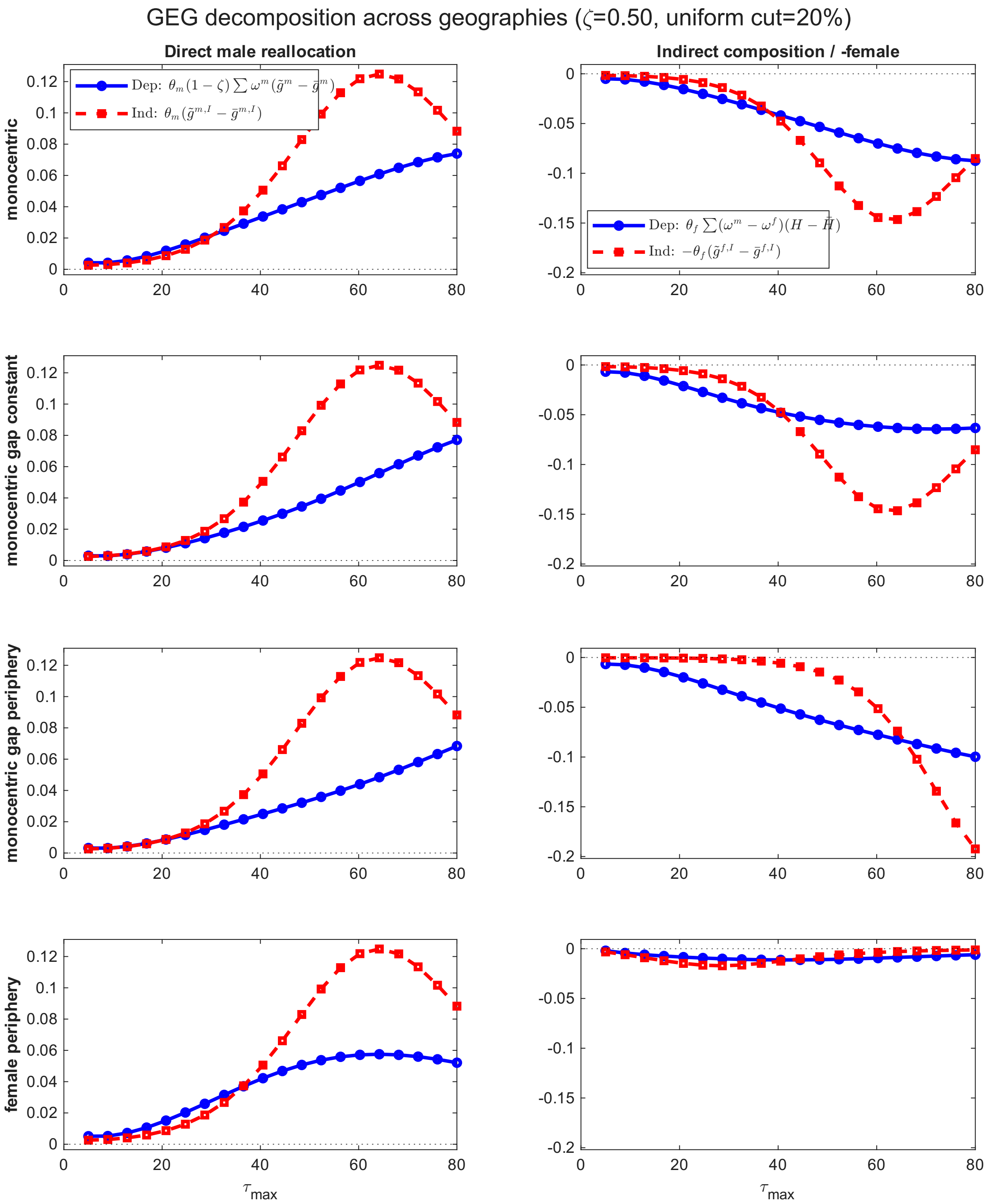


Figure 24: Decomposition of the gender earnings gap response across geographies.

reallocation channel dampens each spouse's sensitivity to their own transit gains. As a result, the GDP growth effects of new transit infrastructure are smaller in the interdependent model than in the independent one. This prediction holds across all four example geographies in Figure 25: in every panel the interdependent curve lies below the independent curve for the same $\tau_{\max}$. This underscores the importance of accounting for dual earner households in welfare and GDP assessments of new transit infrastructure, since neglecting the within household reallocation channel overstates the aggregate income gains.

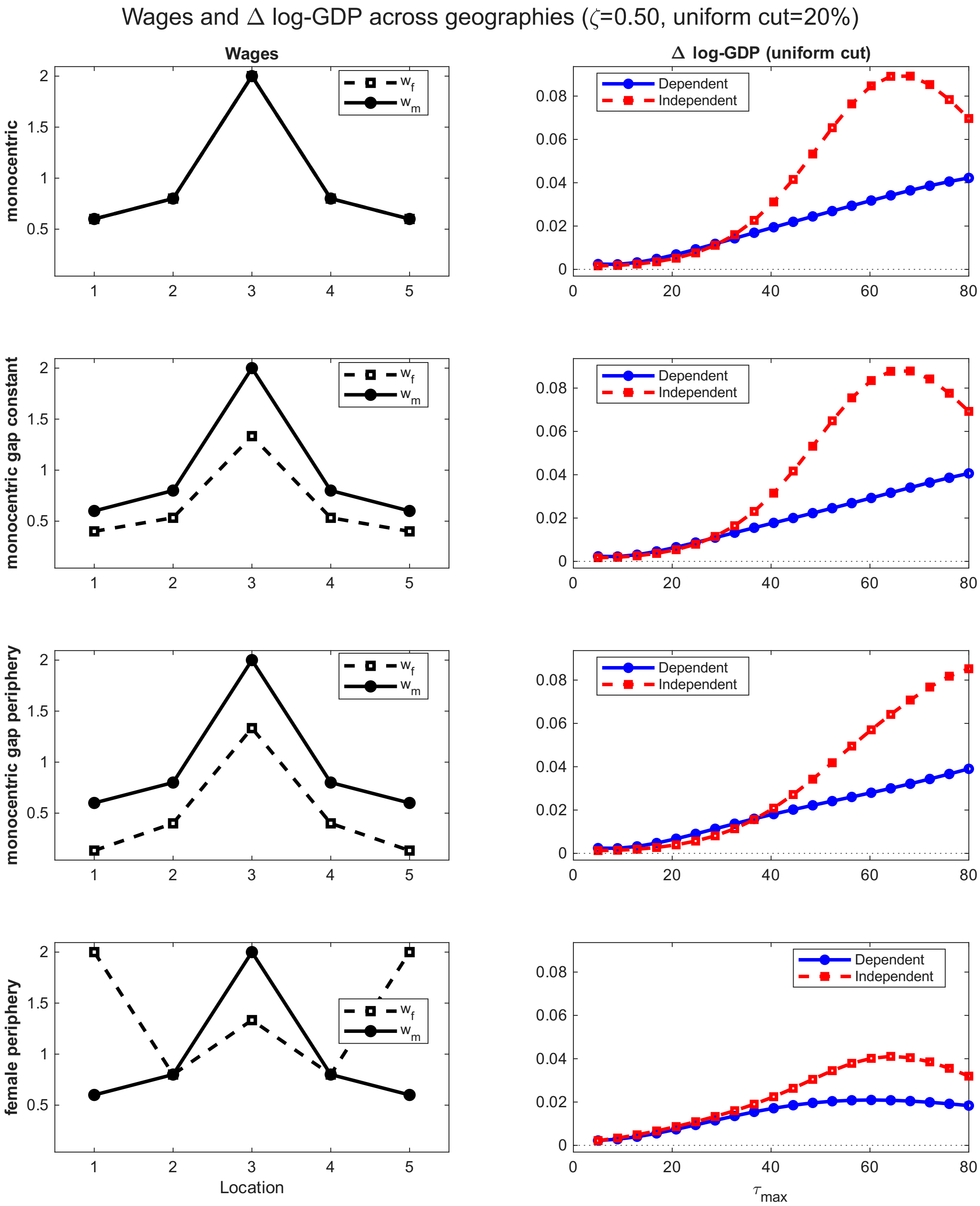


Figure 25: Wages and the aggregate GDP response to a uniform commuting-cost cut, across geographies.

E. Full Model

This appendix presents the full quantitative model that links the empirical analysis to the theory.

E.1. Environment

The city consists of a finite set of locations indexed by $o, d = 1, \ldots, S$. When convenient, I use o to denote residence, i the husband's workplace, and j the wife's workplace. Commuting times between locations are denoted by t_{od}.

There are three household groups: single men, single women, and married couples. A mass N^m of single men and a mass N^f of single women choose where to live and where to work. In addition, a mass N^{mf} of married couples choose a common residence. Within married households, I allow for two arrangements: dual earner households, in which both spouses work, and male-breadwinner households, in which only the husband works. For simplicity, I rule out female-breadwinner households.[42]

Each location o is endowed with a fixed stock of floorspace $\bar{H}_o$. A share $\chi_o \in [0, 1]$ is allocated to residential use, and the remaining share $1 - \chi_o$ is allocated to commercial use. Locations also differ in exogenous amenities for single men, single women, and couples, denoted by $\{u_o^m, u_o^f, u_o^{mf}\}$. Locations also differ in household-production productivity ξ_{o0} when the wife stays at home; in exogenous sectoral productivities $\{A_{os}\}_{s=1}^{K}$; and in a land use wedge ι_o that affects the relative return to commercial floorspace.

Furthermore, each location houses multiple industries which produce a consumption good using labor and commercial floorspace. Each location-by-industry's consumption good is aggregated into a final good with CES preferences. Males and females are imperfect substitutes in the production function. Landowners choose how to allocate a fixed amount of floorspace across residential and commercial use.

E.2. Single households

I begin with singles. To keep the notation distinct from the married commuting block, I use $\varkappa^g$ for the iceberg commuting-cost parameter and ϑ^g for the Fréchet shape parameter governing workplace choice, where $g \in \{m, f\}$ indexes gender.

Preferences. A single worker of gender g who lives in o and works in d chooses private consumption and residential floorspace. After solving this within location consumption problem, indirect utility can be written as

$$\mathcal{V}_{od}^g(\omega) = u_o^g \phi_o^g(\omega) \frac{w_{g,d}\, \epsilon_d^g(\omega)}{P^{\beta^g} r_{R_o}^{1-\beta^g}\, \delta_{od}^g}, \qquad \delta_{od}^g \equiv \exp(\varkappa^g t_{od}), \tag{26}$$

[42]To simplify, in this paper couples are comprised of a male and a female. Another simplifying assumption usually done in the literature on family economics is that the husband always works. In my data, female breadwinner households constitute just 1.3% of all households. This is why I turn off this margin: it does not seem to be quantitatively important but would add greater complexity to the model.

where $w_{g,d}$ is the wage of gender g at destination d, β^g is the expenditure share on the final good, $\epsilon_d^g(\omega)$ is an idiosyncratic taste shock over workplace destinations, and $\phi_o^g(\omega)$ is an idiosyncratic taste shock over residential locations. Hence, singles choose where to work by trading off higher wages against longer commutes, and choose where to live by trading off expected labor market access against housing costs and local amenities.

Expenditure shares. Conditional on the residence-workplace pair (o, d), optimal expenditure is

$$PC_{od}^g = \beta^g w_{g,d}, \qquad r_{R_o} H_{R,od}^g = (1 - \beta^g) w_{g,d}. \tag{27}$$

Commuting choice. Conditional on residence o, singles draw workplace shocks $\epsilon_d^g(\omega)$ iid Fréchet with shape parameter ϑ^g. Standard Fréchet algebra implies

$$\varpi_{d|o}^g = \frac{\left(\frac{w_{g,d}}{\delta_{od}^g}\right)^{\vartheta^g}}{\Psi_o^g}, \qquad \Psi_o^g \equiv \sum_{\ell=1}^{S} \left(\frac{w_{g,\ell}}{\delta_{o\ell}^g}\right)^{\vartheta^g}. \tag{28}$$

The term Ψ_o^g is the commuting market access term for singles of gender g who live in o. It summarizes the attractiveness of all workplace options available from residence o. The parameter ϑ^g governs how sensitive commuting choices are to wage and commute differences across destinations: when the dispersion in workplace tastes is lower, that is, when ϑ^g is higher, commuting choices respond more strongly to differences in net wages.

Residential choice. Before workplace shocks are realized, singles draw residential shocks $\phi_o^g(\omega)$, also iid Fréchet, with shape parameter η^g. Let

$$\Omega_o^g \equiv \left(\Psi_o^g\right)^{1/\vartheta^g}. \tag{29}$$

Then the probability that a single of gender g lives in o is

$$\lambda_o^g = \frac{\left(\frac{u_o^g \Omega_o^g}{P^{\beta^g} r_{R_o}^{1-\beta^g}}\right)^{\eta^g}}{\sum_{n=1}^{S} \left(\frac{u_n^g \Omega_n^g}{P^{\beta^g} r_{R_n}^{1-\beta^g}}\right)^{\eta^g}}. \tag{30}$$

Hence, singles choose residence by trading off local amenities, access to jobs, and housing costs.

Ex ante welfare. After integrating over workplace shocks, utility from residence o is

$$\mathcal{U}_o^g(\omega) = \frac{\gamma_E}{\vartheta^g} + \log\left(\frac{u_o^g \Omega_o^g \phi_o^g(\omega)}{P^{\beta^g} r_{R_o}^{1-\beta^g}}\right), \tag{31}$$

where γ_E is the Euler–Mascheroni constant.

Define

$$\mathcal{R}^g \equiv \left[\sum_{o=1}^{S}\left(\frac{u_o^g \Omega_o^g}{P^{\beta^g} r_{R_o}^{1-\beta^g}}\right)^{\eta^g}\right]^{1/\eta^g}. \tag{32}$$

After integrating over residential shocks, ex ante utility is

$$\mathcal{U}^g = \frac{\gamma_E}{\vartheta^g} + \frac{\gamma_E}{\eta^g} + \log \mathcal{R}^g. \tag{33}$$

E.3. Married households

I now turn to couples. The key novelty is that commuting choices are interdependent within dual earner households because spouses solve a collective household problem with partial pooling.

E.3.1. Dual earner household problem

Preferences and budget set. Conditional on residence o, husband destination i, and wife destination j, let $\epsilon_{m,i}(\omega)$ and $\epsilon_{f,j}(\omega)$ denote the husband- and wife specific commuting shocks. Let $\varsigma_{o1}(\omega)$ denote the preference shock associated with the dual earner arrangement, and let $\varphi_o^{mf}(\omega)$ denote the couple's residential-location shock. The participation and residential shocks are common to both spouses.

The spouses' individual utilities conditional on the dual earner arrangement are

$$\begin{aligned}
V_{oij,1}^m(\omega) = {} & \log\left(u_o^{mf}\frac{C_{m,oij}^{\beta}\left(H_{oij}^R\right)^{1-\beta}}{d_{oi}^m}\right) + \frac{1}{1-\zeta_{ij}}\log \epsilon_{m,i}(\omega) \\
& + \log \varsigma_{o1}(\omega) + \log \varphi_o^{mf}(\omega), \\
V_{oij,1}^f(\omega) = {} & \log\left(u_o^{mf}\frac{C_{f,oij}^{\beta}\left(H_{oij}^R\right)^{1-\beta}}{d_{oj}^f}\right) + \frac{1}{\zeta_{ij}}\log \epsilon_{f,j}(\omega) \\
& + \log \varsigma_{o1}(\omega) + \log \varphi_o^{mf}(\omega).
\end{aligned} \tag{34}$$

Here,

$$d_{oi}^m \equiv \exp\left(\kappa^m t_{oi}\right), \qquad d_{oj}^f \equiv \exp\left(\kappa^f t_{oj}\right). \tag{35}$$

The inverse Pareto weight scaling of the commuting shocks ensures that each spouse specific shock enters the weighted household objective with coefficient one. By contrast, the participation and residential shocks enter both spouses' utilities with coefficient one because they are common to the household.

Although all shocks are displayed together in equation (34), they are realized according to the model's nested timing. The residential shocks are observed when the couple chooses its residence. Conditional on residence, the participation shocks are observed when the household chooses its labor force participation arrangement. Conditional on residence and participation, the commuting shocks are observed when workplace destinations are chosen.

Total expenditure satisfies

$$PC_{m,oij} + PC_{f,oij} + r_{Ro}H^R_{oij} = w_{m,i} + w_{f,j}, \tag{36}$$

and partial pooling imposes

$$PC_{m,oij} \leq vw_{m,i}, \qquad PC_{f,oij} \leq vw_{f,j}. \tag{37}$$

Collective representation. Any efficient allocation can be represented as the solution to a weighted household problem. Defining expenditure units

$$x_{m,oij} \equiv PC_{m,oij}, \qquad x_{f,oij} \equiv PC_{f,oij}, \qquad z_{oij} \equiv r_{Ro}H^R_{oij}, \qquad y_{oij} \equiv w_{m,i} + w_{f,j}, \tag{38}$$

the household solves

$$\begin{aligned} \max_{x_{m,oij},x_{f,oij},z_{oij}} \quad & \widetilde{W}^e_{oij,1}(\omega) \equiv (1-\zeta_{ij})V^m_{oij,1}\left(\frac{x_{m,oij}}{P}, \frac{z_{oij}}{r_{Ro}}; \omega\right) \\ & \qquad + \zeta_{ij}V^f_{oij,1}\left(\frac{x_{f,oij}}{P}, \frac{z_{oij}}{r_{Ro}}; \omega\right) \\ \text{s.t.} \quad & x_{m,oij} + x_{f,oij} + z_{oij} = y_{oij}, \\ & x_{m,oij} \leq vw_{m,i}, \qquad x_{f,oij} \leq vw_{f,j}. \end{aligned} \tag{39}$$

Here, $\zeta_{ij} \in (0,1)$ is the Pareto weight on the wife's utility.

Regimes and conditional housing demand. Let

$$\underline{a}_{ij} \equiv \frac{\beta(1-\zeta_{ij})}{v - \beta(1-\zeta_{ij})}, \qquad \bar{a}_{ij} \equiv \frac{v - \beta\zeta_{ij}}{\beta\zeta_{ij}}. \tag{40}$$

There are three regimes.

If neither pooling constraint binds, the optimal allocation is

$$\left(x^I_{m,oij}, x^I_{f,oij}, z^I_{oij}\right) = \left(\beta(1-\zeta_{ij})y_{oij}, \beta\zeta_{ij}y_{oij}, (1-\beta)y_{oij}\right), \tag{41}$$

which is feasible when

$$\underline{a}_{ij}w_{f,j} \leq w_{m,i} \leq \bar{a}_{ij}w_{f,j}. \tag{42}$$

If the husband's pooling constraint binds, the optimal allocation is

$$\left(x^K_{m,oij}, x^K_{f,oij}, z^K_{oij}\right) = \left(vw_{m,i}, \frac{\beta\zeta_{ij}}{1-\beta+\beta\zeta_{ij}}\left[w_{f,j} + (1-v)w_{m,i}\right], \frac{1-\beta}{1-\beta+\beta\zeta_{ij}}\left[w_{f,j} + (1-v)w_{m,i}\right]\right), \tag{43}$$

which applies when

$$w_{m,i} < \underline{a}_{ij}w_{f,j}.$$

If the wife's pooling constraint binds, the optimal allocation is

$$\left(x^{K'}_{m,oij}, x^{K'}_{f,oij}, z^{K'}_{oij}\right) = \left(\frac{\beta(1-\zeta_{ij})}{1-\beta\zeta_{ij}}\left[w_{m,i} + (1-v)w_{f,j}\right], vw_{f,j}, \frac{1-\beta}{1-\beta\zeta_{ij}}\left[w_{m,i} + (1-v)w_{f,j}\right]\right), \tag{44}$$

which applies when

$$w_{m,i} > \overline{a}_{ij} w_{f,j}.$$

In each regime, residential floorspace demand is

$$H^R_{oij} = \frac{z_{oij}}{r_{Ro}}. \tag{45}$$

The regime specific wage aggregator is

$$\Xi_{ij} = \begin{cases} K_m w_{m,i}^{\beta(1-\zeta_{ij})} \left[w_{f,j} + (1-v) w_{m,i} \right]^{1-\beta+\beta\zeta_{ij}}, & \text{if } w_{m,i} < \underline{a}_{ij} w_{f,j}, \\ K_I \left(w_{m,i} + w_{f,j} \right), & \text{if } \underline{a}_{ij} w_{f,j} \leq w_{m,i} \leq \overline{a}_{ij} w_{f,j}, \\ K_f \left[w_{m,i} + (1-v) w_{f,j} \right]^{1-\beta\zeta_{ij}} w_{f,j}^{\beta\zeta_{ij}}, & \text{if } w_{m,i} > \overline{a}_{ij} w_{f,j}. \end{cases} \tag{46}$$

The constants K_I, K_m, and K_f collect terms that depend only on (β, ζ_{ij}, v).

Deriving household utility. Conditional on (o, i, j), the shocks and the residential amenity do not affect the household's consumption and housing allocation. Substituting the optimal allocation into equation (39) and exponentiating yields

$$W_{oij}(\omega) \equiv \exp\left\{ \widetilde{W}^e_{oij,1}(\omega) \right\} = \frac{u_o^{mf} \Xi_{ij}}{P^\beta r_{Ro}^{1-\beta} \left(d^m_{oi}\right)^{1-\zeta_{ij}} \left(d^f_{oj}\right)^{\zeta_{ij}}} \epsilon_{m,i}(\omega) \epsilon_{f,j}(\omega) \varsigma_{o1}(\omega) \varphi_o^{mf}(\omega). \tag{47}$$

E.3.2. Dual earner commuting

I assume that in dual earner households the wife chooses j first, anticipating the husband's market access term. This ordering is a tractable reduced form device that preserves interdependence and allows male and female commuting elasticities to differ.

At the commuting stage, conditional on residence and the dual earner arrangement, the household first observes the wife's destination shocks $\{\epsilon_{f,j}(\omega)\}_{j=1}^S$ and chooses her workplace j, anticipating the husband's subsequent workplace choice. Conditional on j, the household then observes the husband's destination shocks $\{\epsilon_{m,i}(\omega)\}_{i=1}^S$ and chooses his workplace i.

The residential amenity, the price index, housing costs, the participation shock, and the residential shock are common across workplace pairs. They therefore do not affect the household's commuting choices. I assume that the commuting shocks $\epsilon_{m,i}(\omega)$ are iid Fréchet with shape parameter θ^m, while $\epsilon_{f,j}(\omega)$ are iid Fréchet with shape parameter θ^f.

Conditional on the wife working in j, the husband's commuting probability is

$$\pi^m_{i|o,j} = \frac{\left(\dfrac{\Xi_{ij}}{\left(d^m_{oi}\right)^{1-\zeta_{ij}} \left(d^f_{oj}\right)^{\zeta_{ij}}} \right)^{\theta^m}}{\Phi_{oj}}, \qquad \Phi_{oj} \equiv \sum_{\ell=1}^{S} \left(\frac{\Xi_{\ell j}}{\left(d^m_{o\ell}\right)^{1-\zeta_{\ell j}} \left(d^f_{oj}\right)^{\zeta_{\ell j}}} \right)^{\theta^m}. \tag{48}$$

The term Φ_{oj}^{1/θ^m} is the husband's conditional market access term when the wife works in j.

The wife chooses j anticipating the husband's conditional market access. Her commuting probability in dual earner households is

$$\pi^f_{j|o,1} = \frac{\Phi_{oj}^{\theta^f/\theta^m}}{\sum_{n=1}^{S} \Phi_{on}^{\theta^f/\theta^m}}. \tag{49}$$

The joint commuting probability of the destination pair (i, j) is

$$\Pi_{ij|o} = \pi^m_{i|o,j} \pi^f_{j|o,1}, \tag{50}$$

and the husband's unconditional commuting probability in dual earner households is

$$\pi^m_{i|o,1} = \sum_{j=1}^{S} \Pi_{ij|o}. \tag{51}$$

The dual earner market access term at residence o is

$$\Omega_{o1} \equiv \left(\sum_{j=1}^{S} \Phi_{oj}^{\theta^f/\theta^m} \right)^{1/\theta^f}. \tag{52}$$

The dual earner commuting probabilities are derived in Appendix L.

Conditional on the participation and residential shocks, household utility under the dual earner arrangement is

$$W_{o1}(\omega) = \frac{u_o^{mf} \Omega_{o1}}{P^\beta r_{Ro}^{1-\beta}} \varsigma_{o1}(\omega) \varphi_o^{mf}(\omega). \tag{53}$$

E.3.3. Male-breadwinner households

If the wife does not work, the household derives utility from the husband's labor income and from household production with productivity ξ_{o0}. Let $\varsigma_{o0}(\omega)$ denote the common preference shock associated with the male-breadwinner arrangement. Conditional on residence o and husband destination i, household utility is

$$W_{oi,0}(\omega) = \xi_{o0} \frac{u_o^{mf} w_{m,i}}{P^\beta r_{Ro}^{1-\beta} d_{oi}^m} \epsilon_{m,i}(\omega) \varsigma_{o0}(\omega) \varphi_o^{mf}(\omega). \tag{54}$$

Conditional on residence and the male-breadwinner arrangement, all terms other than the husband's wage, commuting cost, and commuting shock are common across workplace destinations. The husband's commuting probability is therefore

$$\pi^m_{i|o,0} = \frac{\left(\frac{w_{m,i}}{d_{oi}^m} \right)^{\theta^m}}{\Phi_{o0}}, \qquad \Phi_{o0} \equiv \sum_{\ell=1}^{S} \left(\frac{w_{m,\ell}}{d_{o\ell}^m} \right)^{\theta^m}. \tag{55}$$

The breadwinner market access term at residence o is

$$\Omega_{o0} \equiv \Phi_{o0}^{1/\theta^m}. \tag{56}$$

Conditional on the participation and residential shocks, household utility under the male-breadwinner arrangement is

$$W_{o0}(\omega) = \xi_{o0} \frac{u_o^{mf} \Omega_{o0}}{P^\beta r_{Ro}^{1-\beta}} \varsigma_{o0}(\omega) \varphi_o^{mf}(\omega). \tag{57}$$

E.3.4. Female labor force participation

Conditional on residence o, the household compares utility under the dual earner arrangement in equation (53) with utility under the male-breadwinner arrangement in equation (57). The residential amenity, the price index, housing costs, and the residential shock are common across participation arrangements and therefore do not affect the participation choice.

Suppose that $\varsigma_{o1}(\omega)$ and $\varsigma_{o0}(\omega)$ are iid Fréchet with shape parameter ν. The probability that a couple living in o is dual earner is

$$\mu_o = \frac{\Omega_{o1}^{\nu}}{\Omega_{o1}^{\nu} + (\xi_{o0}\Omega_{o0})^{\nu}}. \tag{58}$$

The participation probability is derived in Appendix L.

The marital market access term at residence o is

$$\Omega_o^{mf} = \left[\Omega_{o1}^{\nu} + (\xi_{o0}\Omega_{o0})^{\nu}\right]^{1/\nu}. \tag{59}$$

Conditional on the residential shock, household utility associated with residence o is

$$W_o^{mf}(\omega) = \frac{u_o^{mf}\Omega_o^{mf}}{P^{\beta} r_{Ro}^{1-\beta}} \varphi_o^{mf}(\omega). \tag{60}$$

E.3.5. Residential choice of couples

At the residential-choice stage, couples observe the residential shocks $\{\varphi_o^{mf}(\omega)\}_{o=1}^{S}$. The participation and commuting shocks have not yet been realized, and the value of the corresponding opportunities is summarized by the marital market access term Ω_o^{mf}.

Suppose that the residential shocks are iid Fréchet with shape parameter η^{mf}. The probability that a couple lives in o is

$$\lambda_o^{mf} = \frac{\left(\dfrac{u_o^{mf}\Omega_o^{mf}}{P^{\beta} r_{Ro}^{1-\beta}}\right)^{\eta^{mf}}}{\sum_{n=1}^{S}\left(\dfrac{u_n^{mf}\Omega_n^{mf}}{P^{\beta} r_{Rn}^{1-\beta}}\right)^{\eta^{mf}}}. \tag{61}$$

The residential probability is derived in Appendix L. Couples therefore choose residence by trading off amenities, housing costs, and market access through commuting and labor force participation opportunities.

E.3.6. Ex ante spouses' welfare

Husbands. The ex ante utility of husbands is the expected value of their individual utility evaluated at the residence, participation arrangement, and workplace destinations selected by the household.

To define the husband's utility in male-breadwinner households, let $\bar{\lambda} \in (0,1)$ denote the average Pareto weight on the wife among dual earner households, which I hold fixed in the male-breadwinner arrangement.[43]

The husband's utility in the male-breadwinner arrangement is[44]

$$
\begin{aligned}
V^{m}_{oi,0}(\omega) = \; & \log\left[u^{mf}_{o}\xi_{o0}C^{\beta}_{m,oi,0}\left(H^{R}_{oi,0}\right)^{1-\beta}\right] \\
& + \frac{1}{1-\bar{\lambda}}\log\left(\frac{\epsilon_{m,i}(\omega)}{d^{m}_{oi}}\right) + \log\varsigma_{o0}(\omega) + \log\varphi^{mf}_{o}(\omega).
\end{aligned} \tag{62}
$$

Let

$$
\overline{V}^{m}_{oi,0} \equiv \log\left[u^{mf}_{o}\xi_{o0}C^{\beta}_{m,oi,0}\left(H^{R}_{oi,0}\right)^{1-\beta}\right]. \tag{63}
$$

The husband's expected utility in the male-breadwinner arrangement, after integrating over his commuting shocks but before integrating over the participation and residential shocks, is

$$
\mathcal{U}^{m}_{o0} \equiv \sum_{i=1}^{S}\pi^{m}_{i|o,0}\left[\overline{V}^{m}_{oi,0} + \frac{1}{1-\bar{\lambda}}\left(\frac{\gamma_{\mathrm{E}}}{\theta^{m}} + \log\Omega_{o0} - \log w_{m,i}\right)\right], \tag{64}
$$

where γ_{E} is the Euler–Mascheroni constant. The participation and residential shocks are not yet realized, so the husband's conditional utility in this arrangement is $\mathcal{U}^{m}_{o0} + \log\varsigma_{o0}(\omega) + \log\varphi^{mf}_{o}(\omega)$.

For dual earner households, define

$$
\overline{V}^{m}_{oij,1} \equiv \log\left(u^{mf}_{o}\frac{C^{\beta}_{m,oij}\left(H^{R}_{oij}\right)^{1-\beta}}{d^{m}_{oi}}\right) \tag{65}
$$

and

$$
\mathcal{B}^{m}_{oij} \equiv \frac{\Xi_{ij}}{\left(d^{m}_{oi}\right)^{1-\zeta_{ij}}\left(d^{f}_{oj}\right)^{\zeta_{ij}}}. \tag{66}
$$

The husband's expected utility in the dual earner arrangement, after integrating over both spouses' commuting shocks but before integrating over the participation and residential shocks, is

$$
\begin{aligned}
\mathcal{U}^{m}_{o1} \equiv \; & \sum_{j=1}^{S}\pi^{f}_{j|o,1}\sum_{i=1}^{S}\pi^{m}_{i|o,j}\Bigg[\overline{V}^{m}_{oij,1} \\
& + \frac{1}{1-\zeta_{ij}}\left(\frac{\gamma_{\mathrm{E}}}{\theta^{m}} + \frac{1}{\theta^{m}}\log\Phi_{oj} - \log\mathcal{B}^{m}_{oij}\right)\Bigg].
\end{aligned} \tag{67}
$$

and the husband's conditional utility in this arrangement is $\mathcal{U}^{m}_{o1} + \log\varsigma_{o1}(\omega) + \log\varphi^{mf}_{o}(\omega)$.

The household is a dual earner couple with probability μ_o and a male-breadwinner household with probability $1-\mu_o$. After integrating over the participation shocks, the husband's

[43] This assumption is not needed to solve the model equilibrium; it is used only to divide household consumption and welfare between spouses in male-breadwinner households, where, conditional on residence o and husband destination i, expenditures are allocated as $C_{m,oi,0} = \beta(1-\bar{\lambda})w_{m,i}/P$, $C_{f,oi,0} = \beta\bar{\lambda}w_{m,i}/P$, and $H^{R}_{oi,0} = (1-\beta)w_{m,i}/r_{Ro}$.

[44] The scaling by $1/(1-\bar{\lambda})$ ensures that the commuting component of the weighted household objective is $\log\left(\epsilon_{m,i}(\omega)/d^{m}_{oi}\right)$, so the breadwinner commuting choice continues to depend on $w_{m,i}/d^{m}_{oi}$.

expected utility conditional on residence o, but before integrating over the residential shock, is

$$\mathcal{U}_o^m \equiv \frac{\gamma_E}{\nu} + \log \Omega_o^{mf} + \mu_o \left[\mathcal{U}_{o1}^m - \log \Omega_{o1}\right] + (1-\mu_o)\left[\mathcal{U}_{o0}^m - \log(\xi_{o0}\Omega_{o0})\right], \tag{68}$$

and the husband's conditional utility conditional on residence is $\mathcal{U}_o^m + \log \varphi_o^{mf}(\omega)$.

Finally, define the deterministic attractiveness of residence o as

$$R_o \equiv \frac{u_o^{mf}\Omega_o^{mf}}{P^\beta r_{Ro}^{1-\beta}}, \tag{69}$$

and the corresponding aggregate residential market access term as

$$\mathcal{R}^{mf} \equiv \left[\sum_{o=1}^{S} R_o^{\eta^{mf}}\right]^{1/\eta^{mf}}. \tag{70}$$

The husband's ex ante utility, before the realization of any commuting, participation, or residential shock, is

$$\boxed{\mathcal{U}^m = \frac{\gamma_E}{\eta^{mf}} + \log \mathcal{R}^{mf} + \sum_{o=1}^{S} \lambda_o^{mf}\left[\mathcal{U}_o^m - \log R_o\right].} \tag{71}$$

The derivation of equation (71) is given in Appendix L.

Wives. The ex ante utility of wives is the expected value of their individual utility evaluated at the residence, participation arrangement, and workplace destinations selected by the household.

In the male-breadwinner arrangement, I use the same constant Pareto weight $\bar{\lambda}$ introduced above.

Given the male-breadwinner consumption allocation introduced above, the wife's utility in a male-breadwinner household is

$$\begin{aligned} V_{oi,0}^f(\omega) = {} & \log\left[u_o^{mf}\xi_{o0}C_{f,oi,0}^{\beta}\left(H_{oi,0}^R\right)^{1-\beta}\right] \\ & + \log \varsigma_{o0}(\omega) + \log \varphi_o^{mf}(\omega). \end{aligned} \tag{72}$$

The wife does not commute in this arrangement, so her utility does not contain a commuting shock.

Let

$$\overline{V}_{oi,0}^f \equiv \log\left[u_o^{mf}\xi_{o0}C_{f,oi,0}^{\beta}\left(H_{oi,0}^R\right)^{1-\beta}\right]. \tag{73}$$

After integrating over the husband's commuting shocks, but before integrating over the participation and residential shocks, the wife's expected utility in the male-breadwinner arrangement is

$$\mathcal{U}_{o0}^f \equiv \sum_{i=1}^{S} \pi_{i|o,0}^m \overline{V}_{oi,0}^f. \tag{74}$$

The participation and residential shocks are not yet realized, so the wife's conditional utility in this arrangement is $\mathcal{U}_{o0}^f + \log \varsigma_{o0}(\omega) + \log \varphi_o^{mf}(\omega)$.

For dual earner households, define

$$\overline{V}^{f}_{oij,1} \equiv \log\left(u^{mf}_{o} \frac{C^{\beta}_{f,oij}\left(H^{R}_{oij}\right)^{1-\beta}}{d^{f}_{oj}} \right), \tag{75}$$

and define the household attractiveness of the wife's destination j as

$$\mathcal{B}^{f}_{oj} \equiv \Phi^{1/\theta^{m}}_{oj}. \tag{76}$$

The wife's expected utility in the dual earner arrangement, after integrating over both spouses' commuting shocks but before integrating over the participation and residential shocks, is

$$\begin{aligned}\mathcal{U}^{f}_{o1} \equiv \sum_{j=1}^{S} \pi^{f}_{j|o,1} \sum_{i=1}^{S} \pi^{m}_{i|o,j} \Bigg[& \overline{V}^{f}_{oij,1} \\ & + \frac{1}{\zeta_{ij}} \left(\frac{\gamma_{\mathrm{E}}}{\theta^{f}} + \log \Omega_{o1} - \log \mathcal{B}^{f}_{oj} \right) \Bigg].\end{aligned} \tag{77}$$

and the wife's conditional utility in this arrangement is $\mathcal{U}^{f}_{o1} + \log \varsigma_{o1}(\omega) + \log \varphi^{mf}_{o}(\omega)$.

After integrating over the participation shocks, with the dual earner and male-breadwinner arrangements occurring with probabilities μ_o and $1-\mu_o$, the wife's expected utility conditional on residence o, but before integrating over the residential shock, is

$$\mathcal{U}^{f}_{o} \equiv \frac{\gamma_{\mathrm{E}}}{\nu} + \log \Omega^{mf}_{o} + \mu_o \left[\mathcal{U}^{f}_{o1} - \log \Omega_{o1} \right] + (1-\mu_o) \left[\mathcal{U}^{f}_{o0} - \log(\xi_{o0}\Omega_{o0}) \right], \tag{78}$$

and the wife's conditional utility conditional on residence is $\mathcal{U}^{f}_{o} + \log \varphi^{mf}_{o}(\omega)$.

Using R_o and $\mathcal{R}^{mf}$ defined above, the wife's ex ante utility, before the realization of any commuting, participation, or residential shock, is

$$\boxed{\mathcal{U}^{f} = \frac{\gamma_{\mathrm{E}}}{\eta^{mf}} + \log \mathcal{R}^{mf} + \sum_{o=1}^{S} \lambda^{mf}_{o} \left[\mathcal{U}^{f}_{o} - \log R_o \right].} \tag{79}$$

The derivation of equation (79) is given in Appendix L.

E.4. Labor supply and residential floorspace demand

Let

$$N^{m}_{o} \equiv N^{m}\lambda^{m}_{o}, \qquad N^{f}_{o} \equiv N^{f}\lambda^{f}_{o}, \qquad N^{mf}_{o} \equiv N^{mf}\lambda^{mf}_{o} \tag{80}$$

denote the resident masses at o.

Labor supply to workplaces. Aggregate labor supplied to workplace location d by men is

$$L_{m,d} = \sum_{o=1}^{S} N^{m}_{o} \varpi^{m}_{d|o} + \sum_{o=1}^{S} N^{mf}_{o} (1-\mu_o) \pi^{m}_{d|o,0} + \sum_{o=1}^{S} N^{mf}_{o} \mu_o \sum_{j=1}^{S} \Pi_{dj|o}. \tag{81}$$

Aggregate labor supplied to workplace location d by women is

$$L_{f,d} = \sum_{o=1}^{S} N^{f}_{o} \varpi^{f}_{d|o} + \sum_{o=1}^{S} N^{mf}_{o} \mu_o \pi^{f}_{d|o,1}. \tag{82}$$

Residential floorspace demand. Expected residential demand by singles of gender g living in o is

$$H^g_{R,o} = \sum_{d=1}^{S} \varpi^g_{d|o} \frac{(1-\beta^g) w_{g,d}}{r_{R_o}}. \tag{83}$$

Expected residential demand by dual earner couples living in o is

$$H^{mf,1}_{R,o} = \sum_{i=1}^{S} \sum_{j=1}^{S} \Pi_{ij|o} \frac{z_{oij}}{r_{R_o}}, \tag{84}$$

where z_{oij} is the optimal housing expenditure from the relevant regime in (**??**)–(**??**). Expected residential demand by breadwinner couples living in o is

$$H^{mf,0}_{R,o} = \sum_{i=1}^{S} \pi^m_{i|o,0} \frac{(1-\beta) w_{m,i}}{r_{R_o}}. \tag{85}$$

Hence total residential floorspace demand at o is

$$H^D_{R,o} = N^m_o H^m_{R,o} + N^f_o H^f_{R,o} + N^{mf}_o \left[\mu_o H^{mf,1}_{R,o} + (1-\mu_o) H^{mf,0}_{R,o}\right]. \tag{86}$$

Final good consumption. Expected final good consumption by singles of gender g living in o is

$$C^g_o = \sum_{d=1}^{S} \varpi^g_{d|o} \frac{\beta^g w_{g,d}}{P}. \tag{87}$$

In dual earner households, total expenditure on the final good is the sum of the spouses' private expenditures, $x_{m,oij} + x_{f,oij}$. Hence expected final good consumption by dual earner couples living in o is

$$C^{mf,1}_o = \sum_{i=1}^{S} \sum_{j=1}^{S} \Pi_{ij|o} \frac{x_{m,oij} + x_{f,oij}}{P}. \tag{88}$$

Expected final good consumption by male-breadwinner households living in o is

$$C^{mf,0}_o = \sum_{i=1}^{S} \pi^m_{i|o,0} \frac{\beta w_{m,i}}{P}. \tag{89}$$

Therefore total final good consumption by residents of location o is

$$C^D_o = N^m_o C^m_o + N^f_o C^f_o + N^{mf}_o \left[\mu_o C^{mf,1}_o + (1-\mu_o) C^{mf,0}_o\right]. \tag{90}$$

Aggregating across locations, total household demand for the final good in the city is

$$C^D = \sum_{o=1}^{S} C^D_o. \tag{91}$$

A representative final good producer aggregates the set of location-by-sector varieties $\{Q_{os}\}_{o=1,\dots,S;\, s=1,\dots,K}$ with elasticity of substitution $\sigma_D > 1$. The final good is therefore

$$C = \left[\sum_{o=1}^{S} \sum_{s=1}^{K} Q_{os}^{\frac{\sigma_D - 1}{\sigma_D}}\right]^{\frac{\sigma_D}{\sigma_D - 1}}, \qquad P = \left[\sum_{o=1}^{S} \sum_{s=1}^{K} p_{os}^{1-\sigma_D}\right]^{\frac{1}{1-\sigma_D}}. \tag{92}$$

Cost minimization by the final good producer implies the standard Armington demand system. Given total final good consumption C^D, demand for the variety produced in location o and sector s is

$$Q_{os} = C^D \left(\frac{p_{os}}{P}\right)^{-\sigma_D}, \tag{93}$$

or, in expenditure shares,

$$p_{os}Q_{os} = \left(\frac{p_{os}}{P}\right)^{1-\sigma_D} PC^D. \tag{94}$$

E.5. Production

There are K sectors. In each location-sector pair (o, s), a competitive firm produces a differentiated variety using male labor, female labor, and commercial floorspace. Production is

$$Q_{os} = A_{os}\, M_{os}^{\alpha_s} H_{F,os}^{1-\alpha_s}, \tag{95}$$

where A_{os} is exogenous productivity and the labor composite is

$$M_{os} = \left[\alpha_{ms} L_{m,os}^{\frac{\sigma-1}{\sigma}} + \alpha_{fs} L_{f,os}^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}. \tag{96}$$

The parameter $\sigma > 1$ is the elasticity of substitution between male and female labor, and α_{ms}, α_{fs} govern gender intensities by sector.

Unit costs and factor demand. Let the CES labor cost index at (o, s) be

$$W_{os} = \left[\alpha_{ms}^{\sigma} w_{m,o}^{1-\sigma} + \alpha_{fs}^{\sigma} w_{f,o}^{1-\sigma}\right]^{\frac{1}{1-\sigma}}. \tag{97}$$

Under perfect competition,

$$p_{os} = \frac{1}{A_{os}} \left(\frac{W_{os}}{\alpha_s}\right)^{\alpha_s} \left(\frac{r_{F_o}}{1-\alpha_s}\right)^{1-\alpha_s}, \tag{98}$$

up to a sector specific constant normalization.

The share of sector-s labor expenditure at location o that goes to gender g is

$$\omega_{g,os} = \frac{\alpha_{gs}^{\sigma} w_{g,o}^{1-\sigma}}{\alpha_{ms}^{\sigma} w_{m,o}^{1-\sigma} + \alpha_{fs}^{\sigma} w_{f,o}^{1-\sigma}}. \tag{99}$$

Therefore sector level factor demands satisfy

$$w_{g,o} L_{g,os}^D = \alpha_s \omega_{g,os}\, p_{os} Q_{os}, \qquad r_{F_o} H_{F,os}^D = (1-\alpha_s) p_{os} Q_{os}. \tag{100}$$

Aggregating across sectors, total labor demand by gender and total commercial floorspace demand at location o are

$$L_{g,o}^D \equiv \sum_{s=1}^{K} L_{g,os}^D, \qquad H_{F,o}^D \equiv \sum_{s=1}^{K} H_{F,os}^D. \tag{101}$$

E.6. Housing and land use

The total floorspace stock at location o is fixed at $\bar{H}_o$. A share χ_o is allocated to residential use and a share $1-\chi_o$ to commercial use. Residential and commercial market clearing therefore require

$$\chi_o \bar{H}_o = H^D_{R,o}, \qquad (1-\chi_o)\bar{H}_o = H^D_{F,o} \tag{102}$$

Landowners allocate floorspace to the most profitable use. A unit of residential floorspace earns r_{R_o}, whereas a unit of commercial floorspace earns $(1-\iota_o)r_{F_o}$ because of location specific regulations or wedges. Therefore,

$$\chi_o = \begin{cases} 1, & \text{if } r_{R_o} > (1-\iota_o)r_{F_o}, \\ (0,1), & \text{if } r_{R_o} = (1-\iota_o)r_{F_o}, \\ 0, & \text{if } r_{R_o} < (1-\iota_o)r_{F_o}. \end{cases} \tag{103}$$

E.7. Definition of Equilibrium

Definition 2. Given the model's parameters

$$\left\{\beta^g, \varkappa^g, \vartheta^g, \eta^g, \beta, \kappa^m, \kappa^f, \theta^m, \theta^f, \upsilon, \nu, \alpha_{ms}, \alpha_{fs}, \alpha_s, \sigma, \sigma_D\right\},$$

the mapping from spouses' wage ratios into Pareto weights,

$$\left\{\zeta_{ij}\left(\frac{w_{m,i}}{w_{f,j}}\right)\right\}_{i,j=1,\ldots,S},$$

city population by household group

$$\left\{N^m, N^f, N^{mf}\right\},$$

and exogenous location specific characteristics

$$\left\{\bar{H}_o, \xi_{o0}, u^m_o, u^f_o, u^{mf}_o, A_{os}, t_{od}, \iota_o\right\},$$

the competitive equilibrium of the model is given by the vector

$$\left\{L_{m,o}, L_{f,o}, L^D_{m,o}, L^D_{f,o}, H^D_{R,o}, H^D_{F,o}, w_{m,o}, w_{f,o}, r_{R_o}, r_{F_o}, \chi_o\right\}$$

such that households solve their residence, commuting, and labor force participation problems; firms minimize costs and choose factor demands optimally; labor markets clear,

$$L_{m,o} = L^D_{m,o}, \qquad L_{f,o} = L^D_{f,o},$$

residential and commercial floorspace markets clear,

$$\chi_o \bar{H}_o = H^D_{R,o}, \qquad (1-\chi_o)\bar{H}_o = H^D_{F,o},$$

and the final good market clears.

F. Block 1: Estimation of Commuting Parameters and the Pareto Weight

This appendix describes the details in the estimation procedure summarized in Block 1 of Section 5.2.2. The goal is to recover the commuting parameter vector $\Theta \equiv (\vartheta^m, \vartheta^f, \theta^m, \theta^f, \kappa^f, \zeta_0, \zeta_1)$. These parameters govern how singles, breadwinner households, and dual earner households trade off wages against commute time, and, for dual earners, how the Pareto weight depends on spouses' relative earnings. The overall strategy is a GMM that stacks moments from three groups of gravity-style PPML regressions: single-gender flows, breadwinner flows, and two dual earner flows (husband conditional on the wife's destination, and wife conditional on the husband's destination).

The remainder of this appendix is organized as follows. Section F.1 states the parameter vector together with the identifying restrictions and normalizations. Section F.2 lists the target moments. Section F.3 defines the GMM criterion. Section F.4 describes the weighting matrix and the construction of standard errors. Section F.5 details the numerical algorithm used to estiamte the parameter vector. Section F.6 discusses how each parameter is disciplined by the moment structure. Section F.7 reports the estimates and how successful the model is at fitting Fact 1 and Fact 3 moments.

F.1. Parameters

Restrictions and normalization. I impose two restrictions on commuting parameters. First, the commute-cost scale is common within gender across household types:

$$\varkappa^m = \kappa^m, \qquad \varkappa^f = \kappa^f.$$

Thus, for each gender, the same commute-cost scale governs both singles and married individuals (but not the shape parameters). Second, among married individuals, commuting parameters are the same regardless of whether the spouse also works. In particular, the male commuting parameters are common across male-breadwinner and dual earner households, and the female commuting parameters are common across female-breadwinner and dual earner households. Although female-breadwinner households are not modeled as an endogenous margin in the counterfactual exercises, I retain them in estimation because their commuting flows provide an additional source of information on the married women commuting slope, $\theta^f \kappa^f$, without the spillover effects present in dual earner households from spouses' commute.

Normalization. Following Ahlfeldt et al. (2015), Tsivanidis (2021), and Zárate (2021), I normalize the male commute-cost scale, κ^m, to a fixed value of 0.01. This normalization pins down the overall scale of commuting disutility and allows the remaining commuting shape parameters to be disciplined by the gravity slopes.

Pareto weight. For dual earner households, I allow the intrahousehold allocation rule to depend on spouses' relative earnings, following the collective household literature in which wages and other distribution factors affect the sharing rule (Fortin and Lacroix, 1997; Chiappori et al., 2002). I parameterize the Pareto weight directly as a logit function of the spouses' wage ratio,

$$\zeta_{ij} = \Lambda\left(\zeta_0 + \zeta_1 \log\left(\frac{w_{m,i}}{w_{f,j}}\right)\right), \qquad \Lambda(x) \equiv \frac{e^x}{1+e^x}. \tag{104}$$

This specification is close in spirit to van Klaveren et al. (2008).

Final parameter vector. The full parameter vector estimated in this block is

$$\Theta \equiv \left(\vartheta^m, \vartheta^f, \theta^m, \theta^f, \kappa^f, \zeta_0, \zeta_1\right)'. \tag{105}$$

F.2. Moments

The empirical moments come from three sets of equations: the single commuting equations, the breadwinner equations, and the dual earner equations. These are the same reduced form objects estimated in Facts 1 and 3.

Singles. For singles, I use the commuting probabilities in (28). Let $\pi_{od}^{m,S}$ and $\pi_{od}^{f,S}$ denote the observed commuting shares of single men and single women—the share of single workers of each gender residing in o who commute to d. The estimating equations are

$$\log \pi_{od}^{m,S} = \alpha_o^{m,S} + \delta_d^{m,S} + \gamma_m^S t_{od} + \varepsilon_{od}^{m,S}, \tag{106}$$
$$\log \pi_{od}^{f,S} = \alpha_o^{f,S} + \delta_d^{f,S} + \gamma_f^S t_{od} + \varepsilon_{od}^{f,S}. \tag{107}$$

The corresponding reduced form slopes satisfy

$$\gamma_m^S = -\vartheta^m \varkappa^m, \qquad \gamma_f^S = -\vartheta^f \varkappa^f.$$

The two slopes γ_m^S and γ_f^S enter the GMM criterion as targeted moments.

Breadwinners. For married households with a single earner, I use the husband's commuting probability in male-breadwinner households from (**??**). I also use the analogous expression with genders reversed for wives in female-breadwinner households. Let $\pi_{oi}^{MB,m}$ and $\pi_{oj}^{FB,f}$ denote those observed commuting shares. The estimating equations are

$$\log \pi_{oi}^{MB,m} = \alpha_o^{MB,m} + \delta_i^{MB,m} + \gamma_m^{MB} t_{oi} + \varepsilon_{oi}^{MB,m}, \tag{108}$$
$$\log \pi_{oj}^{FB,f} = \alpha_o^{FB,f} + \delta_j^{FB,f} + \gamma_f^{FB} t_{oj} + \varepsilon_{oj}^{FB,f}. \tag{109}$$

These slopes satisfy

$$\gamma_m^{MB} = -\theta^m \kappa^m, \qquad \gamma_f^{FB} = -\theta^f \kappa^f.$$

The two slopes γ_m^{MB} and γ_f^{FB} also enter the GMM criterion as targeted moments.

Dual earners. The dual earner moments are built from the commuting equations in Section 4.1. The dual earner commuting equations are nonlinear in the wage ratio through the Pareto weight ζ_{ij}, which I parameterize as a logit function of the log wage ratio. I therefore construct model implied dual earner commuting shares for any candidate parameter vector and then estimate on those shares auxiliary PPML regressions projecting commuting shares on commute times and their interactions with the log wage ratio. The resulting coefficients depend on Θ, which I match to their empirical counterparts from Fact 3.

Let $\pi^{m|f}_{oi|j}$ denote the conditional commuting share of husbands who live in o, whose wives work in j, and who themselves work in i. Let $\pi^{f|m}_{oj|i}$ denote the analogous conditional share for wives. The exact theoretical counterparts of these shares follow from (**??**), (**??**), and (**??**).

For husbands, the exact conditional commuting equation is

$$\log \pi^{m|f}_{oi|j} = \underbrace{-\log \Phi_{oj}}_{\alpha^{m|f}_{oj}} + \underbrace{\theta^m \log \Xi_{ij}}_{\delta^{DE,m}_{ij}} - \theta^m \Big[(1-\zeta_{ij})\kappa^m t_{oi} + \zeta_{ij}\kappa^f t_{oj}\Big]. \tag{110}$$

For wives, the exact conditional commuting equation is

$$\log \pi^{f|m}_{oj|i} = \underbrace{-\log \sum_n \Psi_{oni}}_{\alpha^{f|m}_{oi}} + \underbrace{\theta^m \log \Xi_{ij}}_{\delta^{DE,f}_{ij}} + \left(\frac{\theta^f}{\theta^m} - 1\right)\log \Phi_{oj} - \theta^m \Big[(1-\zeta_{ij})\kappa^m t_{oi} + \zeta_{ij}\kappa^f t_{oj}\Big], \tag{111}$$

and integrating over the husband's destination yields the wife's exact unconditional commuting equation,

$$\log \pi^{f,U}_{oj} = \underbrace{-\log\left(\sum_n \Phi_{on}^{\theta^f/\theta^m}\right)}_{\alpha^{f,U}_o} + \frac{\theta^f}{\theta^m}\log \Phi_{oj}, \tag{112}$$

where Φ_{oj} is the husband's market access term defined in (17), and Ψ_{oni} collects the wife's conditional denominator term. These expressions show that the exact dual earner system is nonlinear in commute times and wage ratios through the Pareto weight ζ_{ij}.

I use these equations in three stages.

Stage 1. I estimate in the data three auxiliary PPML equations, which are very similar to the reduced form equations used in Fact 3. Note that in the husband and wife conditional regressions below, a-coefficients multiply the husband's commute time t_{oi} while b-coefficients multiply the wife's commute time t_{oj}. The Z subscript flags the interaction with the log wage ratio $\log(w_{m,i}/w_{f,j})$. The superscript indicates whether the regressor appears in the husband $(m|f)$ or the wife $(f|m)$ conditional regression.[45]

For husbands conditional on the wife's destination, I estimate

$$\log \pi^{m|f}_{oi|j} = \alpha^{m|f}_{oj} + \delta^{DE,m}_{ij} + a^{m|f}_t\, t_{oi} + a^{m|f}_{Zt}\left(\log\left(\tfrac{w_{m,i}}{w_{f,j}}\right) t_{oi}\right) + b^{m|f}_{Zt}\left(\log\left(\tfrac{w_{m,i}}{w_{f,j}}\right) t_{oj}\right) + \varepsilon^{m|f}_{oij}. \tag{113}$$

From this regression I recover the husband's empirical market access term,

$$\log \widehat{\Phi}_{oj} \equiv -\widehat{\alpha}^{m|f}_{oj}, \tag{114}$$

which I then plug in as a regressor in the two wife equations that follow, to be consistent with the model.

[45]Explicitly: $a^{m|f}_t$ is the coefficient on t_{oi} in the husband conditional regression; $a^{m|f}_{Zt}$ is the coefficient on $\log(w_{m,i}/w_{f,j})\, t_{oi}$ in the same regression; $b^{m|f}_{Zt}$ is the coefficient on $\log(w_{m,i}/w_{f,j})\, t_{oj}$. The wife conditional regression uses $b^{f|m}_t$ on t_{oj}, $a^{f|m}_{Zt}$ on $\log(w_{m,i}/w_{f,j})\, t_{oi}$, and $b^{f|m}_{Zt}$ on $\log(w_{m,i}/w_{f,j})\, t_{oj}$.

For wives conditional on the husband's destination, I estimate

$$\log \pi^{f|m}_{oj|i} = \alpha^{f|m}_{oi} + \delta^{DE,f}_{ij} + c^{C}_{\Phi} \log \widehat{\Phi}_{oj} + b^{f|m}_{t}\, t_{oj} + a^{f|m}_{Zt} \left(\log\left(\tfrac{w_{m,i}}{w_{f,j}} \right) t_{oi} \right) + b^{f|m}_{Zt} \left(\log\left(\tfrac{w_{m,i}}{w_{f,j}} \right) t_{oj} \right) + \varepsilon^{f|m}_{oij}. \tag{115}$$

so that the coefficient c^{C}_{Φ} identifies $\theta^f/\theta^m - 1$ from the exact equation (111).

From these estimations, I also recover the fixed effects $\widehat{\alpha}^{m|f}_{oj}$, $\widehat{\delta}^{DE,m}_{ij}$, $\widehat{\alpha}^{f|m}_{oi}$, $\widehat{\delta}^{DE,f}_{ij}$, and $\widehat{\alpha}^{f,U}_{o}$ which are an input for Step 2 below.

For wives unconditional on the husband's destination, I estimate

$$\log \pi^{f,U}_{oj} = \alpha^{f,U}_{o} + c^{U}_{\Phi} \log \widehat{\Phi}_{oj} + \varepsilon^{f,U}_{oj}, \tag{116}$$

so that the coefficient c^{U}_{Φ} identifies θ^f/θ^m from the exact equation (112).

Note that I do not use coefficients $c^{C}_{\Phi} = \theta^f/\theta^m - 1$ and $c^{U}_{\Phi} = \theta^f/\theta^m$ as target moments in the estimation procedure. Instead, I use them as untargeted moments.

Stage 2. Given a candidate parameter vector Θ, I construct the Pareto weight

$$\zeta_{ij}(\Theta) = \Lambda\left(\zeta_0 + \zeta_1 \log\left(\frac{w_{m,i}}{w_{f,j}} \right) \right),$$

and use it to predict model implied dual earner commuting shares, in addition to recovered fixed effects from Step 1. In particular, the model implied husband conditional share is

$$\pi^{m|f,\text{model}}_{oi|j}(\Theta) = \exp\left[\widehat{\alpha}^{m|f}_{oj} + \widehat{\delta}^{DE,m}_{ij} - \theta^m \Big((1-\zeta_{ij}(\Theta))\kappa^m t_{oi} + \zeta_{ij}(\Theta)\kappa^f t_{oj} \Big) \right]. \tag{117}$$

For wives, I use the same market access term recovered from the data, $\log \widehat{\Phi}_{oj}$ and the estimated $\widehat{c}^{C}_{\Phi}$, and construct

$$\pi^{f|m,\text{model}}_{oj|i}(\Theta) = \exp\left[\widehat{\alpha}^{f|m}_{oi} + \widehat{\delta}^{DE,f}_{ij} + \widehat{c}^{C}_{\Phi} \log \widehat{\Phi}_{oj} - \theta^m \Big((1-\zeta_{ij}(\Theta))\kappa^m t_{oi} + \zeta_{ij}(\Theta)\kappa^f t_{oj} \Big) \right]. \tag{118}$$

Stage 3. I estimate on the model implied shares (117) and (118) the same auxiliary PPML equations (113) and (115). Because the auxiliary PPML projection is computed on model implied shares that depend on Θ, each of the resulting coefficients is a function of Θ:

$$\left(a^{m|f}_{t}(\Theta),\, a^{m|f}_{Zt}(\Theta),\, b^{m|f}_{Zt}(\Theta),\, b^{f|m}_{t}(\Theta),\, a^{f|m}_{Zt}(\Theta),\, b^{f|m}_{Zt}(\Theta) \right)$$

These coefficients are the model implied dual earner moments.

F.3. Criterion and loss function

The first four model implied moments are analytical:

$$m^A(\Theta) \equiv \begin{pmatrix} -\vartheta^m \varkappa^m \\ -\vartheta^f \varkappa^f \\ -\theta^m \kappa^m \\ -\theta^f \kappa^f \end{pmatrix}. \tag{119}$$

The remaining six model implied moments, denoted $m^P(\Theta)$, are the projection coefficients obtained from the auxiliary PPML regressions (113)–(115) applied to the model implied shares constructed from the nonlinear model:

$$m^P(\Theta) \equiv \begin{pmatrix} a_t^{m|f}(\Theta) \\ a_{Zt}^{m|f}(\Theta) \\ b_{Zt}^{m|f}(\Theta) \\ b_t^{f|m}(\Theta) \\ a_{Zt}^{f|m}(\Theta) \\ b_{Zt}^{f|m}(\Theta) \end{pmatrix}. \tag{120}$$

The full model implied moment vector is

$$m(\Theta) = \begin{pmatrix} m^A(\Theta) \\ m^P(\Theta) \end{pmatrix}. \tag{121}$$

Define the sample moment function by

$$g(\Theta) \equiv m(\Theta) - \widehat{m}.$$

The estimator solves

$$\widehat{\Theta} = \arg \min_{\Theta:\, \kappa^m = \bar{\kappa}^m} g(\Theta)' W g(\Theta), \tag{122}$$

where $\bar{\kappa}^m = 0.01$ is the normalization imposed on the male commute-cost scale and W is a positive-definite weighting matrix.

F.4. Weighting matrix and standard errors

Let $\widehat{S}$ denote the estimated covariance matrix of the empirical moment vector $\widehat{m}$. This covariance matrix is constructed by stacking the influence functions from the six auxiliary regressions used to estimate the moments: single men, single women, male breadwinners, female breadwinners, husband conditional, and wife conditional. The weighting matrix in (122) is based on $\widehat{S}$.

Let

$$G(\Theta) \equiv \frac{\partial g(\Theta)}{\partial \Theta'}$$

denote the Jacobian of the sample moment function with respect to the structural parameters. The asymptotic covariance matrix of $\widehat{\Theta}$ is given by the sandwich formula

$$\widehat{\mathrm{Var}}(\widehat{\Theta}) = \frac{1}{C} \left(G'WG\right)^{-1} G'W\widehat{S}WG \left(G'WG\right)^{-1}, \tag{123}$$

where C is the number of clusters used in the influence-function calculation. This procedure propagates the uncertainty from the single and breadwinner commuting slopes into the uncertainty of the dual earner parameters because all moments are estimated jointly within the same stacked system.

F.5. Algorithm

The numerical implementation proceeds as follows.

1. Fix the normalization $\kappa^m = \bar{\kappa}^m$.

2. Construct the empirical moment vector $\widehat{m}$ from the reduced form estimates (similar to those in Facts 1 and 3).

3. For each candidate triple $(\kappa^f, \zeta_0, \zeta_1)$, recover the commuting shape parameters $(\vartheta^m, \vartheta^f, \theta^m, \theta^f)$ from the first four commute-slope moments:

$$\vartheta^m = \frac{\widehat{\gamma}^S_m}{-\kappa^m}, \qquad \vartheta^f = \frac{\widehat{\gamma}^S_f}{-\kappa^f}, \qquad \theta^m = \frac{\widehat{\gamma}^{MB}_m}{-\kappa^m}, \qquad \theta^f = \frac{\widehat{\gamma}^{FB}_f}{-\kappa^f}.$$

4. Use $(\kappa^f, \zeta_0, \zeta_1)$ together with the recovered (θ^m, θ^f) and fixed effects to construct model implied shares for dual earner husbands and wives from (117) and (118).

5. Run the auxiliary PPML system on these model implied shares and collect the six model implied dual earner moments.

6. Stack the analytical and projected moments into $m(\Theta)$, compute $g(\Theta) = m(\Theta) - \widehat{m}$, and evaluate the GMM objective $g(\Theta)' W g(\Theta)$.

7. Search over candidate values of $(\kappa^f, \zeta_0, \zeta_1)$, retain the minimizer, and then recover the full parameter vector $\widehat{\Theta}$.

8. Compute the Jacobian $G(\widehat{\Theta})$, combine it with the estimated covariance matrix of the moments, and report standard errors using (123).

9. **Output.** The output of Block 1 is the estimated parameter vector $\widehat{\Theta} = (\widehat{\vartheta}^m, \widehat{\vartheta}^f, \widehat{\theta}^m, \widehat{\theta}^f, \widehat{\kappa}^f, \widehat{\zeta}_0, \widehat{\zeta}_1)'$ with their corresponding standard errors.

A note on the grid search. The grid search is smaller than the dimension of Θ suggests. At each candidate $(\kappa^f, \zeta_0, \zeta_1)$, the single and breadwinner commute slopes are inverted analytically to recover $(\vartheta^m, \vartheta^f, \theta^m, \theta^f)$. Then, I use remaining moments from dual earner commuting equations to estimate $(\kappa^f, \zeta_0, \zeta_1)$. The search is therefore effectively three-dimensional. Even so, I continue to include the single and breadwinner commute slopes as moments in the stacked GMM system.

I design the estimation procedure this way to minimize computation time. Adding an explicit grid over the four commuting shape parameters on top of the three-dimensional $(\kappa^f, \zeta_0, \zeta_1)$ grid would be much slower. Conceptually, the approach is close to calibrating $(\vartheta^m, \vartheta^f, \theta^m, \theta^f)$ directly from the reduced form slopes up to κ_f and then matching the remaining dual earner moments the best way possible. The difference is that by carrying the singles and breadwinner slopes as moments inside the GMM system, I still propagate their sampling variation into the standard errors of $\widehat{\Theta}$.

F.6. Identification

Building on the note above, identification runs in two layers. The singles and breadwinner commute slopes pin down the commuting shape parameters $(\vartheta^m, \vartheta^f, \theta^m, \theta^f)$ given a candidate

κ^f, and the dual earner moments pin down the three search parameters $(\kappa^f, \zeta_0, \zeta_1)$. I take each group in turn.

Singles. The single-men and single-women commute slopes discipline the products

$$\vartheta^m \varkappa^m, \qquad \vartheta^f \varkappa^f.$$

Under the restrictions $\varkappa^m = \kappa^m$ and $\varkappa^f = \kappa^f$, they pin down ϑ^m directly and tie ϑ^f to the free parameter κ^f through $\vartheta^f = \widehat{\gamma}_f^S/(-\kappa^f)$.

Breadwinners. Analogously, the male- and female-breadwinner commute slopes discipline

$$\theta^m \kappa^m, \qquad \theta^f \kappa^f,$$

pinning down θ^m directly and tying θ^f to κ^f through $\theta^f = \widehat{\gamma}_f^{FB}/(-\kappa^f)$. Female-breadwinner households are useful in estimation—even though I do not include them in the counterfactuals—precisely because this slope ties θ^f to κ^f.

Dual earners. The six PPML projection slopes discipline the three search parameters $(\kappa^f, \zeta_0, \zeta_1)$. The intercept ζ_0 shifts the overall level of the Pareto weight and ζ_1 governs how it responds to the spouses' wage ratio. The interaction slopes a_{Zt} and b_{Zt} involving $\log(w_{m,i}/w_{f,j})\, t_{oi}$ and $\log(w_{m,i}/w_{f,j})\, t_{oj}$ capture how the Pareto weight tilts effective commute costs across high- and low relative wage destination pairs, which identifies ζ_1. The main-effect slopes $a_t^{m|f}$ on the husband's commute time and $b_t^{f|m}$ on the wife's commute time, by contrast, pick up the average commute costs effects. Linearizing the Pareto weight around $\log(w_{m,i}/w_{f,j}) = 0$ gives $a_t^{m|f} \approx -\theta^m(1 - \Lambda(\zeta_0))\kappa^m$ and $b_t^{f|m} \approx -\theta^m \Lambda(\zeta_0)\kappa^f$. Because θ^m and κ^m are already pinned at this stage, the relative magnitudes of $a_t^{m|f}$ and $b_t^{f|m}$ pin down the average Pareto weight $\Lambda(\zeta_0)$, which identifies ζ_0. These moments, in particular the wife-commute-time slopes $b_t^{f|m}$ and b_{Zt}, also carry information about κ^f through the $\zeta_{ij}\kappa^f t_{oj}$ term in the model implied flows, which is how κ^f is identified jointly with ζ_0 and ζ_1.

Taken together, the single, breadwinner, and dual earner moments discipline the full parameter vector Θ.

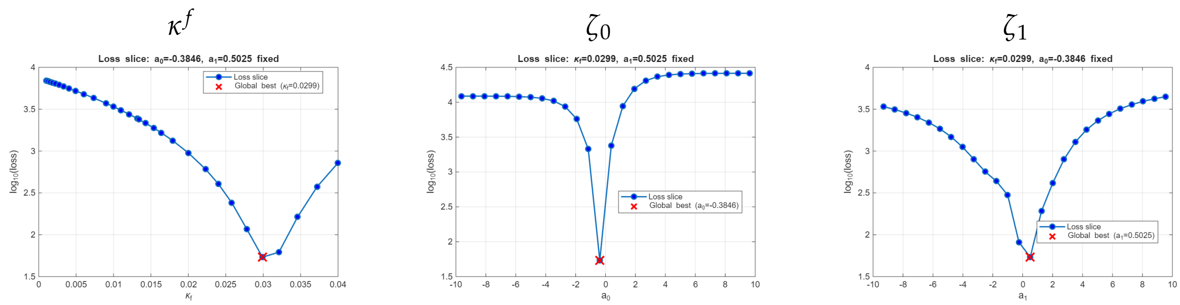
F.7. Estimation results and identification diagnostics

Panel C of Table 3 reports the point estimates and standard errors of the seven elements of $\widehat{\Theta}$. Three patterns are worth highlighting succinctly. First, the estimated female commute-cost scale $\widehat{\kappa}^f \approx 0.03$ is about three times the male normalization $\kappa^m = 0.01$, so iceberg disutility per minute of commute is substantially larger for women than for men. Second, the male Fréchet shapes ($\widehat{\vartheta}^m \approx 5.7$, $\widehat{\theta}^m \approx 4.7$) are about two and a half times the corresponding female shapes ($\widehat{\vartheta}^f \approx 2.2$, $\widehat{\theta}^f \approx 2.1$). This means that women's destination choices are more dispersed than men's. Third, the Pareto weight intercept and slope are $\widehat{\zeta}_0 \approx -0.38$ and $\widehat{\zeta}_1 \approx 0.50$, so at the average wage ratio the household places slightly more weight on the wife ($\Lambda(\widehat{\zeta}_0) \approx 0.41$ for the husband), with the wife's weight rising as the husband's relative wage rises.

The sign of $\widehat{\zeta}_1$ is the opposite of what the standard collective household literature would predict. There, a higher own wage strengthens a spouse's bargaining position through a better outside option, so a higher husband's wage typically raises the husband's Pareto weight and lowers the wife's. In the present setting, however, ζ_{ij} is not a bargaining weight derived from the outside option of dissolving the marriage. Following the discussion in Section 4.1, it is the utility promise required to sustain a given workplace arrangement (i, j) inside the marriage. When the husband's workplace pays much more than the wife's, the arrangement is attractive for the household as a whole but typically forces the wife to accept a longer commute or to work at a lower wage destination, so sustaining that arrangement requires a larger utility promise to her. The wife's Pareto weight therefore *rises* with the husband's relative wage. This direction is already visible in the reduced form patterns of Fact 3 (Section 3.2): the wife's own-commute semi-elasticity becomes more negative as the wage ratio tilts toward the husband, and the husband's becomes less negative. The household, in other words, grows more sensitive to the wife's commute time and prefers shorter commutes for her, which is exactly what a higher Pareto weight on her utility implies, since her commute disutility then carries more weight in the household's joint problem. The positive estimate $\widehat{\zeta}_1 \approx 0.50$ is the structural counterpart of that reduced form pattern.

Identification diagnostics. Figure 26 shows how the GMM loss responds to changes in each of the three search parameters $(\kappa^f, \zeta_0, \zeta_1)$. In each panel, the horizontal axis is the value of one search parameter and the vertical axis is the log of the GMM loss $g(\Theta)'Wg(\Theta)$. At every grid point on the horizontal axis, I hold the other two search parameters fixed at their estimated values and evaluate the loss. A parameter is well identified when its curve has a unique, sharp minimum, and poorly identified when the curve is flat or has multiple minima. The three curves in Figure 26 each have a clean U shape with a single interior minimum at the reported estimate, so the moments discipline all three search parameters.

Figure 26: GMM loss as each search parameter varies.



Notes. Each panel plots log of the GMM loss against one of the three search parameters (κ^f, ζ_0, or ζ_1). At every grid point of the named parameter, the loss is evaluated with the remaining two search parameters held fixed at their estimated values. The red marker in each panel sits at the estimated value of the named parameter.

Moment fit. Table 10 reports, for each of the 12 moments, the empirical value (Data), the model implied value at $\widehat{\Theta}$ (Model), the gap between the two, the empirical standard errors built from the influence functions, and the t-statistic $t = \text{Diff}/\text{SE}$.

Table 10: Block 1 GMM: moment fit at $\widehat{\Theta}$.

Used	Moment	Data	Model	Diff	SE	t
X	γ_m^S	-0.0565	-0.0565	0.0000	0.0029	0.00
X	γ_f^S	-0.0651	-0.0651	0.0000	0.0025	0.00
X	γ_m^{MB}	-0.0466	-0.0466	0.0000	0.0031	0.00
X	γ_f^{FB}	-0.0634	-0.0634	0.0000	0.0026	0.00
	c_Φ^U	5.3177	0.4551	-4.8626	0.0677	-71.84^{***}
	c_Φ^C	-0.5304	-0.5449	-0.0145	0.1662	-0.09
X	$a_t^{m\|f}$	-0.0334	-0.0288	0.0047	0.0043	1.08
X	$a_{Zt}^{m\|f}$	0.0013	0.0051	0.0038	0.0046	0.82
X	$b_{Zt}^{m\|f}$	-0.0033	-0.0164	-0.0131	0.0052	-2.53^{**}
X	$b_t^{f\|m}$	-0.0606	-0.0552	0.0054	0.0041	1.31
X	$a_{Zt}^{f\|m}$	0.0023	0.0049	0.0026	0.0047	0.55
X	$b_{Zt}^{f\|m}$	-0.0042	-0.0152	-0.0110	0.0041	-2.67^{***}

Notes. Used marks moments included in the GMM criterion $g(\Theta)'Wg(\Theta)$. *Diff* is Model − Data. *SE* is the empirical standard error of the moment, built from the influence functions of the auxiliary regressions described in Section F.4. The t-statistic tests the null that the model exactly matches the moment, H_0 : Diff = 0. ${}^{*}p < 0.10$, ${}^{**}p < 0.05$, ${}^{***}p < 0.01$.

The calibration matches nine of the twelve moments within sampling noise. The first four rows, the singles and breadwinner gravity slopes, are matched exactly because the corresponding shape parameters are solved for analytically at every candidate $(\kappa^f, \zeta_0, \zeta_1)$. Of the remaining eight dual earner moments, five are not statistically distinguishable from zero at the 5% level. The three remaining three moments are worth discussing.

The unconditional and conditional Phi-ratios, c_Φ^U and c_Φ^C, are two empirical measures of the husband-to-wife Fréchet shape ratio, recovered from PPML regressions whose fixed-effect structures differ. Under the model, the two are linked by the identity $c_\Phi^U = c_\Phi^C + 1$: the calibration matches both empirical estimates only if they also differ by exactly one. In the data, they do not. The data give $\widehat{c}_\Phi^U - \widehat{c}_\Phi^C \approx 5.85$. Thus, at most one of the two empirical estimates can be reproduced at any $\widehat{\Theta}$. Despite neither of the two moments were targeted, the calibration matches c_Φ^C almost exactly (model -0.545 vs data -0.530, $t \approx -0.09$). The implied c_Φ^U is therefore 0.455, well below the empirical 5.32. The miss suggests the conditional dual earner moments are a more reliable source of identification than the unconditional wife commuting probabilities. In a cross-section like ours, the triple-indexed structure of the dual earner data (origin, husband's workplace, and wife's workplace) admits a rich fixed-effect specification that absorbs much of the unobserved heterogeneity that could otherwise bias the moment. The unconditional c_Φ^U, by contrast, is recovered from a regression with a coarser fixed-effect structure and is therefore more exposed to that endogeneity.

The two remaining significant gaps are in the wife-commute-by-log Z interactions of the dual earner block, $b_{Zt}^{m|f}$ and $b_{Zt}^{f|m}$. The signs agree with the data since both are negative, consistent with the wife's commute-disutility channel becoming more salient as the wage ratio tilts toward the husband. However, the model magnitudes are larger than the data: model values near -0.015 against data values near -0.004. The commuting-flow sensitivity to the wife's commute is therefore more responsive to the spousal wage ratio in the model than in the data. The tension reflects the role of κ^f. The same κ^f scales all three wife-commute moments simultaneously. That is, the level coefficient $b_t^{f|m}$ and the two interactions $b_{Zt}^{m|f}$ and $b_{Zt}^{f|m}$. The GMM optimum balances them, and the chosen $\widehat{\kappa}^f$ ends up matching the level moment $b_t^{f|m}$ closely ($t \approx 1.31$) at the expense of the two interaction moments, which carry larger sampling variance.

G. Block 2: Wage Inversion and Pooling Friction

This appendix describes the details of the wage-inversion procedure summarized in Block 2 of Section 5.2.3. Given the parameters from Blocks 0–1, I recover the vectors of male and female wages $\{w_{m,o}, w_{f,o}\}_{o=1}^{S}$ that clear labor markets at every location, under both the independent and the interdependent commuting models. In the interdependent model, I additionally recover the pooling friction v by matching the gender gap in the Fact 2 interdependence coefficients, $\psi_m - \psi_f$. The propositions below state existence of a wage vector that rationalizes the data as an equilibrium of each model, taking v as given in the interdependent case.

The remainder of this appendix is organized as follows. Section G.1 lists the observables and restrictions used in Block 2. Sections G.2 and G.3 establish existence of a wage vector under the independent and interdependent commuting models, respectively, and Section G.4 discusses uniqueness. Section G.5 explains the identification of the pooling friction v. Section G.6 details the numerical algorithm used to estimate ν. Section G.7 derives the GMM-sandwich standard error for $\widehat{v}$. Section G.8 reports the estimation results and the identification diagnostics. Finally, Section G.9 contains the detailed derivation behind the gross substitution analysis which is relevant for the discussion about uniqueness.

G.1. Observables and restrictions used in Block 2

Throughout this block I treat as observed: residence masses by group $\{N_o^m, N_o^f, N_o^{mf}\}$, the share of dual earner couples at each residence $\{\mu_o\}$, workplace employment by location and industry $\{L_{os}^D\}$, and commute times $\{t_{od}\}$. All commuting-side parameters $\{\varkappa^g, \vartheta^g, \kappa^m, \kappa^f, \theta^m, \theta^f, \zeta_0, \zeta_1\}$ are taken from Block 1 and all production-side parameters $\{\alpha_{sg}, \alpha_s, \sigma\}$ from Block 0. The extensive margin parameter ν, the household-production productivities $\{\xi_{o0}\}$, and the residential elasticities $\{\eta^g, \eta^{mf}\}$ are *not* needed at this stage because residence masses and participation shares enter the labor market clearing system directly as data.

From industry level employment to gender level demand. At this stage I do not observe labor demand $L_{g,os}^D$ by gender at the location-industry level. I only observe total workplace employment $L_{os}^D = L_{m,os}^D + L_{f,os}^D$. Given candidate wages and the production-side parameters, the cost minimization conditions of Section E imply the gender composition of labor demand within each sector,

$$\frac{L_{g,os}^D}{L_{os}^D} = \frac{\alpha_{gs}^{\sigma} w_{g,o}^{-\sigma}}{\alpha_{ms}^{\sigma} w_{m,o}^{-\sigma} + \alpha_{fs}^{\sigma} w_{f,o}^{-\sigma}} \equiv s_{g,os}(w_{m,o}, w_{f,o}). \tag{124}$$

Aggregate labor demand at location o by gender is therefore

$$L_{g,o}^D(w_{m,o}, w_{f,o}) = \sum_{s=1}^{K} s_{g,os}(w_{m,o}, w_{f,o})\, L_{os}^D. \tag{125}$$

Equation (125) is what enters labor market clearing as labor demand in the inversion.

G.2. Existence of the wage vector in the independent commuting model

Proposition 1 (Wage inversion, independent model)**.** *Given observables and parameters as stated above, there exists a vector of location wages $\{w_{m,o}, w_{f,o}\}_{o=1}^{S}$, determined up to a normalization, that clears labor markets at every location under the independent commuting model.*

Proof. Under independent commuting, spouses in dual earner couples choose workplaces independently so their commuting probabilities coincide with the single-gender Fréchet expressions. Labor supplied by gender g to workplace d therefore aggregates into

$$L_{m,d} = \sum_{o=1}^{S} N_o^m \varpi_{d|o}^m + \sum_{o=1}^{S} N_o^{mf}(1-\mu_o)\pi_{d|o,0}^m + \sum_{o=1}^{S} N_o^{mf}\mu_o \varpi_{d|o}^{m,\text{ind}}, \tag{126}$$

$$L_{f,d} = \sum_{o=1}^{S} N_o^f \varpi_{d|o}^f + \sum_{o=1}^{S} N_o^{mf}\mu_o \varpi_{d|o}^{f,\text{ind}}, \tag{127}$$

where $\varpi_{d|o}^g$ is the singles commuting probability in (28), $\pi_{d|o,0}^m$ is the breadwinner commuting probability in (**??**), and $\varpi_{d|o}^{g,\text{ind}}$ denotes the gender-g commuting probability for married dual earner individuals under independence, which takes the same Fréchet form as (28) with shape parameter θ^g and cost scale κ^g.

Stack the excess demand system as $\mathcal{Z} = [\mathcal{Z}_{om}, \mathcal{Z}_{of}]'$ with

$$\mathcal{Z}_{og}(w) = L_{g,o}^D(w_{m,o}, w_{f,o}) - L_{g,o}(w), \qquad g \in \{m, f\}, \quad o = 1, \ldots, S. \tag{128}$$

Index the $2S$ entries of $\mathcal{Z}$ by z so that $z \leq S$ corresponds to male excess demand at location $o = z$ and $z \geq S+1$ to female excess demand at $o = z - S$.

Homogeneity and continuity. The labor demand composition shares $s_{g,os}$ in (124) are homogeneous of degree zero in $(w_{m,o}, w_{f,o})$. Every commuting probability entering (126)–(127) is a ratio of wage powers to the same sum of wage powers, so each probability is homogeneous of degree zero and continuous in $\{w_{m,o}, w_{f,o}\}_o$. Therefore $\mathcal{Z}(w)$ is continuous and homogeneous of degree zero.

Accounting identity. Summing $\mathcal{Z}_{om}$ and $\mathcal{Z}_{of}$ across all locations yields

$$\sum_{o=1}^{S}\left[L_{m,o}^D + L_{f,o}^D\right] = \sum_{o=1}^{S}\sum_{s=1}^{K} L_{os}^D = \sum_{o=1}^{S}\left[L_{m,o} + L_{f,o}\right],$$

where the first equality uses $s_{m,os} + s_{f,os} = 1$ and the second follows from the accounting identity that every worker who lives somewhere also works somewhere. Hence $\sum_\ell \mathcal{Z}_\ell(w) = 0$.

Existence. Since $\mathcal{Z}$ is continuous, homogeneous of degree zero, and satisfies the accounting identity $\sum_\ell \mathcal{Z}_\ell(w) = 0$, standard excess demand arguments apply.[46]

Normalizing wages via $\sum_{o,g} w_{g,o} = 1$, Brouwer's fixed-point theorem yields the existence of $\{w_{m,o}, w_{f,o}\}$ such that $\mathcal{Z}(w) = 0$, determined up to the normalization. □

[46]The existence argument for an excess demand system $\mathcal{Z} : \mathbb{R}_{++}^{2S} \to \mathbb{R}^{2S}$ requires: (i) continuity on strictly positive wages; (ii) homogeneity of degree zero, so only relative wages matter; (iii) the accounting identity $\sum_\ell \mathcal{Z}_\ell(w) = 0$; and, in the usual formulation, (iv) a lower bound $\mathcal{Z}_\ell(w) \geq -\bar{M}_\ell$ for each market ℓ. In the

G.3. Existence of the wage vector in the interdependent commuting model

Proposition 2 (Wage inversion, interdependent model)**.** *Take the same data and parameters as in Proposition 1, augmented with the partial-pooling friction $v \in [1, \infty)$ and the Pareto weight parameters $\{\zeta_0, \zeta_1\}$ of Block 1. There exists a vector of location wages $\{w_{m,o}, w_{f,o}\}_{o=1}^{S}$, determined up to a normalization, that clears labor markets at every location under the interdependent commuting model with partial pooling.*

Proof. Under interdependent commuting, dual earner spouses solve the collective household problem of Section 4.1. Labor supplied by gender g to workplace d is

$$L_{m,d} = \sum_{o=1}^{S} N_o^m \varpi_{d|o}^m + \sum_{o=1}^{S} N_o^{mf}(1-\mu_o)\pi_{d|o,0}^m + \sum_{o=1}^{S} N_o^{mf}\mu_o \sum_{j=1}^{S} \Pi_{dj|o}, \tag{129}$$

$$L_{f,d} = \sum_{o=1}^{S} N_o^f \varpi_{d|o}^f + \sum_{o=1}^{S} N_o^{mf}\mu_o \pi_{d|o,1}^f, \tag{130}$$

where the joint dual earner probability $\Pi_{ij|o} = \pi_{i|o,j}^m \pi_{j|o,1}^f$ is built from the husband's conditional probability (**??**), the wife's marginal (**??**), and the husband's market access term Φ_{oj} defined in (**??**). Both $\pi_{i|o,j}^m$ and $\pi_{j|o,1}^f$ depend on the regime specific wage aggregator $\Xi_{ij}(w_{m,i}, w_{f,j})$ in (**??**) and on the Pareto weight $\zeta_{ij}(w_{m,i}/w_{f,j})$.

Define the excess demand system as in (128), now built from (129)–(130) instead of (126)–(127).

Homogeneity and continuity. The Pareto weight depends only on the ratio $w_{m,i}/w_{f,j}$, so ζ_{ij} is homogeneous of degree zero in wages. Across the three regimes in (**??**), $\Xi_{ij}(w_{m,i}, w_{f,j})$ is homogeneous of degree one, and the regime boundaries $\underline{a}_{ij} w_{f,j}$ and $\overline{a}_{ij} w_{f,j}$ are themselves homogeneous of degree one, so the regime indicator is a homogeneous-of-degree-zero function of the wage ratio. Hence the ratio

$$\frac{\Xi_{ij}^{\theta^m}}{(d_{oi}^m)^{(1-\zeta_{ij})\theta^m} (d_{oj}^f)^{\zeta_{ij}\theta^m}}$$

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present environment, the lower-bound property is immediate because labor supply to any location cannot exceed the finite mass of workers of that gender. To conclude market clearing in every market, it is useful to rule out boundary wage vectors. A standard boundary condition rules out zero wage fixed points by requiring that any market whose wage tends to zero has strictly positive excess demand in the limit. In the present model, partial pooling delivers this condition because labor supply to a zero wage market vanishes, while labor demand remains positive. This is the step that may fail under interdependent commuting with perfect pooling.

Given these properties, homogeneity allows one to normalize wages to $\Delta \equiv \{w \in \mathbb{R}_+^{2S} : \sum_\ell w_\ell = 1\}$ and define the continuous map

$$\Gamma_\ell(w) \equiv \frac{w_\ell + \max\{0, \mathcal{Z}_\ell(w)\}}{\sum_{r=1}^{2S} [w_r + \max\{0, \mathcal{Z}_r(w)\}]}, \qquad \ell = 1, \ldots, 2S,$$

which maps Δ into itself. Brouwer's fixed-point theorem then yields $w^* \in \Delta$ such that $\Gamma(w^*) = w^*$. At such a fixed point, using $\sum_r w_r^* = 1$, $\max\{0, \mathcal{Z}_\ell(w^*)\} = w_\ell^* \sum_r \max\{0, \mathcal{Z}_r(w^*)\}$. The boundary condition rules out zero wages, so $w^* \gg 0$; if $\sum_r \max\{0, \mathcal{Z}_r(w^*)\} > 0$ then $\mathcal{Z}_\ell(w^*) > 0$ for every ℓ, contradicting the accounting identity $\sum_\ell \mathcal{Z}_\ell(w^*) = 0$. Hence $\sum_r \max\{0, \mathcal{Z}_r(w^*)\} = 0$, so $\mathcal{Z}_\ell(w^*) \leq 0$ for all ℓ, and $\sum_\ell \mathcal{Z}_\ell(w^*) = 0$ then gives $\mathcal{Z}_\ell(w^*) = 0$ componentwise. See Mas-Colell, Whinston, and Green (1995, Proposition 17.C.1).

is homogeneous of degree θ^m in wages, which cancels within $\pi^m_{i|o,j}$ because the denominator Φ_{oj} aggregates the same object over alternatives. The same cancellation applies to $\pi^f_{j|o,1}$ via $\Phi_{oj}^{\theta^f/\theta^m}$, and to the breadwinner and singles probabilities as in the independent case. Each commuting probability is therefore homogeneous of degree zero in wages. Continuity across regimes follows from the value matching property of the collective problem: at each regime boundary $w_{m,i} = \underline{a}_{ij} w_{f,j}$ or $w_{m,i} = \bar{a}_{ij} w_{f,j}$ the constrained and interior allocations coincide, so Ξ_{ij} is continuous in $(w_{m,i}, w_{f,j})$ across the piecewise definition.

Accounting identity. The argument replicates the independent-model case because the aggregate accounting identity $\sum_o [L^D_{m,o} + L^D_{f,o}] = \sum_o [L_{m,o} + L_{f,o}]$, equivalently $\sum_\ell \mathcal{Z}_\ell(w) = 0$, holds regardless of which commuting model generates the destination choices.

Existence. Because $\mathcal{Z}$ is homogeneous of degree zero, only relative wages matter, so wages can be normalized to the simplex

$$\Delta \equiv \left\{ w \in \mathbb{R}^{2S}_+ : \sum_{\ell=1}^{2S} w_\ell = 1 \right\}.$$

Consider the continuous mapping

$$\Gamma_\ell(w) \equiv \frac{w_\ell + \max\{0, \mathcal{Z}_\ell(w)\}}{\sum_{r=1}^{2S} \left[w_r + \max\{0, \mathcal{Z}_r(w)\} \right]}, \qquad \ell = 1, \ldots, 2S,$$

which raises the relative wage of markets with positive excess demand and then renormalizes wages back to Δ. Since Γ maps Δ into itself, Brouwer's fixed-point theorem yields a fixed point $w^* \in \Delta$.[47] At such a fixed point, using $\sum_r w^*_r = 1$,

$$\max\{0, \mathcal{Z}_\ell(w^*)\} = w^*_\ell \sum_r \max\{0, \mathcal{Z}_r(w^*)\}.$$

To conclude market clearing in every market, however, it is useful to rule out non-positive wage vectors. The identity above multiplies each excess demand by its own wage, so a market with zero wage carries no force: it could fail to clear while the identity still holds. This is precisely the step that can fail under perfect pooling. When $v \to \infty$, the dual earner problem always remains in the interior regime and $\Xi_{ij} = K_I(w_{m,i} + w_{f,j})$. As a result, if one spouse's wage tends to zero at some destination, that spouse can still be part of an attractive workplace pair because private consumption can be financed out of the partner's income through the shared budget. The corresponding commuting probability therefore does not vanish at the boundary. If labor supply to that market remains large enough, excess demand need not be positive, and the fixed-point map has no force pushing the zero wage back into the interior.

Partial pooling shuts down this mechanism. When $v < \infty$ and $w_{g,d} \to 0$, the constraint $PC_{g,o} \leq v w_{g,d}$ forces the private consumption of the spouse with the vanishing wage to

[47] Brouwer's fixed-point theorem applies because Δ is a nonempty, compact, and convex subset of $\mathbb{R}^{2S}$, and $\Gamma : \Delta \to \Delta$ is continuous. Here continuity of Γ follows from continuity of $\mathcal{Z}$, while $\Gamma(w) \in \Delta$ by construction, since each component is nonnegative and the denominator renormalizes the vector to sum to one. Hence a fixed point $w^* \in \Delta$ exists.

zero. In the binding regimes of (**??**), Ξ_{ij} contains a factor $w_{m,i}^{\beta(1-\zeta_{ij})}$ or $w_{f,j}^{\beta\zeta_{ij}}$, so $\Xi_{ij} \to 0$ as the constrained spouse's wage vanishes. Because commuting probabilities inherit this term, the probability that a dual earner husband or wife commutes to a zero wage destination also converges to zero. Equivalently, partner income can no longer sustain positive labor supply to a zero wage market.

Thus, under partial pooling, a zero wage creates strictly positive excess demand rather than leaving the market inactive. Labor supply to the zero wage market vanishes, while labor demand remains positive because firms always demand inputs whose relative wage becomes arbitrarily low. This delivers the boundary condition: whenever normalized wages approach a boundary point of Δ, any market whose wage tends to zero has strictly positive excess demand. The map Γ then pushes that wage back toward the interior. Hence the Brouwer fixed point is interior, $w^* \gg 0$. With $w^* \gg 0$, if $\sum_r \max\{0, \mathcal{Z}_r(w^*)\} > 0$ then $\mathcal{Z}_\ell(w^*) > 0$ for every ℓ, contradicting the accounting identity $\sum_\ell \mathcal{Z}_\ell(w^*) = 0$; hence $\sum_r \max\{0, \mathcal{Z}_r(w^*)\} = 0$, so $\mathcal{Z}_\ell(w^*) \leq 0$ for all ℓ, and $\sum_\ell \mathcal{Z}_\ell(w^*) = 0$ then gives $\mathcal{Z}_\ell(w^*) = 0$ for every market. The equilibrium wage vector is determined up to a normalization. □

G.4. Uniqueness

Uniqueness, does not follow automatically from the arguments above. In a standard commuting-inversion (e.g., Ahlfeldt et al. (2015)), uniqueness follows because the excess demand system satisfies gross substitution: raising the wage in one workplace reduces excess demand there and raises excess demand elsewhere. That property rules out multiplicity. In the interdependent model, this sufficient condition can fail. A higher male wage at j raises the joint dual earner value of destination pairs involving j, in particular (j, j), through Ξ_{jj}. In some regions of the domain of wages, this increases female labor supply to j. Since labor supply enters $\mathcal{Z}_{f,j} = L^D_{f,j} - L_{f,j}$ with a negative sign, the cross effect of $w_{m,j}$ on $\mathcal{Z}_{f,j}$ is not generally positive. Thus the gross substitution argument used in the standard commuting inversion does not directly apply. This failure does not imply multiplicity: we can get uniqueness if these within household spillovers are weak enough. For example, this can happen when θ^f is small and the wife's destination choice responds only mildly to changes in the husband's wage.

Later, in the appendix, I make this intuition explicit by showing where the ambiguous cross spouse terms enter the derivatives. I derive the relevant derivatives and the resulting ambiguity in Section G.9. In practice, I assess uniqueness numerically by solving the inversion from multiple starting values and checking whether the algorithm converges to the same fixed point.

G.5. Identification of the pooling friction v

In the interdependent model, I also estimate the partial-pooling friction $v \in [1, \infty)$. Recall that the lower bound $v = 1$ corresponds to the case in which only housing is pooled across spouses, while the limit $v \to \infty$ corresponds to full income pooling (see Section 4.1). I pin down v by matching the gender gap in the Fact 2 interdependence coefficients, $\psi_m - \psi_f$.

Targets from the data. Fact 2 defines ψ_m as the coefficient from a two-stage procedure applied to dual earner couples. In the first stage, I estimate by PPML

$$\log \pi^{m|f}_{i|o,j} = \gamma\, t_{oi} + \alpha^{m|f}_{oj} + \delta^{m|f}_{ij} + \varepsilon^{(1)}_{oij}, \tag{131}$$

where $\pi^{m|f}_{i|o,j}$ is the empirical share of husbands who work in i, conditional on residing in o and having a wife who works in j, and $\delta^{m|f}_{ij}$ is a husband-destination-by-wife-destination fixed effect. In the second stage, I collapse across origins o and estimate by OLS

$$\log \pi^{m|f}_{i|j} = \psi_m\, \widehat{\delta}^{m|f}_{ij} + \mathrm{FE}_i + \mathrm{FE}_j + \varepsilon^{(2)}_{ij}. \tag{132}$$

The coefficient ψ_m measures how strongly the husband's conditional commuting distribution responds to the wife's destination, beyond the additive husband- and wife-destination fixed effects. Under full independence, $\psi_m = 0$. Under strong partial pooling, ψ_m approaches one.[48] The wife's analog, ψ_f, is constructed symmetrically.

Model implied counterparts. Let $\widetilde{\pi}^{m|f}_{oij}(v; w(v))$ be the model implied conditional probability from (**??**), evaluated at the zone level wages $w(v)$ from Proposition 2. The inversion is solved at the zone level, which is the level at which the model is defined. However, the empirical moments in Fact 2 are estimated at the municipality level. Therefore, before applying the two-stage procedure to the model, I first aggregate the model implied zone level conditional probabilities to municipality level origin-destination cells. I then compute the model implied coefficients $\psi_m(v)$ and $\psi_f(v)$ by running the same two-stage procedure (131)–(132) on these aggregated model implied probabilities.

In the wife's case, the conditional on the husband's destination is constructed at the zone level from $\pi^{f}_{j|o,1}$ and $\pi^{m}_{i|o,j}$ via Bayes' rule:

$$\widetilde{\pi}^{f|m}_{oji}(v; w(v)) = \frac{\Pi_{ij|o}(v; w(v))}{\sum_{j'=1}^{S} \Pi_{ij'|o}(v; w(v))}. \tag{133}$$

I then aggregate these wife-conditional probabilities to the municipality level before estimating the model implied $\psi_f(v)$.

Estimator. The estimator for v solves

$$\widehat{v} = \arg\min_{v \geq 1} \left[\big(\psi_m(v) - \psi_f(v)\big) - \big(\widehat{\psi}_m - \widehat{\psi}_f\big) \right]^2, \tag{134}$$

Because $\psi_m(v)$ and $\psi_f(v)$ are not available in closed form, I evaluate (134) by grid search, invoking the wage-inversion Proposition 2 at every candidate v.

[48]This interpretation is exact when the model and the empirical moments are computed at the same geographic level. In the model, interdependence is microfounded at the zone level, whereas Fact 2 is estimated after aggregating commuting choices to the municipality level. This aggregation can change the scale of the second stage coefficient ψ. Intuitively, a municipality level destination pair combines many zone level destination pairs. When we collapse to the municipality level, the cells with larger commuting flows receive more weight. If high-flow zone pairs are also the pairs where the wife's workplace is especially informative about the husband's workplace, the municipality level conditional probability can respond more strongly than the underlying zone level relationship. For this reason, the estimated ψ_m need not be mechanically bounded above by one, even though the zone level model nests independence at $\psi_m = 0$ and strong interdependence at $\psi_m = 1$.

G.6. Algorithm

The numerical procedure nests the wage inversion inside the v grid search. I solve the independent model first and use its fixed point as the starting value for every candidate v in the interdependent model.

1. **Independent model.** For each year $y \in \{2007, 2017\}$, initialize wages at $w^{(0)}$. At each iteration k: compute singles, breadwinner, and dual-earner-independent commuting probabilities from the Fréchet expressions. Evaluate model implied labor supply (126)–(127) and labor demand (125). Update $w_{g,o}^{(k+1)} = \rho w_{g,o}^{(k)} + (1-\rho)\, w_{g,o}^{(k)} \cdot \left[L_{g,o}^{D}(w^{(k)}) / L_{g,o}(w^{(k)})\right]$ with damping factor $\rho \in (0,1)$. Normalize wages at each iteration. Iterate until $\max_{g,o} |\mathcal{Z}_{og}(w^{(k)})| / L_{g,o}(w^{(k)})$ falls below tolerance. Denote the fixed point $w^{\mathrm{I},y}$.

2. **Interdependent model, v grid.** Build a grid $\{v^{(1)}, \ldots, v^{(G)}\}$ over $[1, \bar{v}]$ for a large upper bound $\bar{v}$. For each $v^{(g)}$ and year y: initialize at $w^{\mathrm{I},y}$. At each iteration, compute singles and breadwinner commuting probabitilies. For dual earners, recompute ζ_{ij}, classify each (i,j) pair into one of the three regimes in (**??**), evaluate Ξ_{ij}, the husband's conditional $\pi^{m}_{i|o,j}$, the wife's marginal $\pi^{f}_{j|o,1}$, and the joint $\Pi_{ij|o}$. Then, aggregate commuting flows into (129)–(130). Update wages with the same damped rule as in the independent case. Iterate until convergence. Denote the fixed point $w(v^{(g)}; y)$.

3. **Match $\{\psi_m, \psi_f\}$.** For each $v^{(g)}$, evaluate $\psi_m(v^{(g)})$ and $\psi_f(v^{(g)})$ by running the two-stage procedure (131)–(132) on the model implied conditional probabilities at $w(v^{(g)}; 2017)$. Pick $\hat{v}$ according to (134).

4. **Output.** The output of Block 2 consists of $w^{\mathrm{I},2007}, w^{\mathrm{I},2017}$ (independent model), $w(\hat{v}; 2007)$, $w(\hat{v}; 2017)$ (interdependent model), and $\hat{v}$. The fit of $\hat{v}$ is summarized by the realized values of $\psi_m(\hat{v})$ and $\psi_f(\hat{v})$ relative to $\widehat{\psi}_m$ and $\widehat{\psi}_f$.

G.7. Standard error

The grid search delivers a point estimate $\hat{v}$. I report a standard error using the GMM sandwich, mirroring the construction in Appendix F.4. With the single gap moment $g(v) = \big(\psi_m(v) - \psi_f(v)\big) - \big(\widehat{\psi}_m - \widehat{\psi}_f\big)$, the variance is

$$\mathrm{Var}(\hat{v}) = \left(G'G\right)^{-1} G'\,\Omega\, G\left(G'G\right)^{-1}, \tag{135}$$

which, with the scalar Jacobian $G = \partial g / \partial v = \partial(\psi_m - \psi_f)/\partial v$ and the scalar moment variance Ω, reduces to $\mathrm{Var}(\hat{v}) = \Omega / G^2$. I evaluate G numerically around $\hat{v}$. I set $\Omega = \widehat{\mathrm{SE}}_m^2 + \widehat{\mathrm{SE}}_f^2$, the sampling variance of the empirical gap, where $\widehat{\mathrm{SE}}_m$ and $\widehat{\mathrm{SE}}_f$ are the standard errors of the marginal Fact 2 ψ-regressions. I treat the male and female ψ-regressions as independent.[49] Equivalently, $\widehat{\mathrm{SE}}(\hat{v}) = \sqrt{\widehat{\mathrm{SE}}_m^2 + \widehat{\mathrm{SE}}_f^2}\, /\, |G|$.

[49] A fuller treatment would stack the male and female PPML regressions in a single specification with a sex interaction and recover the full 2×2 covariance from the joint cluster-robust variance.

G.8. Estimation results and identification diagnostics

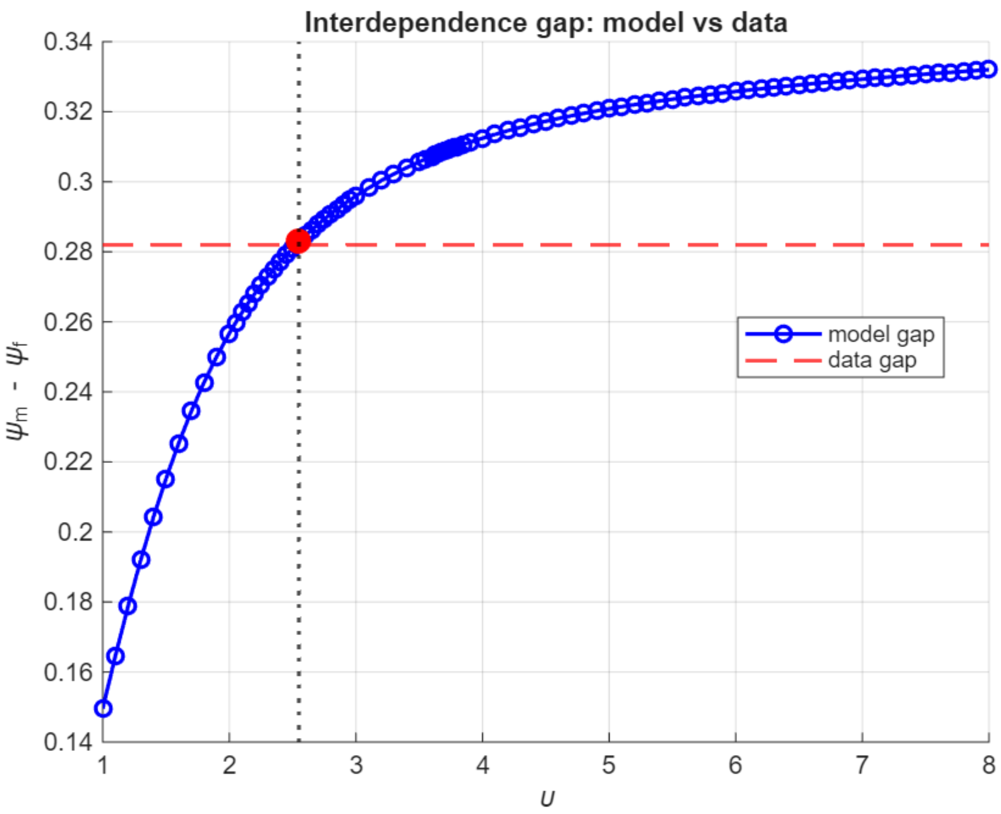
The Block 2 estimate of the pooling friction is $\widehat{v} \approx 2.55$ (SE 0.90), reported in Panel D of Table 3. The estimator targets the gender gap in interdependence. At the optimum the model implies $\psi_m(\widehat{v}) \approx 0.955$ and $\psi_f(\widehat{v}) \approx 0.672$. The resulting gap $\psi_m(\widehat{v}) - \psi_f(\widehat{v}) \approx 0.283$ matches the empirical gap $\widehat{\psi}_m - \widehat{\psi}_f \approx 0.282$, where the underlying empirical coefficients are $\widehat{\psi}_m \approx 0.868$ (SE 0.018) and $\widehat{\psi}_f \approx 0.586$ (SE 0.026).[50] Substantively, the binding constraint splits a spouse's private consumption into a share $1/\widehat{v} \approx 39\%$ that their own wage can fund and a remaining share $(\widehat{v}-1)/\widehat{v} \approx 61\%$ drawn from the partner's wage through household pooling.

Recall the partial-pooling constraint: each spouse's private consumption is capped at v times their own earnings, $PC_k \leq v\, w_k$. The lower bound $v = 1$ corresponds to no pooling—each spouse spends only their own wage on private consumption—and the constraint becomes inactive as $v \to \infty$, recovering full pooling. The estimate $\widehat{v} \approx 2.55$ sits well above the no-pooling boundary.

Identification diagnostic. Figure 27 plots the model implied interdependence gap $\psi_m(v) - \psi_f(v)$ (blue curve) against the empirical gap $\widehat{\psi}_m - \widehat{\psi}_f$ (red dashed line) along the v grid. The model gap is monotone in v and crosses the data gap once, at $\widehat{v} \approx 2.55$ (red marker), which identifies the pooling friction.

[50] Because only the gap is targeted, the individual levels $\psi_m(\widehat{v})$ and $\psi_f(\widehat{v})$ are untargeted moments. The model still reproduces them approximately, overstating each somewhat relative to the empirical $\widehat{\psi}_m$ and $\widehat{\psi}_f$.

Figure 27: Pooling-friction estimation: model implied interdependence gap $\psi_m - \psi_f$ versus the data gap along the v grid.



Notes. The blue curve is the model implied interdependence gap $\psi_m(v) - \psi_f(v)$ along the v grid $\{1.0, 1.1, \ldots, 8.0\}$. The red dashed line is the empirical gap $\widehat{\psi}_m - \widehat{\psi}_f$. The red marker is the optimum $\widehat{v}$, where the model gap equals the data gap.

G.9. Detailed derivation of the gross substitution analysis

This subsection derives the terms behind the uniqueness discussion in Section G.4. To keep the algebra transparent, I first hold the Pareto weight fixed at $\zeta_{ij} = \zeta \in (0,1)$. At the end of the subsection, I discuss how the derivatives change when ζ_{ij} depends on wages through the logit specification used in estimation.

Stacked excess demand. Recall that labor market excess demand is

$$\mathcal{Z}_{og}(w) = L^D_{g,o}(w_{m,o}, w_{f,o}) - L_{g,o}(w), \qquad g \in \{m, f\},\ o = 1, \ldots, S. \tag{136}$$

Stack these objects in a vector $\mathcal{Z}(w)$ of length $2S$. Index a generic market by z, where $z \leq S$ refers to male excess demand at location z, and $z > S$ refers to female excess demand at location $z - S$. Let w_z denote the corresponding wage.

Gross substitution requires

$$\frac{\partial \mathcal{Z}_z(w)}{\partial w_z} < 0, \qquad \frac{\partial \mathcal{Z}_z(w)}{\partial w_{z'}} > 0 \quad \text{for all } z' \neq z. \tag{137}$$

Equivalently, one can take derivatives with respect to log wages and use

$$\frac{\partial \mathcal{Z}_z(w)}{\partial \log w_{z'}} = w_{z'} \frac{\partial \mathcal{Z}_z(w)}{\partial w_{z'}},$$

which preserves the sign because $w_{z'} > 0$.

Derivatives of labor demand. From (124)–(125), labor demand depends only on own-location wages. Differentiating,

$$\frac{\partial \log L^D_{m,o}}{\partial \log w_{m,o}} = -\sigma \sum_{s=1}^{K} \frac{s_{m,os} L^D_{os}}{L^D_{m,o}} \left(1 - s_{m,os}\right) < 0, \tag{138}$$

$$\frac{\partial \log L^D_{m,o}}{\partial \log w_{f,o}} = \sigma \sum_{s=1}^{K} \frac{s_{m,os} L^D_{os}}{L^D_{m,o}} s_{f,os} > 0. \tag{139}$$

The first derivative is negative because a higher male wage makes male labor more expensive at location o, so firms substitute away from men. The second derivative is positive because a higher female wage makes female labor more expensive, so firms substitute toward men. The symmetric expressions hold for $L^D_{f,o}$. Cross-location derivatives are zero because L^D_{os} is observed and $s_{g,os}$ depends only on wages at location o. Thus labor demand satisfies the usual CES substitution property within location.

Derivatives of singles labor supply. Singles' commuting probabilities are standard Fréchet-logit objects. For the singles component of labor supply,

$$L^S_{g,d} \equiv \sum_o N^g_o \varpi^g_{d|o},$$

we have

$$\frac{\partial \log L^S_{m,d}}{\partial \log w_{m,d}} = \sum_o \frac{N^m_o \varpi^m_{d|o}}{L^S_{m,d}} \left(1 - \varpi^m_{d|o}\right) \vartheta^m > 0, \tag{140}$$

$$\frac{\partial \log L^S_{m,d}}{\partial \log w_{m,d'}} = -\sum_o \frac{N^m_o \varpi^m_{d|o}}{L^S_{m,d}} \varpi^m_{d'|o} \vartheta^m < 0, \qquad d' \neq d, \tag{141}$$

$$\frac{\partial \log L^S_{m,d}}{\partial \log w_{f,d'}} = 0 \qquad \forall d'. \tag{142}$$

The own-wage derivative is positive because a higher wage at d makes d more attractive relative to the other workplace options. The cross-destination derivative is negative because a higher wage at another destination d' pulls some single men away from d. Female wages do not enter single men's commuting choices, so the cross gender derivative is zero. The expressions for single women are analogous.

Since labor supply enters excess demand with a minus sign, these supply responses reinforce gross substitution. A higher wage at d raises labor supply to d, which lowers $\mathcal{Z}_{g,d}$. A higher same gender wage at another destination d' lowers labor supply to d, which raises $\mathcal{Z}_{g,d}$.

Derivatives of breadwinner labor supply. The husband breadwinner block has the same structure. Let

$$L^{MB}_{m,d} \equiv \sum_o N^{mf}_o (1-\mu_o)\pi^m_{d|o,0}.$$

Then

$$\frac{\partial \log L^{MB}_{m,d}}{\partial \log w_{m,d}} = \sum_o \frac{N^{mf}_o (1-\mu_o)\pi^m_{d|o,0}}{L^{MB}_{m,d}} (1-\pi^m_{d|o,0})\theta^m > 0, \tag{143}$$

$$\frac{\partial \log L^{MB}_{m,d}}{\partial \log w_{m,d'}} = -\sum_o \frac{N^{mf}_o (1-\mu_o)\pi^m_{d|o,0}}{L^{MB}_{m,d}} \pi^m_{d'|o,0}\theta^m < 0, \qquad d' \neq d, \tag{144}$$

$$\frac{\partial \log L^{MB}_{m,d}}{\partial \log w_{f,d'}} = 0 \qquad \forall d'. \tag{145}$$

A higher male wage at d attracts breadwinner husbands to d, while a higher male wage at another destination d' draws them away from d. Female wages do not enter the male-breadwinner commuting problem. Thus this block also reinforces gross substitution in the male markets.

Dual earner labor supply. The dual earner labor supply is:

$$L^{DE}_{m,d} \equiv \sum_o N^{mf}_o \mu_o \sum_{j=1}^{S} \Pi_{dj|o}, \qquad L^{DE}_{f,d} \equiv \sum_o N^{mf}_o \mu_o \pi^f_{d|o,1},$$

where

$$\Pi_{ij|o} = \pi^m_{i|o,j}\pi^f_{j|o,1}.$$

With fixed ζ, define

$$G_{oij} \equiv \frac{\Xi^{\theta^m}_{ij}}{(d^m_{oi})^{(1-\zeta)\theta^m}(d^f_{oj})^{\zeta\theta^m}}. \tag{146}$$

Then

$$\pi^m_{i|o,j} = \frac{G_{oij}}{\sum_\ell G_{o\ell j}}, \qquad \Phi_{oj} = \sum_\ell G_{o\ell j}, \qquad \pi^f_{j|o,1} = \frac{\Phi^{\theta^f/\theta^m}_{oj}}{\sum_n \Phi^{\theta^f/\theta^m}_{on}}.$$

Let

$$\mathsf{E}^m_{ij} \equiv \frac{\partial \log \Xi_{ij}}{\partial \log w_{m,i}}, \qquad \mathsf{E}^f_{ij} \equiv \frac{\partial \log \Xi_{ij}}{\partial \log w_{f,j}}.$$

For fixed ζ, the elasticities implied by (**??**) are

$$\mathsf{E}^m_{ij} = \begin{cases} \beta(1-\zeta) + (1-\beta+\beta\zeta)\dfrac{(1-v)w_{m,i}}{w_{f,j}+(1-v)w_{m,i}}, & w_{m,i} < \underline{a}\, w_{f,j}, \\ \dfrac{w_{m,i}}{w_{m,i}+w_{f,j}}, & \underline{a}\, w_{f,j} \le w_{m,i} \le \bar{a}\, w_{f,j}, \\ (1-\beta\zeta)\dfrac{w_{m,i}}{w_{m,i}+(1-v)w_{f,j}}, & w_{m,i} > \bar{a}\, w_{f,j}, \end{cases} \tag{147}$$

$$\mathsf{E}^f_{ij} = \begin{cases} (1-\beta+\beta\zeta)\dfrac{w_{f,j}}{w_{f,j}+(1-v)w_{m,i}}, & w_{m,i} < \underline{a}\, w_{f,j}, \\ \dfrac{w_{f,j}}{w_{m,i}+w_{f,j}}, & \underline{a}\, w_{f,j} \le w_{m,i} \le \bar{a}\, w_{f,j}, \\ \beta\zeta + (1-\beta\zeta)\dfrac{(1-v)w_{f,j}}{w_{m,i}+(1-v)w_{f,j}}, & w_{m,i} > \bar{a}\, w_{f,j}. \end{cases} \tag{148}$$

where

$$\underline{a} \equiv \frac{\beta(1-\zeta)}{v-\beta(1-\zeta)}, \qquad \bar{a} \equiv \frac{v-\beta\zeta}{\beta\zeta}.$$

The value matching of the household problem implies that Ξ_{ij}, and hence the commuting probabilities, are continuous across regime boundaries.

Derivative of the husband's conditional probability. Since fixed ζ removes any wage derivative of the commute-cost exponents, the derivatives of the husband's conditional probability depend only on the elasticities of Ξ_{ij}. Recall that

$$\pi^m_{i|o,j} = \frac{G_{oij}}{\sum_{\ell'} G_{o\ell' j}}, \qquad G_{oij} = \frac{\Xi_{ij}^{\theta^m}}{(d^m_{oi})^{(1-\zeta)\theta^m}(d^f_{oj})^{\zeta\theta^m}}.$$

Taking logs,

$$\log \pi^m_{i|o,j} = \log G_{oij} - \log \sum_{\ell'} G_{o\ell' j}.$$

For any wage w, differentiating gives

$$\frac{\partial \log \pi^m_{i|o,j}}{\partial \log w} = \frac{\partial \log G_{oij}}{\partial \log w} - \sum_k \pi^m_{k|o,j}\frac{\partial \log G_{okj}}{\partial \log w}.$$

The first term captures how the wage changes the value of the specific pair (i,j). The second subtracts the average effect of the same wage change across all husband destinations k, holding the wife's destination j fixed.

For male wages,

$$\frac{\partial \log G_{oij}}{\partial \log w_{m,\ell}} = \theta^m \mathsf{E}^m_{ij}\mathbf{1}\{\ell = i\}.$$

Substituting this into the generic derivative above gives

$$\frac{\partial \log \pi^m_{i|o,j}}{\partial \log w_{m,\ell}} = \theta^m\left[\mathsf{E}^m_{ij}\mathbf{1}\{\ell = i\} - \pi^m_{\ell|o,j}\mathsf{E}^m_{\ell j}\right]. \tag{149}$$

A male wage $w_{m,\ell}$ directly raises the value of pair (i,j) only when $\ell = i$. At the same time, it raises the denominator of the husband's conditional choice problem through alternative (ℓ, j).

This denominator effect is weighted by the probability that the husband chooses ℓ, conditional on wife destination j.

For female wages,

$$\frac{\partial \log G_{oij}}{\partial \log w_{f,\ell}} = \theta^m \mathsf{E}^f_{ij} \mathbf{1}\{\ell = j\}.$$

Substituting into the generic derivative gives

$$\frac{\partial \log \pi^m_{i|o,j}}{\partial \log w_{f,\ell}} = \theta^m \mathbf{1}\{\ell = j\} \left[\mathsf{E}^f_{ij} - \sum_k \pi^m_{k|o,j} \mathsf{E}^f_{kj} \right]. \tag{150}$$

A female wage $w_{f,\ell}$ affects the husband's conditional problem only when the wife destination under consideration is $j = \ell$. Conditional on that wife destination, the first term captures how the female wage changes the value of pair (i, j), while the second subtracts the average effect across all possible husband destinations k.

Derivative of the wife's marginal probability. The wife's marginal probability depends on the husband's conditional market access term Φ_{oj}. Recall that

$$\pi^f_{j|o,1} = \frac{\Phi_{oj}^{\theta^f/\theta^m}}{\sum_n \Phi_{on}^{\theta^f/\theta^m}}.$$

Taking logs,

$$\log \pi^f_{j|o,1} = \frac{\theta^f}{\theta^m} \log \Phi_{oj} - \log \sum_n \Phi_{on}^{\theta^f/\theta^m}.$$

For any wage w, differentiating gives

$$\frac{\partial \log \pi^f_{j|o,1}}{\partial \log w} = \frac{\theta^f}{\theta^m} \left[\frac{\partial \log \Phi_{oj}}{\partial \log w} - \sum_n \pi^f_{n|o,1} \frac{\partial \log \Phi_{on}}{\partial \log w} \right].$$

The first term captures how the wage changes the value of wife destination j. The second subtracts the average effect of the same wage change across all wife destinations.

From the husband's conditional problem,

$$\frac{\partial \log \Phi_{oj}}{\partial \log w_{m,\ell}} = \theta^m \pi^m_{\ell|o,j} \mathsf{E}^m_{\ell j},$$

and

$$\frac{\partial \log \Phi_{oj}}{\partial \log w_{f,\ell}} = \theta^m \mathbf{1}\{\ell = j\} \sum_k \pi^m_{k|o,j} \mathsf{E}^f_{kj}.$$

Substituting these expressions into the generic derivative above yields

$$\frac{\partial \log \pi^f_{j|o,1}}{\partial \log w_{m,\ell}} = \theta^f \left[\pi^m_{\ell|o,j} \mathsf{E}^m_{\ell j} - \sum_n \pi^f_{n|o,1} \pi^m_{\ell|o,n} \mathsf{E}^m_{\ell n} \right], \tag{151}$$

$$\frac{\partial \log \pi^f_{j|o,1}}{\partial \log w_{f,\ell}} = \theta^f \left[\mathbf{1}\{\ell = j\} \sum_k \pi^m_{k|o,j} \mathsf{E}^f_{kj} - \pi^f_{\ell|o,1} \sum_k \pi^m_{k|o,\ell} \mathsf{E}^f_{k\ell} \right]. \tag{152}$$

Derivative of the joint probability. Since

$$\Pi_{ij|o} = \pi^m_{i|o,j}\pi^f_{j|o,1},$$

we have

$$d\log\Pi_{ij|o} = d\log\pi^m_{i|o,j} + d\log\pi^f_{j|o,1}.$$

Thus, the derivative of the joint probability combines two margins: the husband's conditional reallocation across destinations i, holding the wife's destination j fixed, and the wife's marginal reallocation across destinations j.

For male wages, combining (149) and (151) gives

$$\begin{aligned}\frac{\partial\log\Pi_{ij|o}}{\partial\log w_{m,\ell}} &= \theta^m\left[\mathsf{E}^m_{ij}\mathbf{1}\{\ell=i\} - \pi^m_{\ell|o,j}\mathsf{E}^m_{\ell j}\right] \\ &\quad + \theta^f\left[\pi^m_{\ell|o,j}\mathsf{E}^m_{\ell j} - \sum_n \pi^f_{n|o,1}\pi^m_{\ell|o,n}\mathsf{E}^m_{\ell n}\right]. \end{aligned}\tag{153}$$

The first bracket is the husband's conditional response. The second bracket is the wife's marginal response: it compares the effect of $w_{m,\ell}$ on wife destination j to the average effect of $w_{m,\ell}$ across all wife destinations.

For female wages, combining (150) and (152) gives

$$\begin{aligned}\frac{\partial\log\Pi_{ij|o}}{\partial\log w_{f,\ell}} &= \theta^m\mathbf{1}\{\ell=j\}\left[\mathsf{E}^f_{ij} - \sum_k \pi^m_{k|o,j}\mathsf{E}^f_{kj}\right] \\ &\quad + \theta^f\left[\mathbf{1}\{\ell=j\}\sum_k \pi^m_{k|o,j}\mathsf{E}^f_{kj} - \pi^f_{\ell|o,1}\sum_k \pi^m_{k|o,\ell}\mathsf{E}^f_{k\ell}\right]. \end{aligned}\tag{154}$$

The first bracket captures how $w_{f,\ell}$ changes the husband's conditional distribution within the wife-destination column $j=\ell$. The second bracket captures how the same wage changes the wife's marginal probability of choosing j, relative to the average effect across wife destinations.

Dual earner female labor supply and the source of the ambiguity. The cleanest way to see the failure of gross substitution is to study female dual earner labor supply at destination d:

$$L^{DE}_{f,d} = \sum_o N^{mf}_o\mu_o\pi^f_{d|o,1}.$$

For any wage w, differentiating this object gives

$$\frac{\partial\log L^{DE}_{f,d}}{\partial\log w} = \sum_o \frac{N^{mf}_o\mu_o\pi^f_{d|o,1}}{L^{DE}_{f,d}}\frac{\partial\log\pi^f_{d|o,1}}{\partial\log w}.$$

Thus the derivative of aggregate female labor supply to d is a weighted average of the residence specific derivatives of the wife's marginal commuting probability.

First consider the own female wage derivative. Applying (152) with $j=d$ and $\ell=d$,

$$\frac{\partial\log\pi^f_{d|o,1}}{\partial\log w_{f,d}} = \theta^f\left[\sum_k \pi^m_{k|o,d}\mathsf{E}^f_{kd} - \pi^f_{d|o,1}\sum_k \pi^m_{k|o,d}\mathsf{E}^f_{kd}\right].$$

Therefore,

$$\frac{\partial \log \pi^f_{d|o,1}}{\partial \log w_{f,d}} = \theta^f \left(1 - \pi^f_{d|o,1}\right) \sum_k \pi^m_{k|o,d} \mathsf{E}^f_{kd}.$$

Aggregating across residences gives

$$\frac{\partial \log L^{DE}_{f,d}}{\partial \log w_{f,d}} = \sum_o \frac{N^{mf}_o \mu_o \pi^f_{d|o,1}}{L^{DE}_{f,d}} \theta^f \left(1 - \pi^f_{d|o,1}\right) \sum_k \pi^m_{k|o,d} \mathsf{E}^f_{kd} > 0. \tag{155}$$

This derivative is positive. A higher female wage at d raises the value of wife destination d. The relevant wage gain is averaged over the husband's possible destinations k, because the wife evaluates destination d through the husband's conditional market access term Φ_{od}. The term $1 - \pi^f_{d|o,1}$ is the usual logit adjustment: the effect is smaller when wife destination d already has a high probability.

Now consider the cross spouse derivative with respect to the male wage at the same destination, $w_{m,d}$. Applying (151) with $j = d$ and $\ell = d$,

$$\frac{\partial \log \pi^f_{d|o,1}}{\partial \log w_{m,d}} = \theta^f \left[\pi^m_{d|o,d} \mathsf{E}^m_{dd} - \sum_n \pi^f_{n|o,1} \pi^m_{d|o,n} \mathsf{E}^m_{dn} \right].$$

Aggregating across residences gives

$$\frac{\partial \log L^{DE}_{f,d}}{\partial \log w_{m,d}} = \sum_o \frac{N^{mf}_o \mu_o \pi^f_{d|o,1}}{L^{DE}_{f,d}} \theta^f \left[\pi^m_{d|o,d} \mathsf{E}^m_{dd} - \sum_n \pi^f_{n|o,1} \pi^m_{d|o,n} \mathsf{E}^m_{dn} \right]. \tag{156}$$

This term is not generally signed. The first term inside brackets captures the effect of a higher male wage at d on the value of the pair (d, d). If a husband working at d is especially attractive when the wife also works at d, then this force raises the wife's probability of choosing d. The second term subtracts the average effect of the same male wage increase across all wife destinations n. This is the denominator effect in the wife's marginal choice: if $w_{m,d}$ also raises the value of other pairs (d, n), then wife destination d does not gain as much relative to the other wife destinations.

The sign therefore depends on whether the pair (d, d) benefits more or less than the average wife destination from the increase in $w_{m,d}$. If

$$\pi^m_{d|o,d} \mathsf{E}^m_{dd} > \sum_n \pi^f_{n|o,1} \pi^m_{d|o,n} \mathsf{E}^m_{dn},$$

then a higher male wage at d raises female labor supply to d. If the inequality is reversed, it lowers female labor supply to d. This is exactly the cross spouse channel: the husband's wage can affect the wife's workplace choice because the household evaluates destination pairs jointly.

This is the source of the gross substitution failure. Female excess demand at d is

$$\mathcal{Z}_{f,d} = L^D_{f,d} - L_{f,d}.$$

Gross substitution with respect to $w_{m,d}$ requires

$$\frac{\partial \mathcal{Z}_{f,d}}{\partial \log w_{m,d}} > 0.$$

The labor demand component has the right sign:

$$\frac{\partial L^D_{f,d}}{\partial \log w_{m,d}} > 0,$$

because when male labor becomes more expensive at d, firms substitute toward female labor. But the labor supply component can also be positive:

$$\frac{\partial L^{DE}_{f,d}}{\partial \log w_{m,d}} > 0.$$

Since labor supply enters excess demand with a minus sign, this second force works against gross substitution:

$$\frac{\partial \mathcal{Z}_{f,d}}{\partial \log w_{m,d}} = \underbrace{\frac{\partial L^D_{f,d}}{\partial \log w_{m,d}}}_{>0} - \underbrace{\frac{\partial L_{f,d}}{\partial \log w_{m,d}}}_{\text{can be }>0}.$$

Hence the total sign is ambiguous. The standard gross substitution proof requires the full expression to be positive, but the interdependent household block can make the supply response large enough to offset the demand response. Thus the standard gross substitution argument does not apply in general.

Why this does not imply multiplicity. The failure of gross substitution is a failure of a sufficient condition, not a proof of multiplicity. Uniqueness can still obtain if the within household spillovers are weak enough. For example, when θ^f is small, the wife's destination choice responds only mildly to changes in the husband's wage, which dampens the cross spouse term in (156). More generally, uniqueness can hold when the standard forces that move labor demand and labor supply back toward market clearing dominate the cross spouse feedback created by interdependent commuting. In the quantitative implementation, I therefore assess uniqueness by solving the inversion from multiple starting values and checking whether the algorithm converges to the same fixed point.

Allowing ζ_{ij} to respond to wages. The derivations above hold ζ fixed. In the estimated model, however,

$$\zeta_{ij} = \Lambda\left(\zeta_0 + \zeta_1 \log \frac{w_{m,i}}{w_{f,j}}\right), \qquad \Lambda(x) = \frac{e^x}{1+e^x}.$$

In that case, changes in wages affect commuting probabilities through an additional channel. For example,

$$\frac{\partial \zeta_{ij}}{\partial \log w_{m,i}} = \zeta_1 \zeta_{ij}(1-\zeta_{ij}), \qquad \frac{\partial \zeta_{ij}}{\partial \log w_{f,j}} = -\zeta_1 \zeta_{ij}(1-\zeta_{ij}).$$

These terms enter the derivatives in two places. First, they change $\log \Xi_{ij}$ because the regime thresholds and the exponents in Ξ_{ij} depend on ζ_{ij}. Second, they change the effective commute-cost term

$$(1-\zeta_{ij}) \log d^m_{oi} + \zeta_{ij} \log d^f_{oj}.$$

Thus, when ζ_{ij} is wage dependent, a change in the male wage affects not only the household wage aggregator, but also the household's relative weight on the husband's and wife's commuting costs. These extra terms can either amplify or dampen the cross spouse spillover depending on the sign of ζ_1 and on whether the change in ζ_{ij} shifts weight toward the spouse whose commute is more costly. Therefore the fixed-ζ derivation identifies the baseline source of the gross substitution failure, while the logit-ζ_{ij} specification adds another wage dependent household-allocation channel that makes the sign ambiguity even less likely to be resolved analytically.

H. Block 3: Extensive Margin and Location Elasticities

This appendix describes the details of Block 3 from Section 5.2.4. The objective is to recover residential amenities $\{u_o^m, u_o^f, u_o^{mf}\}$, the household-productivity shifter $\{\xi_{o0}\}$, and location–industry productivities $\{A_{os}\}$. Inverting the model to recover these objects requires knowledge of the participation parameter ν and the residential location elasticities $\{\eta^m, \eta^f, \eta^{mf}\}$, which I estimate in this section.

The block therefore proceeds in three steps. First, conditional on the wages $\{w_{m,o}, w_{f,o}\}$ and pooling friction $\widehat{v}$ from Block 2, I use firms' first order conditions to invert location–industry productivities $\{A_{os}\}$ in Section H.1. Second, I estimate the female labor force participation Fréchet shape ν (Section H.2) and the residential location elasticities $\{\eta^m, \eta^f, \eta^{mf}\}$ (Section H.4) on the 2007–2017 panel of Lima zones using productivity-side Bartik instruments. Third, given the estimated elasticities, wages, market access terms, and observed dual earner shares, I invert the household-productivity shifter $\{\xi_{o0}\}$ (Section H.3) and residential amenities $\{u_o^m, u_o^f, u_o^{mf}\}$ (Section H.5). The numerical algorithm that stacks these steps is summarized in Section H.6.

Observables and parameters used in Block 3. Throughout this block I treat as observed, at years $y \in \{2007, 2017\}$: workplace employment by location and industry $\{L_{os}\}$, commercial rents $\{r_{F,o}\}$, residential rents $\{r_{R,o}\}$, residential population by household group $\{N_o^m, N_o^f, N_o^{mf}\}$, the dual earner share at each residence $\{\mu_o\}$, and commute times $\{t_{od}\}$. I take as given: from Block 0, the production-side parameters $\{\alpha_{ms}, \alpha_{fs}, \alpha_s, \sigma, \sigma_D\}$ and the housing-expenditure shares $\{\beta^g, \beta\}$; from Block 1, the commuting parameters $\{\varkappa^g, \vartheta^g, \kappa^m, \kappa^f, \theta^m, \theta^f\}$ and the Pareto weight parameters $\{\zeta_0, \zeta_1\}$; and from Block 2, the male and female wage vectors $\{w_{m,o}, w_{f,o}\}$ and the pooling friction $\widehat{v}$.

H.1. Productivity inversion

I recover $\{A_{os}\}$ in closed form from the production-side first order conditions (95)–(100). The only observable that is missing at this stage is the gender composition of employment within each location–industry cell. I impute it from the cost minimization conditions.

Proposition 3 (Closed form productivity inversion)**.** *Given the observables and Block 0–2 inputs stated above, the location–industry productivity vector $\{A_{os}\}$ is determined up to a normalization by the production-side first order conditions. Its closed form expression is*

$$A_{os} = \frac{(W_{os}/\alpha_s)^{\alpha_s}(r_{F,o}/(1-\alpha_s))^{1-\alpha_s}}{X_{os}^{1/(1-\sigma_D)}}, \tag{157}$$

where W_{os} is the CES gender wage composite in (97) *and $X_{os} \equiv p_{os}Q_{os}$ is total sales at location–industry cell (o, s).*

Proof. Four equations pin down A_{os} from the observables. First, within each cell (o, s), cost minimization between male and female labor yields the gender labor-expenditure shares $\omega_{g,os}$ in (99). Because $\omega_{g,os}$ is a share of the wage bill rather than of headcount, I convert it to a quantity share by dividing by the gender wage before applying it to the observed total employment L_{os}. This recovers gender specific employment,

$$L_{g,os} = \frac{\omega_{g,os}/w_{g,o}}{\sum_{h\in\{m,f\}} \omega_{h,os}/w_{h,o}} L_{os} = \frac{\alpha_{gs}^{\sigma} w_{g,o}^{-\sigma}}{\alpha_{ms}^{\sigma} w_{m,o}^{-\sigma} + \alpha_{fs}^{\sigma} w_{f,o}^{-\sigma}} L_{os}, \qquad g \in \{m, f\}. \tag{158}$$

The two expressions coincide because $\omega_{g,os} \propto \alpha_{gs}^{\sigma} w_{g,o}^{1-\sigma}$, so dividing by $w_{g,o}$ leaves $\alpha_{gs}^{\sigma} w_{g,o}^{-\sigma}$; when $w_{m,o} = w_{f,o}$ the quantity share collapses back to $\omega_{g,os}$. Second, summing the labor-bill identity $w_{g,o}L_{g,os} = \alpha_s \omega_{g,os} p_{os} Q_{os}$ from (100) across genders and using $\omega_{m,os} + \omega_{f,os} = 1$ yields total sales at the cell,

$$X_{os} \equiv p_{os}Q_{os} = \frac{w_{m,o}L_{m,os} + w_{f,o}L_{f,os}}{\alpha_s}. \tag{159}$$

Third, final good demand generated by the CES aggregator with elasticity σ_D gives $p_{os}Q_{os} \propto p_{os}^{1-\sigma_D}$, so

$$p_{os} \propto X_{os}^{1/(1-\sigma_D)}, \tag{160}$$

up to a sector level constant absorbed into the numeraire. Fourth, the perfect-competition pricing equation (98) gives

$$p_{os} = \frac{1}{A_{os}} \left(\frac{W_{os}}{\alpha_s}\right)^{\alpha_s} \left(\frac{r_{F,o}}{1-\alpha_s}\right)^{1-\alpha_s}. \tag{161}$$

Combining (160) and (161) and solving for A_{os} yields the closed form expression (157), up to the sector level constant. I normalize productivity at a single location–industry cell, $A_{1,1} \equiv 1$, which pins down the constant and therefore the whole vector $\{A_{os}\}$. □

Intuition. The inversion recovers productivity as the residual that reconciles observed sales with input costs. Equation (157) shows that A_{os} is increasing in total sales X_{os} and, holding sales fixed, increasing in the input costs summarized by the gender wage composite W_{os} and the rent $r_{F,o}$. The reason is that observed sales pin down the cell's output price. A cell that sustains the same sales while paying higher wages and rents must therefore be more productive, so that its unit cost still equals that price.

H.2. Estimation of the female labor force participation Fréchet shape ν

The shape parameter ν governs the elasticity of the dual earner share μ_o with respect to the gap between the dual earner and breadwinner market access measures. From the participation-share expression (**??**),

$$\mu_o = \frac{\Omega_{o1}^{\nu}}{\Omega_{o1}^{\nu} + [\xi_{o0}\Omega_{o0}]^{\nu}},$$

taking logs of the odds yields

$$\log\left(\frac{\mu_o}{1-\mu_o}\right) = \nu\left[\log\Omega_{o1} - \log\Omega_{o0}\right] - \nu\log\xi_{o0}. \tag{162}$$

Differencing (162) across 2007 and 2017 gives

$$\Delta\log\left(\frac{\mu_o}{1-\mu_o}\right) = \nu\,\Delta\mathrm{MA}_o^{mf} - \nu\,\Delta\log\xi_{o0} + \gamma' X_o + \epsilon_o, \tag{163}$$

where

$$\Delta \mathrm{MA}_o^{mf} \equiv \Delta \log \Omega_{o1} - \Delta \log \Omega_{o0} \tag{164}$$

is the change in the dual earner versus breadwinner market access gap. The term $\Delta \log \xi_{o0}$ captures changes in local household productivity that affect the relative attractiveness of the breadwinner arrangement.

Endogeneity concerns. Estimating ν in (163) faces an endogeneity concern: $\Delta \mathrm{MA}_o^{mf}$ depends on changes in wages and commute costs that may be correlated with changes in local household productivity. Household-productivity shocks shift female labor supply and wages, which feed into market access, creating a reverse-causality channel. I address this concern in three ways. First, I instrument $\Delta \mathrm{MA}_o^{mf}$ with a productivity-side Bartik shift share that isolates variation from realized productivity changes at other locations interacted with pre-period commuting exposure. Second, I restrict the estimation sample to zones at least one kilometer from the new Metro and BRT stations, since reduced form evidence in Fact 4 suggests local unobservables respond to proximity to the new infrastructure. Third, I include controls for persistent geographic and economic heterogeneity across zones. These absorb differential trends in local amenities and household productivity that would otherwise threaten the exclusion restriction. The remainder of this section describes each component in turn.

Bartik instrument construction. I construct a productivity-side Bartik instrument that isolates variation from realized changes in productivity across locations and sectors, interacted with pre-period commuting exposure.

Let

$$\Delta \log A_{is} \equiv \log A_{is}^{\text{post}} - \log A_{is}^{\text{pre}} \tag{165}$$

denote the realized change in location–industry productivity between 2007 (pre-Metro) and 2017 (post-Metro).[51]

For each location o and each sector s, I aggregate realized productivity changes at all other locations, weighted by pre-period commuting accessibility:

$$B^{o,s} \equiv \sum_{i=1}^{S} d_{oi}^{-\theta^f} \, \Delta \log A_{is}, \tag{166}$$

where d_{oi} is the pre-period female dual earner commuting friction from o to i, θ^f is the corresponding Fréchet shape, and the weights $d_{oi}^{-\theta^f}$ are normalized to sum to one across i for each o. The weight is large when i is easy to commute to from o and small when it is far. A location more exposed to large productivity changes therefore receives a larger Bartik shock.

The shift share argument requires that productivity shocks at locations i be uncorrelated with the unobservables that drive the outcome at location o. Two concerns arise. First, when

[51]For each year and commuting model, I construct W_{os} using the Block-2 wages and the CES labor aggregator, and compute X_{os} from observed employment and wages using (159). I then apply (157) to recover A_{os} for each location–industry cell. The sector level constant is pinned down by normalizing $A_{1,1} \equiv 1$. The only object that differs across models is the wage vector from Block 2: I evaluate the wage bill under the independent-model wages and under the interdependent-model wages (which depend on $\widehat{v}$). All other inputs are common.

$i = o$, the Bartik sum picks up the productivity change at o itself, which is mechanically linked to the outcome at o through the model's equilibrium (wages, market access, and so on). Second, productivity in zones close to o is likely correlated with unobservables at o through local spillovers. To address both, I drop o and its K nearest commuting neighbors from the Bartik sum. I measure how close another location is to o as the maximum of the inflow and outflow pre-period commuting probabilities between them, rank locations by this measure, and drop the K closest. I set $K = 150$ in the baseline and report robustness to alternative values of K below. Denote the resulting shifter by $B^{o,s,\text{LOO}}$.

The full instrument aggregates the leave-one-out Bartik across sectors using pre-period sectoral employment shares:

$$Z_o^{\nu} \equiv \sum_{s=1}^{K} \frac{L_{os}^{\text{pre}}}{\sum_{s'} L_{os'}^{\text{pre}}} B^{o,s,\text{LOO}}. \tag{167}$$

Sample restriction. I restrict the estimation sample to locations more than one kilometer from the nearest Metro station and more than one kilometer from the nearest BRT station. Fact 4 documents that local outcomes such as population decline with proximity to the new infrastructure. The new stations improve commuting access, which by itself would raise local population, so the fact that population instead falls suggests that amenities or other local unobservables are evolving near the stations in ways that more than offset the gains from better access. Because $\Delta \log \xi_{o0}$ captures precisely these forces, station-adjacent variation would correlate the Bartik shock with the residual in (163) and bias $\widehat{\nu}$ toward zero. Including station-adjacent zones would therefore risk contaminating the IV exclusion restriction by mixing the productivity-driven variation that the Bartik isolates with local responses to the new transit. I set the one-kilometer cutoff as the baseline and report robustness to alternative cutoffs below.

Controls. The exogenous regressors X_o in (163) include polynomial terms up to cubic order in three distances: to the nearest Metro station, to the nearest BRT station, and to the CBD. I also include geographic controls for the location's mean slope, its surface area, and its elevation. To capture the pre-period structure of local commuting, I add the share of commuters who work in the same zone where they live, computed separately for single females, single males, and couples. To capture pre-period economic conditions, I add the dual earner share μ_o^{pre} and the log residential rent. I also include interactions of the distance to Metro and the distance to BRT with the location's mean slope, with its surface area, and with the pre-period log rent. The regression includes a constant. These controls absorb persistent geographic and economic heterogeneity across zones that could covary with the Bartik shock and with differential trends in local amenities and household productivity.

Why not use changes in commute time as the Bartik shock. An alternative would be to weight Metro- and BRT-induced changes in commute times by pre-period commuting shares. Reductions in commute times should generally increase market access and therefore raise residential population. However, if amenities change in response to the new infrastructure, population can go down on net. If this is the case, changes in amenities would be negatively correlated with increases in market access, which would bias the estimation toward zero when commute time changes are used as the shock. Consistent with this view, the reduced form evidence shows that amenities (proxied by residential composition) decline near the new stations. This threatens the identification strategy. In fact, estimates of η using this Bartik instrument tend to be zero or negative.

Robustness to K and the distance cutoff. Table 11 reports the ν point estimate as I vary the leave-out-K-nearest-neighbors radius across the columns (from $K = 0$, plain leave-one-out, up to $K = 250$) and the lower bound on distance to BRT and Metro stations across the rows (from 0 to 3 km). The baseline specification ($K = 150$, $d = 1$ km, $\widehat{\nu} = 1.05$) is shaded. Each cell reports the IV point estimate, the robust standard error in parentheses, and the first stage F-statistic in square brackets. The estimate is robust in sign and order of magnitude across the grid. Holding $d = 1$, $\widehat{\nu}$ moves between 1.01 ($K = 100$) and 1.55 ($K = 0$). Holding $K = 150$, it rises from 0.93 at $d = 0$ to 1.48 at $d = 3$ as the sample drops zones close to the new infrastructure. The baseline sits in the lower part of this range. The first stage F exceeds the conventional rule-of-thumb threshold of 10 (Staiger and Stock, 1997) across the grid (above 50 in all but the most extreme corner) and only weakens where the leave-one-out exclusion radius K is extreme or the distance buffer leaves a small sample.

Table 11: Robustness of ν IV estimate to neighbor-exclusion K and distance-from-stations sample cutoff

Distance cutoff (km)	$K = 0$	$K = 50$	$K = 100$	$K = 150$	$K = 200$	$K = 250$
	1.385	1.098	0.925	0.930	0.904	0.817
	(0.345)	(0.269)	(0.229)	(0.250)	(0.299)	(0.330)
$d = 0$	[74.0]	[118.1]	[143.1]	[140.1]	[117.0]	[97.0]
	1.547	1.182	1.008	**1.051**	1.112	1.157
	(0.262)	(0.210)	(0.183)	(**0.195**)	(0.232)	(0.270)
$d = 1$	[103.4]	[204.0]	[237.2]	**[226.8]**	[180.4]	[143.6]
	1.968	1.488	1.214	1.233	1.257	1.317
	(0.421)	(0.305)	(0.266)	(0.282)	(0.322)	(0.379)
$d = 2$	[53.6]	[130.7]	[148.0]	[141.4]	[119.3]	[93.0]
	2.940	1.974	1.506	1.478	1.399	1.422
	(0.895)	(0.490)	(0.398)	(0.422)	(0.455)	(0.510)
$d = 3$	[18.8]	[64.0]	[81.3]	[75.5]	[69.4]	[58.0]

Notes: Each cell reports the IV estimate of ν (top), robust standard error in parentheses, and first-stage F-statistic in square brackets. Columns vary the number K of nearest commuting neighbors excluded from the leave-one-out productivity Bartik $B^{o,s,\text{LOO}}$. Rows vary the lower bound on distance to BRT and Metro stations imposed on the estimation sample. The instrument is the destination $B^{o,s,\text{LOO}}$ productivity Bartik aggregated across sectors by pre-period employment shares; the endogenous variable is the change in the dual-earner versus breadwinner market-access gap. Controls (24 columns) match the master ν specification. The shaded cell ($K = 150$, $d = 1$ km) is the baseline reported in Panel E of Table 3.

H.3. Inversion of household productivity

Given ν, wages, and market access terms $\{\Omega_{o1}, \Omega_{o0}\}$, the household-productivity shifter ξ_{o0} can be recovered in closed form from the participation equation.

Proposition 4. *Fix ν, Ω_{o1}, and Ω_{o0}. For any observed participation rate $\mu_o \in (0,1)$, there exists a unique $\xi_{o0} > 0$ such that*

$$\mu_o = \frac{\Omega_{o1}^{\nu}}{\Omega_{o1}^{\nu} + (\xi_{o0}\Omega_{o0})^{\nu}}.$$

Moreover, ξ_{o0} is given by

$$\xi_{o0} = \frac{\Omega_{o1}}{\Omega_{o0}} \left(\frac{1-\mu_o}{\mu_o} \right)^{1/\nu}. \tag{168}$$

Proof. Rearranging the participation equation,

$$\frac{\mu_o}{1-\mu_o} = \left(\frac{\Omega_{o1}}{\xi_{o0}\Omega_{o0}} \right)^{\nu}.$$

Taking logs and solving for ξ_{o0} yields (168). Uniqueness follows because the right-hand side is strictly decreasing in ξ_{o0}. □

Intuition. The object ξ_{o0} captures the relative attractiveness of the breadwinner arrangement at location o, reflecting household productivity. For given wages and market access, higher ξ_{o0} reduces the dual earner share μ_o. The inversion therefore assigns higher ξ_{o0} to locations where dual earner participation is low relative to what wages and commuting opportunities would predict.

H.4. Estimation of residential elasticities $\{\eta^m, \eta^f, \eta^{mf}\}$

The residential elasticities are identified from the cross-zone relationship between changes in log residential population and changes in the utility evaluated at the inverted wages and observed rents. From the single residence-choice expression (30),

$$\lambda_o^g \propto \left(\frac{u_o^g \, \Omega_o^g}{P^{\beta^g} r_{R,o}^{1-\beta^g}} \right)^{\eta^g},$$

taking logs and differencing across vintages yields

$$\Delta \log N_o^g = \eta^g \, \Delta U_o^g + \eta^g \, \Delta \log u_o^g + \delta_t^g, \tag{169}$$

where δ_t^g absorbs the denominator of (30) (common across zones) and

$$\Delta U_o^g \equiv \frac{1}{\vartheta^g} \Delta \log \Psi_o^g - (1-\beta^g) \, \Delta \log r_{R,o} \tag{170}$$

is the change in the residential utility index for singles of gender g. The analogous derivation from (**??**) gives the couples' equation

$$\Delta \log N_o^{mf} = \eta^{mf} \, \Delta U_o^{mf} + \eta^{mf} \, \Delta \log u_o^{mf} + \delta_t^{mf}, \tag{171}$$

with

$$\Delta U_o^{mf} \equiv \Delta \log \Omega_o^{mf} - (1-\beta)\,\Delta \log r_{R,o}. \qquad (172)$$

which I can recover after estimating ν.[52]

Treating $\Delta \log u_o^k$ as an unobserved amenity shock absorbed into a residual, the three estimating equations are

$$\Delta \log N_o^k = \eta^k \,\Delta U_o^k + \gamma_k' X_o + \epsilon_o^k, \qquad k \in \{m, f, mf\}. \qquad (173)$$

I estimate (173) separately for the three groups.

Endogeneity concerns. The residual ϵ_o^k contains the change in the unobserved amenity $\log u_o^k$, which is plausibly correlated with ΔU_o^k through demand-side feedback: zones with positive amenity shocks attract workers, reducing local wages and hence ΔU_o^k. I address this concern in the same three ways as for ν. I instrument ΔU_o^k with a productivity-side Bartik shift share, restrict the estimation sample to zones at least one kilometer from the new Metro and BRT stations, and include the same controls. The instrument is the same employment-weighted Bartik shift share used for ν, so only the dependent and endogenous variables differ.

Bartik instrument construction. For all three elasticities I use the same employment-weighted leave-one-out shifter as in the ν regression:

$$Z_o^{\eta^k} \equiv Z_o^{\nu}, \qquad k \in \{m, f, mf\}, \qquad (174)$$

with Z_o^{ν} defined in (167). Using a single instrument for all four parameters keeps the shift share design uniform across the block. The weighted aggregate delivers point estimates above one for all three elasticities, satisfying the lower bound imposed by the Fréchet assumption on η. I report robustness to using each sector specific shifter instead in Panel B of Tables 12–14.

Sample restriction. Same as in Section H.2: I restrict the sample to locations more than one kilometer from the nearest Metro or BRT station. The Fact 4 reasoning—that station-adjacent zones show local responses in population, rents, and household composition, which suggest amenities changes that would correlate with the Bartik shock—applies identically here.

Controls. Identical to those used in the ν regression.

Robustness to K, the distance cutoff, and the instrument. Tables 12, 13, and 14 report the analogous robustness exercise for the three residential elasticities. Each table has two panels. Panel A varies the leave-one-out exclusion radius K used in $B^{o,s,\mathrm{LOO}}$ and the distance-from-stations cutoff, holding the instrument fixed at the employment-weighted Bartik used in the

[52]Recovering Ω_o^{mf} requires $\widehat{\xi}_{o0}$, which I obtain from (168) given $\widehat{\nu}$, the observed dual earner share μ_o, and the model implied market access Ω_{o1} and Ω_{o0}. I construct the latter from Block-2 wages, Block-1 commuting parameters, the estimated pooling friction $\widehat{\upsilon}$, and observed commute times, via (**??**) and (**??**). Plugging $\widehat{\xi}_{o0}$ into (**??**) delivers Ω_o^{mf}. The resulting sequence $\{\xi_{o0}\}$ absorbs changes in social norms or home production conditions across locations.

headline specification. Panel B fixes the baseline ($K = 150, d = 1$) and instead sweeps the instrument across each of the seven sector specific Bartik shifters together with their pre-period employment-weighted aggregate. The shaded cell in each panel marks the baseline reported in Table 3. As in Table 11, each cell reports the IV point estimate, the robust standard error in parentheses, and the first stage F-statistic in square brackets.

Table 12: Robustness of η^f IV estimate

Panel A: $K\times$ distance-cutoff sweep (IV = employment-weighted productivity Bartik)

Distance cutoff (km)	$K = 0$	$K = 50$	$K = 100$	$K = 150$	$K = 200$	$K = 250$
	0.213	0.684	0.689	0.844	0.866	0.763
	(0.113)	(0.197)	(0.180)	(0.197)	(0.211)	(0.193)
$d = 0$	[79.8]	[77.8]	[93.7]	[106.7]	[126.5]	[146.9]
	0.217	0.752	0.790	**1.002**	1.135	1.091
	(0.111)	(0.166)	(0.154)	(**0.162**)	(0.191)	(0.202)
$d = 1$	[138.2]	[128.7]	[140.2]	**[132.5]**	[126.2]	[125.4]
	0.251	0.973	0.981	1.204	1.380	1.376
	(0.138)	(0.211)	(0.190)	(0.200)	(0.244)	(0.267)
$d = 2$	[102.1]	[101.3]	[120.0]	[109.0]	[97.2]	[89.9]
	0.261	1.155	1.114	1.250	1.480	1.479
	(0.167)	(0.251)	(0.215)	(0.208)	(0.264)	(0.291)
$d = 3$	[67.4]	[76.9]	[102.8]	[106.4]	[90.0]	[81.6]

Panel B: IV sweep at $K = 150$, $d = 1$ baseline

IV	sec1	sec2	sec3	sec4	sec5	sec6	sec7	weighted
	1.028	1.316	1.060	0.996	0.948	0.957	1.065	**1.002**
	(0.179)	(0.311)	(0.173)	(0.222)	(0.165)	(0.149)	(0.190)	(**0.162**)
η^f	[102.1]	[43.7]	[122.5]	[79.3]	[115.9]	[156.4]	[100.1]	**[132.5]**

Notes: Each cell reports the IV estimate of η^f (top), robust standard error in parentheses, and first-stage F-statistic in square brackets. Panel A varies K (the leave-one-out nearest-neighbor exclusion radius used in $B^{o,s,\text{LOO}}$) and the distance-from-stations sample cutoff, holding the instrument at the employment-weighted aggregate. Panel B fixes $K = 150$, $d = 1$ and sweeps the instrument across each of the seven sector-specific destination $B^{o,s,\text{LOO}}$ productivity Bartik columns and their employment-weighted aggregate. The seven aggregated sectors for $\{\alpha_{sg}\}$ are, in order: (1) manufacturing (other); (2) manufacturing (textiles); (3) services; (4) financial and business services; (5) wholesale; (6) retail trade; (7) transportation. Shaded cells mark the baseline reported in the main calibration table.

I selected the headline specification subject to three constraints that should hold uniformly across the three elasticities. First, the same instrument, leave-one-out exclusion radius K, and distance buffer should apply to all four parameters. Second, the point estimates should exceed one, which is the lower bound imposed by the Fréchet shape. Third, the first stage F should be high enough to support the IV interpretation. The employment-weighted Bartik with ($K = 150, d = 1$) satisfies all three: it is the same instrument used for ν, so all four parameters share

Table 13: Robustness of η^m IV estimate

Panel A: K× distance-cutoff sweep (IV = employment-weighted productivity Bartik)

Distance cutoff (km)	$K = 0$	$K = 50$	$K = 100$	$K = 150$	$K = 200$	$K = 250$
	0.972	3.643	3.856	3.854	3.305	2.435
	(0.505)	(1.109)	(1.147)	(0.987)	(0.754)	(0.536)
$d = 0$	[34.1]	[24.2]	[26.2]	[39.7]	[68.0]	[107.8]
	0.770	3.280	3.627	**4.260**	4.414	3.732
	(0.412)	(0.743)	(0.790)	(**0.854**)	(0.854)	(0.736)
$d = 1$	[79.0]	[51.7]	[50.0]	[**51.3**]	[60.2]	[77.3]
	0.723	3.486	3.607	4.109	4.676	4.412
	(0.449)	(0.784)	(0.747)	(0.792)	(0.918)	(0.896)
$d = 2$	[72.5]	[53.7]	[61.2]	[60.0]	[57.2]	[61.0]
	0.687	3.764	3.768	3.840	4.539	4.484
	(0.507)	(0.827)	(0.744)	(0.679)	(0.834)	(0.874)
$d = 3$	[63.0]	[54.5]	[69.9]	[81.7]	[71.1]	[68.0]

Panel B: IV sweep at $K = 150$, $d = 1$ baseline

IV	sec1	sec2	sec3	sec4	sec5	sec6	sec7	weighted
	4.533	5.870	4.630	4.957	4.034	3.937	4.684	**4.260**
	(1.032)	(1.926)	(0.964)	(1.391)	(0.851)	(0.740)	(1.083)	(**0.854**)
η^m	[36.4]	[14.6]	[45.3]	[21.8]	[45.5]	[64.8]	[36.3]	[**51.3**]

Notes: Each cell reports the IV estimate of η^m (top), robust standard error in parentheses, and first-stage F-statistic in square brackets. Panel A varies K (the leave-one-out nearest-neighbor exclusion radius used in $B^{o,s,\mathrm{LOO}}$) and the distance-from-stations sample cutoff, holding the instrument at the employment-weighted aggregate. Panel B fixes $K = 150$, $d = 1$ and sweeps the instrument across each of the seven sector-specific destination $B^{o,s,\mathrm{LOO}}$ productivity Bartik columns and their employment-weighted aggregate. The seven aggregated sectors for $\{\alpha_{sg}\}$ are, in order: (1) manufacturing (other); (2) manufacturing (textiles); (3) services; (4) financial and business services; (5) wholesale; (6) retail trade; (7) transportation. Shaded cells mark the baseline reported in the main calibration table.

Table 14: Robustness of η^{mf} IV estimate

Panel A: $K\times$ distance-cutoff sweep (IV = employment-weighted productivity Bartik)

Distance cutoff (km)	$K=0$	$K=50$	$K=100$	$K=150$	$K=200$	$K=250$
	0.763	1.408	1.311	1.584	1.519	1.248
	(0.385)	(0.505)	(0.431)	(0.501)	(0.522)	(0.452)
$d=0$	[27.4]	[39.8]	[52.5]	[57.4]	[63.7]	[74.4]
	0.726	1.557	1.531	**1.999**	2.219	2.138
	(0.341)	(0.380)	(0.330)	(**0.369**)	(0.454)	(0.491)
$d=1$	[40.6]	[58.7]	[72.8]	[**67.7**]	[58.1]	[54.8]
	0.986	2.304	2.159	2.697	2.942	3.040
	(0.511)	(0.624)	(0.498)	(0.551)	(0.690)	(0.805)
$d=2$	[22.5]	[33.6]	[47.8]	[44.6]	[37.4]	[31.7]
	1.180	3.506	2.923	3.295	3.536	3.598
	(0.851)	(1.208)	(0.766)	(0.759)	(0.916)	(1.033)
$d=3$	[8.1]	[13.9]	[27.0]	[29.3]	[26.8]	[23.5]

Panel B: IV sweep at $K=150$, $d=1$ baseline

IV	sec1	sec2	sec3	sec4	sec5	sec6	sec7	weighted
	2.086	3.081	2.039	1.957	1.931	1.927	2.152	**1.999**
	(0.415)	(0.990)	(0.387)	(0.539)	(0.387)	(0.339)	(0.430)	(**0.369**)
η^{mf}	[53.7]	[15.6]	[67.6]	[37.7]	[59.7]	[77.6]	[50.5]	[**67.7**]

Notes: Each cell reports the IV estimate of η^{mf} (top), robust standard error in parentheses, and first-stage F-statistic in square brackets. Panel A varies K (the leave-one-out nearest-neighbor exclusion radius used in $B^{o,s,\text{LOO}}$) and the distance-from-stations sample cutoff, holding the instrument at the employment-weighted aggregate. Panel B fixes $K=150$, $d=1$ and sweeps the instrument across each of the seven sector-specific destination $B^{o,s,\text{LOO}}$ productivity Bartik columns and their employment-weighted aggregate. The seven aggregated sectors for $\{\alpha_{sg}\}$ are, in order: (1) manufacturing (other); (2) manufacturing (textiles); (3) services; (4) financial and business services; (5) wholesale; (6) retail trade; (7) transportation. Shaded cells mark the baseline reported in the main calibration table.

one shift share design, it delivers point estimates above one for all three groups, and its first stages exceed the rule-of-thumb threshold of 10 (Staiger and Stock, 1997) (F between 51 and 132).

The natural alternatives are the seven sector specific shifters. Panel B of Tables 12–14 shows that across these sectors the estimates span 0.95 to 1.32 for single women, 3.94 to 5.87 for single men, and 1.93 to 3.08 for couples. Several individual sectors fall below the Fréchet bound for single women, whereas the employment-weighted aggregate delivers $\widehat{\eta}^f \approx 1.00$, $\widehat{\eta}^m \approx 4.26$, and $\widehat{\eta}^{mf} \approx 2.00$, above one for all three groups while using the same instrument as the ν regression.

Panel A shows that K cannot be reduced below the baseline either. Holding $d = 1$, the point estimate rises with K for all three groups, from values below the Fréchet bound at $K = 0$ (where neighbors are not excluded beyond self) toward the baseline by $K = 150$–200. For single women the estimate is below one at every K smaller than the baseline (0.22, 0.75, and 0.79 at $K = 0$, 50, and 100) and reaches the bound only at the baseline $K = 150$, where $\widehat{\eta}^f = 1.00$, so a smaller exclusion radius would push the estimate below the bound $\eta > 1$. I keep $d = 1$ as the baseline distance buffer, with the sample shrinking as d rises.

H.5. Amenity inversion

The final step of Block 3 recovers residential amenities from the residence-choice equations. At this point, all other objects entering residential choice have already been constructed: wages come from Block 2, $\widehat{\nu}$ is estimated in Section H.2, $\widehat{\xi}_{o0}$ is recovered in Section H.3, and the residential elasticities $(\widehat{\eta}^m, \widehat{\eta}^f, \widehat{\eta}^{mf})$ are estimated in Section H.4. Given these objects, the amenity vectors are recovered in closed form.

Proposition 5 (Closed form amenity inversion)**.** *Given the observables and parameters listed above, the residential amenity vectors $\{u_o^m, u_o^f, u_o^{mf}\}$ are determined up to a group specific normalization by the residence-choice equations. Their closed form expressions are*

$$u_o^g = \frac{\left(N_o^g\right)^{1/\eta^g}}{\Omega_o^g / \left(P^{\beta^g} r_{R,o}^{1-\beta^g}\right)}, \qquad g \in \{m, f\}, \tag{175}$$

$$u_o^{mf} = \frac{\left(N_o^{mf}\right)^{1/\eta^{mf}}}{\Omega_o^{mf} / \left(P^{\beta} r_{R,o}^{1-\beta}\right)}. \tag{176}$$

Here Ω_o^g is the singles' commuting market access, and Ω_o^{mf} is the couples' marital market access.

Proof. For singles of gender g, the residence-choice probability is

$$\lambda_o^g = \frac{\left(u_o^g \Omega_o^g / \left(P^{\beta^g} r_{R,o}^{1-\beta^g}\right)\right)^{\eta^g}}{\sum_n \left(u_n^g \Omega_n^g / \left(P^{\beta^g} r_{R,n}^{1-\beta^g}\right)\right)^{\eta^g}}.$$

Let

$$\Lambda^g \equiv \sum_n \left(u_n^g \Omega_n^g / \left(P^{\beta^g} r_{R,n}^{1-\beta^g}\right)\right)^{\eta^g}.$$

Then

$$\left(u_o^g \Omega_o^g / \left(P^{\beta^g} r_{R,o}^{1-\beta^g}\right)\right)^{\eta^g} = \lambda_o^g \Lambda^g.$$

Solving for u_o^g,

$$u_o^g = (\lambda_o^g \Lambda^g)^{1/\eta^g} \frac{P^{\beta^g} r_{R,o}^{1-\beta^g}}{\Omega_o^g}.$$

The denominator Λ^g is common across all locations within group g, so it is absorbed into the group specific normalization of amenities. Since $\lambda_o^g = N_o^g / N^g$, the total mass N^g is also absorbed into the same normalization. This gives (175).

The couples' residence-choice equation is

$$\lambda_o^{mf} = \frac{\left(u_o^{mf} \Omega_o^{mf} / \left(P^{\beta} r_{R,o}^{1-\beta}\right)\right)^{\eta^{mf}}}{\sum_n \left(u_n^{mf} \Omega_n^{mf} / \left(P^{\beta} r_{R,n}^{1-\beta}\right)\right)^{\eta^{mf}}}.$$

Applying the same argument yields (176).

For couples, the marital market access term is constructed from the participation block:

$$\Omega_o^{mf} = \left[\Omega_{o1}^{\hat{\nu}} + (\hat{\xi}_{o0} \Omega_{o0})^{\hat{\nu}}\right]^{1/\hat{\nu}}. \tag{177}$$

Substituting (177) into the couples' residence equation gives (176). □

Intuition. The inversion assigns amenities to rationalize where households live, after accounting for wages, commuting access, rents, and the estimated taste dispersion parameters. A location with many residents is not necessarily assigned a high amenity. If that location already has high labor market access or low residential rents, the model can explain its population without requiring a large u_o^k. Conversely, a location with many residents despite poor commuting access or high rents must have a high inferred amenity. For couples, the same logic applies, but the relevant access term is the marital value Ω_o^{mf}, which combines the value of dual earner commuting and the value of the breadwinner arrangement through the inverted social norm shifter ξ_{o0}.[53]

H.6. Algorithm

The numerical procedure stacks the closed form inversions around the two IV estimation steps. All intermediate objects are stored separately for the independent and interdependent commuting models.

[53]In the same step I also invert location specific residential and commercial floorspace quality shifters q_o^R and q_o^F so that the floorspace markets in (102) clear at the observed data. Residential and commercial market clearing at data values give

$$q_o^R = \frac{E_{R,o}}{(1-\iota_o)\, r_{R,o}\, \bar{H}_o\, \chi_o}, \qquad q_o^F = \frac{X_o^F}{r_{F,o}\, \bar{H}_o\, (1-\chi_o)},$$

where $E_{R,o}$ is total residential housing expenditure at o and X_o^F is the total firm spending on commercial floorspace at o. The quality shifters are held fixed in counterfactuals alongside $\bar{H}_o$ and ι_o.

1. **Productivity inversion.** For each year in $\{2007, 2017\}$ and each commuting model $M \in \{\mathrm{I}, \mathrm{D}\}$, evaluate the gender labor-expenditure shares $\omega_{g,os}$ at the Block-2 wages w^M and convert them to quantity shares by dividing by the gender wage. Compute gender specific employment $L_{g,os}$ from (158), sales X_{os}, and the composite wage W_{os}. Apply (157), normalize $A_{1,1} \equiv 1$, and store $\{A_{os}(M)\}$.

2. **Shift share construction.** For each commuting model M, compute the productivity changes $\Delta \log A_{is}(M)$ between 2007 and 2017 from (165) using the inverted productivities from Step 1. Evaluate the pre-period female dual earner commuting frictions d_{oi} at 2007 inputs from Block 1. For each location o and sector s, construct the sector specific Bartik shifter $B^{o,s}(M)$ from (166) and the leave-one-out variant $B^{o,s,\mathrm{LOO}}(M)$ by dropping o and its $K = 150$ nearest pre-period commuting neighbors from the sum.

3. **Estimate ν.** For each commuting model M, construct $\Delta \log(\mu_o/(1-\mu_o))$ and $\Delta \mathrm{MA}_o^{mf}(M) = \Delta \log \Omega_{o1}(M) - \Delta \log \Omega_{o0}(M)$. Restrict the sample to locations more than one kilometer from the nearest Metro or BRT station. Estimate (163) by IV, instrumenting $\Delta \mathrm{MA}_o^{mf}(M)$ with the employment-weighted Bartik $Z_o^{\nu}(M)$ defined in (167), including the controls listed in Section H.2. Store $\widehat{\nu}(M)$.

4. **Invert household productivity.** For each year and commuting model M, use $\widehat{\nu}(M)$, μ_o, $\Omega_{o1}(M)$, and $\Omega_{o0}(M)$ to recover $\widehat{\xi}_{o0}(M)$ from (168). Then construct $\Omega_o^{mf}(M)$ from (177).

5. **Estimate residential elasticities.** For each commuting model M and each household group $k \in \{m, f, mf\}$, construct $\Delta \log N_o^k$ and $\Delta U_o^k(M)$ from (170) and (172), using $\widehat{\nu}(M)$ and $\widehat{\xi}_{o0}(M)$ where the couples' utility index requires them. Using the same sample restriction and controls as in Step 3, estimate (173) by IV, instrumenting $\Delta U_o^k(M)$ with the employment-weighted leave-one-out shifter $Z_o^{\eta^k}(M) \equiv Z_o^{\nu}(M)$ defined in (174). Store $\widehat{\eta}^m(M), \widehat{\eta}^f(M), \widehat{\eta}^{mf}(M)$.

6. **Invert residential amenities.** For each year and commuting model M, combine observed residential populations, residential rents, market access, and the estimated residential elasticities $\widehat{\eta}^k(M)$. Apply (175) and (176) to recover $\{u_o^m(M), u_o^f(M), u_o^{mf}(M)\}$.

7. **Output.** The output of Block 3 consists of the productivity matrices $\{A_{os}(M)\}$, the point estimates $(\widehat{\nu}(M), \widehat{\eta}^m(M), \widehat{\eta}^f(M), \widehat{\eta}^{mf}(M))$, the household-productivity shifters $\{\xi_{o0}(M)\}$, and the residential amenity vectors $\{u_o^m(M), u_o^f(M), u_o^{mf}(M)\}$.

I. Untargeted moments

I use the 2017 Census to compute the bilateral commuting probability between any two of Lima's municipalities, separately by household type. I then solve the model at the calibrated parameters and the inverted 2017 exogenous characteristics, aggregate the model implied zone level commuting probabilities up to the same municipalities using resident population weights, and compare the two side by side. Because the model is calibrated on Fact 1 and Fact 3 averages of within couple commute time differences, the bilateral municipality level fit is out-of-sample. I also resolve the model under the alternative *independent* commuting assumption and apply the same aggregation, to give that alternative the best chance of fitting the data.

Figure 28 reports the comparison in logs.[54] Each panel is a binscatter of the municipality pairs. The solid black line is the 45-degree line and the dashed red line is the OLS fit. Points below the 45-degree line are links for which the model overpredicts the Census probability; points above it are links for which the model underpredicts the Census probability. A slope of one, together with an intercept close to zero, would mean that the fitted relationship lines up with the 45-degree benchmark. I therefore use the slope and the distance from the 45-degree line as the main diagnostics. The R^2 is also informative because it measures how much of the variation in Census log probabilities is explained by model log probabilities, but it does not rule out systematic overprediction or underprediction along the 45-degree line.

The interdependent model substantially improves the fit for dual earner households. For wives, the slope rises from 0.462 under independent commuting to 0.555 under interdependent commuting (an improvement of 9.2 percentage points). For husbands, it rises from 0.400 to 0.675, the largest improvement in the figure (27.6 percentage points). For singles, the slope does not improve. It falls from 0.349 to 0.287 for single men and from 0.437 to 0.414 for single women. The single-worker commuting problem is identical under both model structures, so the move comes entirely from the different wage geography recovered under interdependent commuting. That geography captures dual earner workplace pairs much better, at the cost of a slightly worse fit for singles. The R^2 is similar across columns and is not the main object here. The takeaway is that the interdependent model brings the fitted relationship close to the 45-degree benchmark for the dual earner couples that motivate it.

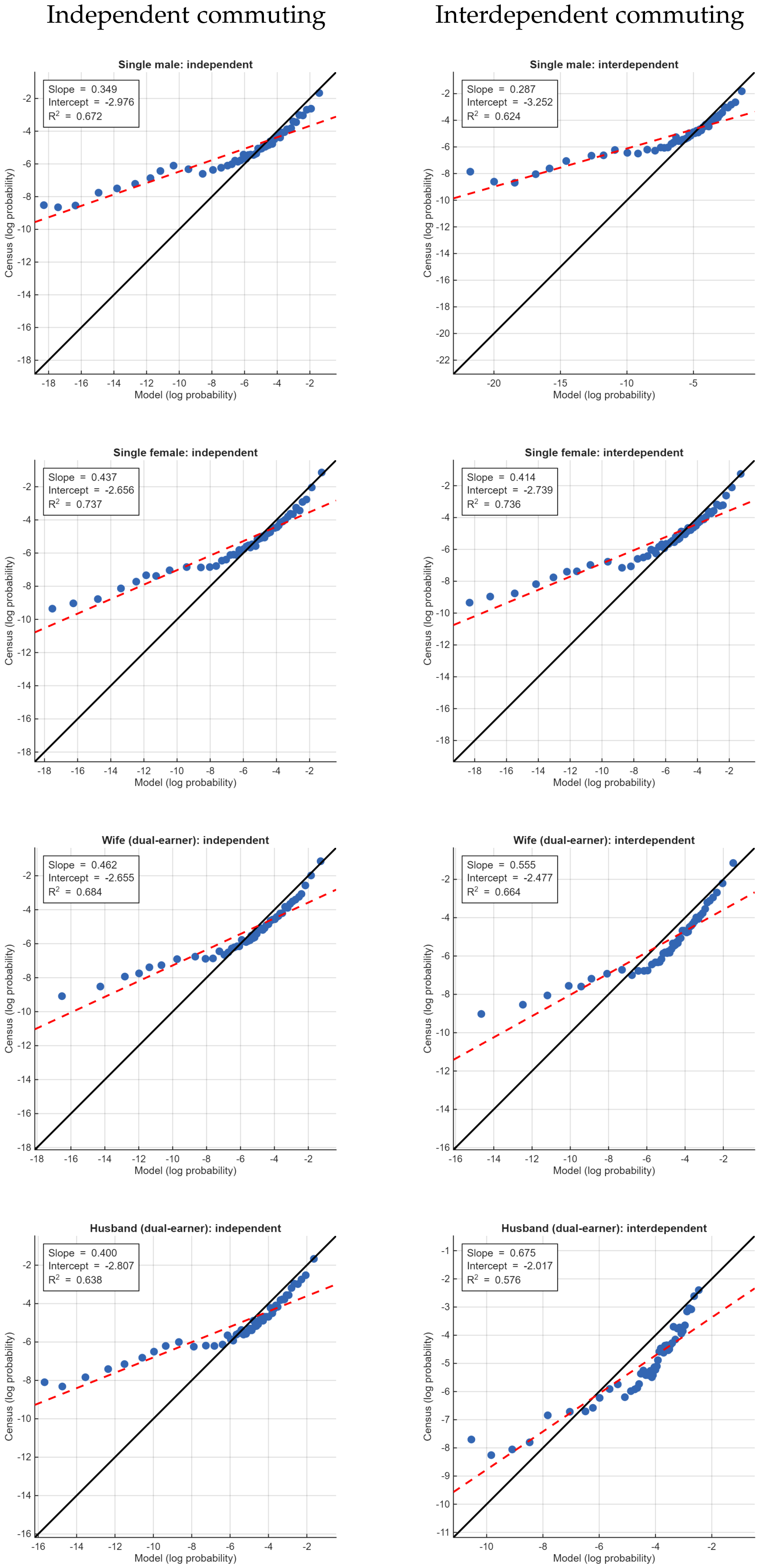
This comparison also addresses a concern about the sequential structure of the dual earner model. Even though I impose an order in the household commuting problem, the interdependent model improves the bilateral fit for both spouses. The improvement is not confined to the spouse who moves in the second stage (or in the first stage for that matter). This suggests that the sequential representation is not mechanically driving fit for only one side of the household. Instead, the model improves on the independent commuting benchmark for both husbands and wives.

A simple way to assess whether sequentiality matters is to impose a common value of θ for married men and married women. With a common θ, the asymmetry introduced by the

[54]Commuting flows with a value of 0 are dropped from the sample.

sequential structure disappears, since the resulting commuting probabilities are isomorphic to those from a model with simultaneous commuting choices. Holding all other parameters fixed and setting the common θ equal to the average within couples, the slope increases from 0.524 to 0.749 for husbands and from 0.298 to 0.374 for wives. These are improvements of 22.5 and 7.6 percentage points, respectively. Thus, the interdependent model improves fit even when commuting choices are effectively simultaneous. However, the gains are smaller than in the baseline specification with spouse specific θ's. This suggests that differences in commuting elasticities across spouses are an important margin (which is only possible with the sequential structure if we desire to keep closed form solutions). This provides an additional reason for allowing the sequential structure in the model. If I instead assumed simultaneous choices while preserving closed form probabilities, I would need to impose a common θ across spouses. That restriction would conflict with Fact 1, which shows that married men and married women have different commute time elasticities, and Fact 3 which suggests that the rate at which commute time translates into commute costs are heterogeneous across spouses.

Figure 28: Over-identification: model vs. Census bilateral commuting probabilities (logs)



Notes. Binscatter of $\log \pi_{ij}$ in the model (x) against the Census 2017 (y), at the municipality level. Rows: single male, single female, dual earner wife, dual earner husband. Pairs with zero probability in either model or data are dropped. Solid line: 45°. Dashed line: OLS fit on the underlying pairs. Slope, intercept, and R^2 reported in each panel.

J. Understanding the reduced form results in light of the model

In the reduced form section, I noted that location choices reflect the overall welfare associated with living in a location and therefore combine several forces. New transit stations can make locations more desirable because they improve commuting access. However, they can also change local amenities, such as crime, foot traffic, or noise. The reduced form evidence presented earlier mixes these two channels. In what follows, I compare the world with improved commute times to the world without the new infrastructure. I then apply the same difference-in-differences specification to model outcomes. I compare locations near constructed stations to locations near yet-to-be-constructed stations, before and after the reduction in commute times.

Figure 29 shows that the model partly reflects the responses observed in the data, but it does not fully match the changes in population counts. In fact, the model predicts more population near the new stations after commute times improve. However, in the data, population falls in those locations relative to comparison locations. The difference suggests that other local fundamentals also changed in the areas where the new infrastructure was built.

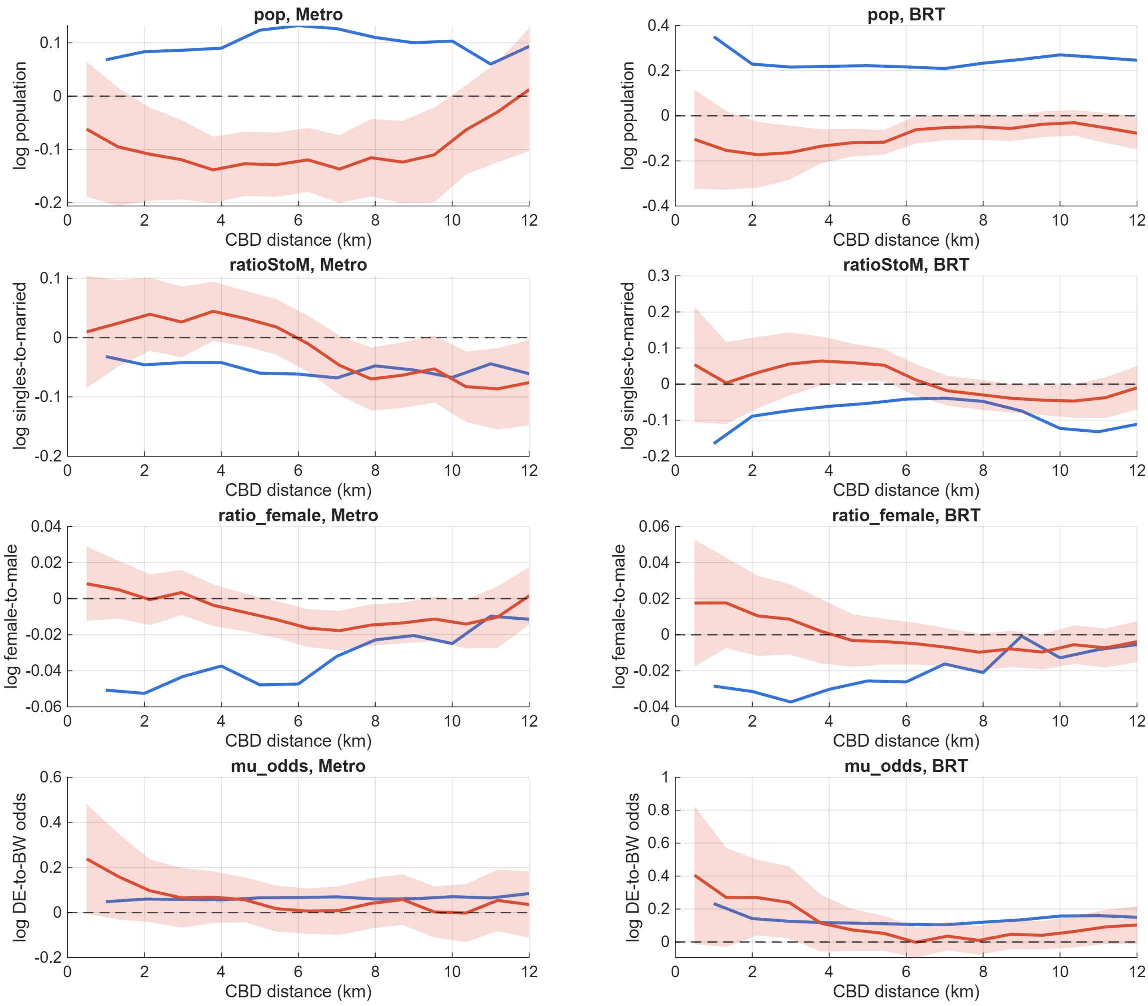


Figure 29: Census outcomes around stations: model versus reduced form, baseline (commute-only) counterfactual.

Notes: Difference-in-differences of census outcomes by distance to the CBD, comparing locations near constructed stations to locations near yet-to-be-constructed stations, before versus after. The model line is the metro-vs-no-metro counterfactual holding 2017 fundamentals fixed. The reduced form line and shaded band are the data estimate with its 95% confidence interval. Metro and BRT are shown in each row.

Figure 30 shows that amenities also changed around the new infrastructure. I recover amenities from the model and estimate the same difference-in-differences specification by distance to the CBD. The results show that amenities worsened near the new stations, especially farther from the center. Better commute times made these locations more desirable, but lower amenities pushed in the opposite direction. This helps explain why the model predicts population gains near the stations while the data show relative population losses.

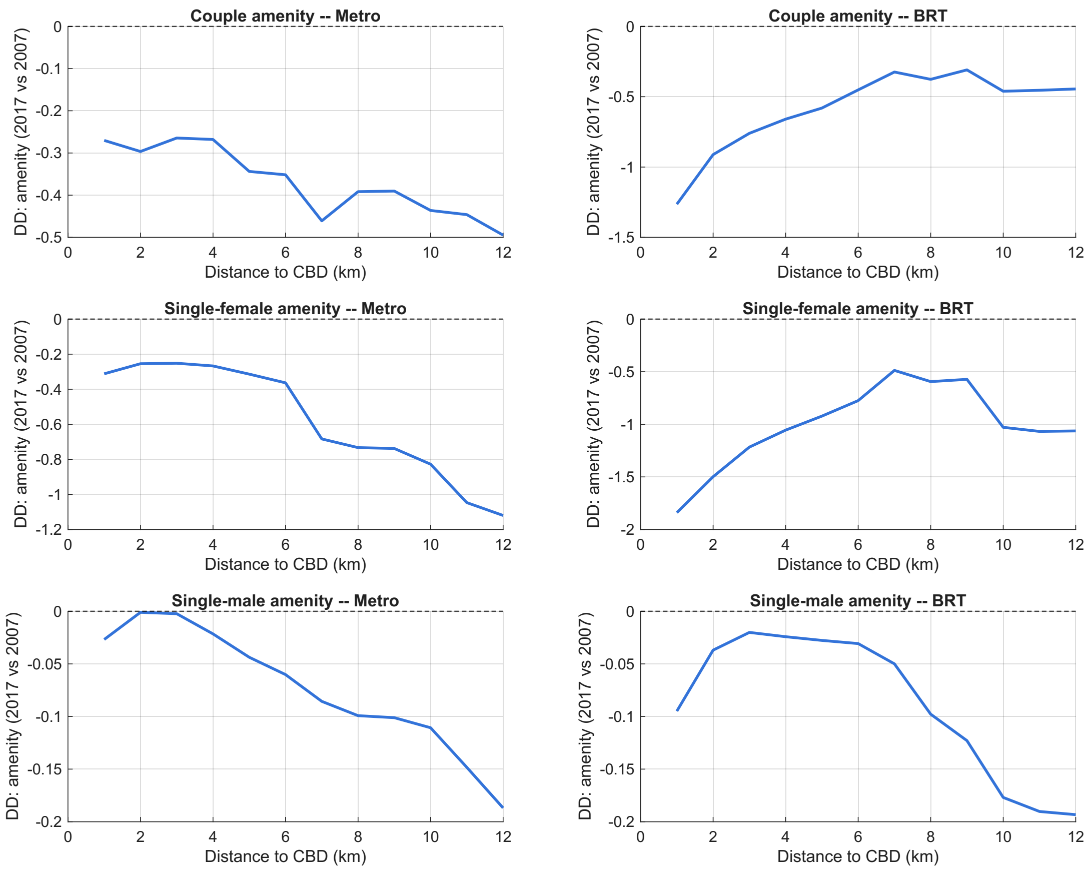


Figure 30: Model-inverted amenities by distance to the CBD.

Notes: Difference-in-differences of the model-inverted exogenous amenities (2007 versus 2017) by distance to the CBD, by group. Model only. A negative gradient near the stations means amenities worsened in treated locations.

Figure 31 shows what happens once I account for this amenity change. The baseline in Figure 29 moves commute times from their 2007 to their 2017 values while holding amenities fixed. In this exercise, I instead compare the 2017 economy to a counterfactual in which *both* commute times and amenities are rolled back to their 2007 values. The resulting comparison combines the transit infrastructure effect that operates through lower commute times with the total change in amenities between 2007 and 2017—not only the portion of the amenity change that the infrastructure itself caused. Because amenities worsened near the new stations, adding this channel lowers predicted population in treated locations, and the model moves much closer to the reduced form estimates for population. This confirms that the mismatch in Figure 29 is mostly explained by amenity changes around the new infrastructure.

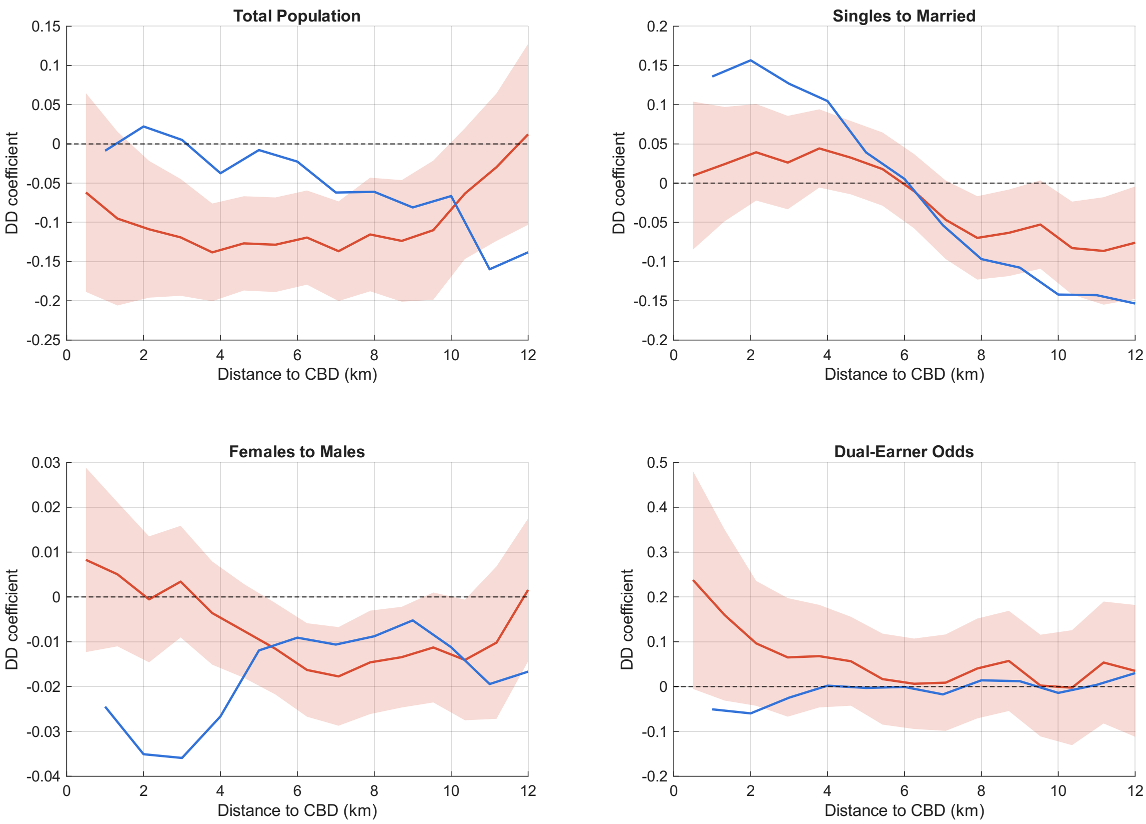


Figure 31: Census outcomes around stations, model versus reduced form, when the comparison also rolls amenities back to 2007.

Notes: As in Figure 29, but the counterfactual now rolls both commute times and amenities back to their 2007 values, so the comparison also includes the 2007–2017 change in amenities. Adding the amenity channel brings the model close to the data.

Figure 32 shows that accounting for this amenity change does not change the model's income response around stations. I estimate the same difference-in-differences specification for the male female income gap, now rolling both commute times and amenities back to their 2007 values in the counterfactual, as in Figure 31. The results are close to the baseline (commute-only) model response. This means that the gender gap effect is not driven by amenity changes around the new stations.

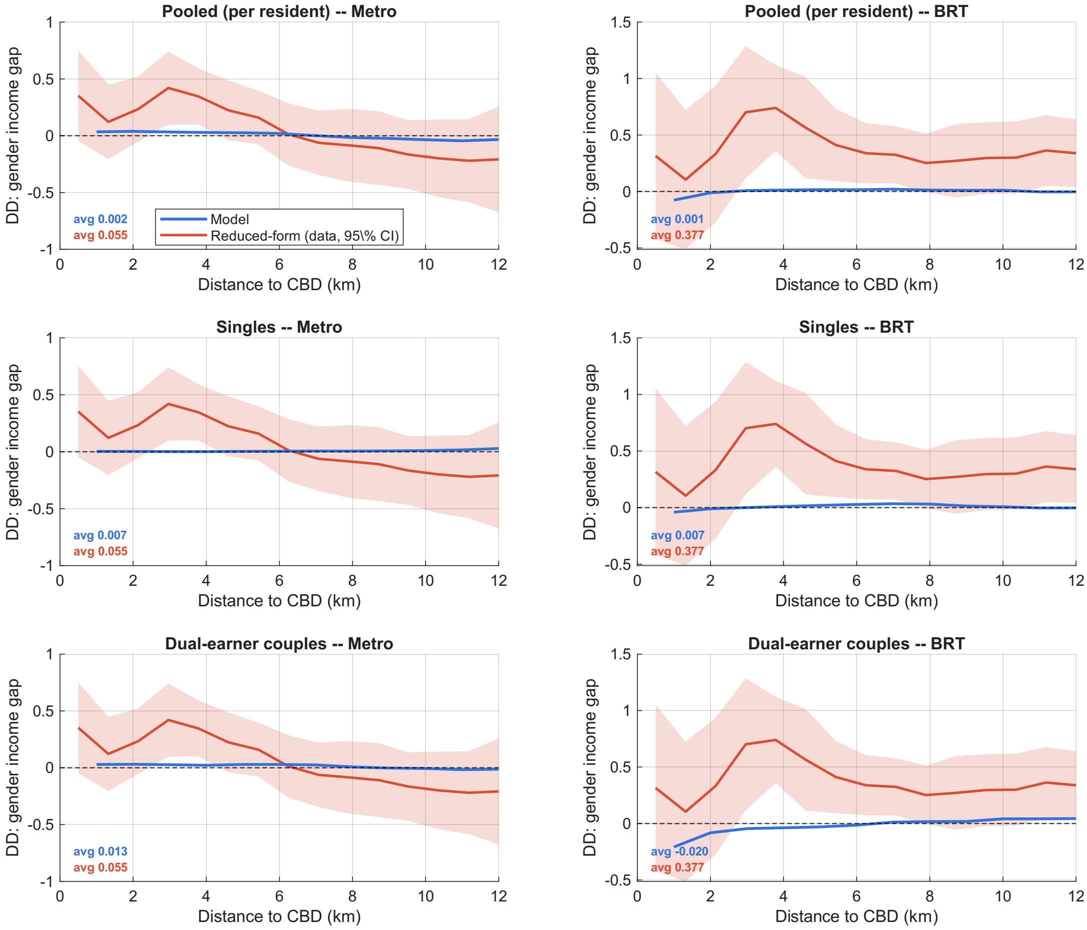
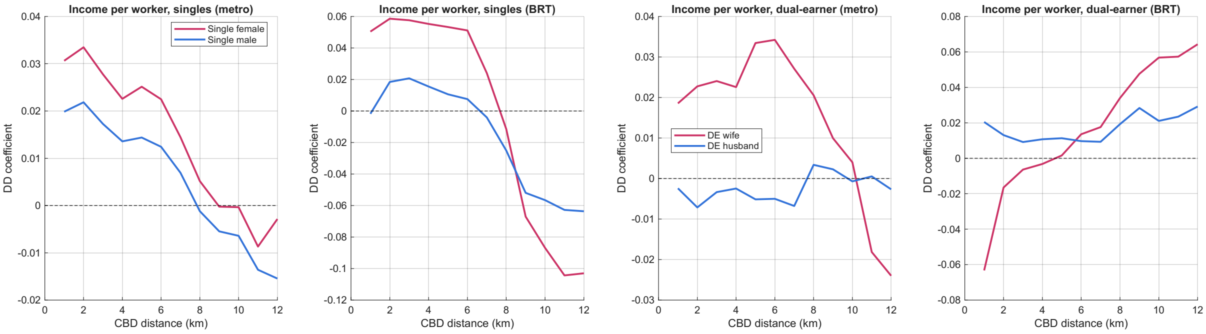


Figure 32: The gender earnings gap around stations is robust to amenities.

Notes: Difference-in-differences of the gender income gap by distance to the CBD. Model under the counterfactual that rolls both commute times and amenities back to 2007, and reduced form (data, 95% confidence interval). The number in the corner is the simple average of the model coefficients across distance bins. The gradient is close to the baseline, so the gender gap response is not driven by amenity changes.

The gender gap by its female and male parts. The gender income-gap DD in the main text (Figure 9) is the difference of two underlying responses, one for women and one for men. Figure 33 reports each separately, by distance to the CBD, for singles and for dual earner couples, under the interdependent model. *[Placeholder: contrast single women with single men and dual earner wives with husbands, note where the female series rises near the stations, and connect the welfare row to the within couple consumption-split result in the main text.]*



(a) Income per worker by gender: singles (left), dual earner couples (right)

(b) Deterministic welfare by gender: singles (left), dual earner couples (right)

Figure 33: Earnings and welfare DD by gender, model only

Notes: DD coefficient against distance to the CBD, rolling window $h = 3$ km, for Line 1 (Metro) and the BRT (each subplot shows both). The female and male series are plotted separately, model only, with no confidence bands. Interdependent model.

K. Completed-network counterfactual

This appendix reports the full income and welfare effects by group for the completed-network exercise. The exercise compares the completed network with the 2017 as-built metro, so it isolates the value of the remaining planned lines on top of Line 1 and the BRT. I report both the interdependent and independent commuting models. The main takeaway parallels the Line 1 and BRT counterfactual: the extension raises real GDP only modestly, by about 0.25 percent under interdependent commuting, yet it narrows the earnings gap for both singles and dual earner couples, which suggests that the remaining lines connect women more directly to better paying jobs rather than working mainly through the joint household choice. Women also gain more in ex ante welfare, so their access improves more broadly. Within dual earner couples, however, husbands still capture a larger share of the realized consumption gain, so the extension improves women's access broadly but does not eliminate the role of within household allocation.

K.1. Income

Table 15 reports the income effects of the extension. The gains run through participation and reallocation rather than through higher pay at existing jobs. Labor force participation rises by 3.40 percent under interdependence and 6.95 percent under independence, more than total labor income, which is roughly flat per worker. Dual earner wives gain the most, raising total earnings by 7.43 percent and earnings per worker by 3.90 percent under interdependence, with comparable gains under independence, 9.03 and 1.94 percent. Dual earner husbands gain less, 4.42 percent total and 0.99 percent per worker. Total earnings in male-breadwinner households fall by 5.51 percent while per worker earnings are roughly flat. This reflects couples switching out of the breadwinner arrangement and into dual earner work as commuting cheapens. Singles see small changes. Single women lose 0.31 percent under interdependence but gain 1.81 percent under independence, and single men lose between 0.6 and 1.3 percent.

The earnings gaps narrow in both models. The dual earner gap falls by 2.80 percent under interdependence and 2.10 under independence, the singles gap by 0.97 and 2.40, and the aggregate gap by 3.57 and 5.55. The similar narrowing across models, and the narrowing for singles in particular, is what distinguishes the extension from Line 1 and the BRT, whose gains for women worked mainly through the joint household choice. Nominal GDP falls by 0.23 percent under interdependence, but real GDP rises by 0.25 percent once the declines in rents and the price index are netted out.

K.2. Welfare

Table 16 splits welfare into the ex ante measure, the option value of the locations a worker can reach before any shock is realized, and the deterministic measure, realized consumption net of commuting after choices are made. Under the ex ante measure every group gains and women gain more. Single women gain 15.86 percent against 6.04 for single men, and married women 14.38 against 7.92 for married men, so both the singles and the married ex ante gaps

Table 15: Completed network, marginal effect (vs the 2017 as-built metro): aggregate outcomes and income by group.

	Interdependent		Independent	
	Δ% total	Δ% p.c.	Δ% total	Δ% p.c.
Panel A: Aggregates				
GDP	−0.23		+1.04	
Total labor income	+0.01	−0.71	+1.13	−0.35
Gender gap, aggregate	−3.57	−1.85	−5.55	−2.12
Rents	−0.66		+0.87	
Price index	−0.43		−0.55	
Labor force participation	+3.40		+6.95	
Panel B: Income by group				
Single females	−0.31	−0.31	+1.81	+1.81
Single males	−1.27	−1.27	−0.63	−0.63
Gender gap, singles	−0.97	−0.97	−2.40	−2.40
Male-breadwinner	−5.51	−0.96	−9.37	+0.03
Dual-earner wife	+7.43	+3.90	+9.03	+1.94
Dual-earner husband	+4.42	+0.99	+6.74	−0.20
Gender gap, dual-earner	−2.80	−2.80	−2.10	−2.10

Notes. Marginal effect of completing the Metro network (Lines 2 and 4) relative to the 2017 as-built system (Line 1 and the BRT), the proportional change between the calibrated economy with and without the completed lines (positive means completing the network raised the outcome). Panel A reports aggregate outcomes and Panel B income by group. The total column is the change in the total and the per capita (p.c.) column is the change per worker; the two coincide for singles and differ on the married rows by the labor force margin. GDP, rents, the price index, and labor force participation carry a total column only. The gender gap rows are the change in male-to-female earnings ($y_{\text{male}}/y_{\text{female}} - 1$), with a negative value meaning the gap narrowed.

narrow, by 8.48 and 5.65 percent. The independent model gives the same qualitative pattern, with the married ex ante gap narrowing by 2.57 percent. Because ex ante welfare measures access, this says the workplaces the extension opens for women are more desirable than the ones Line 1 and the BRT opened, where the married ex ante gap did not narrow.

The deterministic measure tells a different story for couples. Single women now gain on this measure too, by 2.78 percent, unlike under the first project. But among dual earner couples husbands gain far more than wives, 27.87 percent against 10.16, so the dual earner deterministic gap widens by 16.08 percent. This is the household consumption channel. The wife's workplace move raises household resources, and the realized allocation, with the wife's average Pareto weight falling by 1.55 percent, shifts the consumption gain toward the husband. Under independence the channel is muted. Husbands and wives gain similar amounts, 3.66 against 5.15 percent, and the deterministic gap does not widen. The gap still opens under the extension, but by less than the 21.79 percent under Line 1 and the BRT, so the consumption channel operates more weakly here.

Table 16: Completed network, marginal effect (vs the 2017 as-built metro): welfare by group.

	Interdependent	Independent
Panel A: Ex-ante welfare		
Single female	$+15.86$	$+14.85$
Single male	$+6.04$	$+5.81$
Gender gap, singles	-8.48	-7.87
Married woman	$+14.38$	$+8.48$
Married man	$+7.92$	$+5.70$
Gender gap, married	-5.65	-2.57
Panel B: Deterministic per-spouse welfare		
Single female	$+2.78$	$+3.82$
Single male	$+0.71$	$+0.99$
Gender gap, singles	-2.01	-2.73
Male-breadwinner wife	-1.25	$+0.85$
Male-breadwinner husband	$+6.01$	$+3.47$
Gender gap, male-breadwinner	$+7.35$	$+2.60$
Dual-earner wife	$+10.16$	$+5.15$
Dual-earner husband	$+27.87$	$+3.66$
Gender gap, dual-earner	$+16.08$	-1.41
Average Pareto weight	-1.55	*N.A.*

Notes. Percentage change in welfare. Panel A is the ex-ante (nested-Fréchet) utility of each group before any commuting, participation, or residential shock, and the married rows are the husband's and wife's ex-ante utilities. Panel B is the change in deterministic per-spouse welfare. Gender gap rows are the change in the male-to-female ratio, where negative means the gap narrowed. In the independent model the Pareto weight is undefined, so dual-earner spouses are valued at their own earnings and breadwinner consumption uses the average weight. Table 7 decomposes the Panel B changes.

L. Additional Proofs

Married household choice probabilities. This establishes the dual earner commuting probabilities (48)–(51), the breadwinner commuting probability (55), the participation probability (58), and the residential probability (61).

Proof. All four follow from the Fréchet choice formula: if options k have deterministic values $\{a_k\}$ and the agent maximizes $a_k \varepsilon_k$ with ε_k iid Fréchet of shape s ($\Pr(\varepsilon \leq x) = e^{-x^{-s}}$), then k is chosen with probability $a_k^s / \sum_\ell a_\ell^s$, and the maximized value $\max_k a_k \varepsilon_k$ is Fréchet of shape s and scale $\left(\sum_\ell a_\ell^s\right)^{1/s}$, the market access term.

Dual earner commuting. Conditional on residence o, the dual earner arrangement, and the wife's destination j, the only factor in the weighted household objective (39) that varies with the husband's destination i is $\mathcal{B}^m_{oij} \epsilon_{m,i}$, with $\mathcal{B}^m_{oij} = \Xi_{ij} / [(d^m_{oi})^{1-\zeta_{ij}} (d^f_{oj})^{\zeta_{ij}}]$. With $\epsilon_{m,i}$ Fréchet(θ^m), the choice formula gives (48), whose market access term is Φ_{oj}^{1/θ^m}. Anticipating this, the wife chooses j to maximize $\Phi_{oj}^{1/\theta^m} \epsilon_{f,j}$ with $\epsilon_{f,j}$ Fréchet(θ^f). The choice formula gives (49), with market access term Ω_{o1} in (52). Multiplying and summing yields (50) and (51).

Breadwinner commuting. If the wife does not work, the only i-dependent factor in (54) is $(w_{m,i}/d^m_{oi})\epsilon_{m,i}$. With $\epsilon_{m,i}$ Fréchet(θ^m), the choice formula gives (55), with market access term $\Omega_{o0} = \Phi_{o0}^{1/\theta^m}$.

Participation. The dual earner and breadwinner state values (53) and (57) share the common factor $u_o^{mf} / (P^\beta r_{Ro}^{1-\beta})$, so the household's participation choice maximizes between $\Omega_{o1} \varsigma_{o1}$ and $\xi_{o0} \Omega_{o0} \varsigma_{o0}$. With $\varsigma_{o1}, \varsigma_{o0}$ iid Fréchet(ν), the choice formula gives the dual earner share (58), with market access term Ω_o^{mf} in (59).

Residential choice. Across residences the couple maximizes W_o^{mf} in (60), whose o-dependent part is $\left(u_o^{mf} \Omega_o^{mf} / (P^\beta r_{Ro}^{1-\beta})\right) \varphi_o^{mf}$. With φ_o^{mf} Fréchet(η^{mf}), the choice formula gives (61). $\square$

Husband's ex ante utility. The following derivation establishes equation (71).

Proof. Step 1: Husband's utility conditional on the wife's destination. For a dual earner household living in o, with the wife working in j and the husband working in i, the husband's utility is

$$\begin{aligned} V^m_{oij,1}(\omega) = \overline{V}^m_{oij,1} + \frac{1}{1-\zeta_{ij}} \log \epsilon_{m,i}(\omega) \\ + \log \varsigma_{o1}(\omega) + \log \varphi_o^{mf}(\omega). \end{aligned} \tag{178}$$

Fix residence o, the dual earner arrangement, the wife's destination j, and the realizations of the common participation and residential shocks. Let $i^*(\omega)$ denote the husband's destination selected by the household. Define

$$\mathcal{U}^m_{o1|j}\left(\varsigma_{o1}, \varphi_o^{mf}\right) \equiv \mathbb{E}_{\epsilon_m}\left[V^m_{o\,i^*(\omega)\,j,1}(\omega) \mid o, 1, j, \varsigma_{o1}, \varphi_o^{mf}\right]. \tag{179}$$

Step 2: Expanding over the husband's selected destination. By the law of total expectation,

$$\begin{aligned}\mathcal{U}^{m}_{o1|j}\left(\varsigma_{o1},\varphi^{mf}_{o}\right) = {} & \log\varsigma_{o1}+\log\varphi^{mf}_{o} \\ & +\sum_{i=1}^{S}\pi^{m}_{i|o,j}\Bigg[\overline{V}^{m}_{oij,1} \\ & \qquad +\frac{1}{1-\zeta_{ij}}\mathbb{E}\left[\log\epsilon_{m,i}\mid i^{*}=i,o,j\right]\Bigg]. \end{aligned} \tag{180}$$

Step 3: Expected selected commuting shock. Conditional on o and j, the household selects the husband's destination by maximizing

$$\mathcal{B}^{m}_{oij}\epsilon_{m,i}.$$

Let

$$M^{m}_{oj}\equiv\max_{\ell}\left\{\mathcal{B}^{m}_{o\ell j}\epsilon_{m,\ell}\right\}.$$

When destination i is selected,

$$\mathcal{B}^{m}_{oij}\epsilon_{m,i}=M^{m}_{oj},$$

and hence

$$\log\epsilon_{m,i}=\log M^{m}_{oj}-\log\mathcal{B}^{m}_{oij}. \tag{181}$$

This identity generates the subtraction of the deterministic attractiveness of the selected destination.

The Fréchet calculations underlying equation (48) imply

$$\Pr\left(M^{m}_{oj}\leq x\right)=\exp\left(-\Phi_{oj}x^{-\theta^{m}}\right).$$

Moreover, the distribution of M^{m}_{oj} is unchanged conditional on any destination i being selected. Therefore,

$$\mathbb{E}\left[\log M^{m}_{oj}\mid i^{*}=i,o,j\right]=\frac{\gamma_{\mathrm{E}}}{\theta^{m}}+\frac{1}{\theta^{m}}\log\Phi_{oj}.$$

Using equation (181),

$$\mathbb{E}\left[\log\epsilon_{m,i}\mid i^{*}=i,o,j\right]=\frac{\gamma_{\mathrm{E}}}{\theta^{m}}+\frac{1}{\theta^{m}}\log\Phi_{oj}-\log\mathcal{B}^{m}_{oij}. \tag{182}$$

Step 4: Husband's expected utility conditional on the wife's destination. Substituting equation (182) into equation (180) gives

$$\begin{aligned}\mathcal{U}^{m}_{o1|j}\left(\varsigma_{o1},\varphi^{mf}_{o}\right) = {} & \log\varsigma_{o1}+\log\varphi^{mf}_{o} \\ & +\sum_{i=1}^{S}\pi^{m}_{i|o,j}\Bigg[\overline{V}^{m}_{oij,1} \\ & \qquad +\frac{1}{1-\zeta_{ij}}\left(\frac{\gamma_{\mathrm{E}}}{\theta^{m}}+\frac{1}{\theta^{m}}\log\Phi_{oj}-\log\mathcal{B}^{m}_{oij}\right)\Bigg]. \end{aligned} \tag{183}$$

Step 5: Integrating over the wife's commuting shocks. Let $j^{*}(\omega)$ denote the wife's destination selected by the household. The husband's expected utility conditional on residence, dual earner status, and the common participation and residential shocks is

$$\begin{aligned}\mathcal{U}_{o1}^{m}\left(\varsigma_{o1},\varphi_{o}^{mf}\right) &\equiv \mathbb{E}_{\epsilon_f}\left[\mathcal{U}_{o1|j^*(\omega)}^{m}\left(\varsigma_{o1},\varphi_{o}^{mf}\right) \mid o,1\right] \\ &= \sum_{j=1}^{S}\pi_{j|o,1}^{f}\mathcal{U}_{o1|j}^{m}\left(\varsigma_{o1},\varphi_{o}^{mf}\right). \end{aligned} \tag{184}$$

Because the participation and residential shocks are common across wife destinations,

$$\mathcal{U}_{o1}^{m}\left(\varsigma_{o1},\varphi_{o}^{mf}\right) = \mathcal{U}_{o1}^{m} + \log\varsigma_{o1} + \log\varphi_{o}^{mf}, \tag{185}$$

where $\mathcal{U}_{o1}^{m}$ is defined in equation (67).

Step 6: Integrating over the participation shocks. After integrating over commuting shocks, the husband's utility is $\mathcal{U}_{o1}^{m} + \log\varsigma_{o1} + \log\varphi_{o}^{mf}$ in the dual earner arrangement and $\mathcal{U}_{o0}^{m} + \log\varsigma_{o0} + \log\varphi_{o}^{mf}$ in the male-breadwinner arrangement. The household selects the dual earner arrangement when $\Omega_{o1}\varsigma_{o1} > \xi_{o0}\Omega_{o0}\varsigma_{o0}$ and the male-breadwinner arrangement otherwise. Let

$$M_{o}^{p} \equiv \max\left\{\Omega_{o1}\varsigma_{o1},\ \xi_{o0}\Omega_{o0}\varsigma_{o0}\right\}.$$

When the dual earner arrangement is selected, $\log\varsigma_{o1} = \log M_{o}^{p} - \log\Omega_{o1}$. When the male-breadwinner arrangement is selected, $\log\varsigma_{o0} = \log M_{o}^{p} - \log(\xi_{o0}\Omega_{o0})$. The Fréchet distribution of the participation shocks implies

$$\mathbb{E}\left[\log M_{o}^{p} \mid o\right] = \frac{\gamma_{\mathrm{E}}}{\nu} + \log\Omega_{o}^{mf},$$

and this expectation is unchanged conditional on either arrangement being selected. Therefore,

$$\mathbb{E}\left[\log\varsigma_{o1} \mid \text{dual earner}, o\right] = \frac{\gamma_{\mathrm{E}}}{\nu} + \log\Omega_{o}^{mf} - \log\Omega_{o1}, \tag{186}$$

and, analogously, $\mathbb{E}[\log\varsigma_{o0} \mid \text{male-breadwinner}, o] = \gamma_{\mathrm{E}}/\nu + \log\Omega_{o}^{mf} - \log(\xi_{o0}\Omega_{o0})$.

Averaging over the two arrangements, which occur with probabilities μ_o and $1-\mu_o$, gives

$$\mathcal{U}_{o}^{m}\left(\varphi_{o}^{mf}\right) = \mathcal{U}_{o}^{m} + \log\varphi_{o}^{mf}, \tag{187}$$

where $\mathcal{U}_{o}^{m}$ is defined in equation (68).

Step 7: Integrating over the residential shocks. The household chooses residence o by maximizing

$$R_o\varphi_{o}^{mf}.$$

Let

$$M^{R} \equiv \max_{n}\left\{R_n\varphi_{n}^{mf}\right\}.$$

When residence o is selected,

$$\log\varphi_{o}^{mf} = \log M^{R} - \log R_o.$$

The Fréchet distribution of the residential shocks implies

$$\mathbb{E}\left[\log M^{R} \mid o \text{ selected}\right] = \frac{\gamma_{\mathrm{E}}}{\eta^{mf}} + \log\mathcal{R}^{mf}.$$

Hence,

$$\mathbb{E}\left[\log \varphi_o^{mf} \mid o \text{ selected}\right] = \frac{\gamma_{\mathrm{E}}}{\eta^{mf}} + \log \mathcal{R}^{mf} - \log R_o. \tag{188}$$

Finally, averaging the husband's utility in equation (187) over the residential probabilities λ_o^{mf} yields

$$\mathcal{U}^m = \sum_{o=1}^{S} \lambda_o^{mf} \left[\mathcal{U}_o^m + \frac{\gamma_{\mathrm{E}}}{\eta^{mf}} + \log \mathcal{R}^{mf} - \log R_o\right].$$

Using $\sum_o \lambda_o^{mf} = 1$ gives equation (71). □

Wife's ex ante utility. The following derivation establishes equation (79).

Proof. Step 1: Wife's utility conditional on the destination pair. For a dual earner household living in o, with the wife working in j and the husband working in i, the wife's utility is

$$\begin{aligned} V_{oij,1}^f(\omega) = \overline{V}_{oij,1}^f + \frac{1}{\zeta_{ij}} \log \epsilon_{f,j}(\omega) \\ + \log \varsigma_{o1}(\omega) + \log \varphi_o^{mf}(\omega). \end{aligned} \tag{189}$$

Fix residence o, the dual earner arrangement, the wife's selected destination j, and the realizations of the common participation and residential shocks. Let $i^*(\omega)$ denote the husband's subsequent destination selected by the household. Define

$$\mathcal{U}_{o1|j}^f\left(\varsigma_{o1}, \varphi_o^{mf}\right) \equiv \mathbb{E}_{\epsilon_f, \epsilon_m}\left[V_{o\,i^*(\omega)\,j,1}^f(\omega) \mid o, 1, j \text{ selected}, \varsigma_{o1}, \varphi_o^{mf}\right]. \tag{190}$$

Step 2: Expanding over the husband's selected destination. Conditional on the wife selecting j, the husband's commuting shocks are independent of the selected wife shock and determine only the subsequent choice of i. Therefore, by the law of total expectation,

$$\begin{aligned} \mathcal{U}_{o1|j}^f\left(\varsigma_{o1}, \varphi_o^{mf}\right) = {} & \log \varsigma_{o1} + \log \varphi_o^{mf} \\ & + \sum_{i=1}^{S} \pi_{i|o,j}^m \Bigg[\overline{V}_{oij,1}^f \\ & \qquad + \frac{1}{\zeta_{ij}} \mathbb{E}\left[\log \epsilon_{f,j} \mid j \text{ selected}, o, 1\right]\Bigg]. \end{aligned} \tag{191}$$

Step 3: Expected selected commuting shock. The household selects the wife's destination by maximizing

$$\mathcal{B}_{oj}^f \epsilon_{f,j}.$$

Let

$$M_o^f \equiv \max_n \left\{\mathcal{B}_{on}^f \epsilon_{f,n}\right\}.$$

When destination j is selected,

$$\mathcal{B}_{oj}^f \epsilon_{f,j} = M_o^f,$$

and hence

$$\log \epsilon_{f,j} = \log M_o^f - \log \mathcal{B}_{oj}^f. \tag{192}$$

The subtraction therefore arises because the selected commuting shock equals the realized maximum divided by the deterministic attractiveness of the selected destination.

The Fréchet calculations underlying equation (49) imply

$$\Pr\left(M_o^f \leq x\right) = \exp\left[-\Omega_{o1}^{\theta f} x^{-\theta f}\right].$$

Moreover, the distribution of M_o^f is unchanged conditional on any destination j being selected. Therefore,

$$\mathbb{E}\left[\log M_o^f \mid j \text{ selected}, o, 1\right] = \frac{\gamma_{\mathrm{E}}}{\theta^f} + \log \Omega_{o1}.$$

Using equation (192),

$$\mathbb{E}\left[\log \epsilon_{f,j} \mid j \text{ selected}, o, 1\right] = \frac{\gamma_{\mathrm{E}}}{\theta^f} + \log \Omega_{o1} - \log \mathcal{B}_{oj}^f. \tag{193}$$

Step 4: Wife's expected utility conditional on her destination. Substituting equation (193) into equation (191) gives

$$\begin{aligned} \mathcal{U}_{o1|j}^f\left(\varsigma_{o1}, \varphi_o^{mf}\right) = {} & \log \varsigma_{o1} + \log \varphi_o^{mf} \\ & + \sum_{i=1}^{S} \pi_{i|o,j}^m \left[\overline{V}_{oij,1}^f \right. \\ & \left. + \frac{1}{\zeta_{ij}} \left(\frac{\gamma_{\mathrm{E}}}{\theta^f} + \log \Omega_{o1} - \log \mathcal{B}_{oj}^f\right)\right]. \end{aligned} \tag{194}$$

Step 5: Integrating over the wife's commuting shocks. Let $j^*(\omega)$ denote the wife's destination selected by the household. Her expected utility conditional on residence, dual earner status, and the common participation and residential shocks is

$$\begin{aligned} \mathcal{U}_{o1}^f\left(\varsigma_{o1}, \varphi_o^{mf}\right) &\equiv \mathbb{E}_{\epsilon_f}\left[\mathcal{U}_{o1|j^*(\omega)}^f\left(\varsigma_{o1}, \varphi_o^{mf}\right) \mid o, 1\right] \\ &= \sum_{j=1}^{S} \pi_{j|o,1}^f \mathcal{U}_{o1|j}^f\left(\varsigma_{o1}, \varphi_o^{mf}\right). \end{aligned} \tag{195}$$

Because the participation and residential shocks are common across wife destinations,

$$\mathcal{U}_{o1}^f\left(\varsigma_{o1}, \varphi_o^{mf}\right) = \mathcal{U}_{o1}^f + \log \varsigma_{o1} + \log \varphi_o^{mf}, \tag{196}$$

where $\mathcal{U}_{o1}^f$ is defined in equation (77).

Step 6: Integrating over the participation shocks. After integrating over the commuting shocks, the wife's utility is $\mathcal{U}_{o1}^f + \log \varsigma_{o1} + \log \varphi_o^{mf}$ in the dual earner arrangement and $\mathcal{U}_{o0}^f + \log \varsigma_{o0} + \log \varphi_o^{mf}$ in the male-breadwinner arrangement. The household selects the dual earner arrangement when $\Omega_{o1}\varsigma_{o1} > \xi_{o0}\Omega_{o0}\varsigma_{o0}$ and the male-breadwinner arrangement otherwise. As shown in the proof for husbands,

$$\mathbb{E}\left[\log \varsigma_{o1} \mid \text{dual earner}, o\right] = \frac{\gamma_{\mathrm{E}}}{\nu} + \log \Omega_o^{mf} - \log \Omega_{o1}, \tag{197}$$

and, analogously, $\mathbb{E}[\log \varsigma_{o0} \mid \text{male-breadwinner}, o] = \gamma_{\mathrm{E}}/\nu + \log \Omega_o^{mf} - \log(\xi_{o0}\Omega_{o0})$.

Averaging over the two arrangements, which occur with probabilities μ_o and $1-\mu_o$, gives

$$\mathcal{U}_o^f\left(\varphi_o^{mf}\right) = \mathcal{U}_o^f + \log \varphi_o^{mf}, \tag{198}$$

where $\mathcal{U}_o^f$ is defined in equation (78).

Step 7: Integrating over the residential shocks. The household chooses residence o by maximizing

$$R_o \varphi_o^{mf}.$$

As shown in the proof for husbands,

$$\mathbb{E}\left[\log \varphi_o^{mf} \mid o \text{ selected}\right] = \frac{\gamma_{\mathrm{E}}}{\eta^{mf}} + \log \mathcal{R}^{mf} - \log R_o. \tag{199}$$

Finally, averaging the wife's utility in equation (198) over the residential probabilities λ_o^{mf} yields

$$\mathcal{U}^f = \sum_{o=1}^{S} \lambda_o^{mf}\left[\mathcal{U}_o^f + \frac{\gamma_{\mathrm{E}}}{\eta^{mf}} + \log \mathcal{R}^{mf} - \log R_o\right].$$

Using $\sum_o \lambda_o^{mf} = 1$ gives equation (79). □

Ex ante utilities under independent commuting. This part states the spouses' ex ante utilities when dual earner commuting is independent and then establishes (206) and (209).

Under independent commuting, dual earner spouses choose workplaces independently, so each spouse's commuting probabilities coincide with the single-gender Fréchet expressions of (28). The husband chooses i to maximize $(w_{m,i}/d_{oi}^m)\epsilon_{m,i}$ and the wife chooses j to maximize $(w_{f,j}/d_{oj}^f)\epsilon_{f,j}$, separately, so

$$\pi_{i|o,1}^{m,\mathrm{ind}} = \frac{\left(w_{m,i}/d_{oi}^m\right)^{\theta^m}}{\Phi_o^m}, \qquad \Phi_o^m \equiv \sum_{\ell=1}^{S}\left(\frac{w_{m,\ell}}{d_{o\ell}^m}\right)^{\theta^m}, \tag{200}$$

$$\pi_{j|o,1}^{f,\mathrm{ind}} = \frac{\left(w_{f,j}/d_{oj}^f\right)^{\theta^f}}{\Phi_o^f}, \qquad \Phi_o^f \equiv \sum_{\ell=1}^{S}\left(\frac{w_{f,\ell}}{d_{o\ell}^f}\right)^{\theta^f}. \tag{201}$$

The husband's commuting market access term is $(\Phi_o^m)^{1/\theta^m}$ and the wife's is $(\Phi_o^f)^{1/\theta^f}$. Each spouse splits their own wage between own consumption and own housing with the same shares as a single, without pooling resources across the couple. The dual earner market access term therefore combines the two multiplicatively,

$$\Omega_{o1}^{\mathrm{ind}} \equiv \left(\Phi_o^m\right)^{1/\theta^m}\left(\Phi_o^f\right)^{1/\theta^f}, \tag{202}$$

and the dual earner state value mirrors (53) with Ω_{o1} replaced by $\Omega_{o1}^{\mathrm{ind}}$. The participation and marital market access terms reuse (58) and (59) with this substitution,

$$\mu_o^{\mathrm{ind}} = \frac{\left(\Omega_{o1}^{\mathrm{ind}}\right)^{\nu}}{\left(\Omega_{o1}^{\mathrm{ind}}\right)^{\nu} + \left(\xi_{o0}\Omega_{o0}\right)^{\nu}}, \qquad \Omega_o^{mf,\mathrm{ind}} = \left[\left(\Omega_{o1}^{\mathrm{ind}}\right)^{\nu} + \left(\xi_{o0}\Omega_{o0}\right)^{\nu}\right]^{1/\nu}. \tag{203}$$

The male-breadwinner arrangement is unchanged, with market access term $\Omega_{o0} = \Phi_{o0}^{1/\theta^m}$ and scalar split $\bar{\lambda}$. The residence attractiveness R_o in (69), the aggregate term $\mathcal{R}^{mf}$ in (70), and the residential probability λ_o^{mf} in (61) are formed with $\Omega_o^{mf,\mathrm{ind}}$.

Husbands. The deterministic dual earner payoff is $\overline{V}^{m,\text{ind}}_{oi,1} \equiv \log\big(u^{mf}_o C^{\beta}_{m,oi,1}(H^R_{oi,1})^{1-\beta}/d^m_{oi}\big)$, with own-income consumption $C_{m,oi,1} = \beta w_{m,i}/P$ and own housing $H^R_{oi,1} = (1-\beta)w_{m,i}/r_{Ro}$, and the commuting shock enters with coefficient one. After integrating over his own commuting shocks,

$$\mathcal{U}^{m,\text{ind}}_{o1} \equiv \sum_{i=1}^{S} \pi^{m,\text{ind}}_{i|o,1}\left[\overline{V}^{m,\text{ind}}_{oi,1} + \frac{\gamma_{\text{E}}}{\theta^m} + \frac{1}{\theta^m}\log\Phi^m_o - \log\frac{w_{m,i}}{d^m_{oi}}\right], \tag{204}$$

and his conditional dual earner utility is $\mathcal{U}^{m,\text{ind}}_{o1} + \log\varsigma_{o1}(\omega) + \log\varphi^{mf}_o(\omega)$. The breadwinner value $\mathcal{U}^m_{o0}$ of (64) is unchanged. After integrating over the participation shocks, with the dual earner and male-breadwinner arrangements occurring with probabilities μ^{ind}_o and $1-\mu^{\text{ind}}_o$,

$$\mathcal{U}^{m,\text{ind}}_o \equiv \frac{\gamma_{\text{E}}}{\nu} + \log\Omega^{mf,\text{ind}}_o + \mu^{\text{ind}}_o\Big[\mathcal{U}^{m,\text{ind}}_{o1} - \log\Omega^{\text{ind}}_{o1}\Big] + \Big(1-\mu^{\text{ind}}_o\Big)\big[\mathcal{U}^m_{o0} - \log(\xi_{o0}\Omega_{o0})\big], \tag{205}$$

and his conditional utility is $\mathcal{U}^{m,\text{ind}}_o + \log\varphi^{mf}_o(\omega)$. Integrating over the residential shocks,

$$\boxed{\mathcal{U}^{m,\text{ind}} = \frac{\gamma_{\text{E}}}{\eta^{mf}} + \log\mathcal{R}^{mf} + \sum_{o=1}^{S}\lambda^{mf}_o\left[\mathcal{U}^{m,\text{ind}}_o - \log R_o\right].} \tag{206}$$

Wives. Symmetrically, with $\overline{V}^{f,\text{ind}}_{oj,1} \equiv \log\big(u^{mf}_o C^{\beta}_{f,oj,1}(H^R_{oj,1})^{1-\beta}/d^f_{oj}\big)$, $C_{f,oj,1} = \beta w_{f,j}/P$, and $H^R_{oj,1} = (1-\beta)w_{f,j}/r_{Ro}$,

$$\mathcal{U}^{f,\text{ind}}_{o1} \equiv \sum_{j=1}^{S} \pi^{f,\text{ind}}_{j|o,1}\left[\overline{V}^{f,\text{ind}}_{oj,1} + \frac{\gamma_{\text{E}}}{\theta^f} + \frac{1}{\theta^f}\log\Phi^f_o - \log\frac{w_{f,j}}{d^f_{oj}}\right], \tag{207}$$

with the breadwinner value $\mathcal{U}^f_{o0}$ of (74) unchanged, and

$$\mathcal{U}^{f,\text{ind}}_o \equiv \frac{\gamma_{\text{E}}}{\nu} + \log\Omega^{mf,\text{ind}}_o + \mu^{\text{ind}}_o\Big[\mathcal{U}^{f,\text{ind}}_{o1} - \log\Omega^{\text{ind}}_{o1}\Big] + \Big(1-\mu^{\text{ind}}_o\Big)\Big[\mathcal{U}^f_{o0} - \log(\xi_{o0}\Omega_{o0})\Big]. \tag{208}$$

The wife's ex ante utility is

$$\boxed{\mathcal{U}^{f,\text{ind}} = \frac{\gamma_{\text{E}}}{\eta^{mf}} + \log\mathcal{R}^{mf} + \sum_{o=1}^{S}\lambda^{mf}_o\left[\mathcal{U}^{f,\text{ind}}_o - \log R_o\right].} \tag{209}$$

Proof. Consider the husband. Under independence his payoff does not depend on the wife's destination, so the wife-conditioning step of the interdependent proof is vacuous. The household selects the husband's destination by maximizing $\mathcal{B}^{m,\text{ind}}_{oi}\epsilon_{m,i}$ with $\mathcal{B}^{m,\text{ind}}_{oi} = w_{m,i}/d^m_{oi}$. Let $M^m_o \equiv \max_\ell\{\mathcal{B}^{m,\text{ind}}_{o\ell}\epsilon_{m,\ell}\}$. When i is selected, $\log\epsilon_{m,i} = \log M^m_o - \log\mathcal{B}^{m,\text{ind}}_{oi}$. The Fréchet calculation underlying (200) gives $\Pr(M^m_o \le x) = \exp(-\Phi^m_o x^{-\theta^m})$, whose distribution is unchanged conditional on any destination being selected, so $\mathbb{E}[\log M^m_o \mid i^* = i, o, 1] = \gamma_{\text{E}}/\theta^m + \frac{1}{\theta^m}\log\Phi^m_o$. Hence $\mathbb{E}[\log\epsilon_{m,i} \mid i^* = i, o, 1] = \gamma_{\text{E}}/\theta^m + \frac{1}{\theta^m}\log\Phi^m_o - \log(w_{m,i}/d^m_{oi})$, and substituting into the deterministic payoff yields (204). The remaining steps are identical to the proof of (71). Integrating over the participation shocks, which are Fréchet(ν), with the dual earner arrangement selected when $\Omega^{\text{ind}}_{o1}\varsigma_{o1} > \xi_{o0}\Omega_{o0}\varsigma_{o0}$, gives $\mathbb{E}[\log\varsigma_{o1} \mid \text{dual earner}, o] = \gamma_{\text{E}}/\nu + \log\Omega^{mf,\text{ind}}_o - \log\Omega^{\text{ind}}_{o1}$ and the analogous breadwinner expression, which average to (205) across the two arrangements occurring with probabilities μ_o^{ind} and $1 - \mu_o^{\text{ind}}$. Integrating over the residential shocks, which are Fréchet(η^{mf}), gives $\mathbb{E}[\log \varphi_o^{mf} \mid o \text{ selected}] = \gamma_{\mathrm{E}}/\eta^{mf} + \log \mathcal{R}^{mf} - \log R_o$, and averaging over λ_o^{mf} with $\sum_o \lambda_o^{mf} = 1$ yields (206). The wife's derivation is identical after replacing $(m, \theta^m, \Phi_o^m, w_{m,i}, d_{oi}^m)$ by $(f, \theta^f, \Phi_o^f, w_{f,j}, d_{oj}^f)$, her commuting shock entering with coefficient one and her maximand being Fréchet(θ^f) with scale $(\Phi_o^f)^{1/\theta^f}$ directly rather than through the husband's anticipated market access. $\square$

M. Online Supplement: Data Appendix

In this section, I explain the choices I made when processing the datasets available in this setting.

M.1. Population and Household Census, 1993, 2007 and 2017

Household Classification. The primary source of population data is the Population and Household Census of 1993, 2007 and 2017. These were conducted by the National Institute of Statistics and Informatics (INEI). Crucially, these data sets contain the population in each block by gender and civil status. Moreover, the household's roster is observed. Hence, I can observe who lives with whom.

I define a couple as a male and a female adult living in the same two-members-household. In 2017 around 77% of households had two or fewer adult members. In 2007, 75% of households had two or fewer adult members. So, these account for the majority of households. I also include the restriction that at least one of the two should work. For singles, I consider households with one adult member. This lone member should work.

For simplicity, I treat individuals living with two or more additional adults as singles (i.e. as if they were roommates and therefore not sharing their income). I make this simplification because if more than two adults live together, then it is hard to infer who is the couple of whom, and I require this knowledge to compute conditional commute probabilities. However, in some cases, it could potentially be inferred. For example, when the household head lives with his or her partner. Or if only two of the $N > 2$ members report to be married. One can also make some assumptions about the average age gap between the spouses. To avoid further complications, I make the choice of simply treating these individuals as singles. If anything, this choice makes singles to look a bit more alike to couples since I am potentially misidentifying some couples in multi-members households as singles.[55] By considering singles and couples in this simple way I am already improving on what the literature has done. In any case, results are similar if I drop those that report to be married among multi-members households. Finally, my main results consider as adults those between 25 and 65 years old, but results are robust to alternative definitions.

This classification leads to the following distribution of households: (i) single females (31.9% of households in 2017, 29% of households in 2007), (ii) single males (35.9% and 36.1%), (iii) male breadwinner households, that is, households where the wife stays at home while the husband goes to work (13.3% and 17.3%), (iv) female breadwinner households (1.3% and 1.5%), and (v) dual earner households (17.6% and 16.2%). I calculate the number of households for each of these five groups for each block.

For consistency, I use the same classification across databases.

Commuting Flows. In the 2017 Census, employed individuals were asked to provide the municipality in which they work. Using this information, along with data on the locations

[55] Less than 50% of individuals in these households report to be married.

where these individuals reside, I calculate commute probabilities as follows:

$$\pi^g_{j|i} = \frac{N^g_{j|i}}{\sum_k N^g_{k|i}}$$

where $N^g_{j|i}$ is the number of workers of group g that live in municipality i and work in municipality j. For conditional commute probabilities in dual earner households,

$$\pi^g_{j|il} = \frac{N^g_{j|il}}{\sum_k N^g_{k|il}}$$

where $N^g_{j|il}$ is the number of dual earner workers of gender g working in j, living i, and whose partners work in l.

M.2. Economic Census, 2008

The 2008 Economic Census of Peru was conducted by the National Institute of Statistics and Informatics (INEI). This database covers firms of all sizes and all industries, except agriculture and financial services, even those that had not filed their taxes. It focuses exclusively on the firms' operations in 2007. Information and consistency checks were collected by census officers visiting business establishments. To my convenience, the location of the businesses were recorded at a very detailed level, including the specific census block they were located in. I aggregate sectors into seven sectors so that it is easier to match to the Firms' Administrative Data of 2015. In particular, I consider textiles, other manufacture, services, business services, wholesale, retail trade, and transportation services. I aggregate employment counts at the block level for each of the seven industries described above. This way, I get the distribution of economic activity per industry across locations for the first period of the analysis. I scale up employment by the ratio of total employment in Lima according to the Population Census in 2007 to employment in the Economic Census.

M.3. Firms' Administrative Data, 2015

The Ministry of Production used their raw data and classified more than 1 million firms by industry, geographic location, sales, and number of workers. The database is available at this link. The information relevant to my analysis is the one concerning the industrial classification, the geographic location and the number of workers. Firms were classified into seven categories: 0-5 workers, 6-10 workers, 11-20 workers, 21-50 workers, 51-100 workers, 101-200 workers, and more than 200 workers. To compute the number of workers per firm I simply impute the lower bound of each interval.[56] Then, I aggregate these counts at the block level for each of the following broad industries: textiles, other manufacture, services, business services, wholesale, retail trade, and transportation services. This way, I get the distribution of economic activity per industry across locations for the second period of the analysis. I scale up employment by the ratio of total employment in Lima according to the Population Census in 2017 to employment in the Firms' Administrative Data.

[56] Results are similar if I instead use the midpoint between the two bounds.

M.4. Road Network Data and Commute Times

Road network data. I use road network data from Open Street Map. Open Street Map (OSM) is a free, editable map of the world that is created and maintained by a community of volunteer contributors. The project was started in 2004 with the goal of creating a free and open alternative to proprietary mapping services, such as Google Maps. The data in OSM is collected from a variety of sources, including GPS tracks, aerial imagery, and manual surveying. The data is then added to the OSM database and is available for anyone to use. The quality of OpenStreetMap data can vary depending on the region and the level of community engagement. In general, OSM data tends to be more accurate and up-to-date in areas with a high level of community engagement, such as densely populated urban areas and regions with a strong OSM community. In these areas, the map is regularly updated with new data and errors are quickly corrected. Importantly, the OSM data contains information on road type or classification, which I use to impute road speeds.

Commute times. I use the data on road type to input speeds during rush hour according to the following classification: motorway (50 km/h), primary (30 km/h), residential (15 km/h), secondary (20 km/h), and tertiary (10 km/h). This classification is based on a report for the Ministry of Transport and Communications done by the Japan International Cooperation Agency in 2013 (JICA, 2013). Additionally, I manually input speeds in certain roads, the BRT, and the Metro following the same report (see Table 17). Then, I run ArcGis's Network Analyst tool to compute the optimal route from origins to destination to generate an origin-destination matrix at the block level. I utilize this tool to compute commute times in absence of the Metro and the BRT and when the two are functional. Sometimes, I aggregate these times to the municipality level by taking the median value across zones within municipalities. I show that these times are well correlated with times reported in commuting surveys (see Figure 34).

The matrix of commute times used in the analysis is an approximation of actual commute times. Ideally, the routes of public buses and road speeds would be used to calculate how long it takes to travel between locations using public transportation. However, data on bus routes is not readily available in this context. Despite this limitation, the commute elasticities, which link commute times to costs, are still estimated using the approximation of commute times and probabilities. These elasticities already take into account any potential differences between the estimated and actual commute times. For example, if actual commute times are twice as long as the estimated times, the estimated elasticities would be half as large. The only concern would be if the error in the approximation of commute times is correlated with a blocks' location or some characteristic, which could lead to problems.

Table 17: Manually imputed speeds

Road	Speed
Avenida Brasil	15 km/h
Avenida Javier Prado Oeste	15 km/h
Avenida La Marina	15 km/h
Avenida Faustino Sanchez Carrión	15 km/h
Avenida Almirante Miguel Grau	15 km/h
Avenida Abancay	15 km/h
Avenida Arequipa	15 km/h
Avenida de Tomás Marsano	15 km/h
Avenida República de Panamá	15 km/h
Avenida Arica	15 km/h
Avenida Aviación	15 km/h
Avenida República de Venezuela	15 km/h
Avenida Bolognesi	15 km/h
Avenida Prolongación Paseo de la República	15 km/h
Avenida Defensores del Morro	15 km/h
Avenida Universitaria	15 km/h
Metro Lines	45 km/h
Bus Rapid Transit System	35 km/h

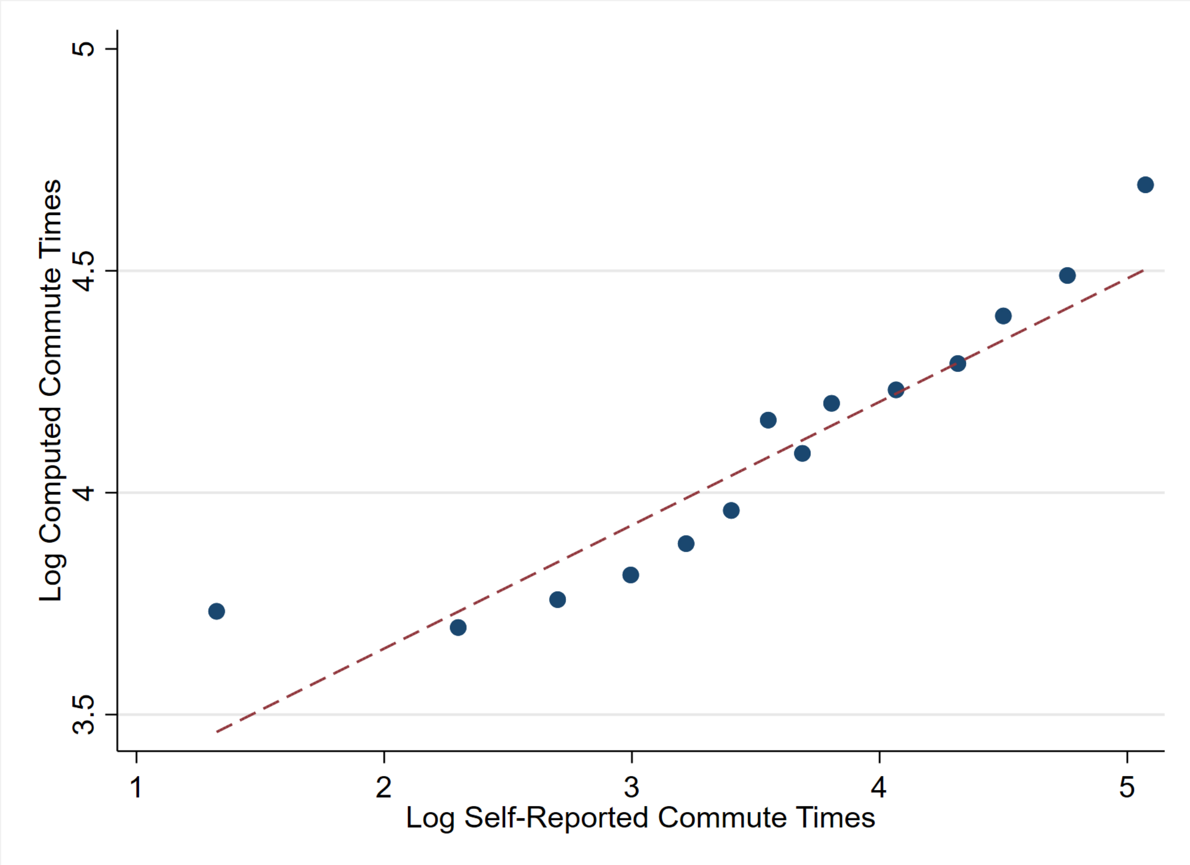


Figure 34: Imputed versus reported commuting times

Notes: Self-reported commute times, at the municipality level, come from the 2010-2018 waves of the Commuting Surveys. Imputed commute times come from the Open Street Road Network data in par with Imputed speeds. Imputed commute times were computed at the zone level, and then aggregated to the municipality level using median values. Scatter was generated using the *binscatter* command in Stata. The correlation between the two measures of commute times is of 53%.

M.5. National Household Survey, 2007-2017

The National Institute of Statistics and Informatics (INEI) in Peru conducts the Peruvian National Household Survey (ENAHO) annually. The sample size of the survey varies each year, with a range of 26,000 to 40,000 households. The survey provides a wealth of information on individual income and demographics, including details on primary and secondary activities such as wages and hours worked. One of the survey's key benefits is that households are geocoded at the cluster level, with each cluster containing about 140 households. This level of granularity allows for accurate assignment of households to census blocks. Additionally, the survey collects valuable information from household heads, such as the estimated rental value of their home, total household income, and expenditures.

I use this data to predict how property values vary with centrality, and other block and dwelling characteristics.[57] Then, I use the estimated coefficients to project property values into census' blocks. I assign data on years 2007-2010 to the first period considered in the

[57]I achieve and R^2 of around 60% across the two periods considered in the analysis.

analysis, and data on years 2014-2017 to the second period. In Figure 35, I compare the log of the average value of predicted rents by municipality to data on the log of property values compiled by the Central Bank of Peru for a subset of municipalities. These averages are computed over years 2014-2017. Figure 35 shows that the two measures are well correlated. Finally, since geocoded data on rents by firms is not available in this setting, I assume that there are no differential rents due to land regulations, and so, the returns to floorspace use in residential and commercial activities are the same, conditional on a block's location.

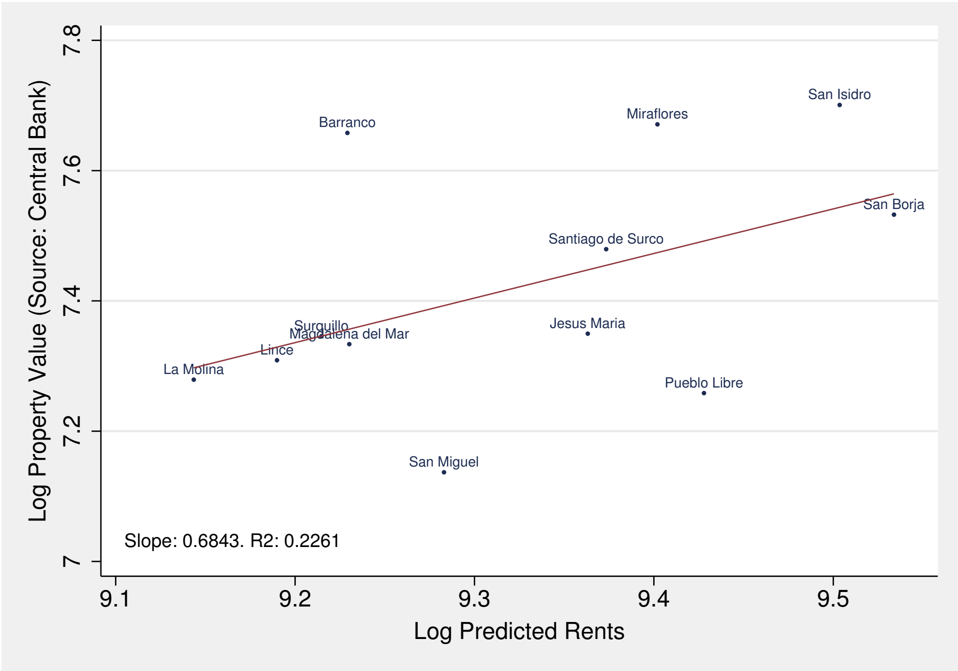


Figure 35: Property Values versus Predicted Rents by Municipality

M.6. Land Use Data, 2020

The Metropolitan Planning Institute collects actual land use data, but it is only available for the year 2020 and not for previous years. Because 2020 is the closest available year, I use the 2020 land use as a proxy for 2017 land use. As a result, when required, I am assuming that land use in 2007 is the same as it was in 2017.[58] Despite this assumption, it does not have any impact on the estimation of any parameters in the model. This is because this data is only used to recover unobservables on the housing side, which is modeled in a straightforward manner following Tsivanidis (2021). These unobsevables are not an input in the estimation of any of the parameters. Additionally, the counterfactual analysis is not affected as I am using 2017 data as a baseline when removing the Metro and BRT from Lima. Therefore, the lack of land use data for 2007 does not affect this exercise. The lack of this data would be of significance if I were to perform a counterfactual analysis using the year 2007 as baseline.

[58] An alternative option would be to use data on floor space zoning, which is available for earlier years.